

# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

Department of Mineral Resources

Lynn D. Helms - Director

North Dakota Industrial Commission

[www.dmr.nd.gov/oilgas/](http://www.dmr.nd.gov/oilgas/)

28600

September 20, 2019

Mr. Jonathon Travis  
Ryan, LLC  
2800 Post Oak Boulevard, Suite 4200  
Houston, TX 77056

**RE: Chalmers #5301 44-24 4T2  
SESE Sec. 24, T.153N., R.101W.  
McKenzie County, North Dakota  
Baker Field  
Well File No. 28600  
STRIPPER WELL DETERMINATION**

Dear Mr. Travis:

Oasis Petroleum North America LLC (Oasis) filed with the North Dakota Industrial Commission – Oil and Gas Division (Commission) on September 4, 2019 an application for a Stripper Well Determination for the above captioned well.

Information contained in the application indicates that the above mentioned well is a stripper well pursuant to statute and rule, and Oasis has elected to designate said well as a stripper well. The well produced from a well depth greater than 10000 feet and was completed after June 30, 2013. During the qualifying period, April 1, 2017 through March 31, 2018, the well produced at a maximum efficient rate or was not capable of exceeding the production threshold. The average daily production from the well was 24.7 barrels of oil per day during this period.

It is therefore determined that the above captioned well qualifies as a “Stripper Well” pursuant to Section 57-51.1-01 of the North Dakota Century Code. This determination is applicable only to the Bakken Pool in and under said well.

The Commission shall have continuing jurisdiction, and shall have the authority to review the matter, and to amend or rescind the determination if such action is supported by additional or newly discovered information. If you have any questions, do not hesitate to contact me.

Sincerely,

  
David J. McCusker  
Petroleum Engineer

Cc: ND Tax Department



# SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)

Well File No.  
**28600**



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <input type="checkbox"/> Notice of Intent                                                                                                       | Approximate Start Date                         |
| <input checked="" type="checkbox"/> Report of Work Done                                                                                         | Date Work Completed<br><b>January 15, 2015</b> |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                         |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Well on pump</b>                               |

|                                                         |                       |                           |              |              |  |
|---------------------------------------------------------|-----------------------|---------------------------|--------------|--------------|--|
| Well Name and Number<br><b>Chalmers 5301 44-24 4T2R</b> |                       |                           |              |              |  |
| Footages                                                | Qtr-Qtr               | Section                   | Township     | Range        |  |
| <b>877 F S L 245 F E L</b>                              | <b>SESE</b>           | <b>24</b>                 | <b>153 N</b> | <b>101 W</b> |  |
| Field<br><b>Baker</b>                                   | Pool<br><b>Bakken</b> | County<br><b>McKenzie</b> |              |              |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

## DETAILS OF WORK

Effective 1/15/2015 the above referenced well is on pump.

End of Tubing: 2-7/8" L-80 tubing @ 10080'

Pump: 2-1/2" x 2.0" x 24' insert pump @ 9918'

|                                                           |                                               |                                         |                          |
|-----------------------------------------------------------|-----------------------------------------------|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b>       |                                               | Telephone Number<br><b>281 404-9652</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>                 |                                               |                                         |                          |
| City<br><b>Houston</b>                                    |                                               | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br>                                             | Printed Name<br><b>Victoria Siemieniewski</b> |                                         |                          |
| Title<br><b>Regulatory Specialist</b>                     | Date<br><b>March 19, 2015</b>                 |                                         |                          |
| Email Address<br><b>vsiemieniewski@oasispetroleum.com</b> |                                               |                                         |                          |

| FOR STATE USE ONLY                           |                                   |
|----------------------------------------------|-----------------------------------|
| <input checked="" type="checkbox"/> Received | <input type="checkbox"/> Approved |
| Date<br><b>4-1-2015</b>                      |                                   |
| By<br>                                       |                                   |
| Title<br><b>JARED THUNE</b>                  | <b>Engineering Technician</b>     |

Well or Facility No  
**28600**

Tight Hole **Yes**

|                                         |                      |                       |
|-----------------------------------------|----------------------|-----------------------|
| Operator                                | Representative       | Rep Phone             |
| <b>OASIS PETROLEUM NORTH AMERICA LL</b> | <b>Kelly Johnson</b> | <b>(701) 580-0524</b> |

|                                                     |             |               |              |                   |                                    |
|-----------------------------------------------------|-------------|---------------|--------------|-------------------|------------------------------------|
| Well Name                                           |             |               |              |                   | Inspector                          |
| <b>CHALMERS 5301 44-24 4T2R</b>                     |             |               |              |                   | <b>Richard Dunn</b>                |
| Well Location                                       | QQ          | Sec           | Twp          | Rng               | County                             |
|                                                     | <b>SESE</b> | <b>24</b>     | <b>153 N</b> | <b>101 W</b>      | <b>MCKENZIE</b>                    |
| Footages                                            | <b>877</b>  | Feet From the | <b>S</b>     | Line              | Field                              |
|                                                     | <b>245</b>  | Feet From the | <b>E</b>     | Line              | <b>BAKER</b>                       |
|                                                     |             |               |              |                   | Pool                               |
|                                                     |             |               |              |                   | <b>BAKKEN</b>                      |
| Date of First Production Through Permanent Wellhead |             |               |              | <b>12/25/2014</b> | <b>This Is Not The First Sales</b> |

|                                      |                          |
|--------------------------------------|--------------------------|
| Purchaser                            | Transporter              |
| <b>OASIS PETROLEUM MARKETING LLC</b> | <b>HILAND CRUDE, LLC</b> |

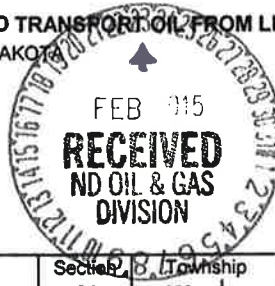
Central Tank Battery Number :

[illegible][illegible]

Start Date **12/25/2014**  
Date Approved **12/25/2014**  
Approved By **Jessica Gilkey**



**AUTHORIZATION TO PURCHASE AND TRANSPORT OIL FROM LEASE -- Form 8**  
INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5698 (03-2000)



|                               |
|-------------------------------|
| Well File No.<br><b>28600</b> |
| NDIC CTB No.<br><b>220407</b> |

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND FOUR COPIES.

|                                                         |                        |                      |                        |                     |                           |
|---------------------------------------------------------|------------------------|----------------------|------------------------|---------------------|---------------------------|
| Well Name and Number<br><b>CHALMERS 5301 44-24 4T2R</b> | Qtr-Qtr<br><b>SESE</b> | Section<br><b>24</b> | Township<br><b>153</b> | Range<br><b>101</b> | County<br><b>Mckenzie</b> |
|---------------------------------------------------------|------------------------|----------------------|------------------------|---------------------|---------------------------|

|                                                      |                                           |                       |
|------------------------------------------------------|-------------------------------------------|-----------------------|
| Operator<br><b>Oasis Petroleum North America LLC</b> | Telephone Number<br><b>(281) 404-9573</b> | Field<br><b>BAKER</b> |
|------------------------------------------------------|-------------------------------------------|-----------------------|

|                                           |                        |                    |                          |
|-------------------------------------------|------------------------|--------------------|--------------------------|
| Address<br><b>1001 Fannin, Suite 1500</b> | City<br><b>Houston</b> | State<br><b>TX</b> | Zip Code<br><b>77002</b> |
|-------------------------------------------|------------------------|--------------------|--------------------------|

|                                                                 |                                           |                            |                                            |
|-----------------------------------------------------------------|-------------------------------------------|----------------------------|--------------------------------------------|
| Name of First Purchaser<br><b>Oasis Petroleum Marketing LLC</b> | Telephone Number<br><b>(281) 404-9627</b> | % Purchased<br><b>100%</b> | Date Effective<br><b>December 25, 2014</b> |
|-----------------------------------------------------------------|-------------------------------------------|----------------------------|--------------------------------------------|

|                                                               |                        |                    |                          |
|---------------------------------------------------------------|------------------------|--------------------|--------------------------|
| Principal Place of Business<br><b>1001 Fannin, Suite 1500</b> | City<br><b>Houston</b> | State<br><b>TX</b> | Zip Code<br><b>77002</b> |
|---------------------------------------------------------------|------------------------|--------------------|--------------------------|

|               |      |       |          |
|---------------|------|-------|----------|
| Field Address | City | State | Zip Code |
|---------------|------|-------|----------|

|                                         |                                           |                             |                                            |
|-----------------------------------------|-------------------------------------------|-----------------------------|--------------------------------------------|
| Transporter<br><b>Hiland Crude, LLC</b> | Telephone Number<br><b>(580) 616-2058</b> | % Transported<br><b>75%</b> | Date Effective<br><b>December 25, 2014</b> |
|-----------------------------------------|-------------------------------------------|-----------------------------|--------------------------------------------|

|                                 |                     |                    |                          |
|---------------------------------|---------------------|--------------------|--------------------------|
| Address<br><b>P.O. Box 3886</b> | City<br><b>Enid</b> | State<br><b>OK</b> | Zip Code<br><b>73702</b> |
|---------------------------------|---------------------|--------------------|--------------------------|

The above named producer authorizes the above named purchaser to purchase the percentage of oil stated above which is produced from the lease designated above until further notice. The oil will be transported by the above named transporter.

|                                                   |             |                |
|---------------------------------------------------|-------------|----------------|
| Other First Purchasers Purchasing From This Lease | % Purchased | Date Effective |
|---------------------------------------------------|-------------|----------------|

|                                                   |             |                |
|---------------------------------------------------|-------------|----------------|
| Other First Purchasers Purchasing From This Lease | % Purchased | Date Effective |
|---------------------------------------------------|-------------|----------------|

|                                                 |               |                |
|-------------------------------------------------|---------------|----------------|
| Other Transporters Transporting From This Lease | % Transported | Date Effective |
|-------------------------------------------------|---------------|----------------|

|                                    |            |                          |
|------------------------------------|------------|--------------------------|
| <b>Prairie Field Services, LLC</b> | <b>25%</b> | <b>December 25, 2014</b> |
|------------------------------------|------------|--------------------------|

|                                                 |               |                |
|-------------------------------------------------|---------------|----------------|
| Other Transporters Transporting From This Lease | % Transported | Date Effective |
|-------------------------------------------------|---------------|----------------|

|  |  |                          |
|--|--|--------------------------|
|  |  | <b>December 25, 2014</b> |
|--|--|--------------------------|

|          |
|----------|
| Comments |
|----------|

|                                                                                                                                |                                        |                                     |
|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-------------------------------------|
| I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records. |                                        | Date<br><b>January 29, 2015</b>     |
| Signature<br>                                                                                                                  | Printed Name<br><b>Brianna Salinas</b> | Title<br><b>Marketing Assistant</b> |

|                               |                                    |                                               |
|-------------------------------|------------------------------------|-----------------------------------------------|
| Above Signature Witnessed By: |                                    |                                               |
| Signature<br>                 | Printed Name<br><b>Dina Barron</b> | Title<br><b>Mktg. Contracts Administrator</b> |

**FOR STATE USE ONLY**

|                                                  |
|--------------------------------------------------|
| Date Approved<br><b>FEB 26 2015</b>              |
| By<br>                                           |
| Title<br><b>Oil &amp; Gas Production Analyst</b> |

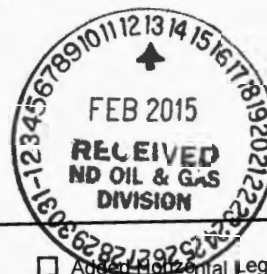




# WELL COMPLETION OR RECOMPLETION REPORT - FORM 6

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 2468 (04-2010)

Well File No.  
**28600**



PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                         |                                   |                                                                            |                                                                                                                                    |
|---------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Designate Type of Completion                            |                                   |                                                                            |                                                                                                                                    |
| <input checked="" type="checkbox"/> Oil Well            | <input type="checkbox"/> EOR Well | <input type="checkbox"/> Recompletion                                      | <input type="checkbox"/> Deepened Well                                                                                             |
| <input type="checkbox"/> Gas Well                       | <input type="checkbox"/> SWD Well | <input type="checkbox"/> Water Supply Well                                 | <input type="checkbox"/> Other:                                                                                                    |
| Well Name and Number<br><b>Chalmers 5301 44-24 4T2R</b> |                                   | Spacing Unit Description<br><b>Sec 24/20 T153N R101W S1/4/20 - 153-100</b> |                                                                                                                                    |
| Operator<br><b>Oasis Petroleum North America</b>        |                                   | Telephone Number<br><b>(281) 404-9591</b>                                  | Field<br><b>Baker</b>                                                                                                              |
| Address<br><b>1001 Fannin, Suite 1500</b>               |                                   | Pool<br><b>Bakken</b>                                                      |                                                                                                                                    |
| City<br><b>Houston</b>                                  | State<br><b>TX</b>                | Zip Code<br><b>77002</b>                                                   | Permit Type<br><input type="checkbox"/> Wildcat <input checked="" type="checkbox"/> Development <input type="checkbox"/> Extension |

## LOCATION OF WELL

|                                                                                                                    |                                             |                                                         |                                  |                                      |                           |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------|----------------------------------|--------------------------------------|---------------------------|
| At Surface<br><b>877 F S L 245 F E L</b>                                                                           | Qtr-Qtr<br><b>SESE</b>                      | Section<br><b>24</b>                                    | Township<br><b>153 N</b>         | Range<br><b>101 W</b>                | County<br><b>McKenzie</b> |
| Spud Date<br><b>April 6, 2014</b>                                                                                  | Date TD Reached<br><b>September 6, 2014</b> | Drilling Contractor and Rig Number<br><b>Nabors B25</b> | KB Elevation (Ft)<br><b>1967</b> | Graded Elevation (Ft)<br><b>1942</b> |                           |
| Type of Electric and Other Logs Run (See Instructions)<br><b>MWD/GR FROM KOP TO TD; CBL FROM INT TD TO SURFACE</b> |                                             |                                                         |                                  |                                      |                           |

## CASING & TUBULARS RECORD (Report all strings set in well)

| Well Bore     | String Type  | Size (Inch) | Top Set (MD Ft) | Depth Set (MD Ft) | Hole Size (Inch) | Weight (Lbs/Ft) | Anchor Set (MD Ft) | Packer Set (MD Ft) | Sacks Cement | Top of Cement |
|---------------|--------------|-------------|-----------------|-------------------|------------------|-----------------|--------------------|--------------------|--------------|---------------|
| Surface Hole  | Surface      | 9 5/8       | 0               | 2095              | 13 1/2           | 36              |                    |                    | 880          | 0             |
| Vertical Hole | Intermediate | 7           | 0               | 11075             | 8 3/4            | 29/32           |                    |                    | 1003         | 2200          |
| Lateral1      | Liner        | 4 1/2       | 10154           | 20938             | 6                | 13.5            |                    |                    |              |               |
|               |              |             |                 |                   |                  |                 |                    |                    |              |               |
|               |              |             |                 |                   |                  |                 |                    |                    |              |               |
|               |              |             |                 |                   |                  |                 |                    |                    |              |               |
|               |              |             |                 |                   |                  |                 |                    |                    |              |               |

## PERFORATION & OPEN HOLE INTERVALS

| Well Bore | Well Bore TD Drillers Depth (MD Ft) | Completion Type | Open Hole/Perforated Interval (MD Ft)<br>Top Bottom | Kick-off Point (MD Ft) | Top of Casing Window (MD Ft) | Date Perf'd or Drilled | Date Isolated | Isolation Method | Sacks Cement |
|-----------|-------------------------------------|-----------------|-----------------------------------------------------|------------------------|------------------------------|------------------------|---------------|------------------|--------------|
| Lateral1  | 20963                               | Perforations    | 11074' 20938                                        | 10121                  |                              | 11/25/2014             |               |                  |              |
|           |                                     |                 | 11075                                               |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |
|           |                                     |                 |                                                     |                        |                              |                        |               |                  |              |

## PRODUCTION

|                                                                                                                                       |                           |                             |                                             |  |                             |                                            |                              |                                                        |                                   |  |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------|---------------------------------------------|--|-----------------------------|--------------------------------------------|------------------------------|--------------------------------------------------------|-----------------------------------|--|
| Current Producing Open Hole or Perforated Interval(s), This Completion, Top and Bottom, (MD Ft)<br><b>Lateral 1- 11074' to 20938'</b> |                           |                             |                                             |  |                             | Name of Zone (If Different from Pool Name) |                              |                                                        |                                   |  |
| Date Well Completed (SEE INSTRUCTIONS)<br><b>December 25, 2014</b>                                                                    |                           |                             | Producing Method<br><b>Flowing</b>          |  | Pumping-Size & Type of Pump |                                            |                              | Well Status (Producing or Shut-In)<br><b>Producing</b> |                                   |  |
| Date of Test<br><b>01/18/2015</b>                                                                                                     | Hours Tested<br><b>24</b> | Choke Size<br><b>22 /64</b> | Production for Test                         |  | Oil (Bbbls)<br><b>831</b>   | Gas (MCF)<br><b>14</b>                     | Water (Bbbls)<br><b>1888</b> | Oil Gravity-API (Cor.)<br><b>°</b>                     | Disposition of Gas<br><b>Sold</b> |  |
| Flowing Tubing Pressure (PSI)                                                                                                         |                           |                             | Flowing Casing Pressure (PSI)<br><b>500</b> |  | Calculated 24-Hour Rate     | Oil (Bbbls)<br><b>831</b>                  | Gas (MCF)<br><b>14</b>       | Water (Bbbls)<br><b>1888</b>                           | Gas-Oil Ratio<br><b>16</b>        |  |

## GEOLOGICAL MARKERS

[illegible]

## PLUG BACK INFORMATION

[illegible]

## CORES CUT

| Top (Ft) | Bottom (Ft) | Formation | Top (Ft) | Bottom (Ft) | Formation |
|----------|-------------|-----------|----------|-------------|-----------|
|          |             |           |          |             |           |
|          |             |           |          |             |           |
|          |             |           |          |             |           |

## Drill Stem Test

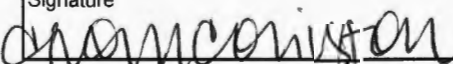
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
|-------------------------|-----------|----------|-------------|--------------|--------|---------|------------------|------------------|
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |
| Test Date               | Formation | Top (Ft) | Bottom (Ft) | BH Temp (°F) | CL ppm | H2S ppm | Shut-in 1 (PSIG) | Shut-in 2 (PSIG) |
| Drill Pipe Recovery     |           |          |             |              |        |         |                  |                  |
| Sample Chamber Recovery |           |          |             |              |        |         |                  |                  |

### Well Specific Stimulations

|                                                                                                                                |                                            |                                |                                                 |                                 |                                                  |                                |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------|-------------------------------------------------|---------------------------------|--------------------------------------------------|--------------------------------|
| Date Stimulated<br><b>11/25/2014</b>                                                                                           | Stimulated Formation<br><b>Three Forks</b> | Top (Ft)<br><b>11074</b>       | Bottom (Ft)<br><b>20938</b>                     | Stimulation Stages<br><b>36</b> | Volume<br><b>98038</b>                           | Volume Units<br><b>Barrels</b> |
| Type Treatment<br><b>Sand Frac</b>                                                                                             | Acid %                                     | Lbs Proppant<br><b>6229694</b> | Maximum Treatment Pressure (PSI)<br><b>8877</b> |                                 | Maximum Treatment Rate (BBLS/Min)<br><b>38.0</b> |                                |
| Details<br><b>40/70 Ceramic: 25340</b><br><b>30/50 Ceramic: 0</b><br><b>40/70 White: 863519</b><br><b>20/40 White: 5340835</b> |                                            |                                |                                                 |                                 |                                                  |                                |
| Date Stimulated                                                                                                                | Stimulated Formation                       | Top (Ft)                       | Bottom (Ft)                                     | Stimulation Stages              | Volume                                           | Volume Units                   |
| Type Treatment                                                                                                                 | Acid %                                     | Lbs Proppant                   | Maximum Treatment Pressure (PSI)                |                                 | Maximum Treatment Rate (BBLS/Min)                |                                |
| Details                                                                                                                        |                                            |                                |                                                 |                                 |                                                  |                                |
| Date Stimulated                                                                                                                | Stimulated Formation                       | Top (Ft)                       | Bottom (Ft)                                     | Stimulation Stages              | Volume                                           | Volume Units                   |
| Type Treatment                                                                                                                 | Acid %                                     | Lbs Proppant                   | Maximum Treatment Pressure (PSI)                |                                 | Maximum Treatment Rate (BBLS/Min)                |                                |
| Details                                                                                                                        |                                            |                                |                                                 |                                 |                                                  |                                |
| Date Stimulated                                                                                                                | Stimulated Formation                       | Top (Ft)                       | Bottom (Ft)                                     | Stimulation Stages              | Volume                                           | Volume Units                   |
| Type Treatment                                                                                                                 | Acid %                                     | Lbs Proppant                   | Maximum Treatment Pressure (PSI)                |                                 | Maximum Treatment Rate (BBLS/Min)                |                                |
| Details                                                                                                                        |                                            |                                |                                                 |                                 |                                                  |                                |
| Date Stimulated                                                                                                                | Stimulated Formation                       | Top (Ft)                       | Bottom (Ft)                                     | Stimulation Stages              | Volume                                           | Volume Units                   |
| Type Treatment                                                                                                                 | Acid %                                     | Lbs Proppant                   | Maximum Treatment Pressure (PSI)                |                                 | Maximum Treatment Rate (BBLS/Min)                |                                |
| Details                                                                                                                        |                                            |                                |                                                 |                                 |                                                  |                                |

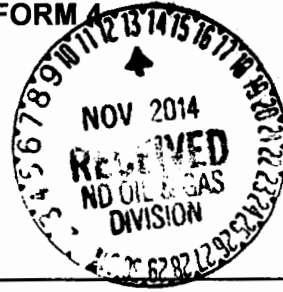
### ADDITIONAL INFORMATION AND/OR LIST OF ATTACHMENTS

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

|                                                                                                                                |                                                                                                 |                                          |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------|
| I hereby swear or affirm that the information provided is true, complete and correct as determined from all available records. | Email Address<br><b>ccovington@oasispetroleum.com</b>                                           | Date<br><b>2/12/15</b>                   |
|                                                                                                                                | Signature<br> | Printed Name<br><b>Chelsea Covington</b> |

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 1**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.

**28600**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>November 12, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                                |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                             |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Change well status to CONFIDENTIAL</b>         |

|                                                         |  |               |           |                 |              |  |  |
|---------------------------------------------------------|--|---------------|-----------|-----------------|--------------|--|--|
| Well Name and Number<br><b>Chalmers 5301 44-24 4T2R</b> |  |               |           |                 |              |  |  |
| Footages                                                |  | Qtr-Qtr       | Section   | Township        | Range        |  |  |
| <b>877 F S L 245 F E L</b>                              |  | <b>SESE</b>   | <b>24</b> | <b>153 N</b>    | <b>101 W</b> |  |  |
| Field                                                   |  | Pool          |           | County          |              |  |  |
| <b>BAKER</b>                                            |  | <b>BAKKEN</b> |           | <b>MCKENZIE</b> |              |  |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Effective immediately, we request CONFIDENTIAL STATUS for the above referenced well.

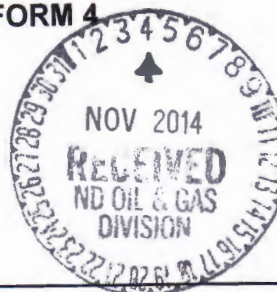
|                                                     |  |                                         |                          |
|-----------------------------------------------------|--|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>281-404-9436</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                         |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br>                                       |  | Printed Name<br><b>Jennifer Swenson</b> |                          |
| Title<br><b>Regulatory Assistant</b>                |  | Date<br><b>November 12, 2014</b>        |                          |
| Email Address<br><b>jswenson@oasispetroleum.com</b> |  |                                         |                          |

| FOR STATE USE ONLY                     |                                              |
|----------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received      | <input checked="" type="checkbox"/> Approved |
| Date<br><b>11/17/14</b>                |                                              |
| By<br>                                 |                                              |
| Title<br><b>Engineering Technician</b> |                                              |



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.

**28600**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>October 30, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                               |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                            |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input type="checkbox"/> Other                | <b>Waiver from tubing/packer requirement</b>      |

|                                                         |                       |                        |                           |                          |                       |
|---------------------------------------------------------|-----------------------|------------------------|---------------------------|--------------------------|-----------------------|
| Well Name and Number<br><b>Chalmers 5301 44-24 4T2R</b> |                       |                        |                           |                          |                       |
| Footages<br><b>877 F S L 245 F E L</b>                  |                       | Qtr-Qtr<br><b>SESE</b> | Section<br><b>24</b>      | Township<br><b>153 N</b> | Range<br><b>101 W</b> |
| Field<br><b>Baker</b>                                   | Pool<br><b>Bakken</b> |                        | County<br><b>McKenzie</b> |                          |                       |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Oasis Petroleum North America LLC requests a variance to NDAC 43-02-03-21 for the tubing/packer requirement: Casing, tubing, and cementing requirements during the completion period immediately following the upcoming fracture stimulation.

The following assurances apply:

1. the well is equipped with new 29# and 32# casing at surface with an API burst rating of 11,220 psi;
2. The Frac design will use a safety factor of 0.85 API burst rating to determine the maximum pressure;
3. Damage to the casing during the frac would be detected immediately by monitoring equipment;
4. The casing is exposed to significantly lower rates and pressures during flowback than during the frac job;
5. The frac fluid and formation fluids have very low corrosion and erosion rates;
6. Production equipment will be installed as soon as possible after the well ceases flowing;
7. A 300# gauge will be installed on the surface casing during the flowback period

|                                                     |  |                                         |                          |
|-----------------------------------------------------|--|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>281-404-9436</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                         |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br>                                       |  | Printed Name<br><b>Jennifer Swenson</b> |                          |
| Title<br><b>Regulatory Assistant</b>                |  | Date<br><b>October 30, 2014</b>         |                          |
| Email Address<br><b>jswenson@oasispetroleum.com</b> |  |                                         |                          |

| FOR STATE USE ONLY                 |                                              |
|------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received  | <input checked="" type="checkbox"/> Approved |
| Date<br><b>November 5, 2014</b>    |                                              |
| By<br>                             |                                              |
| Title<br><b>PETROLEUM ENGINEER</b> |                                              |



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



CTB  
Well File No.  
**220407-01**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>October 15, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                               |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                            |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Central production facility-commingle prod</b> |

|                                       |   |                       |         |                           |                    |
|---------------------------------------|---|-----------------------|---------|---------------------------|--------------------|
| Well Name and Number<br>(see details) |   |                       |         |                           |                    |
| Footages                              |   | Qtr-Qtr               | Section | Township                  | Range              |
| F                                     | L | F                     | L       | <b>20</b>                 | <b>153 N 100 W</b> |
| Field                                 |   | Pool<br><b>Bakken</b> |         | County<br><b>McKenzie</b> |                    |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Oasis Petroleum North America LLC requests approval to commingle oil and gas in a central production facility known as: 5300 19-20 CTB 2 with common ownership for the following wells:

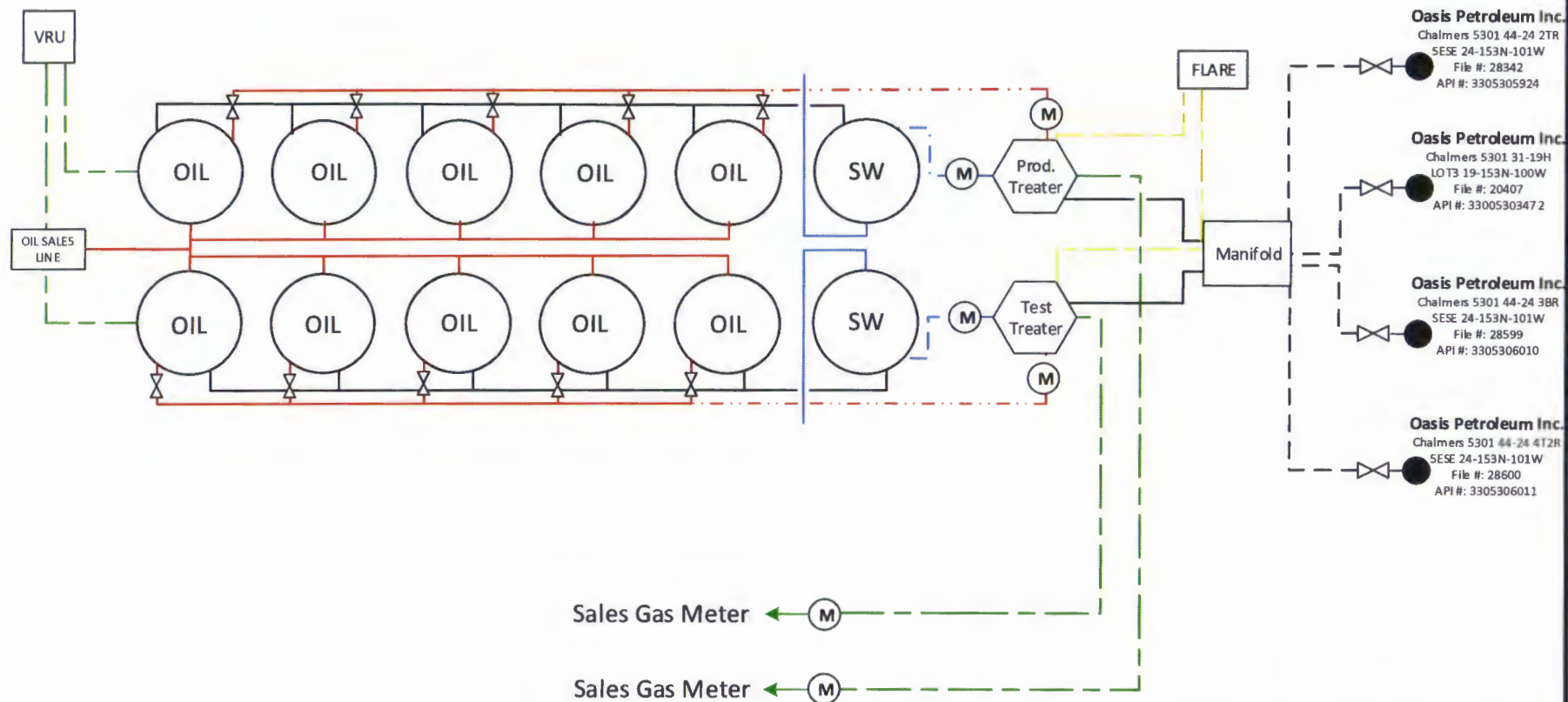
Well File #28342 Chalmes 5301 44-24 2TR SESE Sec 24-153N-R101W API 3305305924  
Well File #20407 Chalmers 5301 31-19H LOT3 Sec 19-153N-101W API 33005303472  
Well File #28599 Chalmers 5301 44-24 3BR SESE Sec 24 153N-101W API 3305306010  
Well File #28600 Chalmers 5301 44-24 4T2R SESE Sec 21 153N-101W API 3305306011

Please find the following attachments:

1. A schematic drawing of the facility which diagrms the testing, treating, routing, and transferring of production. 2. A plat showing the location of the central facility 3. Affidavit of title indicating common ownership. Oasis will allocate production measured at the central production facility to the various wells on the basis of isolated production tests utilizing oil, gas and water meters on a test seperator at the central production facility. Oasis will measure the production from each well separately each month for a minimum of three days. Oasis believes that such allocation will result in an accurate determination of production from each well. Tank vapor gas is being recovered and burned by a 98% DRF enclosed combuster

|                                                       |                                          |                                         |  |
|-------------------------------------------------------|------------------------------------------|-----------------------------------------|--|
| Company<br><b>Oasis Petroleum North America LLC</b>   |                                          | Telephone Number<br><b>281-404-9591</b> |  |
| Address<br><b>1001 Fannin, Suite 1500</b>             |                                          |                                         |  |
| City<br><b>Houston</b>                                | State<br><b>TX</b>                       | Zip Code<br><b>77002</b>                |  |
| Signature<br><i>Chammoniston</i>                      | Printed Name<br><b>Chelsea Covington</b> |                                         |  |
| Title<br><b>Regulatory Assistant</b>                  | Date<br><b>October 13, 2014</b>          |                                         |  |
| Email Address<br><b>ccovington@oasispetroleum.com</b> |                                          |                                         |  |

| FOR STATE USE ONLY                 |                                              |
|------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received  | <input checked="" type="checkbox"/> Approved |
| Date<br><b>10-17-2014</b>          |                                              |
| By<br><i>David Tabor</i>           |                                              |
| Title<br><b>PETROLEUM ENGINEER</b> |                                              |



|                                                                                       |      |       |       |       |
|---------------------------------------------------------------------------------------|------|-------|-------|-------|
|  |      |       |       |       |
| <b>CHALMERS 5300 19-20 CENTRAL TANK<br/>BATTERY 2</b>                                 |      |       |       |       |
| DATE                                                                                  | REV. | BY    | APPR. | SCALE |
| SEPTEMBER 23, 2014                                                                    | 0    | LEE   |       | NA    |
| LOCATION                                                                              |      | FIELD |       |       |
| NORTH DAKOTA                                                                          |      | BAKER |       |       |



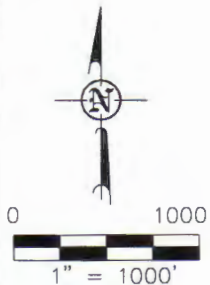
# BATTERY LOCATION PLAT

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"5300 19-20 CTB(A)"

SECTION 24, T153N, R101W, 5TH P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY  
ISSUED AND SEALED BY DARYL D.  
KASEMAN, PLS, REGISTRATION  
NUMBER 3880 LS ON 10/7/14  
AND THE ORIGINAL DOCUMENTS ARE  
STORED AT THE OFFICES OF  
INTERSTATE ENGINEERING, INC.



24

FOUND REBAR  
W/ 2" AC  
LS 2352

R101W  
R100W

AZ 90°03'35"

FOUND REBAR  
W/ 2" AC  
LS 2352

2630.15'

LOT 1

2631.88'

LOT 2

AZ 0°04'59"

2631.68'

LOT 3

AZ 0°05'03"

LOT 4

CHALMERS 5301 44-24 2TR  
CHALMERS 5301 44-24 3BR  
CHALMERS 5301 44-24 4T2R

FOUND STONE  
W/ AC

AZ 90°44'54"

5267.09'

FOUND STONE  
& REBAR

AZ 90°03'03"

2651.37'

FOUND STONE  
& REBAR

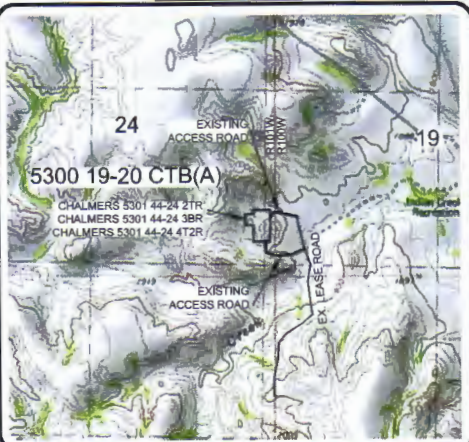
LOT 6

LOT 7

EDGE OF  
LAKE

19

## VICINITY MAP



MONUMENT - RECOVERED

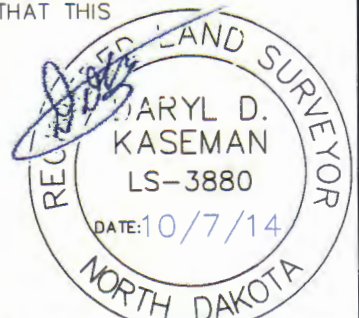
MONUMENT - NOT RECOVERED

STAKED ON 9/3/2013

VERTICAL CONTROL DATUM WAS BASED UPON  
GPS CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST  
OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS  
PLAT CORRECTLY REPRESENTS  
WORK PERFORMED BY ME OR UNDER MY  
SUPERVISION AND IS TRUE AND CORRECT TO  
THE BEST OF MY KNOWLEDGE AND BELIEF.

*[Signature]*



© 2014, INTERSTATE ENGINEERING, INC.

DARYL D. KASEMAN LS-3880

1/5

SHEET NO.



Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
BATTERY LOCATION PLAT  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

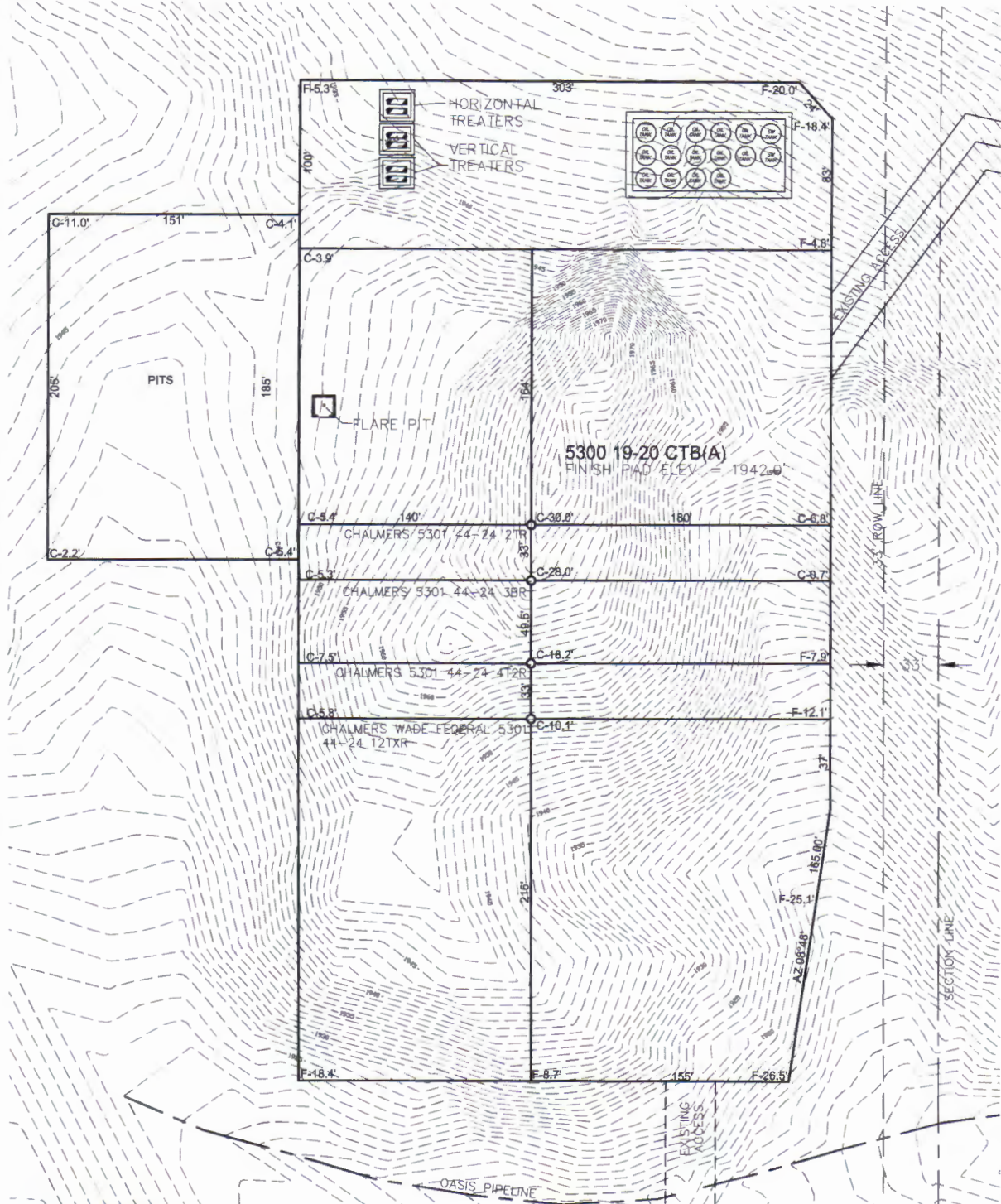
Drawn By: J.D.M. Project No.: S14-08-241  
Checked By: D.D.K. Date: OCT. 2014

| Revision No. | Date | By | Description |
|--------------|------|----|-------------|
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |



THIS DOCUMENT WAS ORIGINALLY ISSUED  
AND SEALED BY DARYL D. KASEMAN,  
P.L.S., REGISTRATION NUMBER 3880 ON  
10/7/14 AND THE ORIGINAL  
DOCUMENTS ARE STORED AT THE  
OFFICES OF INTERSTATE ENGINEERING,  
INC.

**PAD LAYOUT**  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"5300 19-20 CTB(A)"  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



NOTE: All utilities shown are preliminary only, a complete  
utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

2/5

SHEET NO.



Interstate Engineering  
Professionals you need, people you trust

Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com

Other offices in Minneapolis, North Dakota and South Dakota

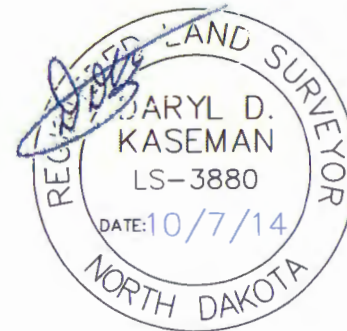
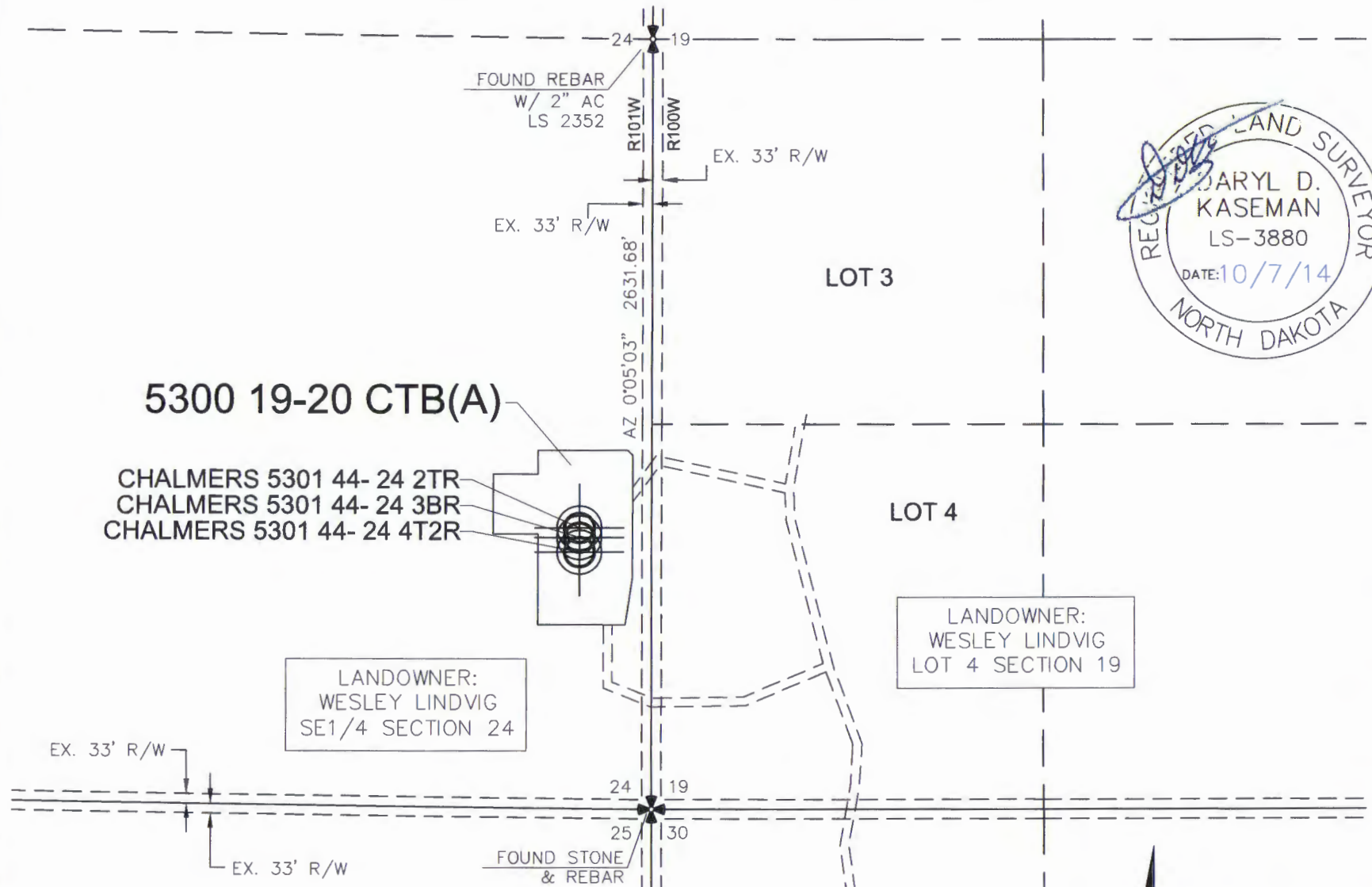
OASIS PETROLEUM NORTH AMERICA, LLC  
PAD LAYOUT  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: 514-09-241  
Checked By: D.D.K. Date: OCT 2014

| Revision | Date | By | Description |
|----------|------|----|-------------|
|          |      |    |             |
|          |      |    |             |
|          |      |    |             |

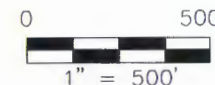
Printed: 10/7/2014 10:01 AM. Path: C:\Users\JD.M\Documents\514-09-241\514-09-241.dwg. Plot: 10/7/2014 4:18 PM. Plotter: eplot

**ACCESS APPROACH**  
**OASIS PETROLEUM NORTH AMERICA, LLC**  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 "5300 19-20 CTB(A)"  
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 10/7/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



© 2014, INTERSTATE ENGINEERING, INC.



**3/5**  
SHEET NO.

Interstate Engineering, Inc.  
 P.O. Box 648  
 425 East Main Street  
 Sidney, Montana 59270  
 Ph (406) 433-5617  
 Fax (406) 433-5618  
 www.interstateeng.com  
 Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
 ACCESS APPROACH  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA  
 Drawn By: J.D.M. Project No: S14-09-241  
 Checked By: D.D.K. Date: OCT. 2014

| Date | By | Description |
|------|----|-------------|
|      |    |             |
|      |    |             |
|      |    |             |
|      |    |             |
|      |    |             |





© 2014, INTERSTATE ENGINEERING, INC.

4/5

SHEET NO.



Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
QUAD LOCATION MAP  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: S14-09-251  
Checked By: D.D.K. Date: OCT 2014

| Revision No. | Date | By | Description |
|--------------|------|----|-------------|
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |

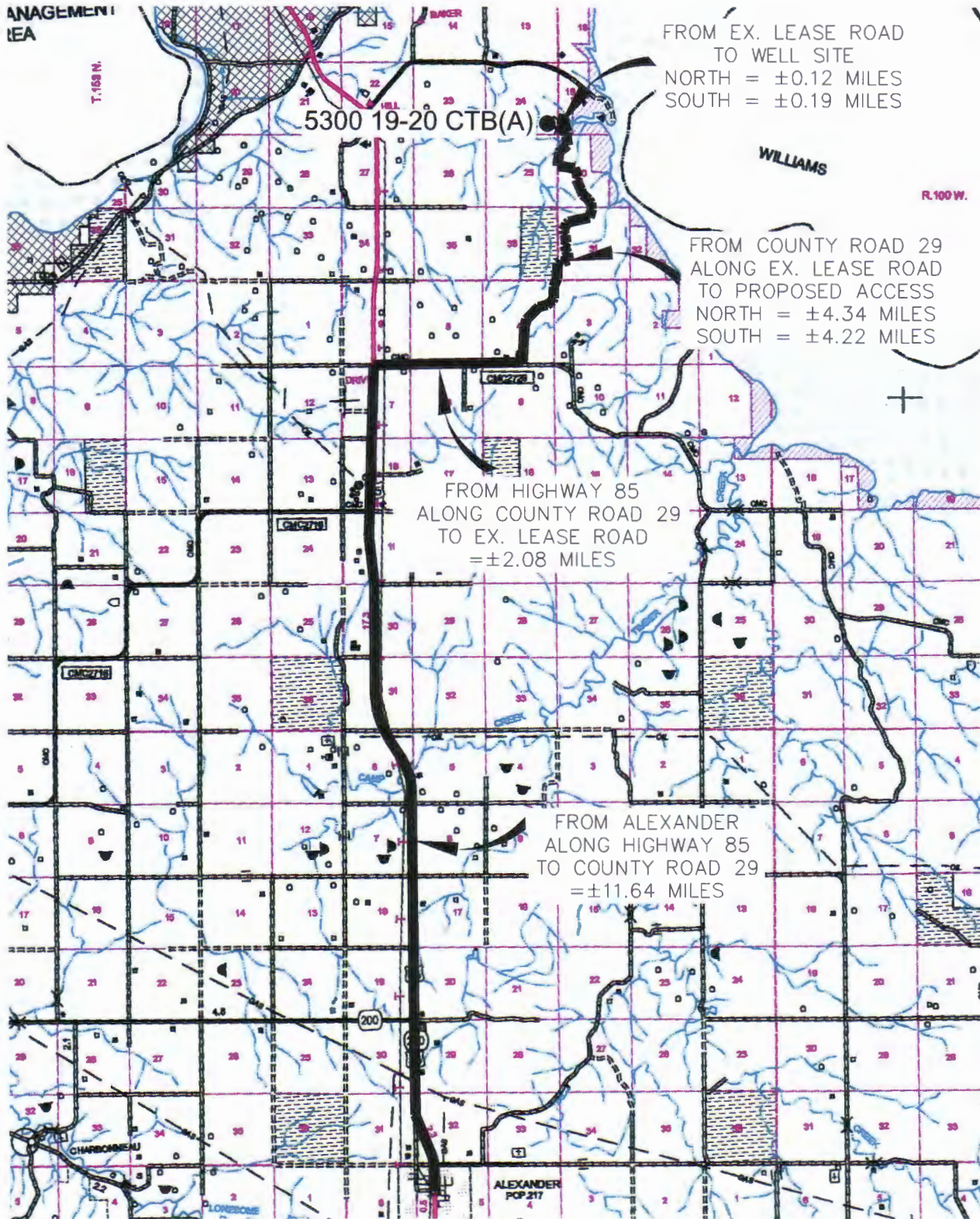


# COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"5300 19-20 CTB(A)"

SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

5/5

SHEET NO.

**INTERSTATE ENGINEERING**  
Professionals you need, people you trust

Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com

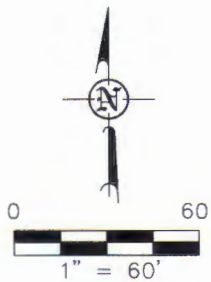
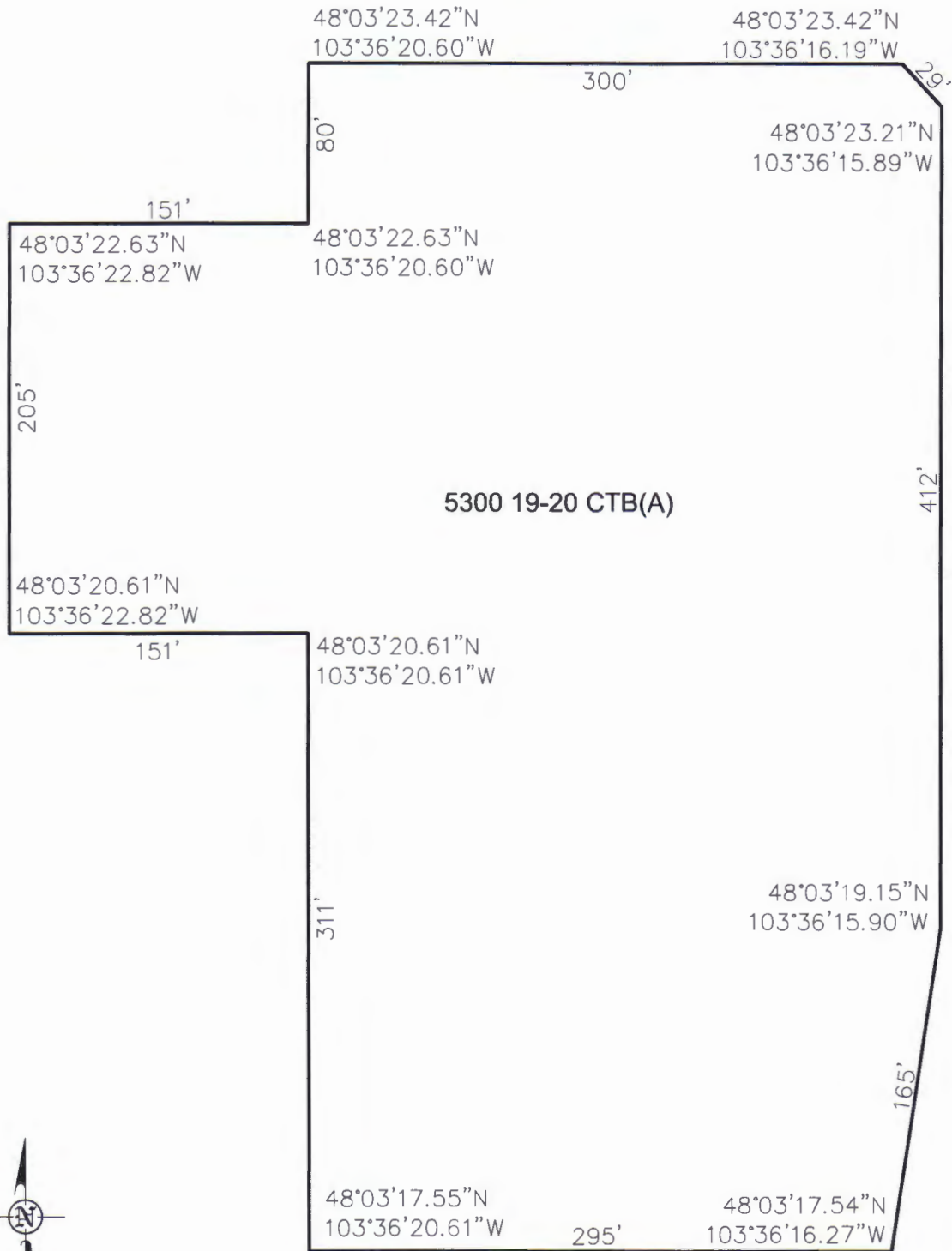
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
COUNTY ROAD MAP  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.D.M. Project No.: S14-09-241  
Checked By: D.D.K. Date: OCT 2014

| Revision No. | Date | By | Description |
|--------------|------|----|-------------|
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |
|              |      |    |             |

# LAT/LONG PAD CORNERS





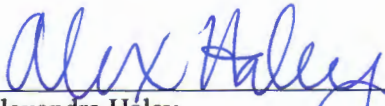
COMMINGLING AFFIDAVIT

STATE OF NORTH DAKOTA   )  
                                          ) ss.  
COUNTY OF WILLIAMS       )

The under signed, Alexandra Haley, of lawful age, being first duly sworn on her oath states that she is a duly authorized agent of Oasis Petroleum North America LLC, and that she has personal knowledge of the facts hereinafter set forth to make this Affidavit.

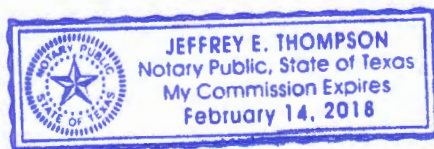
1.     Sections 19 & 20, Township 153 North, Range 100 West, Williams County North Dakota constitute a spacing unit in accordance with the applicable orders for the Bakken pool.
2.     Three wells have been drilled in the spacing unit, which are known as the Chalmers 5300 44-24 2TR, Chalmers 5300 44-24 3BR, and Chalmers 5300 44-24 4T2R.
3.     By NDIC Order 19005 dated May 11, 2012 and recorded in Williams County as Document No. 737164, all oil and gas interest within the aforementioned spacing unit were pooled.
4.     All Working Interests, Royalty Interests and Overriding Royalty Interests in the Chalmers 5300 44-24 2TR, Chalmers 5300 44-24 3BR, and Chalmers 5300 44-24 4T2R will be in common.

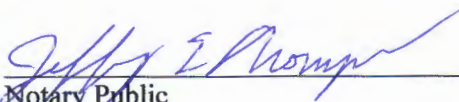
Dated this 15<sup>th</sup> day of September, 2014

  
\_\_\_\_\_  
Alexandra Haley  
Landman

STATE OF TEXAS       )  
                                          ) ss.  
COUNTY OF HARRIS   )

Subscribed to and sworn before me this 15<sup>th</sup> day of September, 2014



  
\_\_\_\_\_  
Notary Public  
State of Texas  
My Commission Expires: 2-14-18

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.

**28600**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

☐ Notice of Intent

Approximate Start Date

☒ Report of Work Done

Date Work Completed

**September 23, 2014**☐ Notice of Intent to Begin a Workover Project that may Qualify  
for a Tax Exemption Pursuant to NDCC Section 57-51.1-03.

Approximate Start Date

☐ Drilling Prognosis☐ Spill Report☐ Redrilling or Repair☐ Shooting☐ Casing or Liner☐ Acidizing☐ Plug Well☐ Fracture Treatment☐ Supplemental History☐ Change Production Method☐ Temporarily Abandon☒ Reclamation☐ OtherReserve pit reclamation

Well Name and Number

**Chalmers 5301 44-24 4T2R**

Footages

**877 F S L****245 F E L**

Qtr-Qtr

**SESE**

Section

**24**

Township

**153 N**

Range

**101 W**

Field

**Baker**

Pool

**Bakken**

County

**McKenzie****24-HOUR PRODUCTION RATE**

| Before |      | After |      |
|--------|------|-------|------|
| Oil    | Bbls | Oil   | Bbls |
| Water  | Bbls | Water | Bbls |
| Gas    | MCF  | Gas   | MCF  |

Name of Contractor(s)

**Neu Construction**

Address

**602 W 9th Street**

City

**Fairview**

State

**MT**

Zip Code

**59221****DETAILS OF WORK**

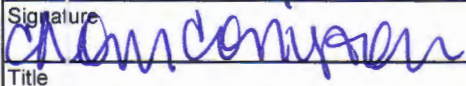
Oasis Petroleum North America LLC plans to reclaim the reserve pit for the above referenced wells as follows:

The NDIC field inspector, Rick Dunn (NDIC) was notified on 09/23/2014

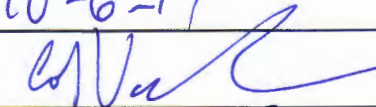
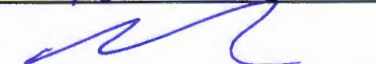
The surface owners, Wes Lindvig was contacted on 09/23/2014

Wes Lindvig 140758 41st Street NW Alexander, ND 58831

Spread material out in pit, cut top edge of liner and fold over cuttings, cover entire pit with line, back fill with clay slope and contour well site to ensure proper drainage

|                                                                                                 |  |                                          |                          |
|-------------------------------------------------------------------------------------------------|--|------------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b>                                             |  | Telephone Number<br><b>281-404-9591</b>  |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>                                                       |  |                                          |                          |
| City<br><b>Houston</b>                                                                          |  | State<br><b>TX</b>                       | Zip Code<br><b>77002</b> |
| Signature<br> |  | Printed Name<br><b>Chelsea Covington</b> |                          |
| Title<br><b>Regulatory Assistant</b>                                                            |  | Date<br><b>September 24, 2014</b>        |                          |
| Email Address<br><b>ccovington@oasispetroleum.com</b>                                           |  |                                          |                          |

**FOR STATE USE ONLY**

|                                                                                                |                                   |
|------------------------------------------------------------------------------------------------|-----------------------------------|
| <input checked="" type="checkbox"/> Received                                                   | <input type="checkbox"/> Approved |
| Date<br><b>10-6-14</b>                                                                         |                                   |
| By<br>    |                                   |
| Title<br> |                                   |





**Oasis Petroleum North America, LLC**

**Chalmers 5301 44-24 4T2R**

**877' FSL & 245' FEL**

**SE SE Section 24, T153N, R101W**

**Baker Field / Three Forks 2nd Bench**

**McKenzie County, North Dakota**

**BOTTOM HOLE LOCATION:**

**281.88' south & 10,434.73' east of surface location or approx.**

**595.12' FSL & 323.64' FEL, SE SE Section 20, T153N, R100W**

**Prepared for:**

Clay Hargett

Oasis Petroleum North America, LLC

1001 Fannin Street, Suite 1500

Houston, Texas 77002

**Prepared by:**

Hannah Thatcher, Daniel Haynes

PO Box 80507; Billings, MT 59108

(406) 259-4124

geology@sunburstconsulting.com

www.sunburstconsulting.com

# WELL EVALUATION

## *Chalmers 5301 44-24 4T2R*



**Figure 1: Nabors B25 drilling the Oasis Petroleum North America Chalmers 5301 44-24 4T2R during July 2014, south of Williston in McKenzie County, North Dakota.  
(Photos by Hannah Thatcher, wellsite geologist)**

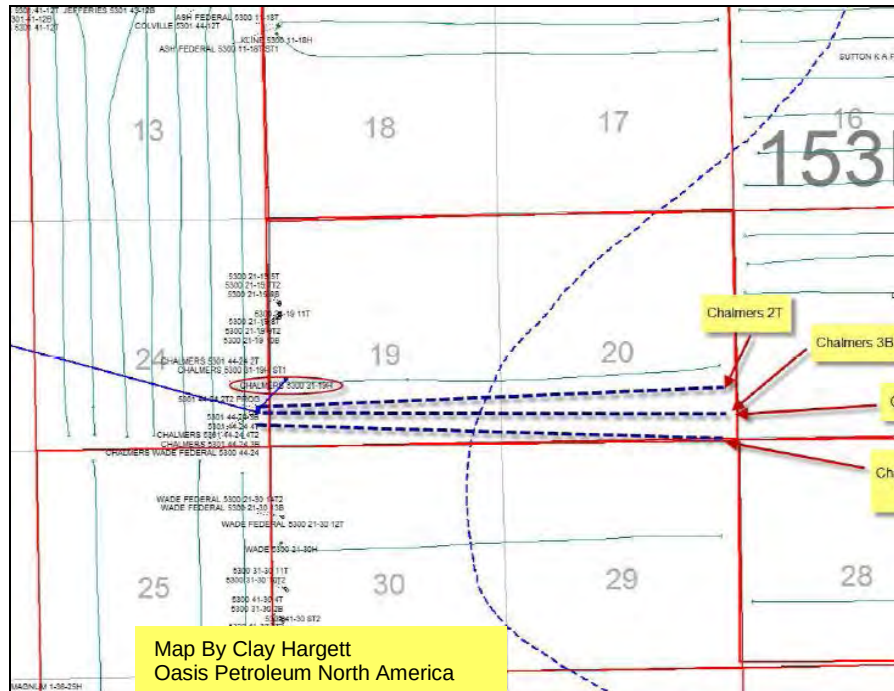
## INTRODUCTION

The Oasis Petroleum North America, LLC Chalmers 5301 44-24 4T2R well [SE SE Section 24, T153N, R101W] is located approximately 31 miles south of Williston, North Dakota. The horizontal well was spud on June 22, 2014 and represents a continuation of Oasis Petroleum's development of the Three Forks Formation within Baker Field. The Chalmers was planned to drill an approximately 9,863' lateral along a proposed azimuth of 91.34°. The well bore will be enhanced for production by multistage fracture stimulation.

## OFFSET CONTROL INFORMATION

Offset well data used for depth correlation during curve operations are found in the 'Structure' section appended to this report. The Chalmers 5301 44-24 3BR and the Chalmers 5301 44-24 2TR were the primary offsets used because they are located on the same pad.





**Figure 2: Offsetting control wells in relation to the Chalmers 44-22 4T2 well.**

## GEOLOGIC EVALUATION

## Methods:

Geologic supervision of the Chalmers 5301 44-24 4T2R was provided by Sunburst Consulting, Inc. with two well site geologists. A digital gas detector and chromatograph were interfaced with a Pason electronic data recorder system. The unit of measure for gas on this well was units (u), which was defined as 100 units equaling 1% methane equivalent in air. The EDR provided drill rate, on-off bottom and pump strokes to the gas detection computer and received total gas information from Sunburst for viewing around location. Lagged samples were caught by the rig crew in 30' intervals from 8,230' MD to 10,870' MD, 10' intervals from 10,870' MD to 11,170' MD, and 30' intervals from 11,170' MD to 20,963' TD. Wet and dry cuttings were examined under a tri-ocular microscope and complete lithologic descriptions and sampling intervals are provided in the lithology document within this evaluation. The set of dry cuttings collected during the duration of the well were sent to the State of North Dakota. Evidence of light hydrocarbons present in the drilling fluid was reported by the gas chromatography equipment and documented on the mud log presented with this report.

### **Zones of Interest:**

The Mission Canyon Formation (Mississippian; Madison Group) was drilled at 9,288' TVD (-7,320') and is comprised largely of light gray lime mudstone followed by gray to tan earthy textured argillaceous lime mudstone. Shows within the Mission Canyon ranged from 30 to 190 units against 9.6 ppg mud.



**Figure 3 & 4: Wet cutting of the Lodgepole Formation (left) and False Bakken sub-interval (right) at 10X.**

The top of the Lodgepole Formation was logged at 9,846' TVD (-7,878'). In general the Lodgepole can be described as a medium to dark gray brown argillaceous lime mudstone with a crystalline texture and trace amounts of disseminated pyrite (Figure 3). The False Bakken (Carrington Shale), drilled at 10,574' TVD (-8,606'), is comprised of very dark brown to black, slightly pyritic shale with an earthy texture, and was found to be soft to firm (Figure 4). Strong hydrocarbon shows in the lower 100' of the Lodgepole were as high as 204 total gas units; suggest that some of the oil and gas from the Upper Bakken Shale may be exploiting fractures thought to exist in the Lower Lodgepole.

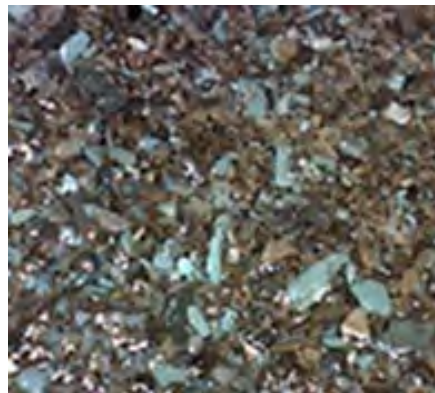
The Bakken Formation (Devonian – Mississippian) has four formal members, an upper and lower black, fissile, organic-rich pyritic shale, separated by an arenaceous limestone, siltstone and silty sandstone middle member. These three members overlay a silty shale or siltstone of the Pronghorn Member toward the basin depositional center. The Upper Bakken Shale was drilled at 10,585' TVD (-8,617') with sample returns typically described as black, carbonaceous, *petroliferous* shale with trace amounts of disseminated pyrite (Figure 5). The Middle Bakken, penetrated at 10,600' TVD (-8,632'), consists of a varying thickness and sequence of interbedded siltstone, limestone and silty sandstone. *Trace spotty light brown oil staining* was present along with gas shows as high as 330 TGU. Penetrated at 10,639' TVD (-8,671'), the Lower Bakken Shale was described as a black to dark brown carbonaceous shale with trace amounts of disseminated pyrite (Figure 6). Gas shows encountered in the Lower Bakken read as high as 548u (C1-C4). The Pronghorn was penetrated at 10,654' TVD (-8,686') and is commonly described as a dark to medium gray siltstone with calcite cement.



**Figure 5 & 6: Wet cuttings of the Upper Bakken Shale (left), Lower Bakken Shale (right) at 10X.**



The Three Forks Formation (Devonian; Kaskaskia Sequence.) represents a regressive sequence deposited in a supratidal sabkha environment. The first bench was drilled at 10,676' TVD (-8,708') and is comprised of a light to medium gray, cream to off white sucrosic dolomite with trace to occasional amounts of disseminated pyrite and light green shale also with trace amounts of disseminated pyrite. The second bench was drilled at 10,706' TVD (-8,738') and is comprised of a light to medium gray, cream to off white dolomite with trace to occasional amounts of disseminated pyrite and light green shale also with trace amounts of disseminated pyrite (Figure 7). The dolomite also contains possible intercrystalline porosity. Another component of the Three Forks is are light gray brown to gray green claystone with dolomite cement and trace amounts of pyrite overlain by an orange to pink and green dolomite. Shows within the Three Forks Formation ranged from 50 to 430 units in a drilling mud of 9.5-10.5 ppg.



**Figure 7: Wet cuttings of the Three Forks 2<sup>nd</sup> Bench dolomite and shale at 10X.**

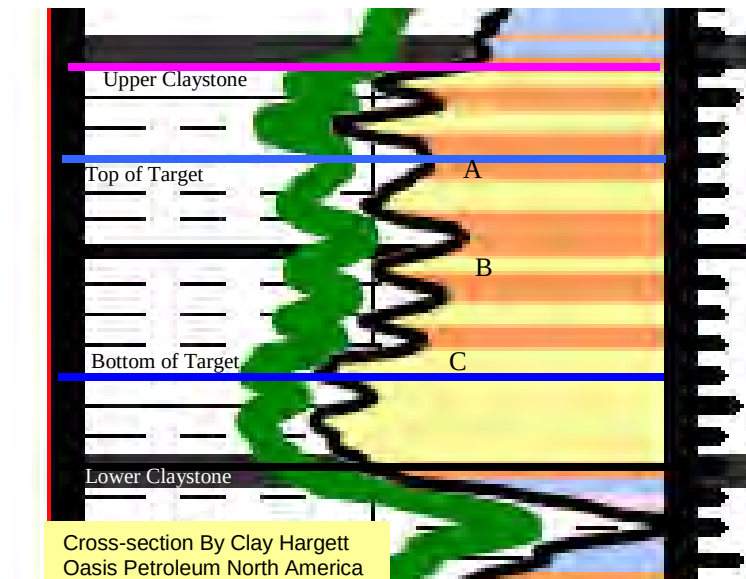
### **Geo-steering:**

Kick-off point for the curve was established from the isopach of the “base last salt” marker to the Three Forks “target” in the offset wells. The Chalmers 5301 44-24 3BR was used as the primary offset through the vertical and curve sections. While drilling the curve, measured gamma ray signatures were compared to those of the three offsets and aided in the landing of the curve. The landing target was confirmed by the depth of the False Bakken, which was consistent with the offset wells. The curve was successfully landed within the Three Forks Formation at a depth of 11,100' MD (10,709' TVD) placing the well bore approximately 3' below the first Claystone member. Directional tools were then pulled out of the hole and a string of 7" casing was set (11,042' MD) and then cemented by Schlumberger.

Samples from the target zone varied in porosity, oil staining, color, cementation, and pyrite content. Stratigraphic location in the target zone was based on these sample observations along with gas shows, drill rates and gamma ray values. Severe doglegs were to be avoided so as to obtain the desired vertical section and aid in a successful completion liner run at TD.

The Oasis Petroleum North America, LLC prospect geologist defined the an initial target zone as an 20' zone that began 6' below the first Claystone member and ended six feet above the second claystone member 32' below the first Claystone member. The target zone consisted of an upper dolomite and shale interval reading 120-150 count gamma (A marker). The center of the target interval was comprised of a cleaner dolomite with lesser amounts of shale reading 90-120

count gamma (B marker). The base of the target zone was characterized by a clean dolomite with trace amounts of shale with gamma readings of 70-90 (C marker). The A-C gamma markers were used for determining depth within the target interval and plotted on the Chalmers 44-22 4T2R dip profile (Figure 9).



**Figure 8: Offset gamma ray profile of the upper Three Forks (0-200 api. scale) \*See dip profile (Figure 9) for marker presentation\***

Using the aforementioned information gathered during drilling, offset log information and the structural data provided from Oasis Petroleum North America, LLC., well site geologists were able to anticipate, and interpret the local apparent dip during the drilling of the Chalmers 44-22 4T2R well. A total depth of 20,963' MD was reached on September 5th, 2014 at 15:00 CDT. The target resultant was 100% within the Three Forks 2nd Bench. The resulting structure of the Three Forks was a fall in TVD of 83' over 9,963' MD; resulting in an overall down dip of  $-0.48^\circ$  as portrayed on the Chalmers 44-22 4T2R dip profile (Figure 9).



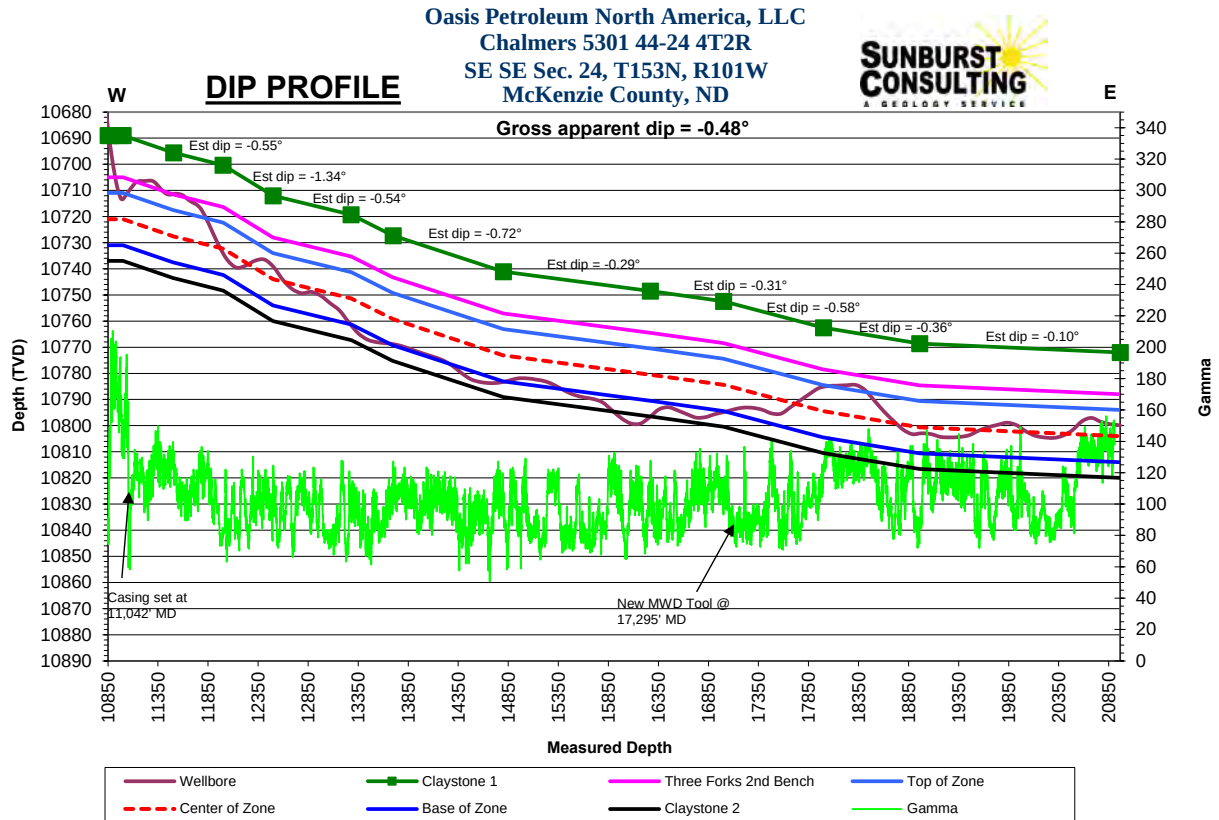


Figure 9: Well profile representing estimated dip value & gamma ray for the Chalmers 44-22 4T2R lateral.

## Hydrocarbons:

Gas monitoring and fluid gains were monitored to evaluate the viability of this reservoir during the drilling of the Chalmers 44-22 4T2R well. In the invert mud system, hydrostatic conditions were maintained near balance. This allowed for gas and fluid gains from the well to be monitored. Gas shows were encountered in vertical, curve and lateral drilling of the well.

Background gas observed during the drilling of the lateral ranged from 45 to 280 units. Invert drilling fluid was used throughout the drilling of the vertical and curve weighing 9.6-10.4 ppg. The lateral was drilled with saline drilling fluid with a mud weight of 9.5-9.6 ppg. Gas shows ranged up to 420 units and connection gases were observed up to 335 units. C1-C4 gas components were observed throughout the course of the lateral. Trip gases were observed as high as 1928 units. Oil shows were very light throughout the lateral ranging from 0-3% in sample. When present it was a *light brown spotty oil stain that yielded a slow to moderate streaming to diffuse light green cut fluorescence.*

## SUMMARY

The Nabors B25 drilling rig successfully drilled a two-section horizontal well bore within the Three Forks Formation at the Chalmers 44-22 4T2R. A net of 9,863' was drilled within the Three Forks. A mud program consisting of diesel invert (9.8–10.4 ppg), during the vertical and curve build sections, and saline based mud (9.5-9.6 ppg), during the lateral maintained stable hole conditions and permitted adequate analysis of gas concentrations.

Projections of dip were successfully used to maintain the well bore in the Three Forks 2nd Bench for 100% of the lateral. Samples from the target consisted of a light to medium gray, cream to off white dolomite with trace to occasional amounts of disseminated pyrite and light green shale also with trace amounts of disseminated pyrite. Intercrystalline porosity was generally seen throughout the entire lateral. Hydrocarbon shows in the target zone were high throughout the lateral. Samples from the ideal zone contained a spotty light brown oil stain.

The Chalmers 44-22 4T2R will be fitted with a 4 ½" production liner and swell packers in preparation for a fracture stimulation that will determine the true commercial value of the well. The well currently awaits fracture stimulation.

Respectfully submitted,

*Hannah Thatcher*

*Daniel Haynes*

Sunburst Consulting, Inc.

September 5, 2014

# WELL DATA SUMMARY

|                                     |                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b><u>OPERATOR:</u></b>             | Oasis Petroleum North America, LLC                                                                                          |
| <b><u>ADDRESS:</u></b>              | 1001 Fannin Street, Suite 1500<br>Houston, Texas 77002                                                                      |
| <b><u>WELL NAME:</u></b>            | Chalmers 5301 44-24 4T2R                                                                                                    |
| <b><u>API #:</u></b>                | 33-053-06011-00-00                                                                                                          |
| <b><u>WELL FILE #:</u></b>          | 28600                                                                                                                       |
| <b><u>SURFACE LOCATION:</u></b>     | 877' FSL & 245' FEL<br>SE SE Section 24, T153N, R101W                                                                       |
| <b><u>FIELD/ OBJECTIVE:</u></b>     | Baker Field / Three Forks 2nd Bench                                                                                         |
| <b><u>COUNTY, STATE</u></b>         | McKenzie County, North Dakota                                                                                               |
| <b><u>BASIN:</u></b>                | Williston                                                                                                                   |
| <b><u>WELL TYPE:</u></b>            | Three Forks Horizontal                                                                                                      |
| <b><u>ELEVATION:</u></b>            | GL: 1,943'<br>KB: 1,968'                                                                                                    |
| <b><u>SPUD/ RE-ENTRY DATE:</u></b>  | Spud: 6/22/2014; Reentered: 7/13/2014                                                                                       |
| <b><u>BOTTOM HOLE LOCATION:</u></b> | 281.88' south & 10,434.73' east of surface location or approx.<br>595.12' FSL & 323.64' FEL, SE SE Section 20, T153N, R100W |
| <b><u>CLOSURE COORDINATES:</u></b>  | Closure Azimuth: 91.55°<br>Closure Distance: 10,438.53'                                                                     |
| <b><u>TOTAL DEPTH / DATE:</u></b>   | 20,963' on Septmeber 5, 2014<br>80% within target interval                                                                  |
| <b><u>TOTAL DRILLING DAYS:</u></b>  | 15 days                                                                                                                     |
| <b><u>CONTRACTOR:</u></b>           | Nabors B25                                                                                                                  |



|                                                  |                                                                                                                                                  |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>PUMPS:</u></b>                             | H &H Triplex (stroke length - 12")                                                                                                               |
| <b><u>TOOLPUSHERS:</u></b>                       | Casey Pippenger, Bruce Walter                                                                                                                    |
| <b><u>FIELD SUPERVISORS:</u></b>                 | Mike Crow, Travis Handran                                                                                                                        |
| <b><u>CHEMICAL COMPANY:</u></b>                  | Fluid Control                                                                                                                                    |
| <b><u>MUD ENGINEER:</u></b>                      | Keith McCarty, Warren Carlson                                                                                                                    |
| <b><u>MUD TYPE:</u></b>                          | Fresh water in surface hole<br>Diesel invert in curve; Salt water in lateral                                                                     |
| <b><u>MUD LOSSES:</u></b>                        | Invert Mud: 425 bbls., Salt Water: Not tracked                                                                                                   |
| <b><u>PROSPECT GEOLOGIST:</u></b>                | Clay Hargett                                                                                                                                     |
| <b><u>WELLSITE GEOLOGISTS:</u></b>               | Hannah Thatcher, Daniel Haynes                                                                                                                   |
| <b><u>GEOSTEERING SYSTEM:</u></b>                | Sunburst Digital Wellsite Geological System                                                                                                      |
| <b><u>ROCK SAMPLING:</u></b>                     | 30' from 8,230' - 10,840'<br>10' from 10,840' -11,170'<br>30' from 11,170' - 20,963' (TD)                                                        |
| <b><u>SAMPLE EXAMINATION:</u></b>                | trinocular microscope & fluoroscope                                                                                                              |
| <b><u>SAMPLE CUTS:</u></b>                       | Trichloroethylene                                                                                                                                |
| <b><u>GAS DETECTION:</u></b>                     | MSI (Mudlogging Systems, Inc.) TGC - total gas with chromatograph<br>Serial Number(s): ML-382                                                    |
| <b><u>DIRECTIONAL DRILLERS:</u></b>              | RPM<br>Mike Crow, Travis Handran                                                                                                                 |
| <b><u>MWD:</u></b>                               | Ryan Directional Service Inc.<br>Dave Unger, Daniel Ogden                                                                                        |
| <b><u>CASING:</u></b>                            | Surface: 13.375" 36# J-55 set to 2,095'<br>Intermediate Surface: 9 5/8" 40# HCL-180 set to 6,035'<br>Intermediate: 7" 32# HCP-110 set to 11,042' |
| <b><u>SAFETY/ H<sub>2</sub>S MONITORING:</u></b> | Oilind Safety                                                                                                                                    |

**KEY OFFSET WELLS:**

**Oasis Petroleum North America**

**Chalmers 5301 44-24 3BR**

SE SE Section 24, T153N, R101W

McKenzie Co., ND

**KB: 1,968'**

**Oasis Petroleum North America**

**Chalmers 5301 44-24 2TR**

SE SE Section 24, T153N, R101W

McKenzie Co., ND

**KB: 1,968'**

**Oasis Petroleum North America**

**Chalmers 5300 31-19H**

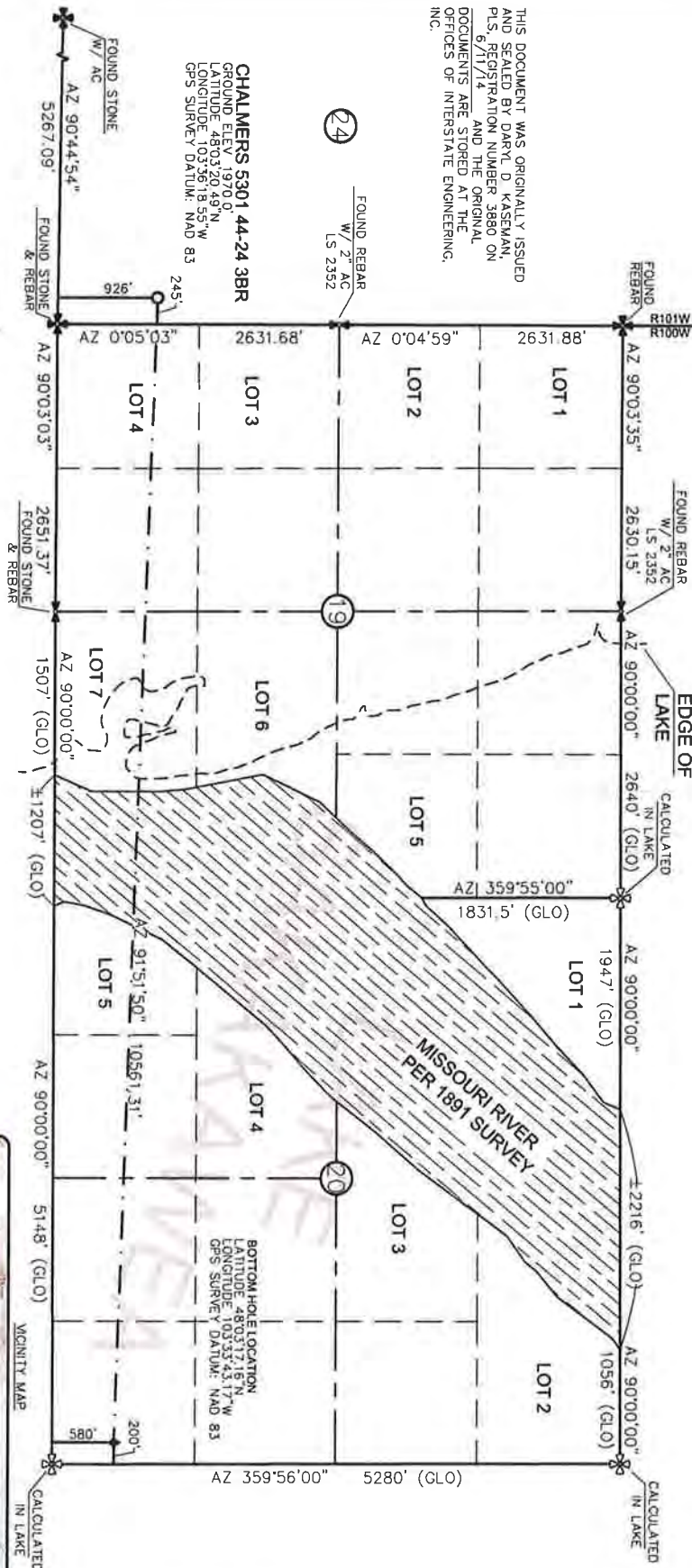
NW SW Sec.19, T153N, R100W

McKenzie Co., ND

**KB: 1,929'**

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS. REGISTRATION NUMBER 3880 ON 6/11/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

**CHALMERS 5301 44-24 3BR**  
GROUND ELEV. 1970.0'  
LATITUDE 48°03'20.49"N  
LONGITUDE 103°36'18.55"W  
GPS SURVEY DATUM: NAD 83



**WELL LOCATION PLAT**  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"CHALMERS 5301 44-24 3BR"  
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

MONUMENT - RECOVERED  
MONUMENT - NOT RECOVERED



STAKED ON 9/3/13  
VERTICAL CONTROL DATUM WAS BASED UPON  
CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'  
THIS SURVEY AND PLAT IS BEING PROVIDED AT THE  
REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY  
THAT THIS PLAT CORRECTLY REPRESENTS WORK  
PERFORMED BY ME OR UNDER MY SUPERVISION  
AND IS TRUE AND CORRECT TO THE BEST OF MY  
KNOWLEDGE AND BELIEF.

DARYL D. KASEMAN LS-3880



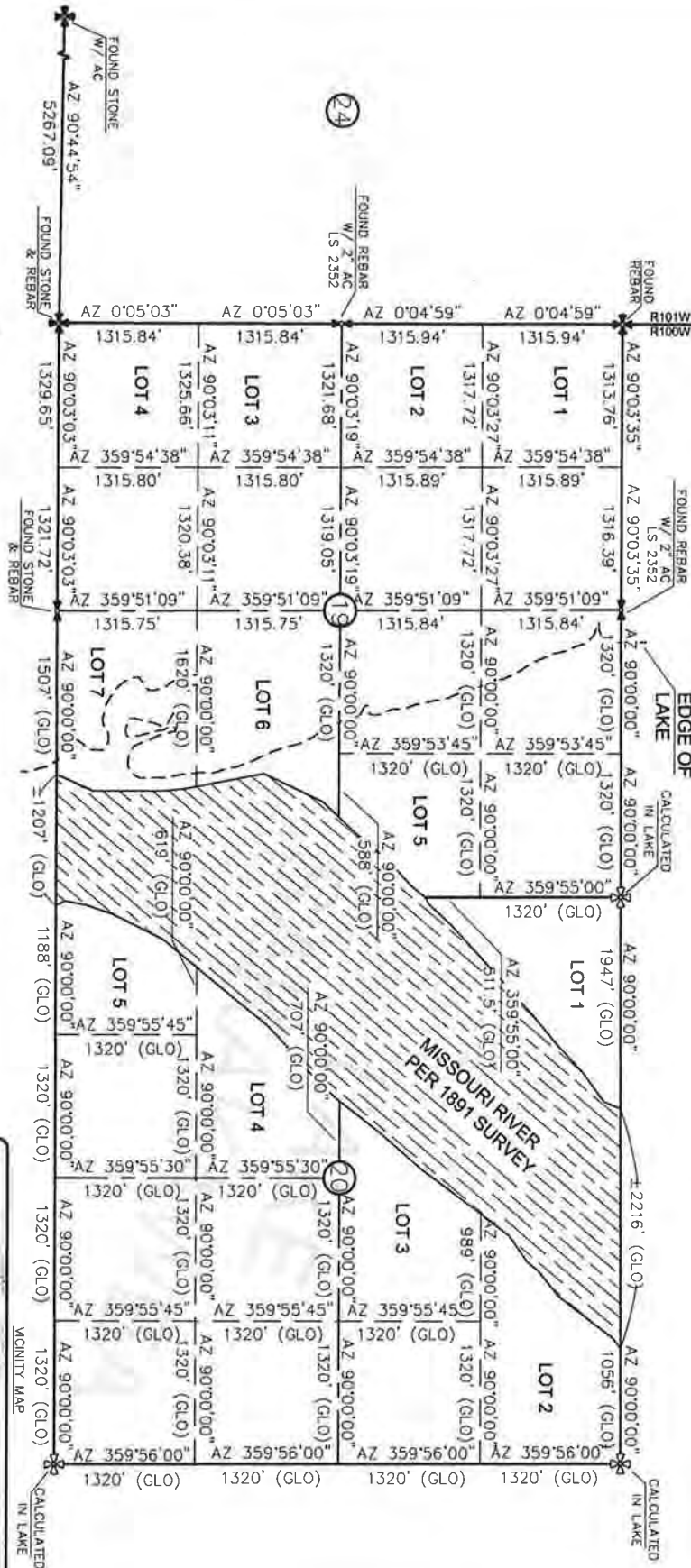
© 2014, INTERSTATE ENGINEERING, INC.

| 1/8<br>SHEET NO.                                                                 | <br>INTERSTATE ENGINEERING<br>Professional and Surveying Services | Interstate Engineering, Inc.<br>P.O. Box 648<br>425 East Main Street<br>Gallatin, Montana 59270<br>Ph: (406) 435-5617<br>Fax: (406) 435-5618<br>www.interstateeng.com<br>Offices in Minnesota, North Dakota and South Dakota | OASIS PETROLEUM NORTH AMERICA, LLC<br>WELL LOCATION PLAT<br>SECTION 24, T153N, R101W<br>MCKENZIE COUNTY, NORTH DAKOTA | <table border="1"> <thead> <tr> <th>Rev</th> <th>Date</th> <th>By</th> <th>Description</th> </tr> </thead> <tbody> <tr> <td>REV 1</td> <td>1/20/14</td> <td>DK</td> <td>CHANGED PAD LAYOUT</td> </tr> <tr> <td>REV 2</td> <td>5/09/14</td> <td>DK</td> <td>ADDED WELL TO PAD</td> </tr> <tr> <td>REV 3</td> <td>5/20/14</td> <td>DK</td> <td>CHANGED NAME, ADDED WELL</td> </tr> <tr> <td>REV 4</td> <td>6/11/14</td> <td>DK</td> <td>ADDED ENCLOSURES</td> </tr> </tbody> </table> | Rev | Date        | By | Description | REV 1 | 1/20/14 | DK | CHANGED PAD LAYOUT | REV 2 | 5/09/14 | DK | ADDED WELL TO PAD | REV 3 | 5/20/14 | DK | CHANGED NAME, ADDED WELL | REV 4 | 6/11/14 | DK | ADDED ENCLOSURES |
|----------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|----|-------------|-------|---------|----|--------------------|-------|---------|----|-------------------|-------|---------|----|--------------------------|-------|---------|----|------------------|
|                                                                                  |                                                                   |                                                                                                                                                                                                                              | Rev                                                                                                                   | Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | By  | Description |    |             |       |         |    |                    |       |         |    |                   |       |         |    |                          |       |         |    |                  |
| REV 1                                                                            | 1/20/14                                                           | DK                                                                                                                                                                                                                           | CHANGED PAD LAYOUT                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |             |    |             |       |         |    |                    |       |         |    |                   |       |         |    |                          |       |         |    |                  |
| REV 2                                                                            | 5/09/14                                                           | DK                                                                                                                                                                                                                           | ADDED WELL TO PAD                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |             |    |             |       |         |    |                    |       |         |    |                   |       |         |    |                          |       |         |    |                  |
| REV 3                                                                            | 5/20/14                                                           | DK                                                                                                                                                                                                                           | CHANGED NAME, ADDED WELL                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |             |    |             |       |         |    |                    |       |         |    |                   |       |         |    |                          |       |         |    |                  |
| REV 4                                                                            | 6/11/14                                                           | DK                                                                                                                                                                                                                           | ADDED ENCLOSURES                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |             |    |             |       |         |    |                    |       |         |    |                   |       |         |    |                          |       |         |    |                  |
| Drawn By: DJS<br>Checked By: DJS<br>Project No.: 315-08-215101<br>Date: MAY 2014 |                                                                   |                                                                                                                                                                                                                              |                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |             |    |             |       |         |    |                    |       |         |    |                   |       |         |    |                          |       |         |    |                  |



**SECTION BREAKDOWN**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 CHALMERS 5301 44-24 3BR

SECTION 24, T153N, R101W, & SECTIONS 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA  
 926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE



MONUMENT - RECOVERED  
 MONUMENT - NOT RECOVERED

0 1000  
 1" = 1000'

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS. REGISTRATION NUMBER 3880 ON 6/11/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

ALL AZIMUTHS ARE BASED ON C.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY -003.



© 2014, INTERSTATE ENGINEERING, INC.

2/8



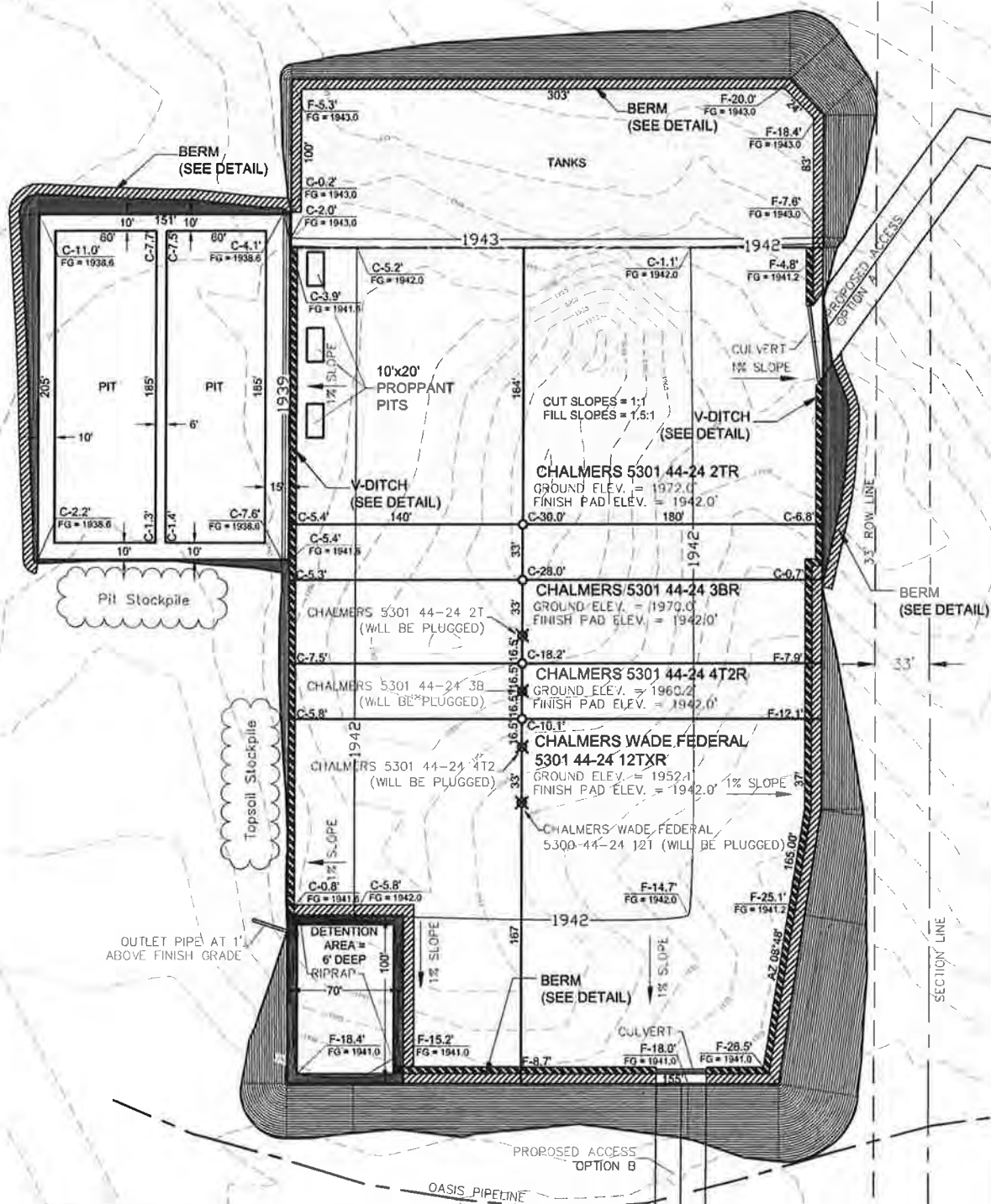
Interstate Engineering, Inc.  
 P.O. Box 548  
 429 East Main Street  
 Sidney, Montana 59270  
 Ph (406) 433-5617  
 Fax (406) 433-5618  
 www.interstateeng.com

OASIS PETROLEUM NORTH AMERICA, LLC  
 SECTION 24, T153N, R101W, & SECTIONS 19 & 20, T153N, R100W  
 MCKENZIE COUNTY, NORTH DAKOTA  
 Drawn By: JAS  
 Checked By: JLS  
 Project No: 512-05235-01  
 Date: MAY 2014

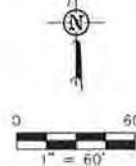
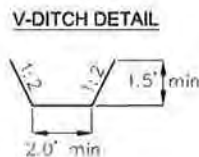
| Rev   | Date    | By  | Description              |
|-------|---------|-----|--------------------------|
| REV 1 | 1/29/14 | SW  | CHANGED AND LAYOUT       |
| REV 2 | 1/29/14 | JAS | ADDED WELL TO PAD        |
| REV 3 | 1/29/14 | JAS | CHANGED NAME, ADDED WELL |
| REV 4 | 4/1/14  | JAS | ADDED CHANGES            |

THIS DOCUMENT WAS ORIGINALLY ISSUED  
AND SEALED BY DARYL D. KASEMAN,  
P.L.S., REGISTRATION NUMBER 3880 ON  
6/11/14 AND THE ORIGINAL  
DOCUMENTS ARE STORED AT THE  
OFFICES OF INTERSTATE ENGINEERING,  
INC.

# **PAD LAYOUT** OASIS PETROLEUM NORTH AMERICA, LLC 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002 "CHALMERS 5301 44-24 3BR" 926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- BERM
- DITCH
- Proposed Contours
- Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

3/8



Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Shelby, Montana 59270  
Ph (408) 433-5617  
Fax (408) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
PAD LAYOUT  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.L.S. Project No.: 813-0923581  
Checked By: J.D.K. Date: MAY 2014

| Revised No. | Date    | By  | Description                 |
|-------------|---------|-----|-----------------------------|
| REV 1       | 5/20/14 | JLS | ON MAPS, PAD LAYOUT         |
| REV 2       | 5/20/14 | JLS | ADDED WELL TO PAD           |
| REV 3       | 5/20/14 | JLS | CHANGED BERM, WIDENED DITCH |
| REV 4       | 6/11/14 | JLS | ADDED DITCHES               |

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



SCALE: 1" = 2 MILE

SHEET NO.



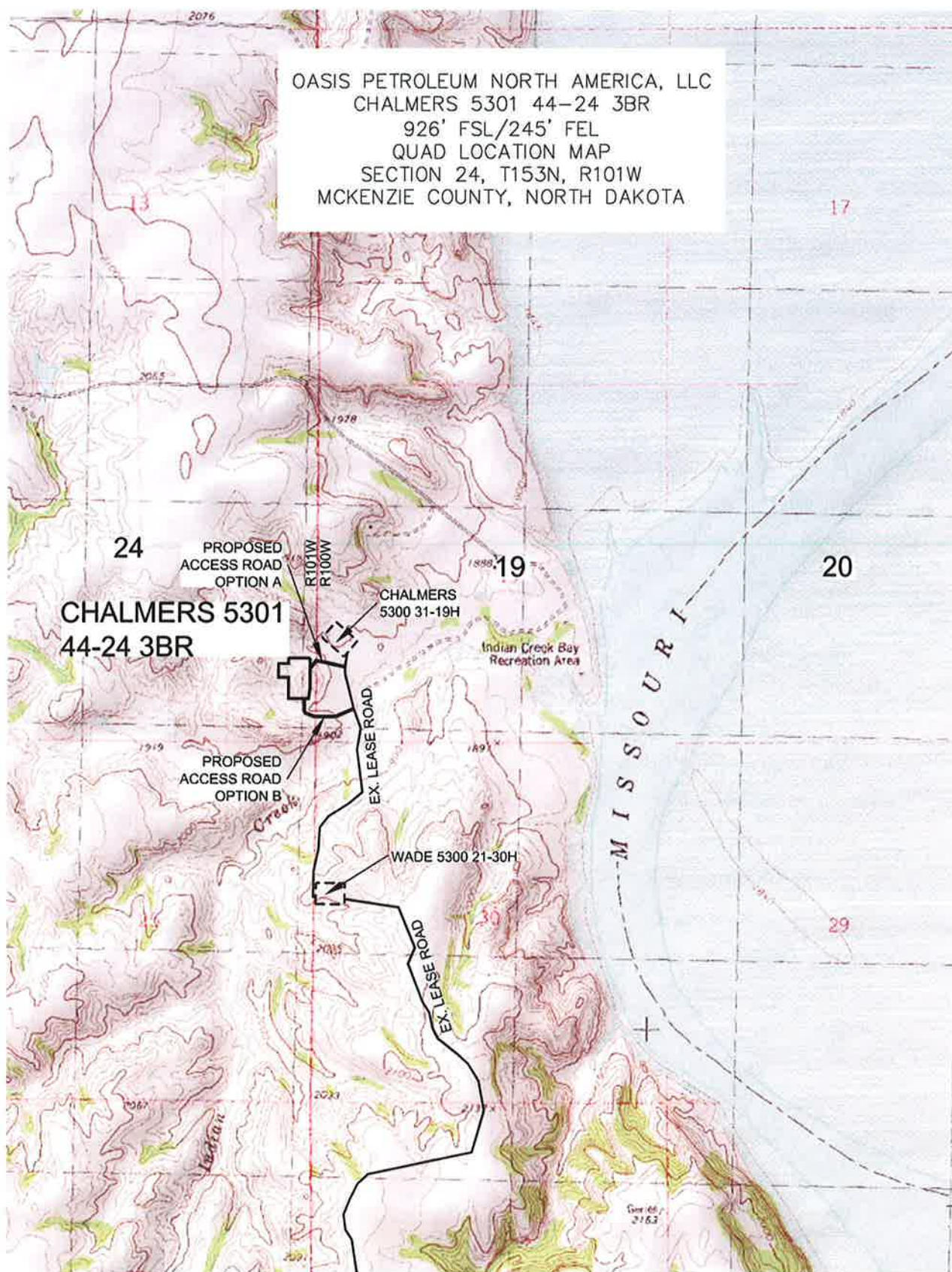
Other offices in Minnesota, North Dakota and South Dakota

|                           |                                   |
|---------------------------|-----------------------------------|
| Drawn By: <u>J.J.S.</u>   | Project No.: <u>S13-09-235.01</u> |
| Checked By: <u>D.D.K.</u> | Date: <u>MAY 2014</u>             |

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 3        | 12/3/13 | JLS | CHANGED PAD LAYOUT       |
| REV 3        | 1/26/14 | BNC | CHANGED PAD LAYOUT       |
| REV 4        | 5/25/14 | JLS | ADDED WELL TO PAD        |
| REV 5        | 5/25/14 | JLS | CHANGED HALL, MOVED WELL |
| REV 6        | 8/11/15 | JSH | ADDED DIMENSIONS         |



OASIS PETROLEUM NORTH AMERICA, LLC  
 CHALMERS 5301 44-24 3BR  
 926' FSL/245' FEL  
 QUAD LOCATION MAP  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

5/8

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.  
 P.O. Box 648  
 425 East Main Street  
 Sidney, Montana 59270  
 Ph (406) 433-5617  
 Fax (406) 433-5618  
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
 QUAD LOCATION MAP  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA

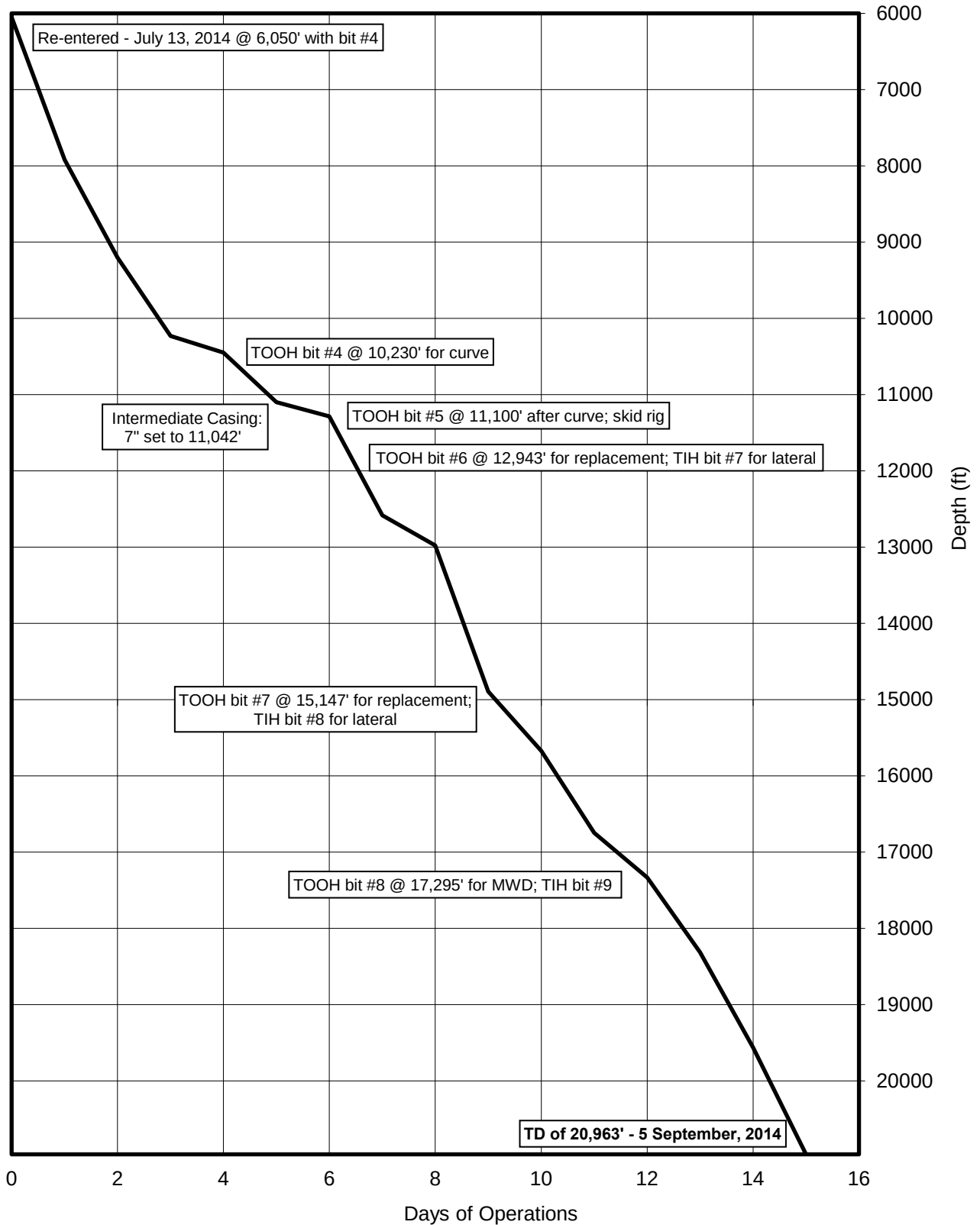
Drawn By: J.J.S. Project No.: S13-09-235-01  
 Checked By: D.D.K. Date: MAY 2014

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 1        | 12/2/13 | JJS | CHANGED PAD LAYOUT       |
| REV 2        | 1/20/14 | DBH | CHANGED PAD LAYOUT       |
| REV 3        | 5/20/14 | JJS | ADDED WELL TO PAD        |
| REV 4        | 6/25/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 5        | 7/1/14  | JJS | ADDED SHADINGS           |

# TIME VS. DEPTH

Oasis Petroleum North America, LLC

Chalmers 5301 44-24 4T2R



# MORNING REPORT SUMMARY

| Day | Date 2014 | Depth (0600 Hrs) | 24 Hr Footage | Bit # | WOB (Klbs) |    | RPM (RT) | WOB (Klbs) |      | RPM (MM) | PP   | SPM |    | GPM | 24 Hr Activity Summary                                                                                                                                                                                                                                                                                                                                 | Formation   |
|-----|-----------|------------------|---------------|-------|------------|----|----------|------------|------|----------|------|-----|----|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|     |           |                  |               |       | RT         | MM |          | RT         | MM   |          |      | 1   | 2  |     |                                                                                                                                                                                                                                                                                                                                                        |             |
| 0   | 7/13      | 6,050'           | -             | -     | -          | -  | -        | -          | -    | -        | -    | -   | -  | -   | -                                                                                                                                                                                                                                                                                                                                                      | -           |
| 1   | 7/14      | 7,920'           | 1870          | 4     | 25         | 10 | 50       | 25         | 10   | 253      | 3300 | 75  | 75 | 517 | TIH, drill out float, formation integrity test, drill F/6050 to 6895, rig service, drill F/6895 to 7920                                                                                                                                                                                                                                                | -           |
| 2   | 7/15      | 9,201'           | 1281          | 4     | 25         | 10 | 50       | 25         | 10   | 253      | 3280 | 75  | 75 | 517 | Drill F/7931 to 8605, service rig, greased crown and blocks, drill F/8605 to 9201                                                                                                                                                                                                                                                                      | -           |
| 3   | 7/16      | 10,230'          | 1029          | 4     | 25         | 10 | 50       | 25         | 10   | 253      | 3400 | 75  | 75 | 517 | Drill F/9165 to 9864, service top drive/greased black jack, lower well/functioned HCR, drill F/9864 to 10230, circulate and condition, TOOH                                                                                                                                                                                                            | -           |
| 4   | 7/17      | 10,447'          | 217           | 5     | 25         | 10 | 50       | 25         | 10   | 253      | 3200 | 75  | 75 | 517 | TOOH, L/D BHA, pre job safety CBL logging rig up and meeting, case hole logs, rig down loggers, P/U BHA, TIH, drill F/10230 to 10447                                                                                                                                                                                                                   | -           |
| 5   | 8/26      | 11,100'          | 653           | 6     | 25         | 10 | 50       | 25         | 10   | 253      | 3200 | 75  | 75 | 517 | Working as directed by operator on previous well, Skid rig, RD flowline, catwalk, derrick assists, water, air to rig, Skid rig, walk, install wellheads, weatherford, remove nightcap, install pack off, Rig up, catwalk, flowline w/ crane truck, Nipple up BOPs                                                                                      | Three Forks |
| 6   | 8/27      | 11,285'          | 185           | 6     | 26         | 40 | 28       | 258        | 2550 | 0        | 80   | 281 | 80 | 281 | Test BOPs, install/remove wear bushing, Pick up BHA, TIH, Service top drive/fix dye blocks/grease blackjack/crown and blocks, TIH, Change rotating head/rubber remove trip nipple install rotating head, Reaming/washing, Drilling cement/float and shoe, Formation integrity test (fit) 9.7 MW @ 1900 pounds for 30 minutes, Drill F/ 11100 to 11,142 | Three Forks |
| 7   | 8/28      | 12,584'          | 1299          | 6     | 17         | 60 | 43       | 138        | 3322 | 0        | 80   | 281 | 80 | 281 | Drill F/ 11271 to 11920, Service top drive, Drills BOP etc./time 5.25 min, Drill F/ 11920 to 12584                                                                                                                                                                                                                                                     | Three Forks |
| 8   | 8/29      | 12,978'          | 394           | 6     | 17         | 50 | 40       | 138        | 2410 | 80       | 0    | 281 | 80 | 281 | Drill F/12585 to 12943/lay down on STD washed HWDP, TOOH/lay down SDS washed HWDP, install/remove wear bushing/pull rotating head/install trip nipple/no flow, TOOH, Pick up BHA, TIH, Change rotating head/rubber/remove trip nipple/install rotating head, TIH, Cut drilling line, TIH, Drill F/12943 to 12978                                       | Three Forks |
| 9   | 8/30      | 14,893'          | 1915          | 7     | 20         | 40 | 43       | 138        | 2893 | 80       | 0    | 281 | 80 | 281 | Drill F/13001-14064, Service top drive/function HCR, Drill F/14064-14893                                                                                                                                                                                                                                                                               | Three Forks |
| 10  | 8/31      | 15,672'          | 779           | 8     | 21         | 50 | 43       | 138        | 3811 | 80       | 0    | 281 | 80 | 281 | Drill F/14893 to 15147, TOOH, Change rotating head/rubber/pull rotating rubber/install trip nipple 0, TOOH, Lay down BHA, Pick up BHA, TIH, Service top drive, TIH, Change rotating head/rubber/remove trip nipple/install rotating head, TIH, Circulate and condition fill pipe and bottoms up, Drill F/15147 to 15672                                | Three Forks |
| 11  | 9/1       | 16,746'          | 1074          | 8     | 20         | 50 | 32       | 131        | 3010 | 76       | 0    | 267 | 76 | 267 | Drill F/ 15778 to 16746                                                                                                                                                                                                                                                                                                                                | Three Forks |
| 12  | 9/2       | 17,332'          | 586           | 9     | 20         | 50 | 32       | 131        | 3010 | 76       | 0    | 267 | 76 | 267 | Drill F/ 16748 to 17295, Service top drive/ck brakes/calibrate martin decker, Drills-BOP etc., Drill F/17295 to 17332, TOOH, Lay down BHA, Pick up BHA, TIH.                                                                                                                                                                                           | Three Forks |
| 13  | 9/3       | 18,315'          | 983           | 9     | 17         | 60 | 45       | 138        | 3390 | 80       | 0    | 281 | 80 | 281 | TIH, Circulate and condition tag bottom, Drill F/17330 to 17641, Service top drive, Drill F/17641 to 17750, Well control BOP drill, Drill F/ 17750 to 18315                                                                                                                                                                                            | Three Forks |
| 14  | 9/4       | 19,563'          | 1248          | 9     | 15         | 55 | 52       | 138        | 3420 | 0        | 80   | 281 | 80 | 281 | Drill F/ 17861 to 18858, Service top drive, Drill F/18858 to 19563                                                                                                                                                                                                                                                                                     | Three Forks |
| 15  | 9/5       | 20,963'          | 1400          | 9     | 14         | 55 | 82       | 138        | 3558 | 0        | 80   | 281 | 80 | 281 | Drill F/19563 to 20963                                                                                                                                                                                                                                                                                                                                 | Three Forks |



# DAILY MUD SUMMARY

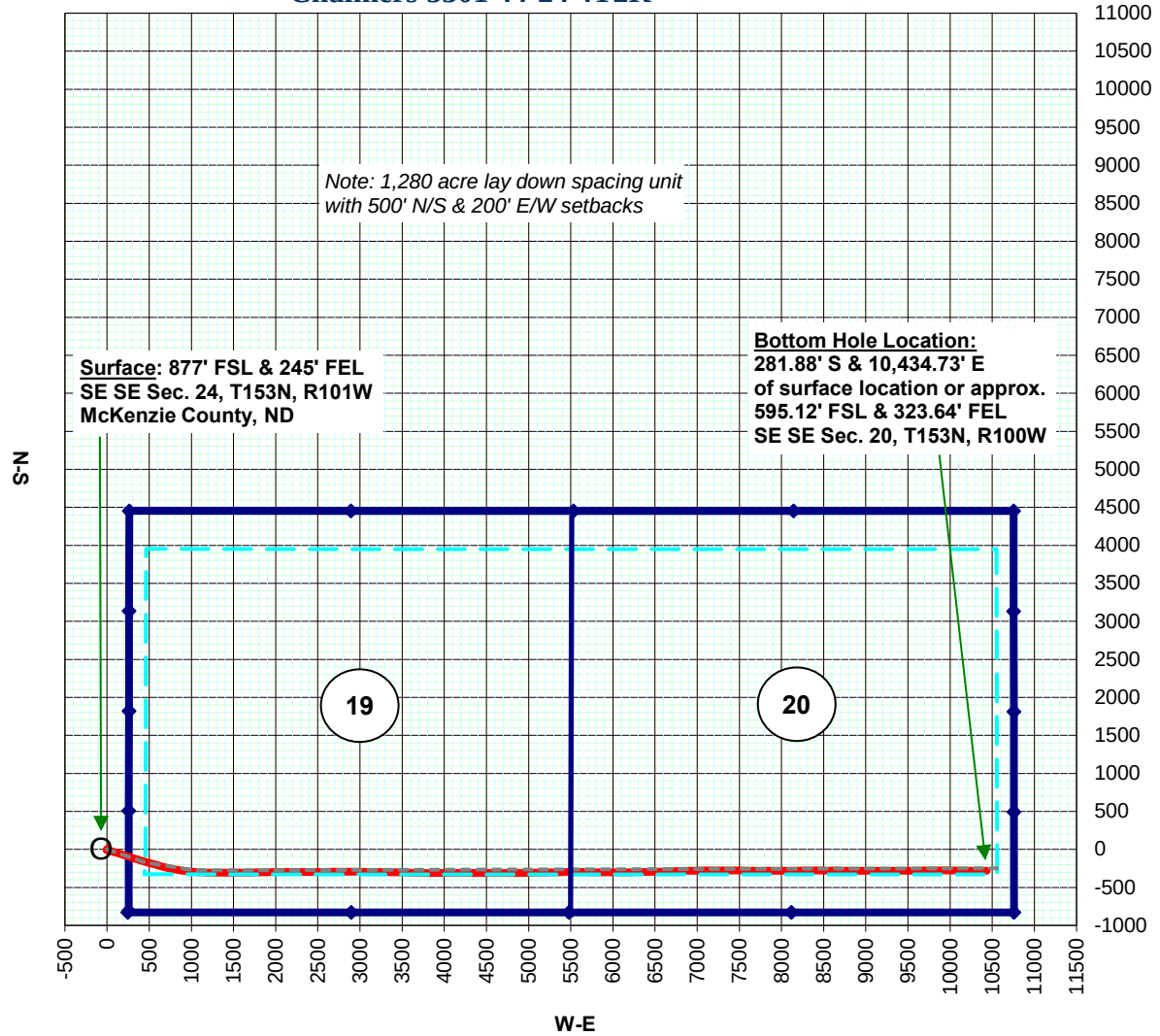
| Date<br>2014 | Mud<br>Depth | Mud<br>WT<br>(ppg)                          | VIS<br>(sec/<br>qt) | PV<br>(cP) | YP (lbs/<br>100 ft <sup>2</sup> ) | Gels<br>(lbs/<br>100 ft <sup>2</sup> ) | 600/ 300 | NAP/ H <sub>2</sub> O<br>(ratio) | NAP/ H <sub>2</sub> O<br>(% by vol) | Cake<br>(API/<br>HTHP) | Cor.<br>Solids<br>(%) | Oil/ H <sub>2</sub> O<br>(%) | Alk | pH  | Excess<br>Lime<br>(lb/bbl) | Cl <sup>-</sup><br>(mg/L) | LGS/ HGS<br>(%) | Salinity<br>(ppm) | ES  | Gain/<br>Loss<br>(bbls) |
|--------------|--------------|---------------------------------------------|---------------------|------------|-----------------------------------|----------------------------------------|----------|----------------------------------|-------------------------------------|------------------------|-----------------------|------------------------------|-----|-----|----------------------------|---------------------------|-----------------|-------------------|-----|-------------------------|
| 07/05        | 80'          | 10.9                                        | 60                  | 20         | 9                                 | 8/13/-                                 | 49/29    | 82.9/17.1                        | 68/14                               | -/3                    | 16.8                  | 68/14                        | 1.5 | -   | 1.9                        | 20k                       | 3.5/13          | 190564            | 620 | -/-                     |
| 07/06        | 1,046'       | 10.9                                        | 60                  | 20         | 9                                 | 8/13/-                                 | 49/29    | 82.9/17.1                        | 68/14                               | -/3                    | 16.8                  | 68/14                        | 1.5 | -   | 1.9                        | 20k                       | 3.5/13          | 190564            | 620 | -/-                     |
| 07/07        | 2,095'       | 10.9                                        | 60                  | 20         | 9                                 | 8/13/-                                 | 49/29    | 82.9/17.1                        | 68/14                               | -/3                    | 16.8                  | 68/14                        | 1.5 | -   | 1.9                        | 20k                       | 3.5/13          | 190564            | 620 | -/-                     |
| 07/08        | 2,095'       | 11.6                                        | 61                  | 21         | 9                                 | 9/14/-                                 | 51/30    | 82.5/17.5                        | 66/14                               | -/2                    | 18.8                  | 66/14                        | 1.5 | -   | 1.9                        | 20k                       | 3.5/15.3        | 190564            | 535 | -/27                    |
| 07/09        | 3,620'       | 11.45                                       | 78                  | 32         | 21                                | 20/25/-                                | 85/53    | 76.5/23.5                        | 62/19                               | -/3                    | 17.8                  | 62/19                        | 2.2 | -   | 2.8                        | 20k                       | 3.4/14.4        | 147829            | 560 | -/27                    |
| 07/10        | 5,620'       | 11.6                                        | 57                  | 27         | 15                                | 15/20/-                                | 69/42    | 78.8/21.                         | 63/17                               | -/3                    | 18.8                  | 63/17                        | 2   | -   | 2.6                        | 20k                       | 4.0/14.8        | 162397            | 580 | -/28                    |
| 07/11        | 5,709'       | 11.6                                        | 55                  | 23         | 14                                | 14/19/-                                | 60/37    | 78.8/21.2                        | 63/17                               | -/3                    | 18.8                  | 63/17                        | 2.1 | -   | 2.7                        | 20k                       | 3.5/15.3        | 162397            | 620 | -/48                    |
| 07/12        | 6,050'       | 11.8                                        | 63                  | 24         | 12                                | 13/18/-                                | 60/36    | 80/20                            | 64/16                               | -/3                    | 18.8                  | 64/16                        | 2   | -   | 2.6                        | 20k                       | 3.5/15.6        | 170813            | 625 | 110/49                  |
| 07/13        | 6,150'       | 9.8                                         | 59                  | 19         | 10                                | 10/15/-                                | 48/29    | 77/23                            | 67/20                               | -/3                    | 11.8                  | 67/20                        | 1.5 | -   | 1.9                        | 20k                       | 3.6/8.5         | 141484            | 500 | 120/-                   |
| 07/14        | 8,213'       | 9.9                                         | 50                  | 20         | 12                                | 11/16                                  | 52/32    | 79.1/20.9                        | 68/18                               | -/2                    | 12.8                  | 68/18                        | 1.9 | -   | 2.5                        | 20k                       | 4/8.5           | 154771            | 660 | 80/63                   |
| 07/15        | 8,106'       | 9.9                                         | 50                  | 20         | 12                                | 11/16                                  | 52/32    | 79.1/20.9                        | 68/18                               | -/2                    | 12.8                  | 68/18                        | 1.9 | -   | 2.5                        | 20k                       | 4/8.5           | 154771            | 660 | 80/63                   |
| 08/24        | 8,170'       | 9.9                                         | 50                  | 20         | 12                                | 11/16                                  | 52/32    | 79.1/20.9                        | 68/18                               | -/2                    | 12.8                  | 68/18                        | 1.9 | -   | 2.5                        | 20k                       | 4/8.5           | 154771            | 660 | 80/63                   |
| 08/25        | 8,312'       | 9.9                                         | 49                  | 18         | 9                                 | 4/7/-                                  | 45/27    | 79.5/20.5                        | 70/18                               | -/3                    | 10.2                  | 70/18                        | 2   | -   | 2.59                       | 30k                       | 3.7/6.5         | 256962            | 652 | -/57                    |
| 08/26        | 11,411'      | Change mud from diesel invert to salt water |                     |            |                                   |                                        |          |                                  |                                     |                        |                       |                              |     |     |                            |                           |                 |                   |     |                         |
| 08/27        | 11,411'      | 9.55                                        | 28                  | 1          | 1                                 | -                                      | 3/2      | -                                | 0/91.6                              | -                      | -                     | -                            | -   | 9   | -                          | 137k                      | 0.1/0           | -                 | -   | -                       |
| 08/29        | 13,317'      | 9.55                                        | 29                  | 1          | 1                                 | -                                      | -3/2     | -                                | 0/91.7                              | -                      | -                     | -                            | -   | 8.5 | -                          | 147k                      | 0.1/0           | -                 | -   | -                       |
| 08/31        | 15,890'      | 9.5                                         | 29                  | 1          | 1                                 | -                                      | 3/2      | -                                | 0/92                                | -                      | -                     | -                            | -   | 8   | -                          | 145k                      | 0.1/0           | -                 | -   | -                       |
| 09/02        | 17,380'      | 9.5                                         | 26                  | 1          | 1                                 | -                                      | 3/2      | -                                | 0/92                                | -                      | -                     | -                            | -   | 8   | -                          | 158k                      | 0.1/0           | -                 | -   | -                       |
| 09/04        | 19,830'      | 9.6                                         | 26                  | 1          | 1                                 | -                                      | 3/2      | -                                | 0/93                                | -                      | -                     | -                            | -   | 8.5 | -                          | 180k                      | 0.1/0           | -                 | -   | -                       |

# BOTTOM HOLE ASSEMBLY RECORD

| Bit Data |            |      |          |        |          |           |         |       |       |            |         |       | Motor Data |             |  | Reason For Removal |
|----------|------------|------|----------|--------|----------|-----------|---------|-------|-------|------------|---------|-------|------------|-------------|--|--------------------|
| Bit #    | Size (in.) | Type | Make     | Model  | Depth In | Depth Out | Footage | Hours | Σ hrs | Vert. Dev. | Make    | Bend  | Rev/Gal    |             |  |                    |
| 1        | 17 1/2     | PDC  | NOV      | -      | 80'      | 2,095'    | 2,015'  | 14    | 14    | Surface    | -       | -     | -          | TD surface  |  |                    |
| 2        | 12 1/4     | PDC  | Atlas    | SG16U  | 2,095'   | 5,672'    | 3,577'  | 30.5  | 44.5  | Vertical   | Bico    | 1.00° | 0.29       | Replacement |  |                    |
| 3        | 12 1/4     | PDC  | Atlas    | SG16U  | 5,672'   | 6,050'    | 378'    | 8.5   | 53    | Vertical   | Hunting | 2.38° | 0.29       | Replacement |  |                    |
| 4        | 8 3/4      | PDC  | Security | MM55D  | 6,050'   | 10,230'   | 4,180'  | 61.5  | 114.5 | Vertical   | NOV     | 2.50° | 0.29       | TD Vertical |  |                    |
| 5        | 8 3/4      | PDC  | Security | MMD55M | 10,230'  | 11,100'   | 870'    | 25.5  | 140   | Curve      | Baker   | 2.50° | 0.29       | TD curve    |  |                    |
| 6        | 6          | PDC  | Security | MMD64D | 11,100'  | 12,943'   | 1,843'  | 23    | 163   | Lateral    | Baker   | 1.50° | 0.49       | Lateral     |  |                    |
| 7        | 6          | PDC  | Baker    | T-406  | 12,943'  | 15,147'   | 2,204'  | 30.5  | 193.5 | Lateral    | Baker   | 1.50° | 0.49       | Lateral     |  |                    |
| 8        | 6          | PDC  | Security | MMD64D | 15,147'  | 17,295'   | 2,148'  | 38.5  | 232   | Lateral    | Baker   | 1.50° | 0.49       | Lateral     |  |                    |
| 9        | 6          | PDC  | Baker    | T307   | 17,295'  | 20,963'   | 3,668'  | 77    | 309   | Lateral    |         |       |            | TD Lateral  |  |                    |

## PLAN VIEW

Oasis Petroleum North America, LLC  
Chalmers 5301 44-24 4T2R

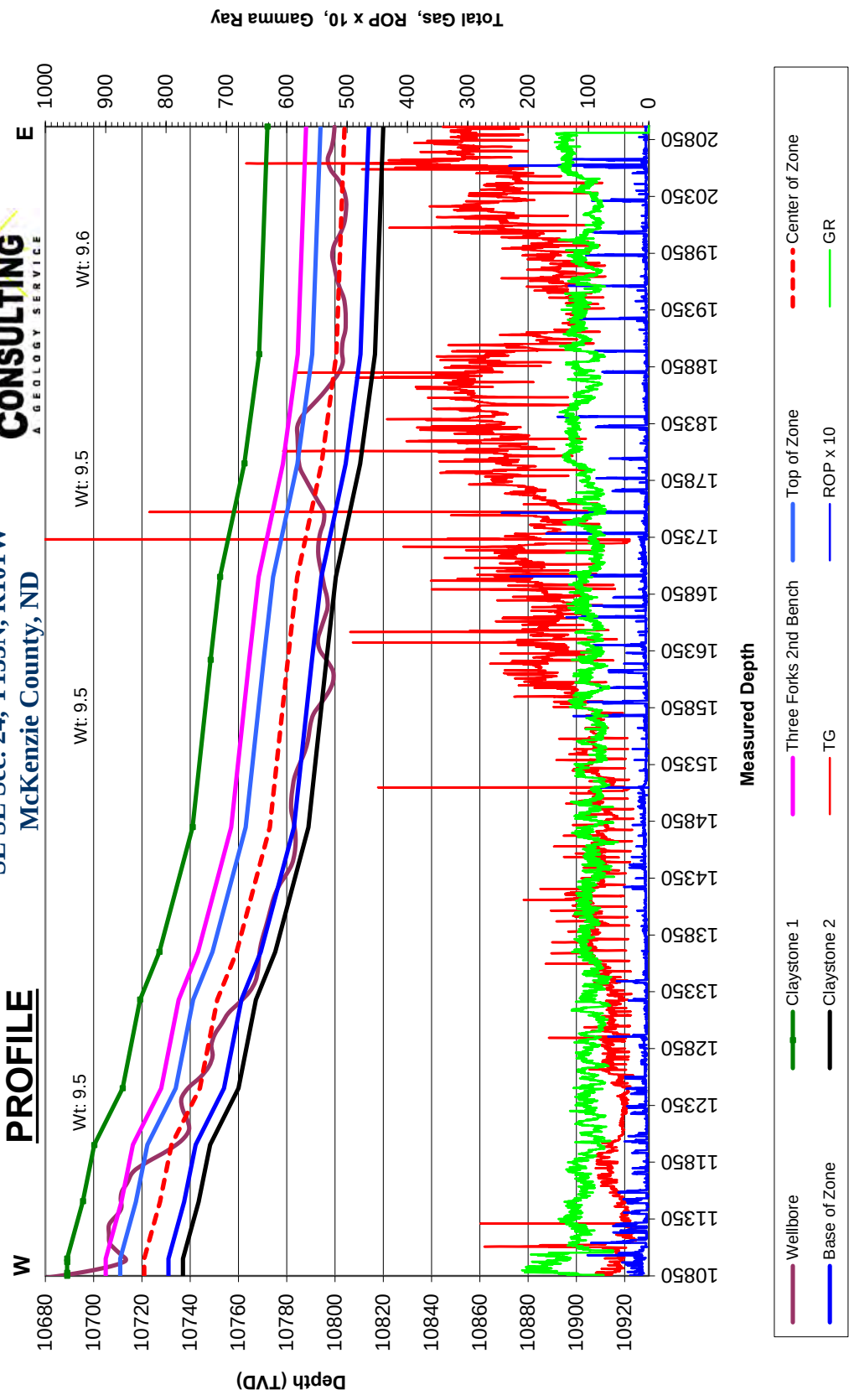






Oasis Petroleum North America, LLC  
Chalmers 5301 44-24 4T2R  
SE SE Sec. 24, T153N, R101W  
McKenzie County, ND

# PROFILE



# FORMATION MARKERS & DIP ESTIMATES

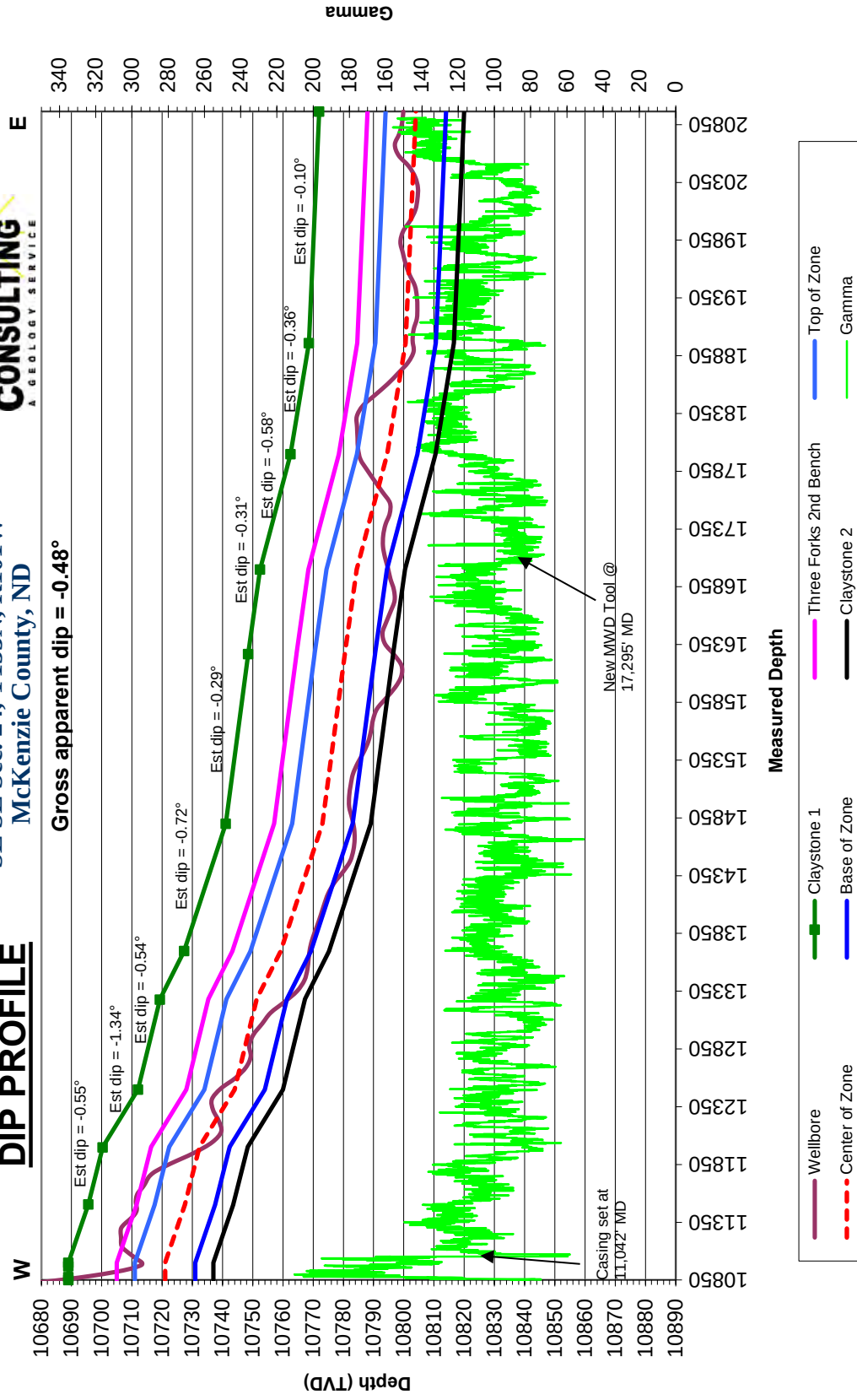
Oasis Petroleum North America, LLC - Chalmers 5301 44-24 4T2R

| Dip Change Points              | MD      | TVD       | TVD diff. | MD diff. | Dip          | Dipping up/down | Type of Marker |
|--------------------------------|---------|-----------|-----------|----------|--------------|-----------------|----------------|
| <b>Marker</b>                  |         |           |           |          |              |                 |                |
| Upper Claystone                | 10,970' | 10,689.00 |           |          |              |                 |                |
| Top of Target                  | 11,000' | 10,689.00 | 0.00      | 30.00    | <b>0.00</b>  | Flat            | Gamma          |
| Upper Claystone                | 11,507' | 10,695.60 | 6.60      | 507.00   | <b>-0.75</b> | Down            | Gamma          |
| Center of Target               | 12,000' | 10,700.30 | 4.70      | 493.00   | <b>-0.55</b> | Down            | Gamma          |
| High Gamma Marker              | 12,500' | 10,712.00 | 11.70     | 500.00   | <b>-1.34</b> | Down            | Gamma          |
| Base of Zone                   | 13,279' | 10,719.30 | 7.30      | 779.00   | <b>-0.54</b> | Down            | Gamma          |
| Base of Zone                   | 13,700' | 10,727.30 | 8.00      | 421.00   | <b>-1.09</b> | Down            | Gamma          |
| Base of Zone                   | 14,800' | 10,741.10 | 13.80     | 1100.00  | <b>-0.72</b> | Down            | Gamma          |
| Lower Claystone                | 16,270' | 10,748.50 | 7.40      | 1470.00  | <b>-0.29</b> | Down            | Gamma          |
| Lower Claystone                | 17,000' | 10,752.40 | 3.90      | 730.00   | <b>-0.31</b> | Down            | Gamma          |
| Top of Zone                    | 18,000' | 10,762.50 | 10.10     | 1000.00  | <b>-0.58</b> | Down            | Gamma          |
| Center of Zone                 | 18,960' | 10,768.60 | 6.10      | 960.00   | <b>-0.36</b> | Down            | Gamma          |
| Target Depth                   | 20,963' | 10,772.00 | 3.40      | 2003.00  | <b>-0.10</b> | Down            | Gamma          |
|                                |         |           |           |          |              |                 |                |
| <b>Gross Dip</b>               |         |           |           |          |              |                 |                |
| Initial Target Contact         | 11,000' | 10,689.00 |           |          |              |                 |                |
| Projected Final Target Contact | 20,963' | 10,772.00 | 83.00     | 9963.00  | <b>-0.48</b> | Down            | Projection     |
|                                |         |           |           |          |              |                 |                |

Oasis Petroleum North America, LLC  
 Chalmers 5301 44-24 4T2R  
 SE SE Sec. 24, T153N, R101W  
 McKenzie County, ND



# DIP PROFILE





&lt;

## SUNBURST CONSULTING, INC.

&gt;

|           |                                    |          |    |
|-----------|------------------------------------|----------|----|
| Operator: | Oasis Petroleum North America, LLC |          |    |
| Well :    | Chalmers 5301 44-24 4T2R           |          |    |
| County:   | McKenzie                           | State:   | ND |
| QQ:       | SE SE                              | Section: | 24 |
| Township: | 153                                | N/S:     | N  |
| Range:    | 101                                | E/W:     | W  |
| Footages: | 877                                | FN/SL:   | S  |
|           | 245                                | FE/WL:   | E  |

|                          |           |
|--------------------------|-----------|
| Kick-off:                | 7/17/2014 |
| Finish:                  | 9/5/2014  |
| Directional Supervision: | Ryan      |

Date: 9/15/2014

Time: 12:36

**F9 to re-calculate**

Minimum Curvature Method (SPE-3362)

Proposed dir: 91.34

[North and East are positive and South and West are negative, relative to surface location]

| No. | MD      | INC  | TRUE<br>AZM | TVD     | N-S   | E-W   | SECT  | DLS/<br>100 |
|-----|---------|------|-------------|---------|-------|-------|-------|-------------|
| Tie | 2031.00 | 0.20 | 159.90      | 2030.88 | 3.22  | 11.37 | 11.37 | 0.58        |
| 1   | 2097.00 | 0.30 | 174.40      | 2096.88 | 2.94  | 11.43 | 11.35 | 0.18        |
| 2   | 2190.00 | 0.40 | 184.10      | 2189.88 | 2.37  | 11.43 | 11.37 | 0.12        |
| 3   | 2284.00 | 0.70 | 186.50      | 2283.87 | 1.48  | 11.34 | 11.30 | 0.32        |
| 4   | 2377.00 | 1.10 | 183.30      | 2376.86 | 0.02  | 11.22 | 11.22 | 0.43        |
| 5   | 2470.00 | 1.40 | 197.20      | 2469.84 | -1.96 | 10.84 | 10.88 | 0.46        |
| 6   | 2564.00 | 1.30 | 220.50      | 2563.81 | -3.86 | 9.80  | 9.89  | 0.59        |
| 7   | 2657.00 | 0.70 | 221.80      | 2656.80 | -5.09 | 8.74  | 8.86  | 0.65        |
| 8   | 2750.00 | 0.70 | 235.90      | 2749.79 | -5.83 | 7.89  | 8.02  | 0.18        |
| 9   | 2843.00 | 0.80 | 219.10      | 2842.78 | -6.65 | 7.01  | 7.16  | 0.26        |
| 10  | 2937.00 | 0.40 | 202.10      | 2936.78 | -7.47 | 6.47  | 6.65  | 0.46        |
| 11  | 3030.00 | 0.10 | 180.50      | 3029.78 | -7.85 | 6.35  | 6.53  | 0.33        |
| 12  | 3123.00 | 0.30 | 36.00       | 3122.78 | -7.73 | 6.49  | 6.67  | 0.41        |
| 13  | 3217.00 | 0.40 | 35.40       | 3216.78 | -7.27 | 6.83  | 7.00  | 0.11        |
| 14  | 3310.00 | 0.40 | 40.90       | 3309.77 | -6.76 | 7.23  | 7.38  | 0.04        |
| 15  | 3403.00 | 0.40 | 38.70       | 3402.77 | -6.26 | 7.64  | 7.79  | 0.02        |
| 16  | 3496.00 | 0.40 | 43.10       | 3495.77 | -5.77 | 8.07  | 8.20  | 0.03        |
| 17  | 3590.00 | 0.60 | 49.80       | 3589.77 | -5.21 | 8.67  | 8.79  | 0.22        |
| 18  | 3683.00 | 0.50 | 48.30       | 3682.76 | -4.63 | 9.34  | 9.45  | 0.11        |
| 19  | 3776.00 | 0.60 | 59.30       | 3775.76 | -4.11 | 10.07 | 10.16 | 0.16        |
| 20  | 3869.00 | 0.60 | 79.70       | 3868.75 | -3.77 | 10.96 | 11.05 | 0.23        |
| 21  | 3963.00 | 0.60 | 96.00       | 3962.75 | -3.74 | 11.94 | 12.02 | 0.18        |
| 22  | 4056.00 | 0.10 | 341.80      | 4055.75 | -3.71 | 12.40 | 12.48 | 0.70        |
| 23  | 4149.00 | 0.40 | 284.50      | 4148.74 | -3.55 | 12.06 | 12.14 | 0.38        |
| 24  | 4243.00 | 0.40 | 262.90      | 4242.74 | -3.51 | 11.41 | 11.49 | 0.16        |
| 25  | 4336.00 | 0.40 | 261.10      | 4335.74 | -3.60 | 10.77 | 10.85 | 0.01        |
| 26  | 4429.00 | 0.60 | 262.60      | 4428.74 | -3.71 | 9.97  | 10.05 | 0.22        |
| 27  | 4522.00 | 0.60 | 261.50      | 4521.73 | -3.85 | 9.00  | 9.09  | 0.01        |
| 28  | 4616.00 | 0.60 | 267.80      | 4615.73 | -3.94 | 8.02  | 8.11  | 0.07        |
| 29  | 4709.00 | 0.70 | 270.00      | 4708.72 | -3.96 | 6.97  | 7.06  | 0.11        |
| 30  | 4802.00 | 0.80 | 264.60      | 4801.71 | -4.02 | 5.75  | 5.85  | 0.13        |
| 31  | 4895.00 | 0.70 | 248.80      | 4894.70 | -4.29 | 4.58  | 4.68  | 0.25        |
| 32  | 4989.00 | 0.70 | 245.30      | 4988.70 | -4.73 | 3.52  | 3.63  | 0.05        |
| 33  | 5082.00 | 0.70 | 239.10      | 5081.69 | -5.26 | 2.52  | 2.64  | 0.08        |
| 34  | 5175.00 | 0.70 | 238.50      | 5174.68 | -5.85 | 1.55  | 1.68  | 0.01        |

&lt;

**SUNBURST CONSULTING, INC.**

&gt;

|           |                                    |          |    |
|-----------|------------------------------------|----------|----|
| Operator: | Oasis Petroleum North America, LLC |          |    |
| Well :    | Chalmers 5301 44-24 4T2R           |          |    |
| County:   | McKenzie                           | State:   | ND |
| QQ:       | SE SE                              | Section: | 24 |
| Township: | 153                                | N/S:     | N  |
| Range:    | 101                                | E/W:     | W  |
| Footages: | 877                                | FN/SL:   | S  |
|           | 245                                | FE/WL:   | E  |

|                          |           |
|--------------------------|-----------|
| Kick-off:                | 7/17/2014 |
| Finish:                  | 9/5/2014  |
| Directional Supervision: | Ryan      |

Date: 9/15/2014

Time: 12:36

**F9 to re-calculate**

Minimum Curvature Method (SPE-3362)

Proposed dir: 91.34

[North and East are positive and South and West are negative, relative to surface location]

| No. | MD      | INC  | TRUE<br>AZM | TVD     | N-S   | E-W   | SECT  | DLS/<br>100 |
|-----|---------|------|-------------|---------|-------|-------|-------|-------------|
| 35  | 5269.00 | 0.60 | 229.80      | 5268.68 | -6.47 | 0.68  | 0.83  | 0.15        |
| 36  | 5362.00 | 0.70 | 241.40      | 5361.67 | -7.05 | -0.19 | -0.03 | 0.18        |
| 37  | 5455.00 | 0.50 | 239.30      | 5454.67 | -7.53 | -1.04 | -0.86 | 0.22        |
| 38  | 5548.00 | 0.20 | 140.70      | 5547.67 | -7.87 | -1.28 | -1.10 | 0.61        |
| 39  | 5642.00 | 0.40 | 148.40      | 5641.66 | -8.27 | -1.01 | -0.81 | 0.22        |
| 40  | 5735.00 | 0.10 | 216.80      | 5734.66 | -8.61 | -0.89 | -0.69 | 0.40        |
| 41  | 5828.00 | 0.10 | 223.00      | 5827.66 | -8.74 | -0.99 | -0.79 | 0.01        |
| 42  | 5922.00 | 0.30 | 1.40        | 5921.66 | -8.55 | -1.04 | -0.84 | 0.40        |
| 43  | 5996.00 | 0.20 | 332.70      | 5995.66 | -8.24 | -1.10 | -0.90 | 0.21        |
| 44  | 6094.00 | 0.10 | 256.00      | 6093.66 | -8.11 | -1.26 | -1.07 | 0.21        |
| 45  | 6187.00 | 0.40 | 183.20      | 6186.66 | -8.46 | -1.35 | -1.16 | 0.41        |
| 46  | 6281.00 | 0.90 | 70.30       | 6280.66 | -8.54 | -0.68 | -0.48 | 1.19        |
| 47  | 6374.00 | 0.50 | 45.80       | 6373.65 | -8.01 | 0.30  | 0.49  | 0.53        |
| 48  | 6467.00 | 0.40 | 132.30      | 6466.65 | -7.94 | 0.83  | 1.02  | 0.67        |
| 49  | 6561.00 | 0.10 | 127.10      | 6560.65 | -8.21 | 1.14  | 1.33  | 0.32        |
| 50  | 6654.00 | 0.70 | 40.30       | 6653.64 | -7.83 | 1.57  | 1.76  | 0.75        |
| 51  | 6747.00 | 0.50 | 38.20       | 6746.64 | -7.08 | 2.19  | 2.36  | 0.22        |
| 52  | 6840.00 | 0.70 | 72.90       | 6839.63 | -6.59 | 2.99  | 3.14  | 0.44        |
| 53  | 6934.00 | 0.70 | 107.60      | 6933.63 | -6.59 | 4.08  | 4.23  | 0.44        |
| 54  | 7027.00 | 0.90 | 11.30       | 7026.62 | -6.05 | 4.77  | 4.91  | 1.29        |
| 55  | 7120.00 | 1.20 | 13.30       | 7119.61 | -4.39 | 5.13  | 5.23  | 0.32        |
| 56  | 7213.00 | 0.60 | 8.30        | 7212.59 | -2.96 | 5.43  | 5.50  | 0.65        |
| 57  | 7307.00 | 0.60 | 26.00       | 7306.59 | -2.03 | 5.71  | 5.76  | 0.20        |
| 58  | 7400.00 | 0.70 | 27.30       | 7399.58 | -1.08 | 6.19  | 6.21  | 0.11        |
| 59  | 7493.00 | 0.40 | 353.30      | 7492.58 | -0.26 | 6.41  | 6.42  | 0.46        |
| 60  | 7587.00 | 0.40 | 7.60        | 7586.58 | 0.39  | 6.42  | 6.41  | 0.11        |
| 61  | 7680.00 | 0.70 | 52.50       | 7679.57 | 1.06  | 6.91  | 6.88  | 0.54        |
| 62  | 7773.00 | 0.70 | 45.20       | 7772.57 | 1.81  | 7.76  | 7.72  | 0.10        |
| 63  | 7866.00 | 1.00 | 26.40       | 7865.56 | 2.93  | 8.53  | 8.46  | 0.44        |
| 64  | 7960.00 | 1.10 | 316.20      | 7959.54 | 4.32  | 8.27  | 8.16  | 1.29        |
| 65  | 8053.00 | 0.80 | 304.20      | 8052.53 | 5.33  | 7.11  | 6.99  | 0.39        |
| 66  | 8146.00 | 0.90 | 263.40      | 8145.52 | 5.61  | 5.85  | 5.72  | 0.65        |
| 67  | 8240.00 | 0.90 | 231.20      | 8239.51 | 5.06  | 4.54  | 4.42  | 0.53        |
| 68  | 8333.00 | 0.90 | 232.10      | 8332.50 | 4.16  | 3.40  | 3.30  | 0.02        |
| 69  | 8426.00 | 1.00 | 211.30      | 8425.49 | 3.01  | 2.40  | 2.33  | 0.38        |

&lt;

## SUNBURST CONSULTING, INC.

&gt;

|           |                                    |          |    |
|-----------|------------------------------------|----------|----|
| Operator: | Oasis Petroleum North America, LLC |          |    |
| Well :    | Chalmers 5301 44-24 4T2R           |          |    |
| County:   | McKenzie                           | State:   | ND |
| QQ:       | SE SE                              | Section: | 24 |
| Township: | 153                                | N/S:     | N  |
| Range:    | 101                                | E/W:     | W  |
| Footages: | 877                                | FN/SL:   | S  |
|           | 245                                | FE/WL:   | E  |

|                          |           |
|--------------------------|-----------|
| Kick-off:                | 7/17/2014 |
| Finish:                  | 9/5/2014  |
| Directional Supervision: | Ryan      |

Date: 9/15/2014

Time: 12:36

**F9 to re-calculate**

Minimum Curvature Method (SPE-3362)

Proposed dir: 91.34

[North and East are positive and South and West are negative, relative to surface location]

| No. | MD       | INC   | TRUE<br>AZM | TVD      | N-S    | E-W    | SECT   | DLS/<br>100 |
|-----|----------|-------|-------------|----------|--------|--------|--------|-------------|
| 70  | 8519.00  | 1.10  | 207.30      | 8518.47  | 1.53   | 1.57   | 1.53   | 0.13        |
| 71  | 8613.00  | 1.30  | 225.20      | 8612.45  | -0.02  | 0.40   | 0.40   | 0.45        |
| 72  | 8706.00  | 0.90  | 191.80      | 8705.43  | -1.48  | -0.50  | -0.47  | 0.79        |
| 73  | 8799.00  | 0.70  | 177.80      | 8798.42  | -2.77  | -0.63  | -0.56  | 0.30        |
| 74  | 8892.00  | 0.90  | 158.70      | 8891.42  | -4.01  | -0.34  | -0.25  | 0.36        |
| 75  | 8986.00  | 1.10  | 174.20      | 8985.40  | -5.60  | 0.02   | 0.15   | 0.36        |
| 76  | 9079.00  | 0.20  | 108.40      | 9078.40  | -6.54  | 0.26   | 0.41   | 1.11        |
| 77  | 9172.00  | 0.70  | 94.80       | 9171.39  | -6.64  | 0.98   | 1.14   | 0.55        |
| 78  | 9266.00  | 0.40  | 78.60       | 9265.39  | -6.62  | 1.88   | 2.03   | 0.36        |
| 79  | 9359.00  | 0.40  | 78.80       | 9358.39  | -6.49  | 2.51   | 2.66   | 0.00        |
| 80  | 9452.00  | 0.50  | 88.40       | 9451.38  | -6.42  | 3.24   | 3.39   | 0.13        |
| 81  | 9545.00  | 0.50  | 103.10      | 9544.38  | -6.50  | 4.04   | 4.19   | 0.14        |
| 82  | 9639.00  | 0.10  | 172.10      | 9638.38  | -6.67  | 4.45   | 4.60   | 0.50        |
| 83  | 9732.00  | 0.50  | 141.90      | 9731.38  | -7.07  | 4.71   | 4.87   | 0.45        |
| 84  | 9825.00  | 0.60  | 71.70       | 9824.37  | -7.24  | 5.42   | 5.59   | 0.69        |
| 85  | 9919.00  | 0.20  | 56.50       | 9918.37  | -7.00  | 6.03   | 6.19   | 0.44        |
| 86  | 10012.00 | 0.20  | 264.90      | 10011.37 | -6.92  | 6.00   | 6.16   | 0.42        |
| 87  | 10105.00 | 0.80  | 197.80      | 10104.37 | -7.55  | 5.64   | 5.82   | 0.80        |
| 88  | 10175.00 | 0.60  | 203.40      | 10174.36 | -8.35  | 5.35   | 5.54   | 0.30        |
| 89  | 10202.00 | 0.70  | 220.60      | 10201.36 | -8.61  | 5.18   | 5.38   | 0.81        |
| 90  | 10233.00 | 2.50  | 125.40      | 10232.35 | -9.14  | 5.61   | 5.82   | 8.57        |
| 91  | 10265.00 | 7.00  | 115.60      | 10264.23 | -10.39 | 7.94   | 8.18   | 14.24       |
| 92  | 10296.00 | 11.30 | 112.60      | 10294.83 | -12.38 | 12.45  | 12.73  | 13.95       |
| 93  | 10327.00 | 16.20 | 111.30      | 10324.93 | -15.12 | 19.28  | 19.63  | 15.84       |
| 94  | 10358.00 | 20.00 | 109.00      | 10354.39 | -18.41 | 28.33  | 28.75  | 12.47       |
| 95  | 10389.00 | 22.60 | 107.50      | 10383.27 | -21.93 | 39.02  | 39.53  | 8.57        |
| 96  | 10420.00 | 25.70 | 106.50      | 10411.56 | -25.63 | 51.15  | 51.74  | 10.09       |
| 97  | 10451.00 | 29.60 | 106.50      | 10439.01 | -29.72 | 64.94  | 65.62  | 12.58       |
| 98  | 10482.00 | 33.40 | 106.00      | 10465.44 | -34.25 | 80.49  | 81.27  | 12.29       |
| 99  | 10513.00 | 37.40 | 106.80      | 10490.70 | -39.32 | 97.71  | 98.61  | 12.99       |
| 100 | 10544.00 | 41.00 | 107.60      | 10514.72 | -45.12 | 116.43 | 117.45 | 11.73       |
| 101 | 10575.00 | 43.70 | 108.60      | 10537.63 | -51.61 | 136.27 | 137.44 | 8.98        |
| 102 | 10607.00 | 46.70 | 109.60      | 10560.18 | -59.04 | 157.72 | 159.06 | 9.63        |
| 103 | 10638.00 | 50.40 | 110.50      | 10580.69 | -67.01 | 179.55 | 181.06 | 12.13       |
| 104 | 10669.00 | 54.20 | 110.30      | 10599.65 | -75.56 | 202.53 | 204.24 | 12.27       |



&lt;

## SUNBURST CONSULTING, INC.

&gt;

|           |                                    |          |    |
|-----------|------------------------------------|----------|----|
| Operator: | Oasis Petroleum North America, LLC |          |    |
| Well :    | Chalmers 5301 44-24 4T2R           |          |    |
| County:   | McKenzie                           | State:   | ND |
| QQ:       | SE SE                              | Section: | 24 |
| Township: | 153                                | N/S:     | N  |
| Range:    | 101                                | E/W:     | W  |
| Footages: | 877                                | FN/SL:   | S  |
|           | 245                                | FE/WL:   | E  |

|                          |           |
|--------------------------|-----------|
| Kick-off:                | 7/17/2014 |
| Finish:                  | 9/5/2014  |
| Directional Supervision: | Ryan      |

Date: 9/15/2014

Time: 12:36

**F9 to re-calculate**

Proposed dir: 91.34

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

| No. | MD       | INC   | TRUE<br>AZM | TVD      | N-S     | E-W     | SECT    | DLS/<br>100 |
|-----|----------|-------|-------------|----------|---------|---------|---------|-------------|
| 105 | 10700.00 | 56.10 | 110.30      | 10617.36 | -84.39  | 226.39  | 228.30  | 6.13        |
| 106 | 10731.00 | 57.10 | 110.20      | 10634.43 | -93.34  | 250.67  | 252.79  | 3.24        |
| 107 | 10762.00 | 60.30 | 110.30      | 10650.53 | -102.51 | 275.52  | 277.84  | 10.33       |
| 108 | 10793.00 | 64.60 | 109.80      | 10664.86 | -111.93 | 301.33  | 303.87  | 13.94       |
| 109 | 10824.00 | 68.70 | 109.20      | 10677.15 | -121.43 | 328.16  | 330.91  | 13.34       |
| 110 | 10855.00 | 71.90 | 108.50      | 10687.60 | -130.85 | 355.77  | 358.74  | 10.54       |
| 111 | 10886.00 | 74.90 | 108.40      | 10696.45 | -140.25 | 383.95  | 387.13  | 9.68        |
| 112 | 10917.00 | 77.80 | 108.40      | 10703.77 | -149.76 | 412.53  | 415.92  | 9.35        |
| 113 | 10948.00 | 80.70 | 108.30      | 10709.55 | -159.35 | 441.44  | 445.04  | 9.36        |
| 114 | 10980.00 | 86.40 | 107.90      | 10713.14 | -169.22 | 471.65  | 475.48  | 17.86       |
| 115 | 11001.00 | 92.70 | 107.10      | 10713.31 | -175.53 | 491.66  | 495.64  | 30.24       |
| 116 | 11042.00 | 93.50 | 107.20      | 10711.09 | -187.61 | 530.78  | 535.03  | 1.97        |
| 117 | 11116.00 | 92.20 | 107.10      | 10707.41 | -209.40 | 601.40  | 606.14  | 1.76        |
| 118 | 11146.00 | 91.00 | 106.20      | 10706.57 | -217.99 | 630.14  | 635.06  | 5.00        |
| 119 | 11177.00 | 89.70 | 106.40      | 10706.38 | -226.69 | 659.89  | 665.01  | 4.24        |
| 120 | 11208.00 | 90.20 | 106.20      | 10706.41 | -235.39 | 689.64  | 694.96  | 1.74        |
| 121 | 11239.00 | 90.10 | 105.00      | 10706.33 | -243.73 | 719.50  | 725.00  | 3.88        |
| 122 | 11270.00 | 90.20 | 103.10      | 10706.25 | -251.25 | 749.57  | 755.24  | 6.14        |
| 123 | 11301.00 | 89.00 | 101.60      | 10706.47 | -257.88 | 779.85  | 785.67  | 6.20        |
| 124 | 11332.00 | 87.50 | 100.60      | 10707.41 | -263.85 | 810.26  | 816.20  | 5.81        |
| 125 | 11363.00 | 87.50 | 100.59      | 10708.76 | -269.54 | 840.70  | 846.77  | 0.03        |
| 126 | 11394.00 | 87.70 | 100.80      | 10710.06 | -275.29 | 871.13  | 877.33  | 0.94        |
| 127 | 11424.00 | 88.40 | 98.40       | 10711.08 | -280.29 | 900.69  | 907.00  | 8.33        |
| 128 | 11455.00 | 89.70 | 98.10       | 10711.60 | -284.74 | 931.37  | 937.77  | 4.30        |
| 129 | 11486.00 | 90.50 | 97.90       | 10711.54 | -289.05 | 962.07  | 968.56  | 2.66        |
| 130 | 11516.00 | 90.50 | 95.20       | 10711.28 | -292.47 | 991.87  | 998.43  | 9.00        |
| 131 | 11547.00 | 89.40 | 95.40       | 10711.31 | -295.34 | 1022.73 | 1029.36 | 3.61        |
| 132 | 11578.00 | 89.70 | 94.70       | 10711.55 | -298.07 | 1053.61 | 1060.29 | 2.46        |
| 133 | 11609.00 | 88.20 | 92.70       | 10712.12 | -300.07 | 1084.54 | 1091.26 | 8.06        |
| 134 | 11640.00 | 88.00 | 92.40       | 10713.15 | -301.44 | 1115.49 | 1122.24 | 1.16        |
| 135 | 11671.00 | 88.70 | 92.60       | 10714.04 | -302.80 | 1146.45 | 1153.22 | 2.35        |
| 136 | 11702.00 | 88.80 | 92.20       | 10714.72 | -304.09 | 1177.41 | 1184.20 | 1.33        |
| 137 | 11732.00 | 88.90 | 92.30       | 10715.32 | -305.27 | 1207.38 | 1214.19 | 0.47        |
| 138 | 11763.00 | 87.30 | 90.60       | 10716.35 | -306.06 | 1238.36 | 1245.17 | 7.53        |
| 139 | 11794.00 | 86.60 | 90.30       | 10718.00 | -306.30 | 1269.31 | 1276.13 | 2.46        |

&lt;

## SUNBURST CONSULTING, INC.

&gt;

|           |                                    |          |    |
|-----------|------------------------------------|----------|----|
| Operator: | Oasis Petroleum North America, LLC |          |    |
| Well :    | Chalmers 5301 44-24 4T2R           |          |    |
| County:   | McKenzie                           | State:   | ND |
| QQ:       | SE SE                              | Section: | 24 |
| Township: | 153                                | N/S:     | N  |
| Range:    | 101                                | E/W:     | W  |
| Footages: | 877                                | FN/SL:   | S  |
|           | 245                                | FE/WL:   | E  |

|                          |           |
|--------------------------|-----------|
| Kick-off:                | 7/17/2014 |
| Finish:                  | 9/5/2014  |
| Directional Supervision: | Ryan      |

Date: 9/15/2014

Time: 12:36

**F9 to re-calculate**

Proposed dir: 91.34

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

| No. | MD       | INC   | TRUE<br>AZM | TVD      | N-S     | E-W     | SECT    | DLS/<br>100 |
|-----|----------|-------|-------------|----------|---------|---------|---------|-------------|
| 140 | 11857.00 | 84.80 | 90.40       | 10722.72 | -306.68 | 1332.13 | 1338.94 | 2.86        |
| 141 | 11950.00 | 85.60 | 89.80       | 10730.50 | -306.84 | 1424.80 | 1431.59 | 1.07        |
| 142 | 12043.00 | 86.90 | 89.60       | 10736.58 | -306.36 | 1517.60 | 1524.35 | 1.41        |
| 143 | 12135.00 | 89.40 | 89.50       | 10739.55 | -305.64 | 1609.54 | 1616.25 | 2.72        |
| 144 | 12228.00 | 91.60 | 89.00       | 10738.74 | -304.42 | 1702.52 | 1709.18 | 2.43        |
| 145 | 12321.00 | 90.70 | 88.60       | 10736.88 | -302.47 | 1795.48 | 1802.07 | 1.06        |
| 146 | 12414.00 | 89.80 | 88.60       | 10736.47 | -300.20 | 1888.45 | 1894.96 | 0.97        |
| 147 | 12507.00 | 86.60 | 88.80       | 10739.39 | -298.09 | 1981.37 | 1987.80 | 3.45        |
| 148 | 12599.00 | 87.30 | 89.20       | 10744.29 | -296.49 | 2073.23 | 2079.59 | 0.88        |
| 149 | 12692.00 | 88.40 | 90.00       | 10747.77 | -295.84 | 2166.16 | 2172.48 | 1.46        |
| 150 | 12786.00 | 89.70 | 90.50       | 10749.33 | -296.25 | 2260.14 | 2266.45 | 1.48        |
| 151 | 12880.00 | 90.90 | 91.10       | 10748.84 | -297.56 | 2354.13 | 2360.44 | 1.43        |
| 152 | 12975.00 | 87.40 | 89.20       | 10750.25 | -297.81 | 2449.10 | 2455.39 | 4.19        |
| 153 | 13069.00 | 89.10 | 89.50       | 10753.12 | -296.75 | 2543.04 | 2549.29 | 1.84        |
| 154 | 13163.00 | 87.50 | 88.50       | 10755.91 | -295.11 | 2636.98 | 2643.16 | 2.01        |
| 155 | 13257.00 | 86.90 | 89.40       | 10760.50 | -293.39 | 2730.86 | 2736.97 | 1.15        |
| 156 | 13350.00 | 88.00 | 89.70       | 10764.64 | -292.66 | 2823.76 | 2829.83 | 1.23        |
| 157 | 13444.00 | 89.00 | 90.30       | 10767.10 | -292.66 | 2917.73 | 2923.77 | 1.24        |
| 158 | 13538.00 | 89.80 | 91.20       | 10768.08 | -293.89 | 3011.71 | 3017.76 | 1.28        |
| 159 | 13631.00 | 89.70 | 91.10       | 10768.49 | -295.75 | 3104.69 | 3110.76 | 0.15        |
| 160 | 13726.00 | 89.70 | 90.80       | 10768.99 | -297.33 | 3199.68 | 3205.75 | 0.32        |
| 161 | 13819.00 | 89.10 | 90.70       | 10769.96 | -298.54 | 3292.66 | 3298.74 | 0.65        |
| 162 | 13913.00 | 89.30 | 91.80       | 10771.27 | -300.60 | 3386.63 | 3392.73 | 1.19        |
| 163 | 14007.00 | 89.20 | 91.50       | 10772.50 | -303.30 | 3480.58 | 3486.72 | 0.34        |
| 164 | 14101.00 | 89.40 | 91.40       | 10773.65 | -305.68 | 3574.55 | 3580.72 | 0.24        |
| 165 | 14195.00 | 89.00 | 91.20       | 10774.96 | -307.81 | 3668.51 | 3674.71 | 0.48        |
| 166 | 14289.00 | 88.50 | 89.80       | 10777.02 | -308.63 | 3762.48 | 3768.67 | 1.58        |
| 167 | 14383.00 | 88.00 | 89.70       | 10779.89 | -308.22 | 3856.44 | 3862.59 | 0.54        |
| 168 | 14476.00 | 89.10 | 90.20       | 10782.24 | -308.14 | 3949.41 | 3955.53 | 1.30        |
| 169 | 14571.00 | 89.60 | 90.30       | 10783.32 | -308.56 | 4044.40 | 4050.51 | 0.54        |
| 170 | 14665.00 | 89.90 | 90.00       | 10783.73 | -308.80 | 4138.40 | 4144.49 | 0.45        |
| 171 | 14758.00 | 90.30 | 90.20       | 10783.56 | -308.96 | 4231.40 | 4237.47 | 0.48        |
| 172 | 14853.00 | 90.60 | 90.40       | 10782.82 | -309.46 | 4326.39 | 4332.45 | 0.38        |
| 173 | 14946.00 | 90.50 | 90.40       | 10781.93 | -310.11 | 4419.39 | 4425.43 | 0.11        |
| 174 | 15040.00 | 89.50 | 89.80       | 10781.93 | -310.28 | 4513.38 | 4519.41 | 1.24        |

&lt;

**SUNBURST CONSULTING, INC.**

&gt;

|           |                                    |          |    |
|-----------|------------------------------------|----------|----|
| Operator: | Oasis Petroleum North America, LLC |          |    |
| Well :    | Chalmers 5301 44-24 4T2R           |          |    |
| County:   | McKenzie                           | State:   | ND |
| QQ:       | SE SE                              | Section: | 24 |
| Township: | 153                                | N/S:     | N  |
| Range:    | 101                                | E/W:     | W  |
| Footages: | 877                                | FN/SL:   | S  |
|           | 245                                | FE/WL:   | E  |

|                          |           |
|--------------------------|-----------|
| Kick-off:                | 7/17/2014 |
| Finish:                  | 9/5/2014  |
| Directional Supervision: | Ryan      |

Date: 9/15/2014

Time: 12:36

**F9 to re-calculate**

Proposed dir: 91.34

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

| No. | MD       | INC   | TRUE<br>AZM | TVD      | N-S     | E-W     | SECT    | DLS/<br>100 |
|-----|----------|-------|-------------|----------|---------|---------|---------|-------------|
| 175 | 15135.00 | 89.90 | 90.40       | 10782.42 | -310.44 | 4608.38 | 4614.38 | 0.76        |
| 176 | 15229.00 | 89.20 | 90.30       | 10783.16 | -311.02 | 4702.38 | 4708.36 | 0.75        |
| 177 | 15323.00 | 88.90 | 89.90       | 10784.72 | -311.18 | 4796.36 | 4802.33 | 0.53        |
| 178 | 15417.00 | 88.70 | 89.10       | 10786.69 | -310.36 | 4890.34 | 4896.26 | 0.88        |
| 179 | 15512.00 | 89.40 | 89.00       | 10788.26 | -308.78 | 4985.31 | 4991.17 | 0.74        |
| 180 | 15606.00 | 89.50 | 88.60       | 10789.17 | -306.82 | 5079.29 | 5085.07 | 0.44        |
| 181 | 15700.00 | 89.70 | 88.40       | 10789.82 | -304.36 | 5173.25 | 5178.96 | 0.30        |
| 182 | 15793.00 | 88.70 | 89.10       | 10791.12 | -302.33 | 5266.22 | 5271.85 | 1.31        |
| 183 | 15887.00 | 87.60 | 88.70       | 10794.16 | -300.52 | 5360.15 | 5365.71 | 1.25        |
| 184 | 15981.00 | 88.70 | 89.70       | 10797.19 | -299.21 | 5454.09 | 5459.60 | 1.58        |
| 185 | 16075.00 | 88.90 | 89.70       | 10799.16 | -298.72 | 5548.07 | 5553.54 | 0.21        |
| 186 | 16169.00 | 91.00 | 90.90       | 10799.24 | -299.21 | 5642.06 | 5647.51 | 2.57        |
| 187 | 16263.00 | 92.00 | 90.20       | 10796.78 | -300.11 | 5736.02 | 5741.47 | 1.30        |
| 188 | 16356.00 | 91.70 | 90.40       | 10793.78 | -300.60 | 5828.97 | 5834.41 | 0.39        |
| 189 | 16450.00 | 89.30 | 90.70       | 10792.96 | -301.50 | 5922.96 | 5928.39 | 2.57        |
| 190 | 16544.00 | 89.20 | 89.60       | 10794.19 | -301.75 | 6016.95 | 6022.36 | 1.17        |
| 191 | 16638.00 | 89.00 | 89.60       | 10795.66 | -301.09 | 6110.93 | 6116.30 | 0.21        |
| 192 | 16731.00 | 89.40 | 88.90       | 10796.96 | -299.88 | 6203.92 | 6209.23 | 0.87        |
| 193 | 16825.00 | 90.90 | 88.90       | 10796.72 | -298.07 | 6297.90 | 6303.14 | 1.60        |
| 194 | 16919.00 | 90.50 | 88.60       | 10795.57 | -296.02 | 6391.87 | 6397.04 | 0.53        |
| 195 | 17012.00 | 90.50 | 89.00       | 10794.76 | -294.07 | 6484.84 | 6489.95 | 0.43        |
| 196 | 17106.00 | 90.70 | 89.00       | 10793.77 | -292.43 | 6578.82 | 6583.86 | 0.21        |
| 197 | 17200.00 | 90.10 | 88.50       | 10793.12 | -290.38 | 6672.80 | 6677.76 | 0.83        |
| 198 | 17295.00 | 89.80 | 87.50       | 10793.20 | -287.07 | 6767.74 | 6772.60 | 1.10        |
| 199 | 17389.00 | 89.30 | 87.50       | 10793.94 | -282.97 | 6861.64 | 6866.39 | 0.53        |
| 200 | 17483.00 | 89.00 | 88.30       | 10795.33 | -279.52 | 6955.57 | 6960.21 | 0.91        |
| 201 | 17576.00 | 90.90 | 89.80       | 10795.41 | -277.98 | 7048.55 | 7053.12 | 2.60        |
| 202 | 17670.00 | 92.10 | 89.70       | 10792.95 | -277.57 | 7142.52 | 7147.05 | 1.28        |
| 203 | 17764.00 | 91.00 | 90.50       | 10790.41 | -277.73 | 7236.48 | 7241.00 | 1.45        |
| 204 | 17858.00 | 92.00 | 90.20       | 10787.95 | -278.31 | 7330.44 | 7334.95 | 1.11        |
| 205 | 17951.00 | 90.80 | 89.90       | 10785.68 | -278.39 | 7423.41 | 7427.89 | 1.33        |
| 206 | 18045.00 | 90.20 | 90.90       | 10784.86 | -279.05 | 7517.41 | 7521.88 | 1.24        |
| 207 | 18138.00 | 90.00 | 90.30       | 10784.69 | -280.02 | 7610.40 | 7614.87 | 0.68        |
| 208 | 18231.00 | 90.20 | 89.90       | 10784.53 | -280.18 | 7703.40 | 7707.85 | 0.48        |
| 209 | 18325.00 | 89.90 | 89.60       | 10784.45 | -279.77 | 7797.40 | 7801.81 | 0.45        |

&lt;

**SUNBURST CONSULTING, INC.**

&gt;

|           |                                    |          |    |
|-----------|------------------------------------|----------|----|
| Operator: | Oasis Petroleum North America, LLC |          |    |
| Well :    | Chalmers 5301 44-24 4T2R           |          |    |
| County:   | McKenzie                           | State:   | ND |
| QQ:       | SE SE                              | Section: | 24 |
| Township: | 153                                | N/S:     | N  |
| Range:    | 101                                | E/W:     | W  |
| Footages: | 877                                | FN/SL:   | S  |
|           | 245                                | FE/WL:   | E  |

|                          |           |
|--------------------------|-----------|
| Kick-off:                | 7/17/2014 |
| Finish:                  | 9/5/2014  |
| Directional Supervision: | Ryan      |

Date: 9/15/2014

Time: 12:36

**F9 to re-calculate**Proposed dir: **91.34**

Minimum Curvature Method (SPE-3362)

[North and East are positive and South and West are negative, relative to surface location]

| No. | MD       | INC   | TRUE<br>AZM | TVD      | N-S     | E-W      | SECT     | DLS/<br>100 |
|-----|----------|-------|-------------|----------|---------|----------|----------|-------------|
| 210 | 18418.00 | 87.60 | 90.40       | 10786.48 | -279.77 | 7890.37  | 7894.76  | 2.62        |
| 211 | 18512.00 | 87.40 | 89.90       | 10790.58 | -280.02 | 7984.28  | 7988.65  | 0.57        |
| 212 | 18606.00 | 87.60 | 90.10       | 10794.68 | -280.02 | 8078.19  | 8082.53  | 0.30        |
| 213 | 18699.00 | 88.10 | 89.50       | 10798.17 | -279.69 | 8171.12  | 8175.43  | 0.84        |
| 214 | 18793.00 | 87.90 | 89.50       | 10801.45 | -278.87 | 8265.06  | 8269.32  | 0.21        |
| 215 | 18887.00 | 89.90 | 90.10       | 10803.25 | -278.55 | 8359.04  | 8363.27  | 2.22        |
| 216 | 18982.00 | 90.50 | 90.40       | 10802.92 | -278.96 | 8454.04  | 8458.25  | 0.71        |
| 217 | 19076.00 | 89.10 | 90.00       | 10803.25 | -279.29 | 8548.03  | 8552.23  | 1.55        |
| 218 | 19171.00 | 89.60 | 89.70       | 10804.33 | -279.04 | 8643.03  | 8647.19  | 0.61        |
| 219 | 19266.00 | 90.20 | 90.60       | 10804.49 | -279.29 | 8738.03  | 8742.17  | 1.14        |
| 220 | 19361.00 | 90.00 | 90.50       | 10804.33 | -280.20 | 8833.02  | 8837.16  | 0.24        |
| 221 | 19455.00 | 90.80 | 90.50       | 10803.67 | -281.02 | 8927.01  | 8931.14  | 0.85        |
| 222 | 19550.00 | 91.30 | 90.50       | 10801.93 | -281.85 | 9021.99  | 9026.12  | 0.53        |
| 223 | 19644.00 | 90.30 | 90.00       | 10800.62 | -282.26 | 9115.98  | 9120.09  | 1.19        |
| 224 | 19739.00 | 90.80 | 89.70       | 10799.71 | -282.01 | 9210.98  | 9215.05  | 0.61        |
| 225 | 19833.00 | 90.20 | 90.40       | 10798.89 | -282.09 | 9304.97  | 9309.03  | 0.98        |
| 226 | 19927.00 | 88.70 | 89.80       | 10799.79 | -282.26 | 9398.97  | 9403.00  | 1.72        |
| 227 | 20020.00 | 88.40 | 88.70       | 10802.14 | -281.04 | 9491.93  | 9495.90  | 1.23        |
| 228 | 20114.00 | 89.50 | 89.50       | 10803.86 | -279.56 | 9585.90  | 9589.81  | 1.45        |
| 229 | 20208.00 | 89.70 | 89.00       | 10804.52 | -278.33 | 9679.89  | 9683.75  | 0.57        |
| 230 | 20302.00 | 90.10 | 89.70       | 10804.68 | -277.27 | 9773.88  | 9777.69  | 0.86        |
| 231 | 20396.00 | 91.00 | 90.80       | 10803.78 | -277.68 | 9867.87  | 9871.67  | 1.51        |
| 232 | 20490.00 | 91.60 | 90.40       | 10801.65 | -278.66 | 9961.84  | 9965.63  | 0.77        |
| 233 | 20584.00 | 92.50 | 90.00       | 10798.29 | -278.99 | 10055.78 | 10059.55 | 1.05        |
| 234 | 20678.00 | 89.00 | 90.70       | 10797.06 | -279.56 | 10149.75 | 10153.52 | 3.80        |
| 235 | 20772.00 | 89.20 | 90.50       | 10798.53 | -280.55 | 10243.74 | 10247.50 | 0.30        |
| 236 | 20866.00 | 89.70 | 90.30       | 10799.44 | -281.20 | 10337.73 | 10341.48 | 0.57        |
| 237 | 20963.00 | 89.80 | 90.50       | 10799.86 | -281.88 | 10434.73 | 10438.47 | 0.23        |



# FORMATION TOPS & STRUCTURAL RELATIONSHIPS

| Operator:<br>Well Name:<br>Location: | Subject Well:                                                                                                                                             |                         |                                |                                 |                |                       |                        |                 |      |     |     | Offset Wells:                        |                                      |                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------|---------------------------------|----------------|-----------------------|------------------------|-----------------|------|-----|-----|--------------------------------------|--------------------------------------|-----------------------------------|
|                                      | Oasis Petroleum North America, LLC<br>Chalmers 5301 44-24 4T2R<br>877' FSL & 245' FEL<br>SE SE Section 24, T153N, R101W<br>GL: 1,943' Sub: 25' KB: 1,968' |                         |                                |                                 |                |                       |                        |                 |      |     |     |                                      |                                      |                                   |
|                                      | Elevation:                                                                                                                                                |                         |                                |                                 |                |                       |                        |                 |      |     |     | Dip To<br>Chalmers 5301<br>44-24 3BR | Dip To<br>Chalmers 5301<br>44-24 2TR | Dip To<br>Chalmers<br>5300 31-19H |
| Formation/<br>Marker                 | Prog.<br>Top                                                                                                                                              | Prog.<br>Datum<br>(MSL) | Driller's<br>Depth Top<br>(MD) | Driller's<br>Depth Top<br>(TVD) | Datum<br>(MSL) | Interval<br>Thickness | Thickness<br>to Target | Dip To<br>Prog. |      |     |     |                                      |                                      |                                   |
| Kibbey Lime                          | 8,269'                                                                                                                                                    | -6,301'                 | 8,267'                         | 8,266'                          | -6,298'        | 172'                  | 2,443'                 | 3'              | 2'   | -2' | 13' |                                      |                                      |                                   |
| Charles                              | 8,433'                                                                                                                                                    | -6,465'                 | 8,439'                         | 8,438'                          | -6,470'        | 545'                  | 2,271'                 | -5'             | 1'   | -9' | 5'  |                                      |                                      |                                   |
| UB                                   | 8,982'                                                                                                                                                    | -7,014'                 | 8,982'                         | 8,983'                          | -7,015'        | 84'                   | 1,726'                 | -1'             | 4'   | -1' | 34' |                                      |                                      |                                   |
| Base Last Salt                       | 9,062'                                                                                                                                                    | -7,094'                 | 9,068'                         | 9,067'                          | -7,099'        | 46'                   | 1,642'                 | -5'             | 4'   | -5' | 31' |                                      |                                      |                                   |
| Ratcliffe                            | 9,109'                                                                                                                                                    | -7,141'                 | 9,114'                         | 9,113'                          | -7,145'        | 175'                  | 1,596'                 | -4'             | 4'   | -4' | 34' |                                      |                                      |                                   |
| Mission Canyon                       | 9,286'                                                                                                                                                    | -7,318'                 | 9,289'                         | 9,288'                          | -7,320'        | 558'                  | 1,421'                 | -2'             | 3'   | -2' | 29' |                                      |                                      |                                   |
| Lodgepole                            | 9,845'                                                                                                                                                    | -7,877'                 | 9,847'                         | 9,846'                          | -7,878'        | 109'                  | 863'                   | -1'             | 3'   | -1' | 37' |                                      |                                      |                                   |
| Lodgepole A                          | 9,952'                                                                                                                                                    | -7,984'                 | 9,956'                         | 9,955'                          | -7,987'        | 136'                  | 754'                   | -3'             | -7'  | -2' | 7'  |                                      |                                      |                                   |
| Lodgepole B                          | 10,093'                                                                                                                                                   | -8,125'                 | 10,092'                        | 10,091'                         | -8,123'        | 68'                   | 618'                   | 2'              | -38' | 2'  | 25' |                                      |                                      |                                   |
| Lodgepole C                          | 10,180'                                                                                                                                                   | -8,212'                 | 10,160'                        | 10,159'                         | -8,191'        | 115'                  | 550'                   | 21'             | -68' | 0'  | 28' |                                      |                                      |                                   |
| Lodgepole D                          | 10,283'                                                                                                                                                   | -8,315'                 | 10,275'                        | 10,274'                         | -8,306'        | 147'                  | 435'                   | 9'              | 4'   | 1'  | 16' |                                      |                                      |                                   |
| Lodgepole E                          | 10,428'                                                                                                                                                   | -8,460'                 | 10,432'                        | 10,421'                         | -8,453'        | 72'                   | 288'                   | 7'              | 8'   | 0'  | 15' |                                      |                                      |                                   |
| Lodgepole F                          | 10,501'                                                                                                                                                   | -8,533'                 | 10,518'                        | 10,493'                         | -8,525'        | 81'                   | 216'                   | 8'              | 4'   | -1' | 15' |                                      |                                      |                                   |
| False Bakken                         | 10,577'                                                                                                                                                   | -8,609'                 | 10,626'                        | 10,574'                         | -8,606'        | 11'                   | 135'                   | 3'              | 10'  | 3'  | 21' |                                      |                                      |                                   |
| Upper Bakken                         | 10,585'                                                                                                                                                   | -8,617'                 | 10,642'                        | 10,585'                         | -8,617'        | 15'                   | 124'                   | 0'              | 7'   | 0'  | 19' |                                      |                                      |                                   |
| Middle Bakken                        | 10,602'                                                                                                                                                   | -8,634'                 | 10,668'                        | 10,600'                         | -8,632'        | 39'                   | 109'                   | 2'              | 12'  | 2'  | 22' |                                      |                                      |                                   |
| Lower Bakken                         | 10,644'                                                                                                                                                   | -8,676'                 | 10,740'                        | 10,639'                         | -8,671'        | 15'                   | 70'                    | 5'              | 8'   | 5'  | 14' |                                      |                                      |                                   |
| Pronghorn                            | 10,655'                                                                                                                                                   | -8,687'                 | 10,800'                        | 10,654'                         | -8,686'        | 22'                   | 55'                    | 1'              | 4'   | 1'  | 9'  |                                      |                                      |                                   |
| Threeforks 1st Bench                 | 10,669'                                                                                                                                                   | -8,701'                 | 10,816'                        | 10,676'                         | -8,708'        | 16'                   | 33'                    | -7'             | -1'  | -2' | 3'  |                                      |                                      |                                   |
| Claystone 1                          | 10,690'                                                                                                                                                   | -8,722'                 | 10,876'                        | 10,692'                         | -8,724'        | 14'                   | 17'                    | -2'             | 10'  | 10' | 15' |                                      |                                      |                                   |
| Three Forks 2nd Bench                | 10,702'                                                                                                                                                   | -8,734'                 | 10,973'                        | 10,706'                         | -8,738'        | 6'                    | 3'                     | -4'             | 8'   | 8'  | 14' |                                      |                                      |                                   |
| Target Top                           | 10,710'                                                                                                                                                   | -8,742'                 | 10,999'                        | 10,712'                         | -8,744'        | -3'                   | -3'                    | -2'             | 12'  | 12' | 18' |                                      |                                      |                                   |
| Landing Target                       | 10,715'                                                                                                                                                   | -8,747'                 | 11,100'                        | 10,709'                         | -8,741'        | -                     | 0'                     | 6'              | 20'  | 20' | 26' |                                      |                                      |                                   |

# CONTROL DATA

| Operator:<br>Well Name:<br>Location: | Oasis Petroleum North America<br>Chalmers 5301 44-24 3BR<br>SE SE Section 24, T153N, R101W<br>McKenzie Co., ND<br>shares pad with subject well<br>KB: 1,968' |              |                |                       |                        | Oasis Petroleum North America<br>Chalmers 5301 44-24 2TR<br>SE SE Section 24, T153N, R101W<br>McKenzie Co., ND<br>shares pad with subject well<br>KB: 1,968' |                |                       |                        |              | Oasis Petroleum North America<br>Chalmers 5300 31-19H<br>NW SW Sec.19, T153N, R100W<br>McKenzie Co., ND<br>0.16 miles northeast of subject well<br>KB: 1,929' |                       |                        |  |  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------|-----------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------|------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------|--|--|
| Elevation:                           | Formation/<br>Zone                                                                                                                                           | E-Log<br>Top | Datum<br>(MSL) | Interval<br>Thickness | Thickness<br>to Target | E-Log<br>Top                                                                                                                                                 | Datum<br>(MSL) | Interval<br>Thickness | Thickness<br>to Target | E-Log<br>Top | Datum<br>(MSL)                                                                                                                                                | Interval<br>Thickness | Thickness<br>to Target |  |  |
|                                      | Kibbey Lime                                                                                                                                                  | 8,268'       | -6,300'        | 171'                  | 2,461'                 | 8,264'                                                                                                                                                       | -6,296'        | 165'                  | 2,465'                 | 8,240'       | -6,311'                                                                                                                                                       | 164'                  | 2,456'                 |  |  |
|                                      | Charles                                                                                                                                                      | 8,439'       | -6,471'        | 548'                  | 2,290'                 | 8,429'                                                                                                                                                       | -6,461'        | 553'                  | 2,300'                 | 8,404'       | -6,475'                                                                                                                                                       | 574'                  | 2,292'                 |  |  |
|                                      | UB                                                                                                                                                           | 8,987'       | -7,019'        | 84'                   | 1,742'                 | 8,982'                                                                                                                                                       | -7,014'        | 80'                   | 1,747'                 | 8,978'       | -7,049'                                                                                                                                                       | 81'                   | 1,718'                 |  |  |
|                                      | Base Last Salt                                                                                                                                               | 9,071'       | -7,103'        | 46'                   | 1,658'                 | 9,062'                                                                                                                                                       | -7,094'        | 47'                   | 1,667'                 | 9,059'       | -7,130'                                                                                                                                                       | 49'                   | 1,637'                 |  |  |
|                                      | Ratcliffe                                                                                                                                                    | 9,117'       | -7,149'        | 174'                  | 1,612'                 | 9,109'                                                                                                                                                       | -7,141'        | 177'                  | 1,620'                 | 9,108'       | -7,179'                                                                                                                                                       | 170'                  | 1,588'                 |  |  |
|                                      | Mission Canyon                                                                                                                                               | 9,291'       | -7,323'        | 558'                  | 1,438'                 | 9,286'                                                                                                                                                       | -7,318'        | 559'                  | 1,443'                 | 9,278'       | -7,349'                                                                                                                                                       | 566'                  | 1,418'                 |  |  |
|                                      | Lodgepole                                                                                                                                                    | 9,849'       | -7,881'        | 99'                   | 880'                   | 9,845'                                                                                                                                                       | -7,877'        | 108'                  | 884'                   | 9,844'       | -7,915'                                                                                                                                                       | 79'                   | 852'                   |  |  |
|                                      | Lodgepole A                                                                                                                                                  | 9,948'       | -7,980'        | 105'                  | 781'                   | 9,953'                                                                                                                                                       | -7,985'        | 140'                  | 776'                   | 9,923'       | -7,994'                                                                                                                                                       | 154'                  | 773'                   |  |  |
|                                      | Lodgepole B                                                                                                                                                  | 10,053'      | -8,085'        | 38'                   | 676'                   | 10,093'                                                                                                                                                      | -8,125'        | 66'                   | 636'                   | 10,077'      | -8,148'                                                                                                                                                       | 71'                   | 619'                   |  |  |
|                                      | Lodgepole C                                                                                                                                                  | 10,091'      | -8,123'        | 187'                  | 638'                   | 10,159'                                                                                                                                                      | -8,191'        | 116'                  | 570'                   | 10,148'      | -8,219'                                                                                                                                                       | 103'                  | 548'                   |  |  |
|                                      | Lodgepole D                                                                                                                                                  | 10,278'      | -8,310'        | 151'                  | 451'                   | 10,275'                                                                                                                                                      | -8,307'        | 146'                  | 454'                   | 10,251'      | -8,322'                                                                                                                                                       | 146'                  | 445'                   |  |  |
|                                      | Lodgepole E                                                                                                                                                  | 10,429'      | -8,461'        | 68'                   | 300'                   | 10,421'                                                                                                                                                      | -8,453'        | 71'                   | 308'                   | 10,397'      | -8,468'                                                                                                                                                       | 72'                   | 299'                   |  |  |
|                                      | Lodgepole F                                                                                                                                                  | 10,497'      | -8,529'        | 87'                   | 232'                   | 10,492'                                                                                                                                                      | -8,524'        | 85'                   | 237'                   | 10,469'      | -8,540'                                                                                                                                                       | 87'                   | 227'                   |  |  |
|                                      | False Bakken                                                                                                                                                 | 10,584'      | -8,616'        | 8'                    | 145'                   | 10,577'                                                                                                                                                      | -8,609'        | 8'                    | 152'                   | 10,556'      | -8,627'                                                                                                                                                       | 9'                    | 140'                   |  |  |
|                                      | Upper Bakken                                                                                                                                                 | 10,592'      | -8,624'        | 20'                   | 137'                   | 10,585'                                                                                                                                                      | -8,617'        | 17'                   | 144'                   | 10,565'      | -8,636'                                                                                                                                                       | 18'                   | 131'                   |  |  |
|                                      | Middle Bakken                                                                                                                                                | 10,612'      | -8,644'        | 35'                   | 117'                   | 10,602'                                                                                                                                                      | -8,634'        | 42'                   | 127'                   | 10,583'      | -8,654'                                                                                                                                                       | 31'                   | 113'                   |  |  |
|                                      | Lower Bakken                                                                                                                                                 | 10,647'      | -8,679'        | 11'                   | 82'                    | 10,644'                                                                                                                                                      | -8,676'        | 11'                   | 85'                    | 10,614'      | -8,685'                                                                                                                                                       | 10'                   | 82'                    |  |  |
|                                      | Pronghorn                                                                                                                                                    | 10,658'      | -8,690'        | 17'                   | 71'                    | 10,655'                                                                                                                                                      | -8,687'        | 19'                   | 74'                    | 10,624'      | -8,695'                                                                                                                                                       | 16'                   | 72'                    |  |  |
|                                      | Threeforks 1st Bench                                                                                                                                         | 10,675'      | -8,707'        | 27'                   | 54'                    | 10,674'                                                                                                                                                      | -8,706'        | 28'                   | 55'                    | 10,640'      | -8,711'                                                                                                                                                       | 28'                   | 56'                    |  |  |
|                                      | Claystone 1                                                                                                                                                  | 10,702'      | -8,734'        | 12'                   | 27'                    | 10,702'                                                                                                                                                      | -8,734'        | 12'                   | 27'                    | 10,668'      | -8,739'                                                                                                                                                       | 13'                   | 28'                    |  |  |
|                                      | Three Forks 2nd Bench                                                                                                                                        | 10,714'      | -8,746'        | 10'                   | 15'                    | 10,714'                                                                                                                                                      | -8,746'        | 10'                   | 15'                    | 10,681'      | -8,752'                                                                                                                                                       | 10'                   | 15'                    |  |  |
|                                      | Target Top                                                                                                                                                   | 10,724'      | -8,756'        | 5'                    | 5'                     | 10,724'                                                                                                                                                      | -8,756'        | 5'                    | 5'                     | 10,691'      | -8,762'                                                                                                                                                       | 5'                    | 5'                     |  |  |
|                                      | Landing Target                                                                                                                                               | 10,729'      | -8,761'        | -                     | 0'                     | 10,729'                                                                                                                                                      | -8,761'        | -                     | 0'                     | 10,696'      | -8,767'                                                                                                                                                       | -                     | 0'                     |  |  |

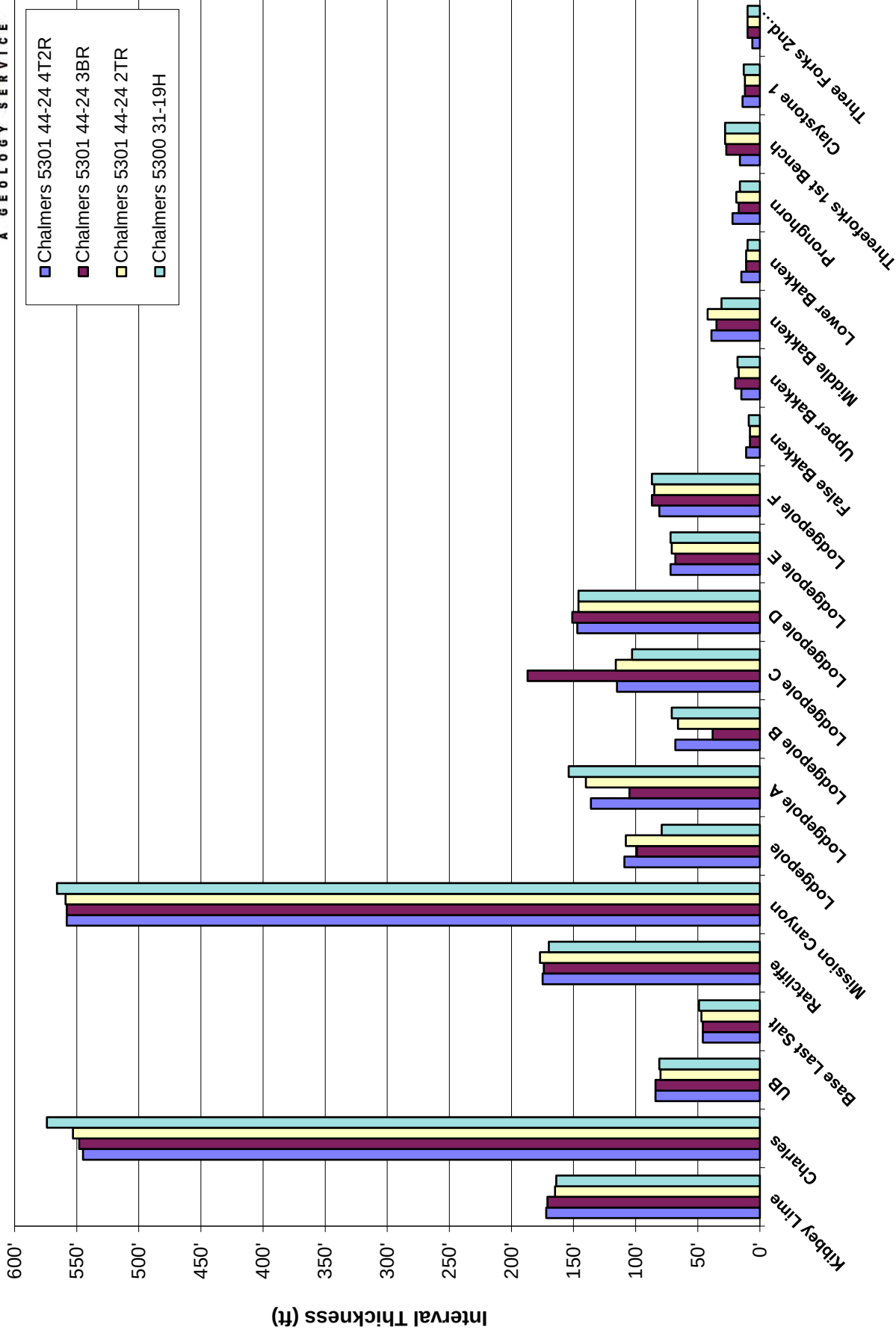
# LANDING PROJECTION

| Formation/ Zone:      | Proposed Top of Target From: |                         |                      | Average of Offset Wells |
|-----------------------|------------------------------|-------------------------|----------------------|-------------------------|
|                       | Chalmers 5301 44-24 3BR      | Chalmers 5301 44-24 2TR | Chalmers 5300 31-19H |                         |
| Kibbey Lime           | 10,727'                      | 10,731'                 | 10,722'              | 10,727'                 |
| Charles               | 10,728'                      | 10,738'                 | 10,730'              | 10,732'                 |
| UB                    | 10,725'                      | 10,730'                 | 10,701'              | 10,719'                 |
| Base Last Salt        | 10,725'                      | 10,734'                 | 10,704'              | 10,721'                 |
| Ratcliffe             | 10,725'                      | 10,733'                 | 10,701'              | 10,720'                 |
| Mission Canyon        | 10,726'                      | 10,731'                 | 10,706'              | 10,721'                 |
| Lodgepole             | 10,726'                      | 10,730'                 | 10,698'              | 10,718'                 |
| Lodgepole A           | 10,736'                      | 10,731'                 | 10,728'              | 10,732'                 |
| Lodgepole B           | 10,767'                      | 10,727'                 | 10,710'              | 10,735'                 |
| Lodgepole C           | 10,797'                      | 10,729'                 | 10,707'              | 10,744'                 |
| Lodgepole D           | 10,725'                      | 10,728'                 | 10,719'              | 10,724'                 |
| Lodgepole E           | 10,721'                      | 10,729'                 | 10,720'              | 10,723'                 |
| Lodgepole F           | 10,725'                      | 10,730'                 | 10,720'              | 10,725'                 |
| False Bakken          | 10,719'                      | 10,726'                 | 10,714'              | 10,720'                 |
| Upper Bakken          | 10,722'                      | 10,729'                 | 10,716'              | 10,722'                 |
| Middle Bakken         | 10,717'                      | 10,727'                 | 10,713'              | 10,719'                 |
| Lower Bakken          | 10,721'                      | 10,724'                 | 10,721'              | 10,722'                 |
| Pronghorn             | 10,725'                      | 10,728'                 | 10,726'              | 10,726'                 |
| Threeforks 1st Bench  | 10,730'                      | 10,731'                 | 10,732'              | 10,731'                 |
| Claystone 1           | 10,719'                      | 10,719'                 | 10,720'              | 10,719'                 |
| Three Forks 2nd Bench | 10,721'                      | 10,721'                 | 10,721'              | 10,721'                 |
| Target Top            | 10,717'                      | 10,717'                 | 10,717'              | 10,717'                 |
| Landing Target        | 10,709'                      | 10,709'                 | 10,709'              | 10,709'                 |

Current Landing Target (3' below the base of the Claystone 1): **10,709'**

# INTERVAL THICKNESS

Oasis Petroleum North America, LLC - Chalmers 5301 44-24 4T2R

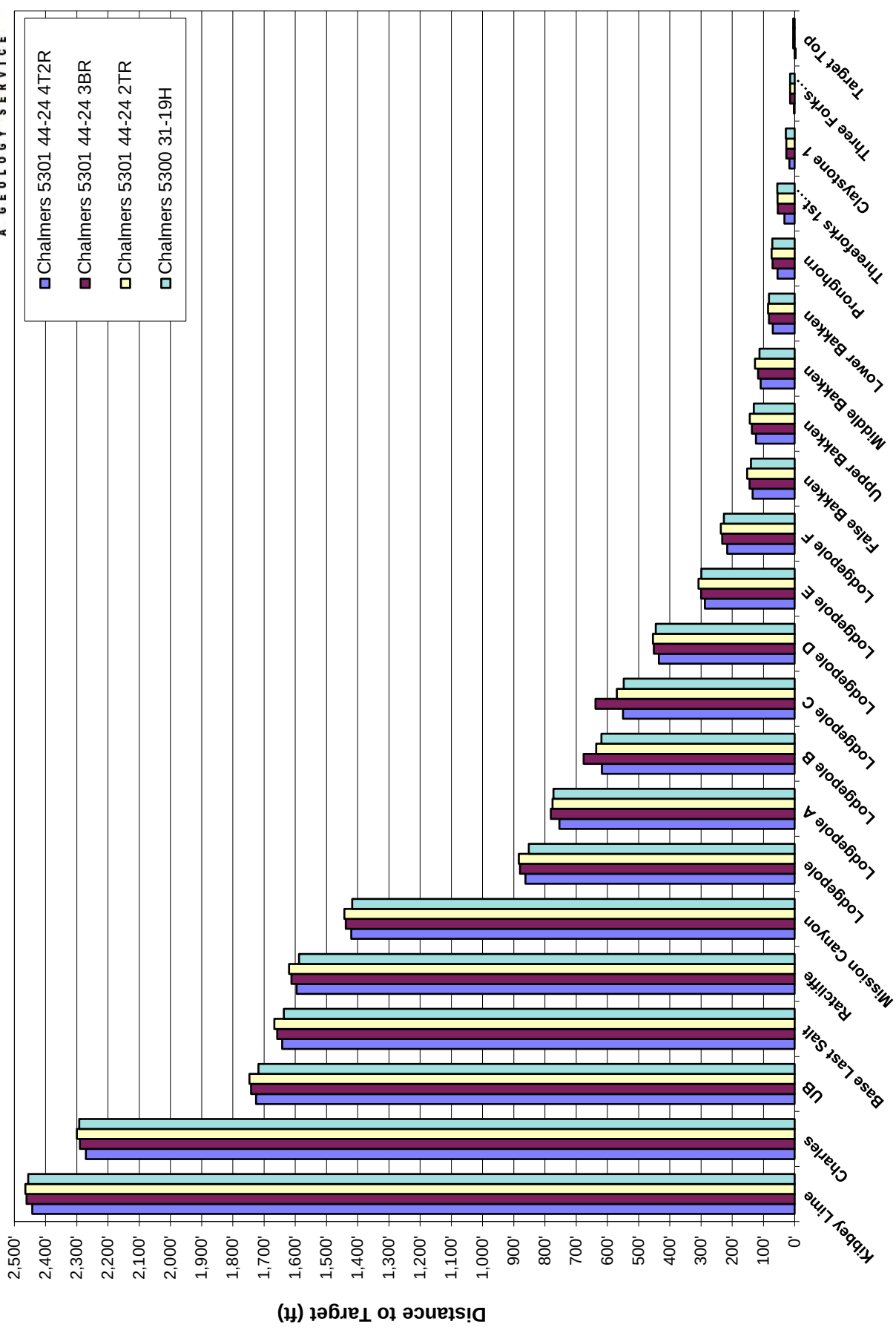






# ISOPACH TO TARGET

Oasis Petroleum North America, LLC - Chalmers 5301 44-24 4T2R



## **LITHOLOGY**

*Rig crews caught lagged samples in 30' intervals under the supervision of Sunburst geologists. A detailed list of sampling intervals is included in the well data summary page. Sample or gamma ray marker tops have been inserted in the sample descriptions below for reference. Samples were examined wet and dry under a trinocular microscope. The drilling fluid was diesel-based invert from surface casing to intermediate casing, while salt water drilling fluid was used throughout the lateral. Sample collection began at 8,230'.*

### **Drilling in the Kibbey Formation**

8230-8260 SILTSTONE: dark orange to light brown, tan, pink, soft, sub blocky, calcite cement, poorly cemented; rare ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: tan to off white, very fine grained, sub rounded, moderately sorted, calcite cement, poorly cemented

### **Kibbey "Lime" [8,267' MD, 8,266' TVD (-6,298')]**

8260-8290 SILTSTONE: orange to light brown, tan, gray, soft, sub blocky, calcite cement, poorly cemented; rare ANHYDRITE: off white, light pink, soft, amorphous texture; trace SILTY SANDSTONE: tan to off white, very fine grained, sub round, moderately sorted, calcite cement, poorly cemented

8290-8320 SILTSTONE: dark orange to light brown, tan, pink, soft, sub blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8320-8350 SILTSTONE: dark orange to light brown, tan, pink, soft, sub blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8350-8380 SILTSTONE: dark orange to light brown, tan, pink, soft, sub blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

8380-8410 SILTSTONE: dark orange to light brown, tan, pink, soft, sub blocky, calcite cement, poorly cemented; occasional ANHYDRITE: off white, soft, amorphous texture; trace SILTY SANDSTONE: as above

### **The First Charles Salt [8,439' MD, 8,438' TVD (-6,470')]**

8410-8440 SALT: translucent, microcrystalline, anhedral, crystalline texture, hard, frosted; trace LIMESTONE: mudstone, light to medium gray to gray brown, microcrystalline, firm to hard, argillaceous in part, dense, chalky texture, no visible porosity

8440-8470 SALT: translucent, microcrystalline, anhedral, crystalline texture, hard, frosted; trace LIMESTONE: mudstone, light to medium gray to gray brown, microcrystalline, firm to hard, argillaceous in part, dense, chalky texture, no visible porosity

8470-8500 SALT: translucent, microcrystalline, anhedral, crystalline texture, hard, frosted; trace LIMESTONE: mudstone, light to medium gray to gray brown, microcrystalline, firm to hard, argillaceous in part, dense, chalky texture, no visible porosity

8500-8530 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8530-8560 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8560-8590 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8590-8620 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8620-8650 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8650-8680 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light to medium gray, rare gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8680-8710 SALT: translucent, rare milky, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, medium gray, occasional gray brown, trace light gray, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8710-8740 SALT: translucent, rare milky, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, medium gray, occasional gray brown, trace light gray, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8740-8770 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light gray, trace gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8770-8800 SALT: translucent, rare frosted, microcrystalline, anhedral, crystalline texture, hard; occasional LIMESTONE: mudstone, light gray, trace gray brown, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8800-8830 LIMESTONE: mudstone, light gray, trace gray brown, rare off white, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8830-8860 LIMESTONE: mudstone, light gray, trace gray brown, rare off white, microcrystalline, firm to hard, argillaceous in part, dense, earthy texture, no visible porosity

8860-8890 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: off white, soft, massive, amorphous, no visible porosity

8890-8920 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

8920-8950 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

8950-8980 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

**UB [8,982' MD, 8,983' TVD (-7,015')]**

8980-9010 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

9010-9040 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

**Base Last Salt [9,068' MD, 9,067' TVD (-7,099')]**

9040-9070 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, occasional light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, rare ANHYDRITE: off white to white, soft, massive, amorphous, no visible porosity, trace SALT: as above

9070-9100 SALT: translucent, rare milky, crystalline, hard, euhedral; occasional ARGILLACEOUS LIMESTONE: mudstone, cream to tan, rare light to medium gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity

**Ratcliffe [9,114' MD, 9,113' TVD (-7,145')]**

9100-9130 ARGILLACEOUS LIMESTONE: mudstone, cream to tan, rare brown, trace gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity; trace ANHYDRITE: off white, soft, massive, amorphous, no visible porosity

9130-9160 ARGILLACEOUS LIMESTONE: mudstone, cream to tan, rare brown, trace gray, slightly mottled, microcrystalline, firm, dense to banded, earthy texture, no visible porosity; trace ANHYDRITE: off white, soft, massive, amorphous, no visible porosity

9160-9190 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9190-9220 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9220-9250 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9250-9280 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

**Mission Canyon [9,289' MD, 9,288' TVD (-7,320')]**

9280-9310 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9310-9340 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9340-9370 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9370-9400 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9400-9430 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9430-9460 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9460-9490 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9490-9520 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above



9520-9550 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9550-9580 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9580-9610 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9610-9640 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9640-9670 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9670-9700 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9700-9730 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9730-9760 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9760-9790 ARGILLACEOUS LIMESTONE: mudstone, cream, tan, light gray, microcrystalline, firm, dense to banded, earthy texture, no visible porosity, occasional ANHYDRITE: as above

9790-9820 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, microcrystalline, firm to friable, rare algal material; trace ANHYDRITE: off white, massive, soft, amorphous

**Lodgepole [9,847' MD, 9,846' TVD (-7,878')]**

9820-9850 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, microcrystalline, firm to friable, rare algal material; trace ANHYDRITE: off white, massive, soft, amorphous

9850-9880 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture, trace algal material

9880-9910 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture, trace algal material

9910-9940 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture, trace algal material

9940-9970 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture

9970-10000 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture

10000-10030 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture

10030-10060 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture; trace ANHYDRITE: off white, massive, soft, amorphous

10060-10090 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture; trace ANHYDRITE: off white, massive, soft, amorphous

10090-10120 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture; trace ANHYDRITE: off white, massive, soft, amorphous

10120-10150 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture; trace ANHYDRITE: off white, massive, soft, amorphous

10150-10180 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, trace dark gray, microcrystalline, firm to friable, earthy texture; trace ANHYDRITE: off white, massive, soft, amorphous

10180-10230 ARGILLACEOUS LIMESTONE: mudstone, light to medium gray brown, tan, microcrystalline, firm to friable, rare algal material; rare ANHYDRITE: off white, massive, soft, amorphous

10210-10240 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10240-10270 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10270-10300 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10300-10330 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10330-10360 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10360-10390 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10390-10420 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10420-10450 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10450-10480 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10480-10510 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10510-10540 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10540-10570 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

10570-10600 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

**False Bakken [10,626' MD, 10,574' TVD (-8,606')]**

10600-10630 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

**Upper Bakken Shale [10,642' MD, 10,585' TVD (-8,617')]**

10630-10660 ARGILLACEOUS LIMESTONE: mudstone, medium gray to dark gray, microcrystalline, firm, banded, dense, trace anhydrite, trace disseminated pyrite, no visible porosity

**Middle Bakken [10,668' MD, 10,600' TVD (-8,632')]**

10660-10690 SHALE: black to very dark brown, firm, blocky, earthy texture, disseminated pyrite, petroliferous, carbonaceous, no visible porosity, abundant brown even oil stain

10690-10720 SILTSTONE: light gray to light gray brown, sub blocky, calcite cemented, trace disseminated pyrite, possible intergranular porosity; occasional SILTY SANDSTONE: as above; trace SHALE: as above, trace spotty light brown oil stain

**Lower Bakken Shale [10,740' MD, 10,639' TVD (-8,671')]**

10720-10750 SILTSTONE: light gray to light gray brown, sub blocky, calcite cemented, trace disseminated pyrite, possible intergranular porosity; occasional SILTY SANDSTONE: as above; trace SHALE: as above, trace spotty light brown oil stain

10750-10780 SHALE: black to very dark brown, firm, blocky, earthy texture, disseminated pyrite, petroliferous, carbonaceous, no visible porosity, abundant brown even oil stain

**Pronghorn [10,800' MD, 10,654' TVD (-8,686')]**

10780-10810 SILTSTONE: as above; trace SHALE: black to dark brown, firm to hard, blocky, earthy texture, carbonaceous, rare disseminated pyrite

**Three Forks First Bench [10,816' MD, 10,676' TVD (-8,708')]**

10810-10840 SILTSTONE: as above; trace SHALE: black to brown, firm, blocky, earthy texture, carbonaceous, rare disseminated pyrite

10840-10870 ARGILLACEOUS DOLOMITE: mudstone, medium gray to gray brown, pale grain, cream to tan, fine crystalline, sucrosic texture, firm to friable, rare disseminated pyrite, silty in part, sandy in part, possible intercrystalline porosity, trace spotty light brown oil stain; rare SHALE: as above; trace DOLOMITE: mudstone, tan, pink, friable, earthy texture, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain

**Claystone 1 [10,876' MD, 10,692' TVD (-8,724')]**

10870-10880 ARGILLACEOUS DOLOMITE: mudstone, medium gray to gray brown, pale grain, cream to tan, fine crystalline, sucrosic texture, firm to friable, rare disseminated pyrite, silty in part, sandy in part, possible intercrystalline porosity, trace spotty light brown oil stain; rare SHALE: as above; trace DOLOMITE: as above, trace spotty light brown oil stain

10880-10890 ARGILLACEOUS DOLOMITE: mudstone, medium gray to gray brown, pale grain, cream to tan, fine crystalline, sucrosic texture, firm to friable, rare disseminated pyrite, silty in part, sandy in part, possible intercrystalline porosity, trace spotty light brown oil stain; rare SHALE: as above; trace DOLOMITE: as above, trace spotty light brown oil stain

10890-10900 ARGILLACEOUS DOLOMITE: mudstone, medium gray to gray brown, pale grain, cream to tan, fine crystalline, sucrosic texture, firm to friable, rare disseminated pyrite, silty in part, sandy in part, possible intercrystalline porosity, trace spotty light brown oil stain; rare SHALE: as above; trace DOLOMITE: as above, trace spotty light brown oil stain

10900-10910 ARGILLACEOUS DOLOMITE: as above; occasional CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; rare SHALE: as above

10910-10920 ARGILLACEOUS DOLOMITE: as above; occasional CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; rare SHALE: as above

10920-10930 ARGILLACEOUS DOLOMITE: as above; occasional CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; rare SHALE: as above

10930-10940 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity, trace spotty light brown oil stain

10940-10950 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity, trace spotty light brown oil stain

10950-10960 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

10960-10970 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

### **Three Forks Second Bench [10,973' MD, 10,706' TVD (-8,738')]**

10970-10980 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

10980-10990 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

10990-11000 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11000-11010 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11010-11020 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11020-11030 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11030-11040 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11040-11050 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain



11050-11060 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11060-11070 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain

11070-11080 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain

11080-11090 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11090-11100 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11100-11110 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11110-11140 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11140-11170 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain

11170-11200 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, fast weak diffuse pale green cut fluorescence

11200-11230 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; rare SHALE: as above, fast weak diffuse pale green cut fluorescence

11230-11260 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; rare SHALE: as above, fast weak diffuse pale green cut fluorescence

11260-11290 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; rare SHALE: as above, fast weak diffuse pale green cut fluorescence

11290-11320 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; rare SHALE: as above, fast weak diffuse pale green cut fluorescence

11320-11350 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; rare SHALE: as above, fast weak diffuse pale green cut fluorescence

11350-11380 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; rare SHALE: as above, fast weak diffuse pale green cut fluorescence

11770-11800 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; trace SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, no visible porosity, fast weak diffuse pale green cut fluorescence

11800-11830 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; trace SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, no visible porosity, fast weak diffuse pale green cut fluorescence

11830-11860 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; trace SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, no visible porosity, fast weak diffuse pale green cut fluorescence

11860-11890 CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity; occasional DOLOMITE: as above; trace SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, no visible porosity, fast weak diffuse pale green cut fluorescence

11890-11920 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

11920-11950 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

11950-11980 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

11980-12010 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

12010-12040 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

12040-12070 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above; trace CLAYSTONE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

12070-12100 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

12100-12130 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

12130-12160 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

12490-12520 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; common SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence





13210-13240 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast moderately cloudy pale green cut fluorescence

13570-13600 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; rare SHALE: as above, trace spotty light brown oil stain, fast moderately cloudy pale green cut fluorescence

13930-13960 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14290-14320 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14320-14350 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14350-14380 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14380-14410 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14410-14440 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14440-14470 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14470-14500 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14500-14530 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14530-14560 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14560-14590 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14590-14620 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14620-14650 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14650-14680 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence



14710-14740 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14770-14800 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14830-14860 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14860-14890 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14890-14920 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak cloudy pale green cut fluorescence

14920-14950 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

14950-14980 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

14980-15010 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15010-15040 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15070-15100 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15130-15160 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15190-15220 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15250-15280 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15310-15340 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15370-15400 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

15760-15790 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, moderately contaminated with lube

15790-157820 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, weak contaminated with lube

15820-15850 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, weak contaminated with lube

15850-15880 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, weak contaminated with lube

15880-15910 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, weak contaminated with lube

15910-15940 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; trace CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity, trace spotty light brown oil stain,, weak contaminated with lube

15940-15970 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; trace CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity, trace spotty light brown oil stain,, weak contaminated with lube, weak contaminated with lube

15970-16000 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; trace CLAYSTONE: light gray, medium gray to tan, occasional off white, massive, fine crystalline, dolomitic cement, trace disseminated pyrite, no visible porosity, trace spotty light brown oil stain,, weak contaminated with lube, weak contaminated with lube

16000-16030 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, weak contaminated with lube

16030-16060 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; rare CLAYSTONE: as above, trace spotty light brown oil stain, weak contaminated with lube

16060-16090 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; occasional CLAYSTONE: as above, trace spotty light brown oil stain, weak contaminated with lube

16090-16120 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; common CLAYSTONE: as above, trace spotty light brown oil stain, weak contaminated with lube

16120-16150 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above; common CLAYSTONE: as above, trace spotty light brown oil stain, weak contaminated with lube

16510-16540 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, weak contaminated with lube



16960-16990 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, moderately weak cloudy pale green cut fluorescence

17380-17410 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, weak contaminated with lube

17830-17860 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, moderately contaminated with lube

18280-18310 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, moderately contaminated with lube

18700-18730 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence



18760-18790 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

18820-18850 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

18880-18910 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

18940-18970 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

18970-19000 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

19000-19030 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

19030-19060 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

19060-19090 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

19420-19450 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

19780-19810 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, fast weak diffuse pale green cut fluorescence

20200-20230 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, heavy contaminated with lube

20650-20680 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, heavy contaminated with lube

20680-20710 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: as above, trace spotty light brown oil stain, heavy contaminated with lube

20710-20740 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube

20740-20770 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube

20770-20800 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube

20800-20830 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube

20830-20860 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube

20860-20890 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube

20890-20920 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube

20920-20963 DOLOMITE: mudstone, tan, pink, friable, earthy texture, rare sucrosic texture, hard to firm, silty in part, sandy in part, trace disseminated pyrite, possible intercrystalline porosity, trace spotty light brown oil stain; occasional SHALE: light green, light gray, friable, earthy texture, trace disseminated pyrite, trace spotty light brown oil stain, heavy contaminated with lube





**RYAN DIRECTIONAL SERVICES, INC.**  
A NABORS COMPANY

19510 Oil Center Blvd  
Houston, TX 77073  
Bus 281.443.1414  
Fax 281.443.1676

Monday, September 08, 2014

State of North Dakota

Subject: **Surveys**

Re: **Oasis**  
**Chalmers 5301 44-24 4T2R**  
**McKenzie, ND**

Enclosed, please find the original and one copy of the survey performed on the above-referenced well by Ryan Directional Services, Inc.. Other information required by your office is as follows:

| <i>Surveyor Name</i> | <i>Surveyor Title</i> | <i>Borehole Number</i> | <i>Start Depth</i> | <i>End Depth</i> | <i>Start Date</i> | <i>End Date</i> | <i>Type of</i> | <i>TD Straight Line Projection</i> |
|----------------------|-----------------------|------------------------|--------------------|------------------|-------------------|-----------------|----------------|------------------------------------|
| Ogden, Daniel        | MWD Operator          | O.H.                   | 0'                 | 2095'            | 06/22/14          | 06/23/14        | MWD            | 2095'                              |
| Ogden, Daniel        | MWD Operator          | O.H.                   | 2095'              | 11042'           | 07/06/14          | 07/18/14        | MWD            | 11042'                             |
| Ogden, Daniel        | MWD Operator          | O.H.                   | 11042'             | 20898'           | 08/26/14          | 09/05/14        | MWD            | 20963'                             |

A certified plat on which the bottom hole location is oriented both to the surface location and to the lease lines (or unit lines in case of pooling) is attached to the survey report. If any other information is required please contact the undersigned at the letterhead address or phone number.

---

**Hudson, Douglas**  
Well Planner



**RYAN DIRECTIONAL SERVICES, INC.**

A NABORS COMPANY

Ryan Directional Services, Inc.

19510 Oil Center Blvd.

Houston, Texas 77073

Bus: 281.443.1414

Fax: 281.443.1676

Tuesday, June 24, 2014

State of North Dakota

County of McKenzie Co.

Subject: **Survey Certification Letter**

Survey Company: Ryan Directional Services, Inc.

Job Number: 7789

Survey Job Type: Ryan MWD

Customer: Oasis Petroleum N.A. LLC

Well Name: Chalmers 5301 44-24 4T2R

Rig Name: Nabors B-25

Surface: 48°03'20.01", -103°36'18.55"

A.P.I. No: 33-053-06011

Location: McKenzie Co., North Dakota

RKB Height: 25'

Distance to Bit: 64'

| Surveyor Name | Surveyor Title | Borehole Number | Start Depth | End Depth | Start Date | End Date | Type of | TD Straight Line Projection |
|---------------|----------------|-----------------|-------------|-----------|------------|----------|---------|-----------------------------|
| Daniel Ogden  | MWD Supervisor | OH              | 100'        | 2095'     | 06/22/14   | 06/23/14 | MWD     | 2095'                       |

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

  
**Daniel Ogden**

MWD Supervisor

Ryan Directional Services, Inc.



**RYAN DIRECTIONAL SERVICES, INC.**

A NABORS COMPANY

Ryan Directional Services, Inc.

19510 Oil Center Blvd.

Houston, Texas 77073

Bus: 281.443.1414

Fax: 281.443.1676

Friday, July 18, 2014

State of North Dakota

County of McKenzie

Subject: **Survey Certification Letter**

Survey Company: Ryan Directional Services, Inc.  
Job Number: 7830  
Survey Job Type: Ryan MWD  
Customer: Oasis Petroleum North America LLC  
Well Name: Chalmers 5301 44-24 4T2R  
Rig Name: Nabors B-25

Surface: 153N-100W-19/20  
A.P.I. No: 33-053-06011  
Location: McKenzie, North Dakota  
RKB Height: 25'  
Distance to Bit: 59'

|               |                |          |       |        |          |          |      |            |  | TD Straight |
|---------------|----------------|----------|-------|--------|----------|----------|------|------------|--|-------------|
| Surveyor Name | Surveyor Title | Borehole | Start | End    | Start    | End      | Type | Line       |  |             |
|               |                | Number   | Depth | Depth  | Date     | Date     | of   | Projection |  |             |
| Daniel Ogden  | MWD Supervisor | OH       | 2095' | 11042' | 07/06/14 | 07/18/14 | MWD  | 11100'     |  |             |

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

  
**Daniel Ogden**

MWD Supervisor

Ryan Directional Services, Inc.



Ryan Directional Services, Inc.  
19510 Oil Center Blvd.  
Houston, Texas 77073  
Bus: 281.443.1414  
Fax: 281.443.1676

Saturday, September 06, 2014

State of North Dakota  
County of McKenzie

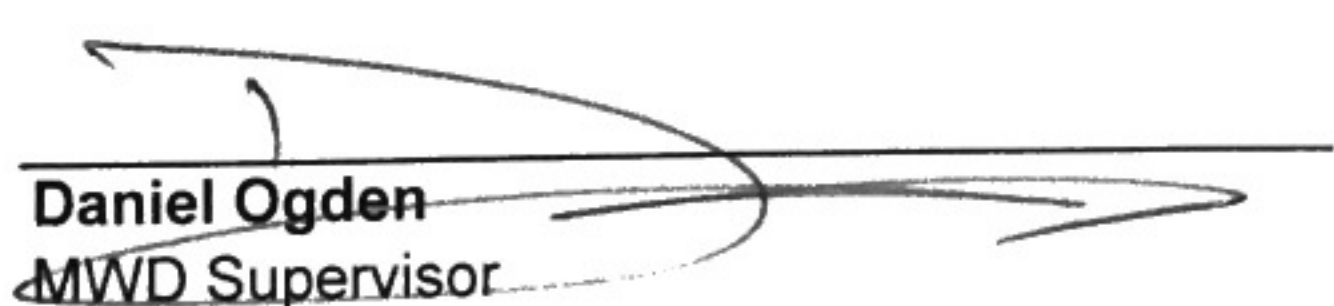
Subject: **Survey Certification Letter**

Survey Company: Ryan Directional Services, Inc.  
Job Number: 7944  
Survey Job Type: Ryan MWD  
Customer: Oasis Petroleum North America LLC  
Well Name: Chalmers 5301 44-24 4T2R  
Rig Name: Nabors B-25

Surface: 48°03'20.49", -103°36'18.55"  
A.P.I. No: 33-053-06011  
Location: McKenzie, ND  
RKB Height: 25'  
Distance to Bit: 65'

| Surveyor Name | Surveyor Title | Borehole Number | Start Depth | End Depth | Start Date | End Date | Type of | TD Straight Line Projection |
|---------------|----------------|-----------------|-------------|-----------|------------|----------|---------|-----------------------------|
| Daniel Ogden  | MWD Supervisor | OH              | 11100'      | 20963'    | 08/26/14   | 09/05/14 | MWD     | 20963'                      |

The data and calculations for this survey have been checked by me and conform to the calibration standards and operational procedures set forth by Ryan Directional Services, Inc. I am authorized and qualified to review the data, calculations and these reports; the reports represents true and correct Directional Surveys of this well based on the original data, the minimum curvature method, corrected to True North and obtained at the well site.

  
Daniel Ogden  
MWD Supervisor  
Ryan Directional Services, Inc.



Report #: **1**  
Date: **23-Jun-14**



**RYAN DIRECTIONAL SERVICES, INC.**  
A NABORS COMPANY

Ryan Job # **7789**

**SURVEY REPORT**

Customer: **Oasis Petroleum N.A. LLC**  
Well Name: **Chalmers 5301 44-24 4T2R**  
Rig #: **Nabors B-25**  
API #: **33-053-06011**  
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**  
Directional Drillers: **RPM**  
Survey Corrected To: **True North**  
Vertical Section Direction: **90**  
Total Correction: **8.27**  
Temperature Forecasting Model (Chart Only): **Logarithmic**

| Survey #                       | MD   | Inc  | Azm    | Temp    | TVD     | VS    | N/S   | E/W   | DLS  |
|--------------------------------|------|------|--------|---------|---------|-------|-------|-------|------|
| Tie in to Conductor Well Depth |      |      |        |         |         |       |       |       |      |
| Tie In                         | 0    | 0.00 | 0.00   | 0.00    | 0.00    | 0.00  | 0.00  | 0.00  | 0.00 |
| 1                              | 184  | 0.60 | 158.50 | 73.00   | 184.00  | 0.35  | -0.90 | 0.35  | 0.33 |
| 2                              | 215  | 0.70 | 159.80 | 73.00   | 214.99  | 0.48  | -1.23 | 0.48  | 0.33 |
| 3                              | 308  | 0.40 | 49.50  | 73.00   | 307.99  | 0.92  | -1.55 | 0.92  | 0.99 |
| 4                              | 402  | 1.00 | 53.20  | 73.00   | 401.98  | 1.83  | -0.84 | 1.83  | 0.64 |
| 5                              | 495  | 0.90 | 67.40  | 77.00   | 494.97  | 3.15  | -0.08 | 3.15  | 0.27 |
| 6                              | 588  | 0.90 | 62.50  | 77.00   | 587.96  | 4.47  | 0.54  | 4.47  | 0.08 |
| 7                              | 680  | 0.70 | 38.20  | 77.00   | 679.95  | 5.46  | 1.32  | 5.46  | 0.42 |
| 8                              | 772  | 0.90 | 43.90  | 77.00   | 771.94  | 6.31  | 2.28  | 6.31  | 0.23 |
| 9                              | 865  | 1.20 | 62.40  | 78.00   | 864.93  | 7.68  | 3.26  | 7.68  | 0.48 |
| 10                             | 957  | 0.90 | 46.20  | 78.00   | 956.91  | 9.06  | 4.20  | 9.06  | 0.46 |
| 11                             | 1049 | 0.90 | 52.10  | 80.00   | 1048.90 | 10.15 | 5.15  | 10.15 | 0.10 |
| 12                             | 1142 | 0.20 | 322.00 | 86.00   | 1141.90 | 10.62 | 5.72  | 10.62 | 0.99 |
| 13                             | 1234 | 0.30 | 356.90 | 87.00   | 1233.90 | 10.51 | 6.09  | 10.51 | 0.19 |
| 14                             | 1326 | 0.30 | 166.10 | 89.00   | 1325.90 | 10.56 | 6.10  | 10.56 | 0.65 |
| 15                             | 1418 | 0.50 | 200.60 | 91.00   | 1417.89 | 10.47 | 5.49  | 10.47 | 0.33 |
| 16                             | 1510 | 0.30 | 211.00 | 93.00   | 1509.89 | 10.21 | 4.91  | 10.21 | 0.23 |
| 17                             | 1604 | 0.20 | 213.80 | 93.00   | 1603.89 | 9.99  | 4.56  | 9.99  | 0.11 |
| 18                             | 1697 | 0.40 | 135.90 | 96.00   | 1696.89 | 10.13 | 4.19  | 10.13 | 0.44 |
| 19                             | 1790 | 0.40 | 147.50 | 96.00   | 1789.89 | 10.53 | 3.68  | 10.53 | 0.09 |
| 20                             | 1883 | 0.10 | 64.30  | 100.00  | 1882.89 | 10.77 | 3.44  | 10.77 | 0.43 |
| 21                             | 1977 | 0.40 | 109.90 | 86.00   | 1976.88 | 11.16 | 3.37  | 11.16 | 0.36 |
| 22                             | 2031 | 0.20 | 159.90 | 86.00   | 2030.88 | 11.37 | 3.22  | 11.37 | 0.58 |
| 23                             | 2097 | 0.30 | 174.40 | 93.00   | 2096.88 | 11.42 | 2.94  | 11.42 | 0.18 |
| 24                             | 2190 | 0.40 | 184.10 | 5548.00 | 2189.88 | 11.42 | 2.37  | 11.42 | 0.12 |
| 25                             | 2284 | 0.70 | 186.50 | 100.00  | 2283.88 | 11.33 | 1.47  | 11.33 | 0.32 |
| 26                             | 2377 | 1.10 | 183.30 | 102.00  | 2376.87 | 11.22 | 0.02  | 11.22 | 0.43 |
| 27                             | 2470 | 1.40 | 197.20 | 105.00  | 2469.84 | 10.83 | -1.96 | 10.83 | 0.46 |
| 28                             | 2564 | 1.30 | 220.50 | 109.00  | 2563.82 | 9.80  | -3.87 | 9.80  | 0.59 |
| 29                             | 2657 | 0.70 | 221.80 | 111.00  | 2656.80 | 8.73  | -5.09 | 8.73  | 0.65 |
| 30                             | 2750 | 0.70 | 235.90 | 114.00  | 2749.80 | 7.89  | -5.84 | 7.89  | 0.18 |
| 31                             | 2843 | 0.80 | 219.10 | 116.00  | 2842.79 | 7.01  | -6.66 | 7.01  | 0.26 |
| 32                             | 2937 | 0.40 | 202.10 | 120.00  | 2936.78 | 6.47  | -7.47 | 6.47  | 0.46 |
| 33                             | 3030 | 0.10 | 180.50 | 120.00  | 3029.78 | 6.35  | -7.85 | 6.35  | 0.33 |
| 34                             | 3123 | 0.30 | 36.00  | 123.00  | 3122.78 | 6.49  | -7.74 | 6.49  | 0.41 |
| 35                             | 3217 | 0.40 | 35.40  | 125.00  | 3216.78 | 6.82  | -7.27 | 6.82  | 0.11 |
| 36                             | 3310 | 0.40 | 40.90  | 129.00  | 3309.78 | 7.22  | -6.76 | 7.22  | 0.04 |
| 37                             | 3403 | 0.40 | 38.70  | 131.00  | 3402.78 | 7.64  | -6.26 | 7.64  | 0.02 |
| 38                             | 3496 | 0.40 | 43.10  | 132.00  | 3495.77 | 8.06  | -5.77 | 8.06  | 0.03 |
| 39                             | 3590 | 0.60 | 49.80  | 132.00  | 3589.77 | 8.66  | -5.21 | 8.66  | 0.22 |
| 40                             | 3683 | 0.50 | 48.30  | 134.00  | 3682.77 | 9.34  | -4.63 | 9.34  | 0.11 |
| 41                             | 3776 | 0.60 | 59.30  | 134.00  | 3775.76 | 10.06 | -4.11 | 10.06 | 0.16 |
| 42                             | 3869 | 0.60 | 79.70  | 136.00  | 3868.76 | 10.96 | -3.78 | 10.96 | 0.23 |
| 43                             | 3963 | 0.60 | 96.00  | 140.00  | 3962.75 | 11.93 | -3.74 | 11.93 | 0.18 |
| 44                             | 4056 | 0.10 | 341.80 | 145.00  | 4055.75 | 12.39 | -3.71 | 12.39 | 0.70 |
| 45                             | 4149 | 0.40 | 284.50 | 147.00  | 4148.75 | 12.05 | -3.55 | 12.05 | 0.38 |
| 46                             | 4243 | 0.40 | 262.90 | 149.00  | 4242.75 | 11.41 | -3.51 | 11.41 | 0.16 |
| 47                             | 4336 | 0.40 | 261.10 | 149.00  | 4335.74 | 10.77 | -3.60 | 10.77 | 0.01 |
| 48                             | 4429 | 0.60 | 262.60 | 150.00  | 4428.74 | 9.96  | -3.72 | 9.96  | 0.22 |
| 49                             | 4522 | 0.60 | 261.50 | 152.00  | 4521.74 | 9.00  | -3.85 | 9.00  | 0.01 |
| 50                             | 4616 | 0.60 | 267.80 | 154.00  | 4615.73 | 8.02  | -3.94 | 8.02  | 0.07 |
| 51                             | 4709 | 0.70 | 270.00 | 156.00  | 4708.72 | 6.96  | -3.96 | 6.96  | 0.11 |
| 52                             | 4802 | 0.80 | 264.60 | 156.00  | 4801.72 | 5.75  | -4.02 | 5.75  | 0.13 |
| 53                             | 4895 | 0.70 | 248.80 | 156.00  | 4894.71 | 4.57  | -4.29 | 4.57  | 0.25 |
| 54                             | 4989 | 0.70 | 245.30 | 158.00  | 4988.70 | 3.52  | -4.74 | 3.52  | 0.05 |
| 55                             | 5082 | 0.70 | 239.10 | 159.00  | 5081.69 | 2.51  | -5.27 | 2.51  | 0.08 |
| 56                             | 5175 | 0.70 | 238.50 | 159.00  | 5174.69 | 1.54  | -5.85 | 1.54  | 0.01 |
| 57                             | 5269 | 0.60 | 229.80 | 156.00  | 5268.68 | 0.68  | -6.47 | 0.68  | 0.15 |
| 58                             | 5362 | 0.70 | 241.40 | 156.00  | 5361.68 | -0.20 | -7.06 | -0.20 | 0.18 |
| 59                             | 5455 | 0.50 | 239.30 | 156.00  | 5454.67 | -1.04 | -7.54 | -1.04 | 0.22 |
| 60                             | 5548 | 0.20 | 140.70 | 158.00  | 5547.67 | -1.29 | -7.87 | -1.29 | 0.61 |
| 61                             | 5642 | 0.40 | 148.40 | 154.00  | 5641.67 | -1.01 | -8.28 | -1.01 | 0.22 |
| 62                             | 5735 | 0.10 | 216.80 | 161.00  | 5734.67 | -0.89 | -8.62 | -0.89 | 0.40 |
| 63                             | 5828 | 0.10 | 223.00 | 163.00  | 5827.67 | -1.00 | -8.74 | -1.00 | 0.01 |
| 64                             | 5922 | 0.30 | 1.40   | 163.00  | 5921.67 | -1.05 | -8.56 | -1.05 | 0.40 |
| 65                             | 5996 | 0.20 | 332.70 | 165.00  | 5995.67 | -1.10 | -8.25 | -1.10 | 0.21 |

Report #: **1**  
Date: **23-Jun-14**



**RYAN DIRECTIONAL SERVICES, INC.**  
A NABORS COMPANY

Ryan Job # **7789**

**SURVEY REPORT**

Customer: **Oasis Petroleum N.A. LLC**  
Well Name: **Chalmers 5301 44-24 4T2R**  
Rig #: **Nabors B-25**  
API #: **33-053-06011**  
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**  
Directional Drillers: **RPM**  
Survey Corrected To: **True North**  
Vertical Section Direction: **90**  
Total Correction: **8.27**  
Temperature Forecasting Model (Chart Only): **Logarithmic**

| Survey # | MD    | Inc   | Azm    | Temp   | TVD      | VS     | N/S     | E/W    | DLS   |
|----------|-------|-------|--------|--------|----------|--------|---------|--------|-------|
| 66       | 6094  | 0.10  | 256.00 | 129.00 | 6093.67  | -1.26  | -8.12   | -1.26  | 0.21  |
| 67       | 6187  | 0.40  | 183.20 | 134.00 | 6186.67  | -1.36  | -8.46   | -1.36  | 0.41  |
| 68       | 6281  | 0.90  | 70.30  | 140.00 | 6280.66  | -0.68  | -8.54   | -0.68  | 1.19  |
| 69       | 6374  | 0.50  | 45.80  | 143.00 | 6373.65  | 0.30   | -8.01   | 0.30   | 0.53  |
| 70       | 6467  | 0.40  | 132.30 | 147.00 | 6466.65  | 0.83   | -7.95   | 0.83   | 0.67  |
| 71       | 6561  | 0.10  | 127.10 | 149.00 | 6560.65  | 1.14   | -8.22   | 1.14   | 0.32  |
| 72       | 6654  | 0.70  | 40.30  | 152.00 | 6653.65  | 1.57   | -7.83   | 1.57   | 0.75  |
| 73       | 6747  | 0.50  | 38.20  | 156.00 | 6746.64  | 2.19   | -7.08   | 2.19   | 0.22  |
| 74       | 6840  | 0.70  | 72.90  | 159.00 | 6839.64  | 2.98   | -6.59   | 2.98   | 0.44  |
| 75       | 6934  | 0.70  | 107.60 | 159.00 | 6933.63  | 4.08   | -6.60   | 4.08   | 0.44  |
| 76       | 7027  | 0.90  | 11.30  | 163.00 | 7026.63  | 4.76   | -6.05   | 4.76   | 1.29  |
| 77       | 7120  | 1.20  | 13.30  | 165.00 | 7119.61  | 5.13   | -4.39   | 5.13   | 0.32  |
| 78       | 7213  | 0.60  | 8.30   | 168.00 | 7212.60  | 5.42   | -2.96   | 5.42   | 0.65  |
| 79       | 7307  | 0.60  | 26.00  | 170.00 | 7306.59  | 5.71   | -2.03   | 5.71   | 0.20  |
| 80       | 7400  | 0.70  | 27.30  | 172.00 | 7399.59  | 6.18   | -1.09   | 6.18   | 0.11  |
| 81       | 7493  | 0.40  | 353.30 | 170.00 | 7492.58  | 6.41   | -0.26   | 6.41   | 0.46  |
| 82       | 7587  | 0.40  | 7.60   | 172.00 | 7586.58  | 6.41   | 0.39    | 6.41   | 0.11  |
| 83       | 7680  | 0.70  | 52.50  | 172.00 | 7679.58  | 6.90   | 1.06    | 6.90   | 0.54  |
| 84       | 7773  | 0.70  | 45.20  | 174.00 | 7772.57  | 7.76   | 1.80    | 7.76   | 0.10  |
| 85       | 7866  | 1.00  | 26.40  | 176.00 | 7865.56  | 8.52   | 2.93    | 8.52   | 0.44  |
| 86       | 7960  | 1.10  | 316.20 | 176.00 | 7959.55  | 8.26   | 4.32    | 8.26   | 1.29  |
| 87       | 8053  | 0.80  | 304.20 | 177.00 | 8052.54  | 7.11   | 5.33    | 7.11   | 0.39  |
| 88       | 8146  | 0.90  | 263.40 | 179.00 | 8145.53  | 5.85   | 5.61    | 5.85   | 0.65  |
| 89       | 8240  | 0.90  | 231.20 | 181.00 | 8239.51  | 4.54   | 5.06    | 4.54   | 0.53  |
| 90       | 8333  | 0.90  | 232.10 | 185.00 | 8332.50  | 3.39   | 4.15    | 3.39   | 0.02  |
| 91       | 8426  | 1.00  | 211.30 | 185.00 | 8425.49  | 2.39   | 3.01    | 2.39   | 0.38  |
| 92       | 8519  | 1.10  | 207.30 | 186.00 | 8518.47  | 1.56   | 1.52    | 1.56   | 0.13  |
| 93       | 8613  | 1.30  | 225.20 | 188.00 | 8612.45  | 0.39   | -0.03   | 0.39   | 0.45  |
| 94       | 8706  | 0.90  | 191.80 | 190.00 | 8705.44  | -0.51  | -1.49   | -0.51  | 0.79  |
| 95       | 8799  | 0.70  | 177.80 | 194.00 | 8798.43  | -0.63  | -2.77   | -0.63  | 0.30  |
| 96       | 8892  | 0.90  | 158.70 | 197.00 | 8891.42  | -0.35  | -4.02   | -0.35  | 0.36  |
| 97       | 8986  | 1.10  | 174.20 | 199.00 | 8985.41  | 0.01   | -5.60   | 0.01   | 0.36  |
| 98       | 9079  | 0.20  | 108.40 | 201.00 | 9078.40  | 0.26   | -6.54   | 0.26   | 1.11  |
| 99       | 9172  | 0.70  | 94.80  | 197.00 | 9171.40  | 0.98   | -6.64   | 0.98   | 0.55  |
| 100      | 9266  | 0.40  | 78.60  | 199.00 | 9265.39  | 1.87   | -6.62   | 1.87   | 0.36  |
| 101      | 9359  | 0.40  | 78.80  | 201.00 | 9358.39  | 2.51   | -6.50   | 2.51   | 0.00  |
| 102      | 9452  | 0.50  | 88.40  | 201.00 | 9451.39  | 3.23   | -6.42   | 3.23   | 0.13  |
| 103      | 9545  | 0.50  | 103.10 | 201.00 | 9544.38  | 4.03   | -6.50   | 4.03   | 0.14  |
| 104      | 9639  | 0.10  | 172.10 | 203.00 | 9638.38  | 4.44   | -6.68   | 4.44   | 0.50  |
| 105      | 9732  | 0.50  | 141.90 | 206.00 | 9731.38  | 4.70   | -7.08   | 4.70   | 0.45  |
| 106      | 9825  | 0.60  | 71.70  | 206.00 | 9824.38  | 5.42   | -7.24   | 5.42   | 0.69  |
| 107      | 9919  | 0.20  | 56.50  | 208.00 | 9918.37  | 6.02   | -7.00   | 6.02   | 0.44  |
| 108      | 10012 | 0.20  | 264.90 | 208.00 | 10011.37 | 6.00   | -6.92   | 6.00   | 0.42  |
| 109      | 10105 | 0.80  | 197.80 | 210.00 | 10104.37 | 5.64   | -7.56   | 5.64   | 0.80  |
| 110      | 10175 | 0.60  | 203.40 | 210.00 | 10174.37 | 5.34   | -8.36   | 5.34   | 0.30  |
| 111      | 10202 | 0.70  | 220.60 | 188.00 | 10201.36 | 5.18   | -8.61   | 5.18   | 0.81  |
| 112      | 10233 | 2.50  | 125.40 | 192.00 | 10232.35 | 5.60   | -9.15   | 5.60   | 8.57  |
| 113      | 10265 | 7.00  | 115.60 | 195.00 | 10264.24 | 7.93   | -10.40  | 7.93   | 14.24 |
| 114      | 10296 | 11.30 | 112.60 | 195.00 | 10294.83 | 12.44  | -12.38  | 12.44  | 13.95 |
| 115      | 10327 | 16.20 | 111.30 | 199.00 | 10324.94 | 19.28  | -15.12  | 19.28  | 15.84 |
| 116      | 10358 | 20.00 | 109.00 | 203.00 | 10354.40 | 28.32  | -18.42  | 28.32  | 12.47 |
| 117      | 10389 | 22.60 | 107.50 | 203.00 | 10383.28 | 39.02  | -21.94  | 39.02  | 8.57  |
| 118      | 10420 | 25.70 | 106.50 | 203.00 | 10411.56 | 51.15  | -25.64  | 51.15  | 10.09 |
| 119      | 10451 | 29.60 | 106.50 | 203.00 | 10439.02 | 64.94  | -29.72  | 64.94  | 12.58 |
| 120      | 10482 | 33.40 | 106.00 | 206.00 | 10465.44 | 80.49  | -34.25  | 80.49  | 12.29 |
| 121      | 10513 | 37.40 | 106.80 | 206.00 | 10490.71 | 97.71  | -39.33  | 97.71  | 12.99 |
| 122      | 10544 | 41.00 | 107.60 | 206.00 | 10514.73 | 116.42 | -45.12  | 116.42 | 11.73 |
| 123      | 10575 | 43.70 | 108.60 | 206.00 | 10537.64 | 136.27 | -51.61  | 136.27 | 8.98  |
| 124      | 10607 | 46.70 | 109.60 | 206.00 | 10560.18 | 157.72 | -59.05  | 157.72 | 9.63  |
| 125      | 10638 | 50.40 | 110.50 | 206.00 | 10580.70 | 179.54 | -67.02  | 179.54 | 12.13 |
| 126      | 10669 | 54.20 | 110.30 | 210.00 | 10599.65 | 202.53 | -75.56  | 202.53 | 12.27 |
| 127      | 10700 | 56.10 | 110.30 | 210.00 | 10617.37 | 226.39 | -84.39  | 226.39 | 6.13  |
| 128      | 10731 | 57.10 | 110.20 | 210.00 | 10634.43 | 250.67 | -93.35  | 250.67 | 3.24  |
| 129      | 10762 | 60.30 | 110.30 | 210.00 | 10650.53 | 275.51 | -102.52 | 275.51 | 10.33 |
| 130      | 10793 | 64.60 | 109.80 | 210.00 | 10664.87 | 301.33 | -111.93 | 301.33 | 13.94 |



Report #: **1**  
Date: **23-Jun-14**



**RYAN DIRECTIONAL SERVICES, INC.**  
A NABORS COMPANY

Ryan Job # **7789**

**SURVEY REPORT**

Customer: **Oasis Petroleum N.A. LLC**  
Well Name: **Chalmers 5301 44-24 4T2R**  
Rig #: **Nabors B-25**  
API #: **33-053-06011**  
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**  
Directional Drillers: **RPM**  
Survey Corrected To: **True North**  
Vertical Section Direction: **90**  
Total Correction: **8.27**  
Temperature Forecasting Model (Chart Only): **Logarithmic**

| Survey # | MD    | Inc   | Azm    | Temp   | TVD      | VS      | N/S     | E/W     | DLS   |
|----------|-------|-------|--------|--------|----------|---------|---------|---------|-------|
| 131      | 10824 | 68.70 | 109.20 | 212.00 | 10677.15 | 328.15  | -121.43 | 328.15  | 13.34 |
| 132      | 10855 | 71.90 | 108.50 | 212.00 | 10687.60 | 355.77  | -130.86 | 355.77  | 10.54 |
| 133      | 10886 | 74.90 | 108.40 | 212.00 | 10696.46 | 383.95  | -140.26 | 383.95  | 9.68  |
| 134      | 10917 | 77.80 | 108.40 | 212.00 | 10703.77 | 412.53  | -149.76 | 412.53  | 9.35  |
| 135      | 10948 | 80.70 | 108.30 | 212.00 | 10709.55 | 441.43  | -159.35 | 441.43  | 9.36  |
| 136      | 10980 | 86.40 | 107.90 | 212.00 | 10713.15 | 471.64  | -169.23 | 471.64  | 17.86 |
| 137      | 11001 | 92.70 | 107.10 | 212.00 | 10713.31 | 491.66  | -175.54 | 491.66  | 30.24 |
| 138      | 11042 | 93.50 | 107.20 | 212.00 | 10711.09 | 530.78  | -187.61 | 530.78  | 1.97  |
| 139      | 11116 | 92.20 | 107.10 | 212.00 | 10707.42 | 601.40  | -209.40 | 601.40  | 1.76  |
| 140      | 11146 | 91.00 | 106.20 | 213.00 | 10706.58 | 630.13  | -217.99 | 630.13  | 5.00  |
| 141      | 11177 | 89.70 | 106.40 | 210.00 | 10706.39 | 659.88  | -226.69 | 659.88  | 4.24  |
| 142      | 11208 | 90.20 | 106.20 | 208.00 | 10706.42 | 689.64  | -235.40 | 689.64  | 1.74  |
| 143      | 11239 | 90.10 | 105.00 | 208.00 | 10706.33 | 719.49  | -243.73 | 719.49  | 3.88  |
| 144      | 11270 | 90.20 | 103.10 | 208.00 | 10706.25 | 749.57  | -251.26 | 749.57  | 6.14  |
| 145      | 11301 | 89.00 | 101.60 | 210.00 | 10706.47 | 779.85  | -257.89 | 779.85  | 6.20  |
| 146      | 11332 | 87.50 | 100.60 | 213.00 | 10707.42 | 810.25  | -263.85 | 810.25  | 5.81  |
| 147      | 11363 | 87.50 | 100.61 | 212.00 | 10708.77 | 840.69  | -269.55 | 840.69  | 0.03  |
| 148      | 11394 | 87.70 | 100.80 | 212.00 | 10710.07 | 871.13  | -275.31 | 871.13  | 0.89  |
| 149      | 11424 | 88.40 | 98.40  | 213.00 | 10711.09 | 900.69  | -280.31 | 900.69  | 8.33  |
| 150      | 11455 | 89.70 | 98.10  | 212.00 | 10711.60 | 931.36  | -284.75 | 931.36  | 4.30  |
| 151      | 11486 | 90.50 | 97.90  | 212.00 | 10711.55 | 962.06  | -289.07 | 962.06  | 2.66  |
| 152      | 11516 | 90.50 | 95.20  | 213.00 | 10711.29 | 991.86  | -292.49 | 991.86  | 9.00  |
| 153      | 11547 | 89.40 | 95.40  | 213.00 | 10711.31 | 1022.73 | -295.35 | 1022.73 | 3.61  |
| 154      | 11578 | 89.70 | 94.70  | 215.00 | 10711.56 | 1053.60 | -298.08 | 1053.60 | 2.46  |
| 155      | 11609 | 88.20 | 92.70  | 217.00 | 10712.12 | 1084.53 | -300.08 | 1084.53 | 8.06  |
| 156      | 11640 | 88.00 | 92.40  | 219.00 | 10713.15 | 1115.48 | -301.46 | 1115.48 | 1.16  |
| 157      | 11671 | 88.70 | 92.60  | 219.00 | 10714.04 | 1146.44 | -302.81 | 1146.44 | 2.35  |
| 158      | 11702 | 88.80 | 92.10  | 217.00 | 10714.72 | 1177.41 | -304.08 | 1177.41 | 1.64  |
| 159      | 11732 | 88.90 | 92.30  | 219.00 | 10715.32 | 1207.38 | -305.23 | 1207.38 | 0.75  |
| 160      | 11763 | 87.30 | 90.60  | 219.00 | 10716.35 | 1238.35 | -306.02 | 1238.35 | 7.53  |
| 161      | 11794 | 86.60 | 90.30  | 222.00 | 10718.00 | 1269.31 | -306.26 | 1269.31 | 2.46  |
| 162      | 11857 | 84.80 | 90.40  | 224.00 | 10722.72 | 1332.12 | -306.64 | 1332.12 | 2.86  |
| 163      | 11950 | 85.60 | 89.80  | 222.00 | 10730.51 | 1424.80 | -306.81 | 1424.80 | 1.07  |
| 164      | 12043 | 86.90 | 89.60  | 222.00 | 10736.59 | 1517.59 | -306.32 | 1517.59 | 1.41  |
| 165      | 12135 | 89.40 | 89.50  | 226.00 | 10739.56 | 1609.54 | -305.60 | 1609.54 | 2.72  |
| 166      | 12228 | 91.60 | 89.00  | 226.00 | 10738.75 | 1702.52 | -304.38 | 1702.52 | 2.43  |
| 167      | 12321 | 90.70 | 88.60  | 228.00 | 10736.88 | 1795.48 | -302.43 | 1795.48 | 1.06  |
| 168      | 12414 | 89.80 | 88.60  | 230.00 | 10736.47 | 1888.45 | -300.16 | 1888.45 | 0.97  |
| 169      | 12507 | 86.60 | 88.80  | 233.00 | 10739.39 | 1981.37 | -298.05 | 1981.37 | 3.45  |
| 170      | 12599 | 87.30 | 89.20  | 237.00 | 10744.29 | 2073.22 | -296.45 | 2073.22 | 0.88  |
| 171      | 12692 | 88.40 | 90.00  | 240.00 | 10747.78 | 2166.15 | -295.80 | 2166.15 | 1.46  |
| 172      | 12786 | 89.70 | 90.50  | 242.00 | 10749.34 | 2260.14 | -296.21 | 2260.14 | 1.48  |
| 173      | 12880 | 90.90 | 91.10  | 244.00 | 10748.85 | 2354.12 | -297.52 | 2354.12 | 1.43  |
| 174      | 12975 | 87.40 | 89.20  | 237.00 | 10750.25 | 2449.09 | -297.77 | 2449.09 | 4.19  |
| 175      | 13069 | 89.10 | 89.50  | 240.00 | 10753.13 | 2543.04 | -296.71 | 2543.04 | 1.84  |
| 176      | 13163 | 87.50 | 88.50  | 240.00 | 10755.91 | 2636.98 | -295.07 | 2636.98 | 2.01  |
| 177      | 13257 | 86.90 | 89.40  | 235.00 | 10760.51 | 2730.85 | -293.35 | 2730.85 | 1.15  |
| 178      | 13350 | 88.00 | 89.70  | 239.00 | 10764.64 | 2823.75 | -292.62 | 2823.75 | 1.23  |
| 179      | 13444 | 89.00 | 90.30  | 237.00 | 10767.10 | 2917.72 | -292.62 | 2917.72 | 1.24  |
| 180      | 13538 | 89.80 | 91.20  | 240.00 | 10768.09 | 3011.71 | -293.85 | 3011.71 | 1.28  |
| 181      | 13631 | 89.70 | 91.10  | 242.00 | 10768.49 | 3104.69 | -295.71 | 3104.69 | 0.15  |
| 182      | 13726 | 89.70 | 90.80  | 242.00 | 10768.99 | 3199.67 | -297.29 | 3199.67 | 0.32  |
| 183      | 13819 | 89.10 | 90.70  | 244.00 | 10769.97 | 3292.66 | -298.51 | 3292.66 | 0.65  |
| 184      | 13913 | 89.30 | 91.80  | 244.00 | 10771.28 | 3386.62 | -300.56 | 3386.62 | 1.19  |
| 185      | 14007 | 89.20 | 91.50  | 246.00 | 10772.51 | 3480.58 | -303.26 | 3480.58 | 0.34  |
| 186      | 14101 | 89.40 | 91.40  | 246.00 | 10773.66 | 3574.54 | -305.64 | 3574.54 | 0.24  |
| 187      | 14195 | 89.00 | 91.20  | 242.00 | 10774.97 | 3668.51 | -307.77 | 3668.51 | 0.48  |
| 188      | 14289 | 88.50 | 89.80  | 240.00 | 10777.02 | 3762.48 | -308.59 | 3762.48 | 1.58  |
| 189      | 14383 | 88.00 | 89.70  | 242.00 | 10779.89 | 3856.43 | -308.18 | 3856.43 | 0.54  |
| 190      | 14476 | 89.10 | 90.20  | 240.00 | 10782.24 | 3949.40 | -308.10 | 3949.40 | 1.30  |
| 191      | 14571 | 89.60 | 90.30  | 242.00 | 10783.32 | 4044.39 | -308.52 | 4044.39 | 0.54  |
| 192      | 14665 | 89.90 | 90.00  | 242.00 | 10783.73 | 4138.39 | -308.76 | 4138.39 | 0.45  |
| 193      | 14758 | 90.30 | 90.20  | 244.00 | 10783.57 | 4231.39 | -308.93 | 4231.39 | 0.48  |
| 194      | 14853 | 90.60 | 90.40  | 244.00 | 10782.82 | 4326.39 | -309.42 | 4326.39 | 0.38  |
| 195      | 14946 | 90.50 | 90.40  | 246.00 | 10781.93 | 4419.38 | -310.07 | 4419.38 | 0.11  |

Report #: **1**  
Date: **23-Jun-14**



**RYAN DIRECTIONAL SERVICES, INC.**  
A NABORS COMPANY

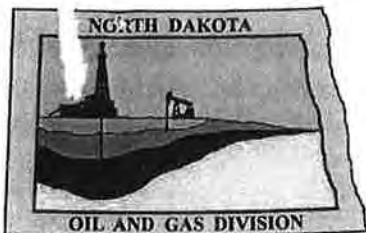
Ryan Job # **7789**

**SURVEY REPORT**

Customer: **Oasis Petroleum N.A. LLC**  
Well Name: **Chalmers 5301 44-24 4T2R**  
Rig #: **Nabors B-25**  
API #: **33-053-06011**  
Calculation Method: **Minimum Curvature Calculation**

MWD Operator: **Daniel Ogden**  
Directional Drillers: **RPM**  
Survey Corrected To: **True North**  
Vertical Section Direction: **90**  
Total Correction: **8.27**  
Temperature Forecasting Model (Chart Only): **Logarithmic**

| Survey #   | MD    | Inc   | Azm   | Temp   | TVD      | VS       | N/S     | E/W      | DLS  |
|------------|-------|-------|-------|--------|----------|----------|---------|----------|------|
| 196        | 15040 | 89.50 | 89.80 | 246.00 | 10781.93 | 4513.38  | -310.24 | 4513.38  | 1.24 |
| 197        | 15135 | 89.90 | 90.40 | 249.00 | 10782.43 | 4608.38  | -310.40 | 4608.38  | 0.76 |
| 198        | 15229 | 89.20 | 90.30 | 249.00 | 10783.17 | 4702.37  | -310.98 | 4702.37  | 0.75 |
| 199        | 15323 | 88.90 | 89.90 | 251.00 | 10784.72 | 4796.36  | -311.14 | 4796.36  | 0.53 |
| 200        | 15417 | 88.70 | 89.10 | 251.00 | 10786.69 | 4890.33  | -310.32 | 4890.33  | 0.88 |
| 201        | 15512 | 89.40 | 89.00 | 253.00 | 10788.27 | 4985.31  | -308.75 | 4985.31  | 0.74 |
| 202        | 15606 | 89.50 | 88.60 | 253.00 | 10789.17 | 5079.28  | -306.78 | 5079.28  | 0.44 |
| 203        | 15700 | 89.70 | 88.40 | 253.00 | 10789.83 | 5173.25  | -304.32 | 5173.25  | 0.30 |
| 204        | 15793 | 88.70 | 89.10 | 251.00 | 10791.13 | 5266.21  | -302.29 | 5266.21  | 1.31 |
| 205        | 15887 | 87.60 | 88.70 | 249.00 | 10794.16 | 5360.15  | -300.48 | 5360.15  | 1.25 |
| 206        | 15981 | 88.70 | 89.70 | 251.00 | 10797.19 | 5454.09  | -299.17 | 5454.09  | 1.58 |
| 207        | 16075 | 88.90 | 89.60 | 253.00 | 10799.16 | 5548.06  | -298.60 | 5548.06  | 0.24 |
| 208        | 16169 | 91.00 | 90.90 | 253.00 | 10799.25 | 5642.06  | -299.01 | 5642.06  | 2.63 |
| 209        | 16263 | 92.00 | 90.20 | 255.00 | 10796.78 | 5736.02  | -299.91 | 5736.02  | 1.30 |
| 210        | 16356 | 91.70 | 90.40 | 253.00 | 10793.78 | 5828.97  | -300.40 | 5828.97  | 0.39 |
| 211        | 16450 | 89.30 | 90.70 | 255.00 | 10792.96 | 5922.95  | -301.30 | 5922.95  | 2.57 |
| 212        | 16544 | 89.20 | 89.60 | 257.00 | 10794.19 | 6016.94  | -301.55 | 6016.94  | 1.17 |
| 213        | 16638 | 89.00 | 89.60 | 255.00 | 10795.67 | 6110.93  | -300.89 | 6110.93  | 0.21 |
| 214        | 16731 | 89.40 | 88.90 | 255.00 | 10796.97 | 6203.91  | -299.67 | 6203.91  | 0.87 |
| 215        | 16825 | 90.90 | 88.90 | 257.00 | 10796.72 | 6297.89  | -297.87 | 6297.89  | 1.60 |
| 216        | 16919 | 90.50 | 88.60 | 257.00 | 10795.57 | 6391.86  | -295.82 | 6391.86  | 0.53 |
| 217        | 17012 | 90.50 | 89.00 | 255.00 | 10794.76 | 6484.84  | -293.87 | 6484.84  | 0.43 |
| 218        | 17106 | 90.70 | 89.00 | 257.00 | 10793.78 | 6578.82  | -292.23 | 6578.82  | 0.21 |
| 219        | 17200 | 90.10 | 88.50 | 258.00 | 10793.12 | 6672.79  | -290.18 | 6672.79  | 0.83 |
| 220        | 17295 | 89.80 | 87.50 | 255.00 | 10793.20 | 6767.73  | -286.86 | 6767.73  | 1.10 |
| 221        | 17389 | 89.30 | 88.40 | 253.00 | 10793.94 | 6861.67  | -283.50 | 6861.67  | 1.10 |
| 222        | 17483 | 89.00 | 88.30 | 255.00 | 10795.34 | 6955.62  | -280.80 | 6955.62  | 0.34 |
| 223        | 17576 | 90.90 | 89.80 | 253.00 | 10795.42 | 7048.60  | -279.25 | 7048.60  | 2.60 |
| 224        | 17670 | 92.10 | 89.70 | 255.00 | 10792.96 | 7142.56  | -278.84 | 7142.56  | 1.28 |
| 225        | 17764 | 91.00 | 90.50 | 253.00 | 10790.41 | 7236.53  | -279.01 | 7236.53  | 1.45 |
| 226        | 17858 | 92.00 | 90.20 | 253.00 | 10787.95 | 7330.49  | -279.58 | 7330.49  | 1.11 |
| 227        | 17951 | 90.80 | 89.90 | 255.00 | 10785.68 | 7423.46  | -279.66 | 7423.46  | 1.33 |
| 228        | 18045 | 90.20 | 90.90 | 255.00 | 10784.86 | 7517.46  | -280.32 | 7517.46  | 1.24 |
| 229        | 18138 | 90.00 | 90.30 | 255.00 | 10784.70 | 7610.45  | -281.29 | 7610.45  | 0.68 |
| 230        | 18231 | 90.20 | 89.90 | 257.00 | 10784.54 | 7703.45  | -281.46 | 7703.45  | 0.48 |
| 231        | 18325 | 89.90 | 89.60 | 255.00 | 10784.45 | 7797.45  | -281.05 | 7797.45  | 0.45 |
| 232        | 18418 | 87.60 | 90.20 | 255.00 | 10786.48 | 7890.42  | -280.88 | 7890.42  | 2.56 |
| 233        | 18512 | 87.40 | 89.90 | 255.00 | 10790.58 | 7984.33  | -280.96 | 7984.33  | 0.38 |
| 234        | 18606 | 87.60 | 90.10 | 257.00 | 10794.68 | 8078.24  | -280.96 | 8078.24  | 0.30 |
| 235        | 18699 | 88.10 | 89.50 | 258.00 | 10798.17 | 8171.17  | -280.64 | 8171.17  | 0.84 |
| 236        | 18793 | 87.90 | 89.50 | 255.00 | 10801.45 | 8265.11  | -279.82 | 8265.11  | 0.21 |
| 237        | 18887 | 89.90 | 90.10 | 257.00 | 10803.26 | 8359.09  | -279.49 | 8359.09  | 2.22 |
| 238        | 18982 | 90.50 | 90.40 | 258.00 | 10802.93 | 8454.09  | -279.91 | 8454.09  | 0.71 |
| 239        | 19076 | 89.10 | 90.00 | 258.00 | 10803.25 | 8548.08  | -280.24 | 8548.08  | 1.55 |
| 240        | 19171 | 89.60 | 89.70 | 258.00 | 10804.33 | 8643.08  | -279.99 | 8643.08  | 0.61 |
| 241        | 19266 | 90.20 | 90.60 | 258.00 | 10804.50 | 8738.07  | -280.24 | 8738.07  | 1.14 |
| 242        | 19361 | 90.00 | 90.50 | 258.00 | 10804.33 | 8833.07  | -281.15 | 8833.07  | 0.24 |
| 243        | 19455 | 90.80 | 90.50 | 258.00 | 10803.68 | 8927.06  | -281.97 | 8927.06  | 0.85 |
| 244        | 19550 | 91.30 | 90.50 | 258.00 | 10801.93 | 9022.04  | -282.80 | 9022.04  | 0.53 |
| 245        | 19644 | 90.30 | 90.00 | 258.00 | 10800.62 | 9116.03  | -283.21 | 9116.03  | 1.19 |
| 246        | 19739 | 90.80 | 89.70 | 260.00 | 10799.71 | 9211.03  | -282.96 | 9211.03  | 0.61 |
| 247        | 19833 | 90.20 | 90.40 | 258.00 | 10798.89 | 9305.02  | -283.04 | 9305.02  | 0.98 |
| 248        | 19927 | 88.70 | 89.80 | 260.00 | 10799.79 | 9399.01  | -283.20 | 9399.01  | 1.72 |
| 249        | 20020 | 88.40 | 89.70 | 258.00 | 10802.15 | 9491.98  | -282.80 | 9491.98  | 0.34 |
| 250        | 20114 | 89.50 | 89.50 | 260.00 | 10803.87 | 9585.96  | -282.14 | 9585.96  | 1.19 |
| 251        | 20208 | 89.70 | 89.00 | 262.00 | 10804.52 | 9679.95  | -280.91 | 9679.95  | 0.57 |
| 252        | 20302 | 90.10 | 89.70 | 260.00 | 10804.69 | 9773.95  | -279.85 | 9773.95  | 0.86 |
| 253        | 20396 | 91.00 | 90.80 | 262.00 | 10803.79 | 9867.94  | -280.26 | 9867.94  | 1.51 |
| 254        | 20490 | 91.60 | 90.40 | 264.00 | 10801.65 | 9961.91  | -281.24 | 9961.91  | 0.77 |
| 255        | 20584 | 92.50 | 90.00 | 258.00 | 10798.29 | 10055.85 | -281.57 | 10055.85 | 1.05 |
| 256        | 20678 | 89.00 | 90.70 | 260.00 | 10797.06 | 10149.82 | -282.14 | 10149.82 | 3.80 |
| 257        | 20772 | 89.20 | 90.50 | 262.00 | 10798.54 | 10243.81 | -283.13 | 10243.81 | 0.30 |
| 258        | 20866 | 89.70 | 90.30 | 262.00 | 10799.44 | 10337.80 | -283.78 | 10337.80 | 0.57 |
| 259        | 20898 | 89.80 | 90.50 | 262.00 | 10799.58 | 10369.80 | -284.01 | 10369.80 | 0.70 |
| Projection | 20963 | 89.80 | 90.50 | 262.00 | 10799.81 | 10434.79 | -284.57 | 10434.79 | 0.00 |



# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

## Department of Mineral Resources

Lynn D. Helms - Director

## North Dakota Industrial Commission

www.dmr.nd.gov/oilgas

28600

BRANDI TERRY  
OASIS PETROLEUM NORTH AMERICA LLC  
1001 FANNIN STE 1500  
HOUSTON, TX 77002 USA

Date: 6/17/2014

**RE: CORES AND SAMPLES**

Well Name: **CHALMERS 5301 44-24 4T2R** Well File No.: **28600**  
Location: **SESE 24-153-101** County: **MCKENZIE**  
Permit Type: **Development - HORIZONTAL**  
Field: **BAKER** Target Horizon: **THREE FORKS B2**

Dear BRANDI TERRY:

North Dakota Century Code Section 38-08-04 provides for the preservation of cores and samples and their shipment to the State Geologist when requested. The following is required on the above referenced well:

- 1) All cores, core chips and samples must be submitted to the State Geologist as provided for under North Dakota Century Code: Section 38-08-04 and North Dakota Administrative Code: Section 43-02-03-38.1.
- 2) Samples: The Operator is to begin collecting sample drill cuttings no lower than the:  
**Base of the Last Charles Salt**
  - Sample cuttings shall be collected at:
    - o 30' maximum intervals through all vertical and build sections.
    - o 100' maximum intervals through any horizontal sections.
  - Samples must be washed, dried, placed in standard sample envelopes (3" x 4.5"), packed in the correct order into standard sample boxes (3.5" x 5.25" x 15.25").
  - Samples boxes are to be carefully identified with a label that indicates the operator, well name, well file number, American Petroleum Institute (API) number, location and depth of samples; and forwarded in to the state core and sample library within 30 days of the completion of drilling operations.
- 3) Cores: Any cores cut shall be preserved in correct order, boxed in standard core boxes (4.5", 4.5", 35.75"), and the entire core forwarded to the state core and samples library within 180 days of completion of drilling operations. Any extension of time must have approval on a Form 4 Sundry Notice.

All cores, core chips, and samples must be shipped, prepaid, to the state core and samples library at the following address:

**ND Geological Survey Core Library  
2835 Campus Road, Stop 8156  
Grand Forks, ND 58202**

North Dakota Century Code Section 38-08-16 allows for a civil penalty for any violation of Chapter 38 08 not to exceed \$12,500 for each offense, and each day's violation is a separate offense.

Sincerely

  
Stephen Fried  
Geologist

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.  
**28400**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                               |                                                                                  |                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>June 1, 2014</b> | <input checked="" type="checkbox"/> Drilling Prognosis                           | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                           | <input type="checkbox"/> Redrilling or Repair                                    | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                        | <input type="checkbox"/> Casing or Liner                                         | <input type="checkbox"/> Acidizing                |
|                                                                                                                                                 |                                               | <input type="checkbox"/> Plug Well                                               | <input type="checkbox"/> Fracture Treatment       |
|                                                                                                                                                 |                                               | <input type="checkbox"/> Supplemental History                                    | <input type="checkbox"/> Change Production Method |
|                                                                                                                                                 |                                               | <input type="checkbox"/> Temporarily Abandon                                     | <input type="checkbox"/> Reclamation              |
|                                                                                                                                                 |                                               | <input checked="" type="checkbox"/> Other <b>Waiver to rule Rule 43-02-03-31</b> |                                                   |

|                                                         |               |                 |              |              |  |
|---------------------------------------------------------|---------------|-----------------|--------------|--------------|--|
| Well Name and Number<br><b>Chalmers 5301 44-24 4T2R</b> |               |                 |              |              |  |
| Footages                                                | Qtr-Qtr       | Section         | Township     | Range        |  |
| <b>877 F S L 245 F E L</b>                              | <b>SESE</b>   | <b>24</b>       | <b>153 N</b> | <b>101 W</b> |  |
| Field                                                   | Pool          | County          |              |              |  |
|                                                         | <b>Bakken</b> | <b>McKenzie</b> |              |              |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

Oasis Petroleum respectfully requests a waiver to Rule 43-02-03-31 in regards to running open hole logs for the above referenced well. Justification for this request is as follows:

Oasis Petroleum/ Chalmers 5300 31-19H (NDIC 20407) located within one mile of the subject well

If this exception is approved, Oasis Petroleum will run a CBL on the intermediate string, and we will also run GR to surface. Oasis Petroleum will also submit two digital copies of each cased hole log and a copy of the mud log containing MWD GR.

|                                                     |  |                                         |                          |
|-----------------------------------------------------|--|-----------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>281-404-9563</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                         |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                      | Zip Code<br><b>77002</b> |
| Signature<br><i>Heather McCowan</i>                 |  | Printed Name<br><b>Heather McCowan</b>  |                          |
| Title<br><b>Regulatory Assistant</b>                |  | Date<br><b>May 28, 2014</b>             |                          |
| Email Address<br><b>hmccowan@oasispetroleum.com</b> |  |                                         |                          |

| FOR STATE USE ONLY                          |                                              |
|---------------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received           | <input checked="" type="checkbox"/> Approved |
| Date<br><b>6-11-2014</b>                    |                                              |
| By<br><i>Stephen Fried</i>                  |                                              |
| Title<br><b>Stephen Fried<br/>Geologist</b> |                                              |



**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)



Well File No.  
**28600**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>June 18, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                            |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                         |

|                                                             |                                                   |
|-------------------------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis                 | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair               | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner                    | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well                          | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History               | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon                | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other <b>Change BHL</b> |                                                   |

|                                                         |               |                 |          |       |  |
|---------------------------------------------------------|---------------|-----------------|----------|-------|--|
| Well Name and Number<br><b>Chalmers 5301 44-24 4T2R</b> |               |                 |          |       |  |
| Footages                                                | Qtr-Qtr       | Section         | Township | Range |  |
| <b>877 F S L 245 F E L SESE 24 153 N 101 W</b>          |               |                 |          |       |  |
| Field                                                   | Pool          | County          |          |       |  |
|                                                         | <b>Bakken</b> | <b>McKenzie</b> |          |       |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                       |      |       |          |
|-----------------------|------|-------|----------|
| Name of Contractor(s) |      |       |          |
| Address               | City | State | Zip Code |

**DETAILS OF WORK**

**Oasis Petroleum respectfully requests permission to make the following changes to the above referenced well:**

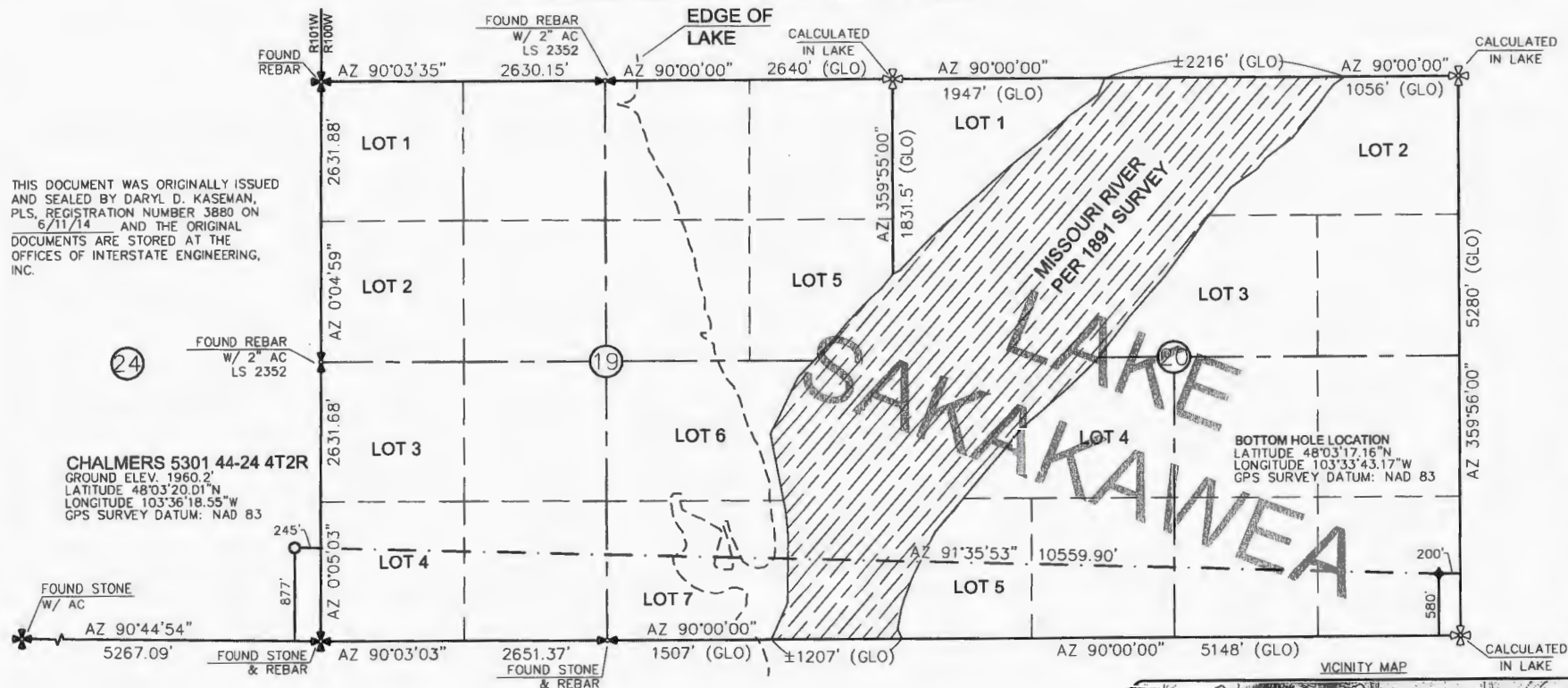
**Latitude: 48, 03', 20.01" N**

**Plats: Please ensure plats attached are on file for the APD.**

|                                                     |                                         |
|-----------------------------------------------------|-----------------------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> | Telephone Number<br><b>281-404-9563</b> |
| Address<br><b>1001 Fannin, Suite 1500</b>           |                                         |
| City<br><b>Houston</b>                              | State<br><b>TX</b>                      |
| Signature<br><i>Heather McCowan</i>                 | Printed Name<br><b>Heather McCowan</b>  |
| Title<br><b>Regulatory Assistant</b>                | Date<br><b>June 18, 2014</b>            |
| Email Address<br><b>hmccowan@oasispetroleum.com</b> |                                         |

| FOR STATE USE ONLY                            |                                   |
|-----------------------------------------------|-----------------------------------|
| <input checked="" type="checkbox"/> Received  | <input type="checkbox"/> Approved |
| Date<br><b>6/23/14</b>                        |                                   |
| By<br><i>Anthony Ebele</i>                    |                                   |
| Title<br><b>Petroleum Resource Specialist</b> |                                   |

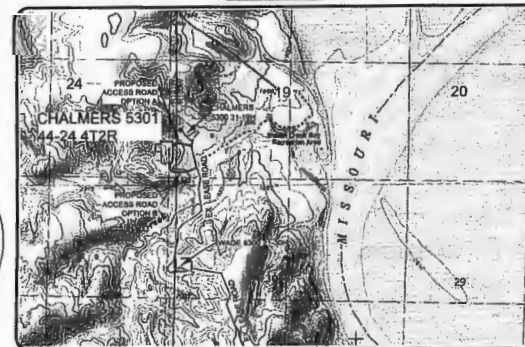
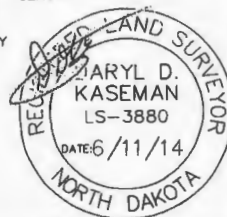
**WELL LOCATION PLAT**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 "CHALMERS 5301 44-24 4T2R"  
 877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



⊕ - MONUMENT - RECOVERED  
 ⊗ - MONUMENT - NOT RECOVERED

STAKED ON 9/3/13  
 VERTICAL CONTROL DATUM WAS BASED UPON  
 CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'  
 THIS SURVEY AND PLAT IS BEING PROVIDED AT THE  
 REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY  
 THAT THIS PLAT CORRECTLY REPRESENTS WORK  
 PERFORMED BY ME OR UNDER MY SUPERVISION  
 AND IS TRUE AND CORRECT TO THE BEST OF MY  
 KNOWLEDGE AND BELIEF.

*[Signature]*  
 DARYL D. KASEMAN LS-3880



© 2014, INTERSTATE ENGINEERING, INC.

| Revision | No.     | Date | Description              |
|----------|---------|------|--------------------------|
| REV 1    | 1/29/14 | BM   | CHANGED AND LAYOUT       |
| REV 2    | 5/29/14 | LS   | ADDED WELL TO PLAT       |
| REV 3    | 6/11/14 | LS   | CHANGED NAME, ADDED WELL |
| REV 4    | 6/11/14 | LS   | ADDED CHALMERS 5301      |
| REV 5    | 6/11/14 | LS   | ADDED CHALMERS 5301      |

OASIS PETROLEUM NORTH AMERICA, LLC  
 WELL LOCATION PLAT  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA  
 Drawn By: J.J.S.  
 Project No.: 115-25-25-02  
 Checked By: D.D.K.

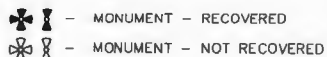
Interstate Engineering, Inc.  
 P.O. Box 644  
 425 East Main Street  
 Sidney, Montana 59270  
 Ph: (408) 433-5617  
 Fax: (408) 433-5618  
 www.interstateeng.com  
 Other offices in Montana, North Dakota and South Dakota



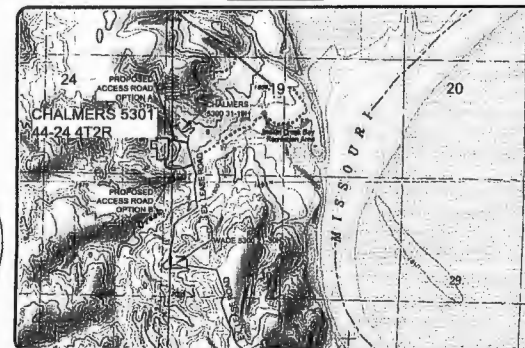
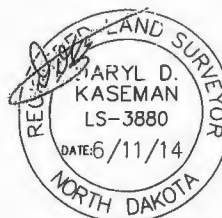
1/8  
 SHEET NO.



SECTION 24, T153N, R101W, & SECTION 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY -0°03'.



| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 3        | 1/20/14 | BWH | CHANGED P&Q LAYOUT       |
| REV 4        | 5/08/14 | JLS | ADDED WELL TO PAD        |
| REV 5        | 5/20/14 | JLS | CHANGED NAME, MOVED WELL |
| REV 6        | 6/11/14 | JSM | ADDED DIMENSIONS         |

|                                                                                                                   |                                         |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| OASIS PETROLEUM NORTH AMERICA, LLC<br>SECTION BREAKDOWN<br>SECTION 24, T15N, R10W, & SECTIONS 19 & 20, T15N, R10W |                                         |
| MCKENZIE COUNTY, NORTH DAKOTA                                                                                     |                                         |
| Drawn By: JJS<br>Checked By: DDK                                                                                  | Project No.: S13-06-25-07<br>Date: 2014 |

**Interstate Engineering, Inc.**  
P.O. Box 948  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5817  
Fax (406) 433-5818  
[www.interstateeng.com](http://www.interstateeng.com)



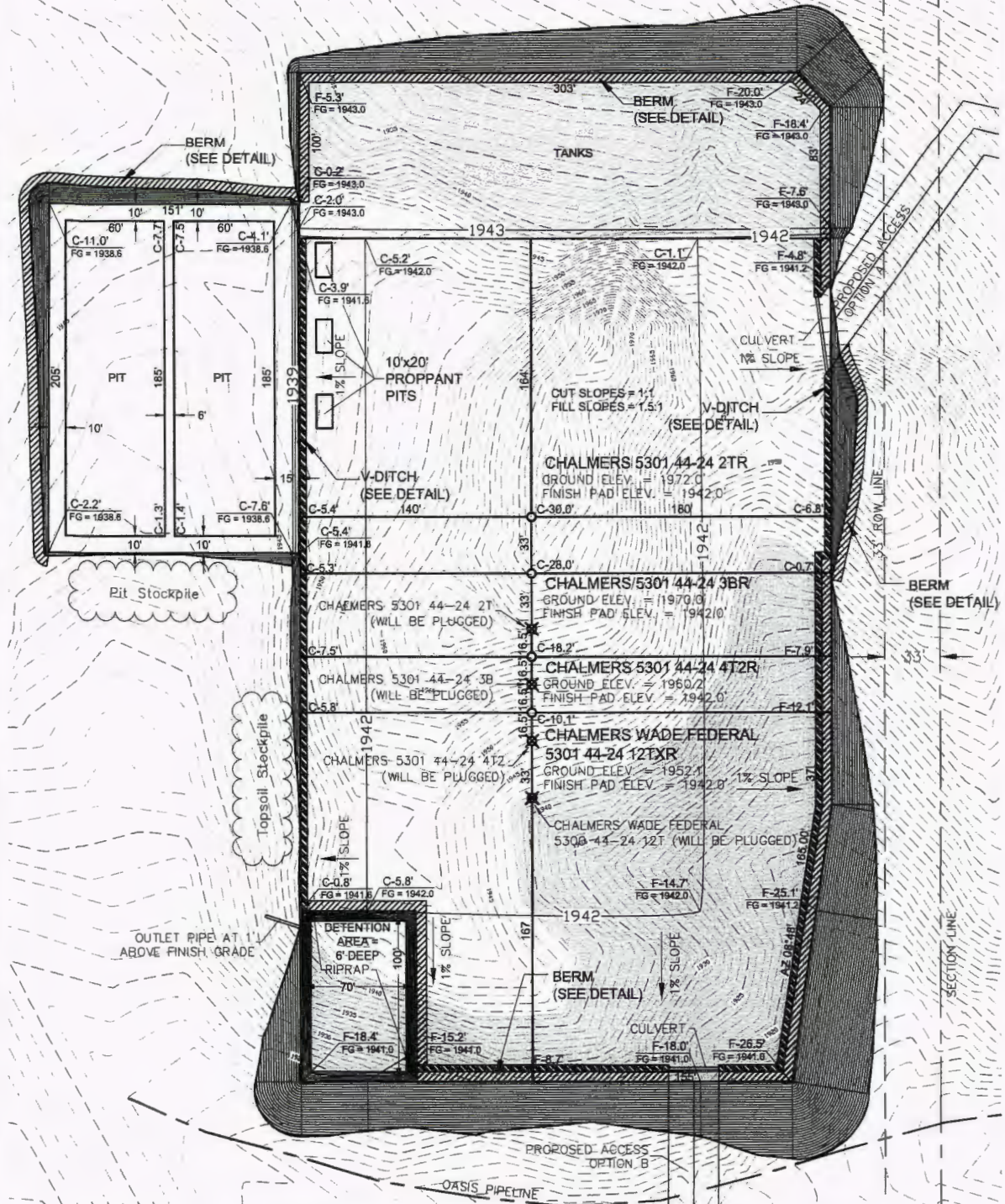
**INTERSTATE  
ENGINEERING**

Productions are your need, outside the box

2/8

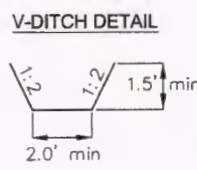
THIS DOCUMENT WAS ORIGINALLY ISSUED  
AND SEALED BY DARYL D. KASEMAN,  
P.L.S., REGISTRATION NUMBER 3880 ON  
6/11/14 AND THE ORIGINAL  
DOCUMENTS ARE STORED AT THE  
OFFICES OF INTERSTATE ENGINEERING,  
INC.

# **PAD LAYOUT** OASIS PETROLEUM NORTH AMERICA, LLC 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002 "CHALMERS 5301 44-24 4T2R" 877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



RECORDED  
DARYL D. KASEMAN  
LS-3880  
DATE: 6/11/14  
NORTH DAKOTA

- BERM
- DITCH
- Proposed Contours
- - - Original Contours



NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

3/8  
SHEET NO.



Interstate Engineering, Inc.  
P.O. Box 948  
425 East Main Street  
Sidney, Montana 59720  
Ph (408) 433-5617  
Fax (408) 433-5610  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota

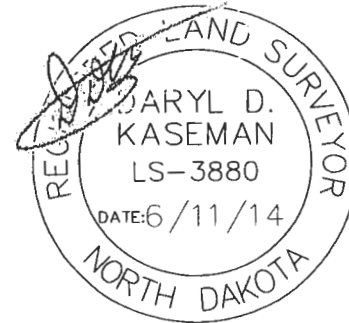
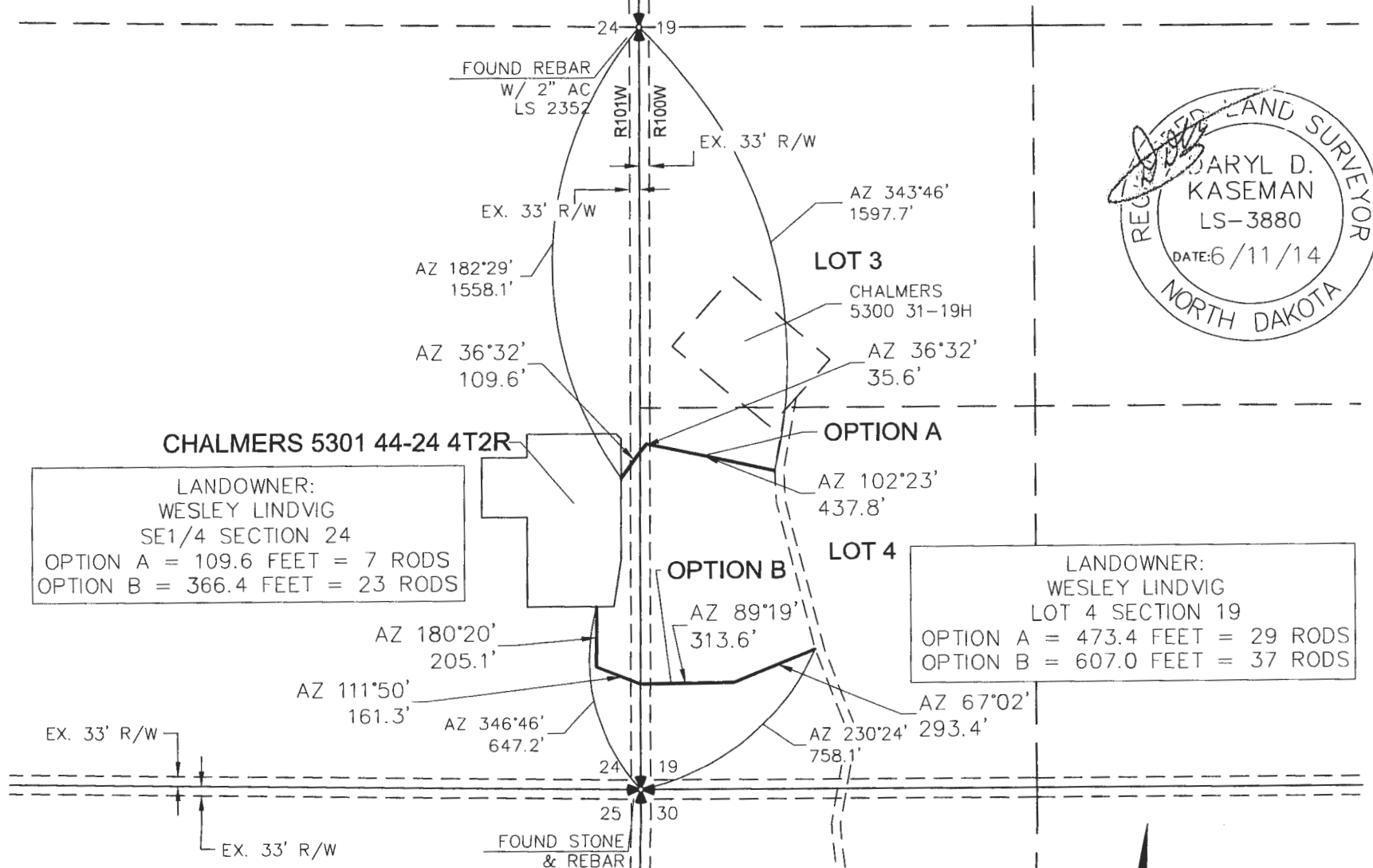
OASIS PETROLEUM NORTH AMERICA, LLC  
PAD LAYOUT  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: 515-08-236.02  
Checked By: D.D.K. Date: 10/14

| Revision | Date    | By  | Description              |
|----------|---------|-----|--------------------------|
| REV 3    | 1/20/14 | BW  | CHANGED PAD LAYOUT       |
| REV 4    | 5/09/14 | JAS | ADDED WELL TO PAD        |
| REV 5    | 5/20/14 | JAS | CHANGED NAME, MOVED WELL |
| REV 6    | 6/11/14 | JDM | ADDED DIMENSIONS         |

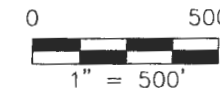


**ACCESS APPROACH**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 "CHALMERS 5301 44-24 4T2R"  
 877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 6/11/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



© 2014, INTERSTATE ENGINEERING, INC.

Interstate Engineering, Inc.  
 P.O. Box 648  
 425 East Main Street  
 Sidney, Montana 59270  
 Ph (406) 433-5617  
 Fax (406) 433-5618  
 www.interstateeng.com  
 Other offices in Minnesota, North Dakota and South Dakota



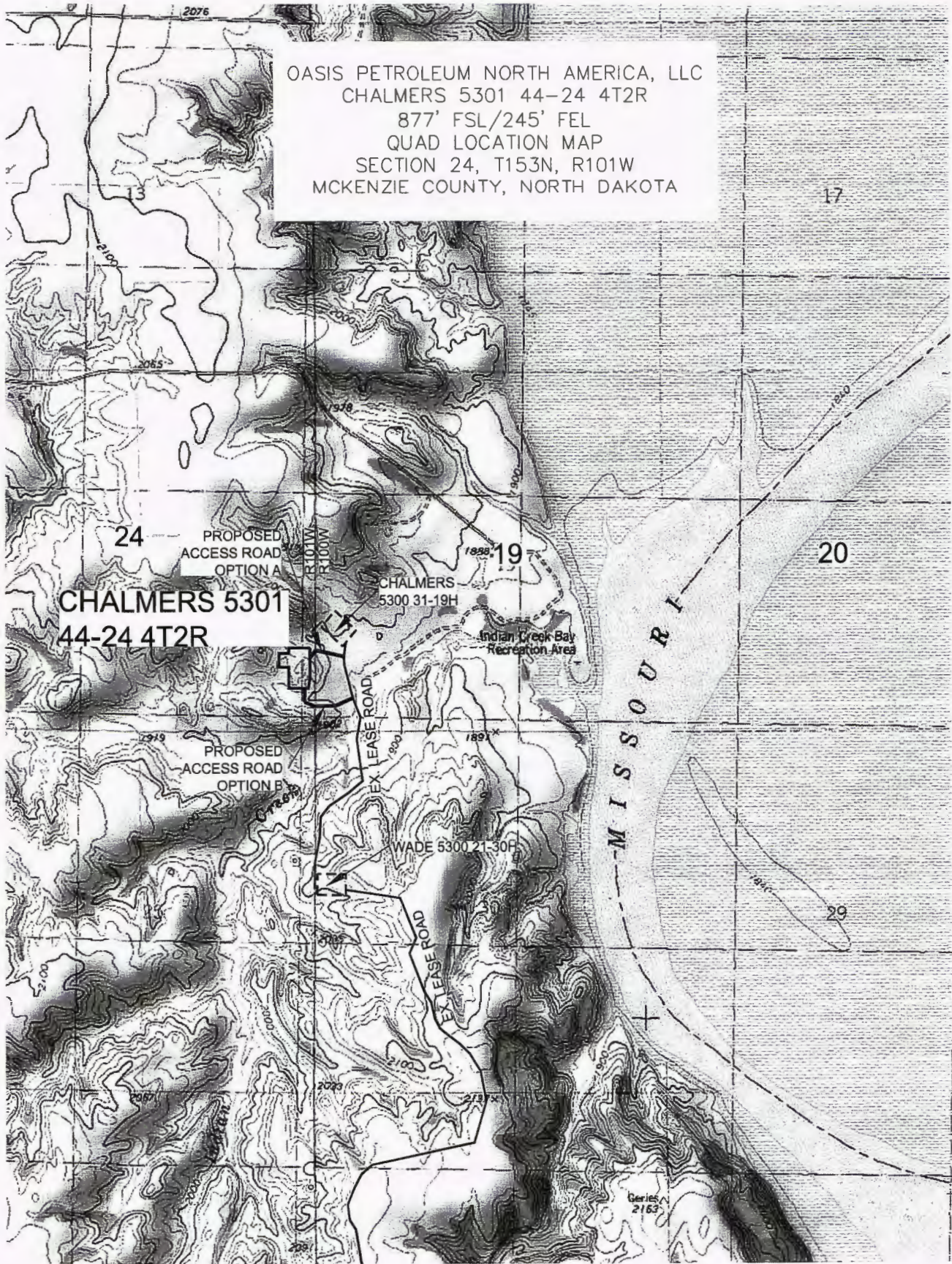
**4/8**  
 SHEET NO.

| Revision | Date     | By | Description            |
|----------|----------|----|------------------------|
| REV 1    | 12/5/13  | JS | CHANGED PAD LAYOUT     |
| REV 2    | 12/5/13  | BM | CHANGED PAD LAYOUT     |
| REV 3    | 1/7/2014 | JS | ADDED WELL TO PAD      |
| REV 4    | 5/29/14  | JS | ADDED NAME, MOVED WELL |
| REV 5    | 5/29/14  | JS | ADDED NAME, MOVED WELL |
| REV 6    | 6/11/14  | JS | ADDED DIMENSIONS       |

Drawn By: J.J.S.  
 Checked By: D.D.K.  
 Date: MAY 2014



OASIS PETROLEUM NORTH AMERICA, LLC  
 CHALMERS 5301 44-24 4T2R  
 877' FSL/245' FEL  
 QUAD LOCATION MAP  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

5/8

SHEET NO.



**INTERSTATE  
ENGINEERING**

Professionals you need, people you trust

Interstate Engineering, Inc.  
 P.O. Box 648  
 425 East Main Street  
 Sidney, Montana 59270  
 Ph (406) 433-5617  
 Fax (406) 433-5618  
 www.interstateeng.com  
 Other offices in Minnesota, North Dakota and South Dakota

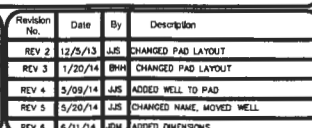
OASIS PETROLEUM NORTH AMERICA, LLC  
 QUAD LOCATION MAP  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.02  
 Checked By: D.D.K. Date: MAY 2014

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 1        | 12/5/13 | JJS | CHANGED PAD LAYOUT       |
| REV 2        | 1/20/14 | BPH | CHANGED PAD LAYOUT       |
| REV 3        | 5/28/14 | JJS | ADDED WELL TO PAD        |
| REV 4        | 5/20/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 5        | 6/11/14 | LDW | ADDED DIMENSIONS         |



877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



# CROSS SECTIONS

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5301 44-24 4T2R"

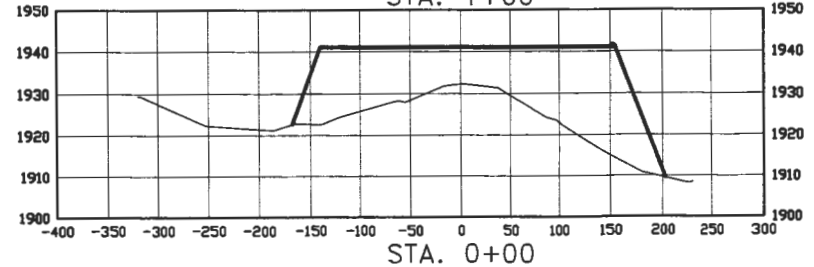
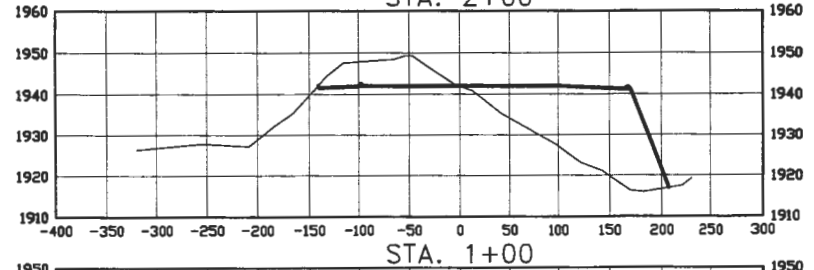
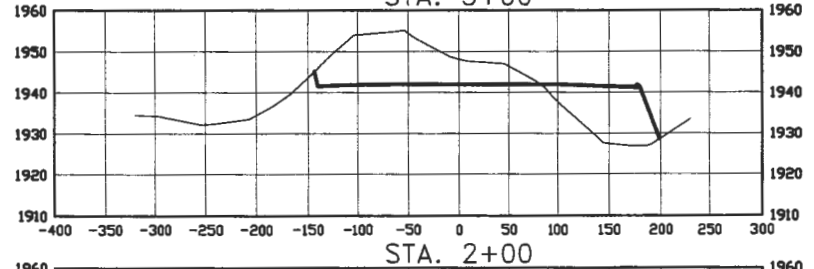
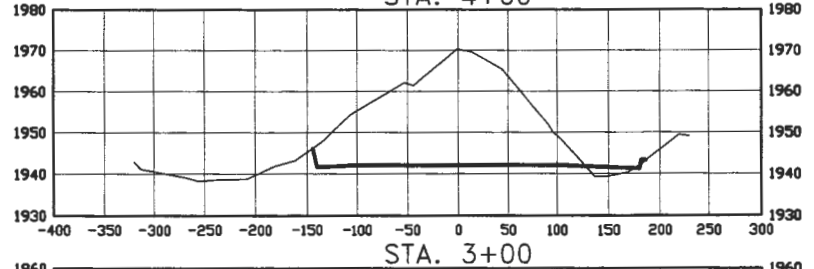
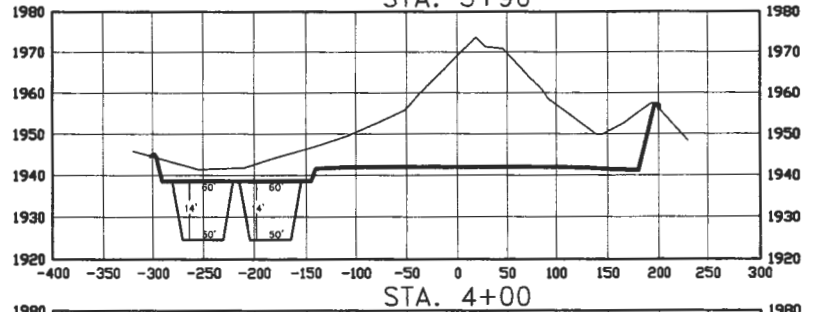
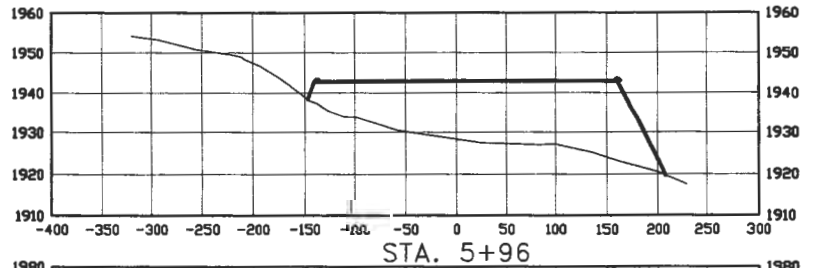
877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE

SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED  
AND SEALED BY DARYL D. KASEMAN,  
PLS, REGISTRATION NUMBER 3880 ON  
6/11/14 AND THE ORIGINAL  
DOCUMENTS ARE STORED AT THE  
OFFICES OF INTERSTATE ENGINEERING,  
INC.



SCALE  
HORIZ 1"=180'  
VERT 1"=45'



© 2014, INTERSTATE ENGINEERING, INC.

7/8  
SHEET NO.



Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
PAD CROSS SECTIONS  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No: S13-09-235.02  
Checked By: D.D.K. Date: MAY 2014

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 2        | 12/5/13 | JJS | CHANGED PAD LAYOUT       |
| REV 3        | 1/25/14 | BSH | CHANGED PAD LAYOUT       |
| REV 4        | 5/09/14 | JJS | ADDED WELL TO PAD        |
| REV 5        | 5/20/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 6        | 6/11/14 | JDM | ADDED DIMENSIONS         |



**WELL LOCATION SITE QUANTITIES**  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"CHALMERS 5301 44-24 4T2R"  
877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

WELL SITE ELEVATION 1960.2  
WELL PAD ELEVATION 1942.0

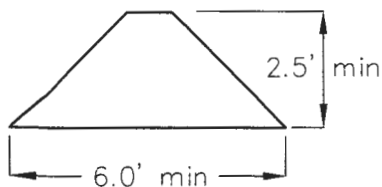
|                         |                   |
|-------------------------|-------------------|
| EXCAVATION              | 55,136            |
| PLUS PIT                | <u>9,450</u>      |
|                         | 64,586            |
| EMBANKMENT              | 37,558            |
| PLUS SHRINKAGE (30%)    | <u>11,267</u>     |
|                         | 48,825            |
| STOCKPILE PIT           | 9,450             |
| STOCKPILE TOP SOIL (6") | 4,770             |
| BERMS                   | 1,711 LF = 554 CY |
| DITCHES                 | 1,029 LF = 157 CY |
| DETENTION AREA          | 1,428 CY          |
| STOCKPILE MATERIAL      | 2,572             |
| DISTURBED AREA FROM PAD | 5.91 ACRES        |

NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)  
CUT END SLOPES AT 1:1  
FILL END SLOPES AT 1.5:1

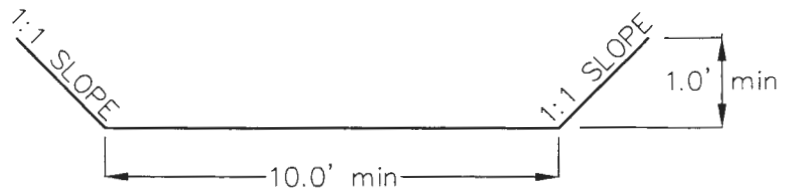
WELL SITE LOCATION

877' FSL  
245' FEL

BERM DETAIL



DITCH DETAIL



© 2014, INTERSTATE ENGINEERING, INC.

| <div style="font-size: 2em; font-weight: bold;">8/8</div> <div style="font-size: 0.8em;">SHEET NO.</div> | <div style="font-weight: bold; font-size: 0.8em;">INTERSTATE ENGINEERING</div> <div style="font-size: 0.7em;">Professionals you need, people you trust</div> | Interstate Engineering, Inc.<br>P.O. Box 648<br>425 East Main Street<br>Sidney, Montana 59270<br>Ph (406) 433-5617<br>Fax (406) 433-5618<br>www.interstateeng.com<br><small>Other offices in Minnesota, North Dakota and South Dakota</small> |                          | OASIS PETROLEUM NORTH AMERICA, LLC<br>QUANTITIES<br>SECTION 24, T153N, R101W<br>MCKENZIE COUNTY, NORTH DAKOTA<br>Drawn By: J.J.S.      Project No.: S13-09-235.02<br>Checked By: D.D.K.      Date: MAY 2014 |             | <table border="1" style="font-size: 0.7em;"> <thead> <tr> <th>Revision No.</th> <th>Date</th> <th>By</th> <th>Description</th> </tr> </thead> <tbody> <tr> <td>REV 1</td> <td>12/5/13</td> <td>JJS</td> <td>CHANGED PAD LAYOUT</td> </tr> <tr> <td>REV 2</td> <td>1/20/14</td> <td>BRH</td> <td>CHANGED PAD LAYOUT</td> </tr> <tr> <td>REV 3</td> <td>5/08/14</td> <td>JJS</td> <td>ADDED WELL TO PAD</td> </tr> <tr> <td>REV 4</td> <td>5/20/14</td> <td>JJS</td> <td>CHANGED NAME, MOVED WELL</td> </tr> <tr> <td>REV 5</td> <td>6/11/14</td> <td>JDM</td> <td>ADDED DIMENSIONS</td> </tr> </tbody> </table> |  | Revision No. | Date | By | Description | REV 1 | 12/5/13 | JJS | CHANGED PAD LAYOUT | REV 2 | 1/20/14 | BRH | CHANGED PAD LAYOUT | REV 3 | 5/08/14 | JJS | ADDED WELL TO PAD | REV 4 | 5/20/14 | JJS | CHANGED NAME, MOVED WELL | REV 5 | 6/11/14 | JDM | ADDED DIMENSIONS |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------|------|----|-------------|-------|---------|-----|--------------------|-------|---------|-----|--------------------|-------|---------|-----|-------------------|-------|---------|-----|--------------------------|-------|---------|-----|------------------|
|                                                                                                          |                                                                                                                                                              | Revision No.                                                                                                                                                                                                                                  | Date                     | By                                                                                                                                                                                                          | Description |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |              |      |    |             |       |         |     |                    |       |         |     |                    |       |         |     |                   |       |         |     |                          |       |         |     |                  |
| REV 1                                                                                                    | 12/5/13                                                                                                                                                      | JJS                                                                                                                                                                                                                                           | CHANGED PAD LAYOUT       |                                                                                                                                                                                                             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |              |      |    |             |       |         |     |                    |       |         |     |                    |       |         |     |                   |       |         |     |                          |       |         |     |                  |
| REV 2                                                                                                    | 1/20/14                                                                                                                                                      | BRH                                                                                                                                                                                                                                           | CHANGED PAD LAYOUT       |                                                                                                                                                                                                             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |              |      |    |             |       |         |     |                    |       |         |     |                    |       |         |     |                   |       |         |     |                          |       |         |     |                  |
| REV 3                                                                                                    | 5/08/14                                                                                                                                                      | JJS                                                                                                                                                                                                                                           | ADDED WELL TO PAD        |                                                                                                                                                                                                             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |              |      |    |             |       |         |     |                    |       |         |     |                    |       |         |     |                   |       |         |     |                          |       |         |     |                  |
| REV 4                                                                                                    | 5/20/14                                                                                                                                                      | JJS                                                                                                                                                                                                                                           | CHANGED NAME, MOVED WELL |                                                                                                                                                                                                             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |              |      |    |             |       |         |     |                    |       |         |     |                    |       |         |     |                   |       |         |     |                          |       |         |     |                  |
| REV 5                                                                                                    | 6/11/14                                                                                                                                                      | JDM                                                                                                                                                                                                                                           | ADDED DIMENSIONS         |                                                                                                                                                                                                             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |              |      |    |             |       |         |     |                    |       |         |     |                    |       |         |     |                   |       |         |     |                          |       |         |     |                  |
|                                                                                                          |                                                                                                                                                              |                                                                                                                                                                                                                                               |                          |                                                                                                                                                                                                             |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |              |      |    |             |       |         |     |                    |       |         |     |                    |       |         |     |                   |       |         |     |                          |       |         |     |                  |

**SUNDRY NOTICES AND REPORTS ON WELLS - FORM 4**

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 5749 (09-2006)

Well File No. **28600**

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                                                                                                                 |                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <input checked="" type="checkbox"/> Notice of Intent                                                                                            | Approximate Start Date<br><b>June 1, 2014</b> |
| <input type="checkbox"/> Report of Work Done                                                                                                    | Date Work Completed                           |
| <input type="checkbox"/> Notice of Intent to Begin a Workover Project that may Qualify for a Tax Exemption Pursuant to NDCC Section 57-51.1-03. | Approximate Start Date                        |

|                                               |                                                   |
|-----------------------------------------------|---------------------------------------------------|
| <input type="checkbox"/> Drilling Prognosis   | <input type="checkbox"/> Spill Report             |
| <input type="checkbox"/> Redrilling or Repair | <input type="checkbox"/> Shooting                 |
| <input type="checkbox"/> Casing or Liner      | <input type="checkbox"/> Acidizing                |
| <input type="checkbox"/> Plug Well            | <input type="checkbox"/> Fracture Treatment       |
| <input type="checkbox"/> Supplemental History | <input type="checkbox"/> Change Production Method |
| <input type="checkbox"/> Temporarily Abandon  | <input type="checkbox"/> Reclamation              |
| <input checked="" type="checkbox"/> Other     | <b>Suspension of Drilling</b>                     |

|                                                         |  |               |           |                 |              |  |  |
|---------------------------------------------------------|--|---------------|-----------|-----------------|--------------|--|--|
| Well Name and Number<br><b>Chalmers 5301 44-24 4T2R</b> |  |               |           |                 |              |  |  |
| Footages                                                |  | Qtr-Qtr       | Section   | Township        | Range        |  |  |
| <b>877 F S L 245 F E L</b>                              |  | <b>SESE</b>   | <b>24</b> | <b>153 N</b>    | <b>101 W</b> |  |  |
| Field                                                   |  | Pool          |           | County          |              |  |  |
| <b>Bakken</b>                                           |  | <b>Bakken</b> |           | <b>McKENZIE</b> |              |  |  |

| 24-HOUR PRODUCTION RATE |      |       |      |
|-------------------------|------|-------|------|
| Before                  |      | After |      |
| Oil                     | Bbls | Oil   | Bbls |
| Water                   | Bbls | Water | Bbls |
| Gas                     | MCF  | Gas   | MCF  |

|                                                          |      |       |          |
|----------------------------------------------------------|------|-------|----------|
| Name of Contractor(s)<br><b>Advanced Energy Services</b> |      |       |          |
| Address                                                  | City | State | Zip Code |

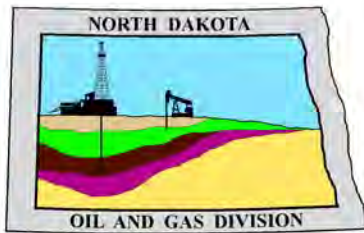
**DETAILS OF WORK**

Oasis Petroleum North America LLC requests permission for suspension of drilling for up to 90 days for the referenced well under NDAC 43-02-03-55. Oasis Petroleum North America LLC intends to drill the surface hole with freshwater based drilling mud and set surface casing with a small drilling rig and move off within 3 to 5 days. The casing will be set at a depth pre-approved by the NDIC per the Application for Permit to Drill NDAC 43-02-03-21. No saltwater will be used in the drilling and cementing operations of the surface casing. Once the surface casing is cemented, a plug or mechanical seal will be placed at the top of the casing to prevent any foreign matter from getting into the well. A rig capable of drilling to TD will move onto the location within the 90 days previously outlined to complete the drilling and casing plan as per the APD. The undersigned states that this request for suspension of drilling operations in accordance with the Subsection 4 of Section 43-02-03-55 of the NDAC, is being requested to take advantage of the cost savings and time savings of using an initial rig that is smaller than the rig necessary to drill a well to total depth but is not intended to alter or extend the terms and conditions of, or suspend any obligation under, any oil and gas lease with acreage in or under the spacing or drilling unit for the above-referenced well. Oasis Petroleum North America LLC understands NDAC 43-02-03-31 requirements regarding confidentiality pertaining to this permit. The drilling pit will be fenced immediately after construction if the well pad is located in a pasture (NDAC 43-02-03-19 & 19.1). Oasis Petroleum North America LL will plug and abandon the well and reclaim the well site if the well is not drilled by the larger rotary rig within 90 days after spudding the well with the smaller drilling rig.

Notify NDIC inspector **Richard Dunn** at 701-770-3554 with spud and TD info.

|                                                     |  |                                           |                          |
|-----------------------------------------------------|--|-------------------------------------------|--------------------------|
| Company<br><b>Oasis Petroleum North America LLC</b> |  | Telephone Number<br><b>(281) 404-9563</b> |                          |
| Address<br><b>1001 Fannin, Suite 1500</b>           |  |                                           |                          |
| City<br><b>Houston</b>                              |  | State<br><b>TX</b>                        | Zip Code<br><b>77002</b> |
| Signature<br>                                       |  | Printed Name<br><b>Heather McCowan</b>    |                          |
| Title<br><b>Regulatory Assistant</b>                |  | Date<br><b>May 28, 2014</b>               |                          |
| Email Address<br><b>hmccowan@oasispetroleum.com</b> |  |                                           |                          |

| FOR STATE USE ONLY                            |                                              |
|-----------------------------------------------|----------------------------------------------|
| <input type="checkbox"/> Received             | <input checked="" type="checkbox"/> Approved |
| Date<br><b>6/11/14</b>                        |                                              |
| By<br><b>Nathaniel Erbele</b>                 |                                              |
| Title<br><b>Petroleum Resource Specialist</b> |                                              |



# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

## Department of Mineral Resources

Lynn D. Helms - Director

## North Dakota Industrial Commission

[www.oilgas.nd.gov](http://www.oilgas.nd.gov)

June 11, 2014

Heather McCowan  
Regulatory Assistant  
OASIS PETROLEUM NORTH AMERICA LLC  
1001 Fannin Suite 1500  
Houston, TX 77002

**RE: HORIZONTAL WELL  
CHALMERS 5301 44-24 4T2R  
SESE Section 24-153N-101W  
McKenzie County  
Well File # 28600**

Dear Heather:

Pursuant to Commission Order No. 23752, approval to drill the above captioned well is hereby given. The approval is granted on the condition that all portions of the well bore not isolated by cement, be no closer than the **500' setback** from the north & south boundaries and **200' setback** from the east & west boundaries within the 1280 acre spacing unit consisting of Sections 19 & 20 T153N R100W.

**PERMIT STIPULATIONS: DUE TO STREAM ADJACENT TO THE WELL SITE, A DIKE IS REQUIRED SURROUNDING THE ENTIRE LOCATION. In cases where a spacing unit is accessed from an off-site drill pad, an affidavit must be provided affirming that the surface owner of the multi-well pad agrees to accept burial on their property of the cuttings generated from drilling the well(s) into an offsite spacing/drilling unit. Effective June 1, 2014, a covered leak-proof container (with placard) for filter sock disposal must be maintained on the well site beginning when the well is spud, and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. OASIS PETRO NO AMER must contact NDIC Field Inspector Richard Dunn at 701-770-3554 prior to setting conductor.**

### Drilling pit

NDAC 43-02-03-19.4 states that "a pit may be utilized to bury drill cuttings and solids generated during well drilling and completion operations, providing the pit can be constructed, used and reclaimed in a manner that will prevent pollution of the land surface and freshwaters. Reserve and circulation of mud system through earthen pits are prohibited. All pits shall be inspected by an authorized representative of the director prior to lining and use. Drill cuttings and solids must be stabilized in a manner approved by the director prior to placement in a cuttings pit."

### Form 1 Changes & Hard Lines

Any changes, shortening of casing point or lengthening at Total Depth must have prior approval by the NDIC. The proposed directional plan is at a legal location. The minimum legal coordinate from the well head at casing point is: 445' east. Also, based on the azimuth of the proposed lateral the maximum legal coordinates from the well head are: 377' south and 10558' east.



Heather McCowan

June 11, 2014

Page 2

### **Location Construction Commencement (Three Day Waiting Period)**

Operators shall not commence operations on a drill site until the 3rd business day following publication of the approved drilling permit on the NDIC - OGD Daily Activity Report. If circumstances require operations to commence before the 3rd business day following publication on the Daily Activity Report, the waiting period may be waived by the Director. Application for a waiver must be by sworn affidavit providing the information necessary to evaluate the extenuating circumstances, the factors of NDAC 43-02-03-16.2 (1), (a)-(f), and any other information that would allow the Director to conclude that in the event another owner seeks revocation of the drilling permit, the applicant should retain the permit.

### **Permit Fee & Notification**

Payment was received in the amount of \$100 via credit card .The permit fee has been received. It is requested that notification be given immediately upon the spudding of the well. This information should be relayed to the Oil & Gas Division, Bismarck, via telephone. The following information must be included: Well name, legal location, permit number, drilling contractor, company representative, date and time of spudding. Office hours are 8:00 a.m. to 12:00 p.m. and 1:00 p.m. to 5:00 p.m. Central Time. Our telephone number is (701) 328-8020, leave a message if after hours or on the weekend.

### **Survey Requirements for Horizontal, Horizontal Re-entry, and Directional Wells**

NDAC Section 43-02-03-25 (Deviation Tests and Directional Surveys) states in part (that) the survey contractor shall file a certified copy of all surveys with the director free of charge within thirty days of completion. Surveys must be submitted as one electronic copy, or in a form approved by the director. However, the director may require the directional survey to be filed immediately after completion if the survey is needed to conduct the operation of the director's office in a timely manner. Certified surveys must be submitted via email in one adobe document, with a certification cover page to [certsurvey@nd.gov](mailto:certsurvey@nd.gov).

Survey points shall be of such frequency to accurately determine the entire location of the well bore.

Specifically, the Horizontal and Directional well survey frequency is 100 feet in the vertical, 30 feet in the curve (or when sliding) and 90 feet in the lateral.

### **Surface casing cement**

Tail cement utilized on surface casing must have a minimum compressive strength of 500 psi within 12 hours, and tail cement utilized on production casing must have a minimum compressive strength of 500 psi before drilling the plug or initiating tests.

### **Logs**

NDAC Section 43-02-03-31 requires the running of (1) a suite of open hole logs from which formation tops and porosity zones can be determined, (2) a Gamma Ray Log run from total depth to ground level elevation of the well bore, and (3) a log from which the presence and quality of cement can be determined (Standard CBL or Ultrasonic cement evaluation log) in every well in which production or intermediate casing has been set, this log must be run prior to completing the well. All logs run must be submitted free of charge, as one digital TIFF (tagged image file format) copy and one digital LAS (log ASCII) formatted copy. Digital logs may be submitted on a standard CD, DVD, or attached to an email sent to [digitallogs@nd.gov](mailto:digitallogs@nd.gov)

Thank you for your cooperation.

Sincerely,

Nathaniel Erbele

Petroleum Resource Specialist



# APPLICATION FOR PERMIT TO DRILL HORIZONTAL WELL - FORM 1H

INDUSTRIAL COMMISSION OF NORTH DAKOTA  
OIL AND GAS DIVISION  
600 EAST BOULEVARD DEPT 405  
BISMARCK, ND 58505-0840  
SFN 54269 (08-2005)

PLEASE READ INSTRUCTIONS BEFORE FILLING OUT FORM.  
PLEASE SUBMIT THE ORIGINAL AND ONE COPY.

|                                                      |                                      |                                                         |                                             |
|------------------------------------------------------|--------------------------------------|---------------------------------------------------------|---------------------------------------------|
| Type of Work<br><b>New Location</b>                  | Type of Well<br><b>Oil &amp; Gas</b> | Approximate Date Work Will Start<br><b>6 / 1 / 2013</b> | Confidential Status<br><b>No</b>            |
| Operator<br><b>OASIS PETROLEUM NORTH AMERICA LLC</b> |                                      |                                                         | Telephone Number<br><b>281-404-9563</b>     |
| Address<br><b>1001 Fannin Suite 1500</b>             |                                      | City<br><b>Houston</b>                                  | State<br><b>TX</b> Zip Code<br><b>77002</b> |

☒ Notice has been provided to the owner of any permanently occupied dwelling within 1,320 feet.

☒ This well is not located within five hundred feet of an occupied dwelling.

## WELL INFORMATION (If more than one lateral proposed, enter data for additional laterals on page 2)

|                                                                                                    |                           |                                                                                                         |                                                               |                                                                               |                                                                                                                 |                                                                                                    |                                     |
|----------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------|
| Well Name<br><b>CHALMERS</b>                                                                       |                           |                                                                                                         |                                                               | Well Number<br><b>5301 44-24 4T2R</b>                                         |                                                                                                                 |                                                                                                    |                                     |
| Surface Footages<br><b>877 F S L 245 F E L</b>                                                     |                           | Qtr-Qtr<br><b>SESE</b>                                                                                  | Section<br><b>24</b>                                          | Township<br><b>153 N</b>                                                      | Range<br><b>101 W</b>                                                                                           | County<br><b>McKenzie</b>                                                                          |                                     |
| Longstring Casing Point Footages<br><b>727 F S L 204 F W L</b>                                     |                           | Qtr-Qtr<br><b>SWSW</b>                                                                                  | Section<br><b>19</b>                                          | Township<br><b>153 N</b>                                                      | Range<br><b>100 W</b>                                                                                           | County<br><b>McKenzie</b>                                                                          |                                     |
| Longstring Casing Point Coordinates From Well Head<br><b>150 S From WH 449 E From WH</b>           |                           | Azimuth<br><b>108.5 °</b>                                                                               | Longstring Total Depth<br><b>11006 Feet MD 10737 Feet TVD</b> |                                                                               |                                                                                                                 |                                                                                                    |                                     |
| Bottom Hole Footages From Nearest Section Line<br><b>580 F S L 224 F E L</b>                       |                           | Qtr-Qtr<br><b>SESE</b>                                                                                  | Section<br><b>20</b>                                          | Township<br><b>153 N</b>                                                      | Range<br><b>100 W</b>                                                                                           | County<br><b>Williams</b>                                                                          |                                     |
| Bottom Hole Coordinates From Well Head<br><b>297 S From WH 10534 E From WH</b>                     |                           | KOP Lateral 1<br><b>10259 Feet MD</b>                                                                   | Azimuth Lateral 1<br><b>90 °</b>                              |                                                                               | Estimated Total Depth Lateral 1<br><b>21106 Feet MD 10807 Feet TVD</b>                                          |                                                                                                    |                                     |
| Latitude of Well Head<br><b>48 ° 03 ' 20.5 "</b>                                                   |                           | Longitude of Well Head<br><b>-103 ° 36 ' 18.55 "</b>                                                    |                                                               | NAD Reference<br><b>NAD83</b>                                                 |                                                                                                                 | Description of (Subject to NDIC Approval)<br><b>Spacing Unit: Sections 19 &amp; 20 T153N R100W</b> |                                     |
| Ground Elevation<br><b>1960 Feet Above S.L.</b>                                                    |                           | Acres in Spacing/Drilling Unit<br><b>1280</b>                                                           |                                                               | Spacing/Drilling Unit Setback Requirement<br><b>500 Feet N/S 200 Feet E/W</b> |                                                                                                                 | Industrial Commission Order<br><b>23752</b>                                                        |                                     |
| North Line of Spacing/Drilling Unit<br><b>10489 Feet</b>                                           |                           | South Line of Spacing/Drilling Unit<br><b>10513 Feet</b>                                                |                                                               | East Line of Spacing/Drilling Unit<br><b>5280 Feet</b>                        |                                                                                                                 | West Line of Spacing/Drilling Unit<br><b>5263 Feet</b>                                             |                                     |
| Objective Horizons<br><b>Three Forks - B2</b>                                                      |                           |                                                                                                         |                                                               |                                                                               |                                                                                                                 | Pierre Shale Top<br><b>1918</b>                                                                    |                                     |
| Proposed<br>Surface Casing                                                                         | Size<br><b>13 - 3/8 "</b> | Weight<br><b>54 Lb./Ft.</b>                                                                             | Depth<br><b>2018 Feet</b>                                     | Cement Volume<br><b>967 Sacks</b>                                             | <b>NOTE: Surface hole must be drilled with fresh water and surface casing must be cemented back to surface.</b> |                                                                                                    |                                     |
| Proposed<br>Longstring Casing                                                                      | Size<br><b>7 - "</b>      | Weight(s)<br><b>32 Lb./Ft.</b>                                                                          | Longstring Total Depth<br><b>11006 Feet MD 10737 Feet TVD</b> |                                                                               | Cement Volume<br><b>898 Sacks</b>                                                                               | Cement Top<br><b>3841 Feet</b>                                                                     | Top Dakota Sand<br><b>5341 Feet</b> |
| Base Last Charles Salt (If Applicable)<br><b>9091 Feet</b>                                         |                           | <b>NOTE: Intermediate or longstring casing string must be cemented above the top Dakota Group Sand.</b> |                                                               |                                                                               |                                                                                                                 |                                                                                                    |                                     |
| Proposed Logs<br><b>Triple Combo: KOP to Kibby GR/Res to BSC GR to surf CND through the Dakota</b> |                           |                                                                                                         |                                                               |                                                                               |                                                                                                                 |                                                                                                    |                                     |
| Drilling Mud Type (Vertical Hole - Below Surface Casing)<br><b>Invert</b>                          |                           |                                                                                                         |                                                               | Drilling Mud Type (Lateral)<br><b>Salt Water Gel</b>                          |                                                                                                                 |                                                                                                    |                                     |
| Survey Type in Vertical Portion of Well<br><b>MWD Every 100 Feet</b>                               |                           | Survey Frequency: Build Section<br><b>30 Feet</b>                                                       |                                                               | Survey Frequency: Lateral<br><b>90 Feet</b>                                   |                                                                                                                 | Survey Contractor<br><b>Ryan</b>                                                                   |                                     |

**NOTE: A Gamma Ray log must be run to ground surface and a CBL must be run on intermediate or longstring casing string if set.**

Surveys are required at least every 30 feet in the build section and every 90 feet in the lateral section of a horizontal well. Measurement inaccuracies are not considered when determining compliance with the spacing/drilling unit boundary setback requirement except in the following scenarios: 1) When the angle between the well bore and the respective boundary is 10 degrees or less; or 2) If Industry standard methods and equipment are not utilized. Consult the applicable field order for exceptions.

If measurement inaccuracies are required to be considered, a 2° MWD measurement inaccuracy will be applied to the horizontal portion of the well bore. This measurement inaccuracy is applied to the well bore from KOP to TD.

**REQUIRED ATTACHMENTS:** Certified surveyor's plat, horizontal section plat, estimated geological tops, proposed mud/cementing plan, directional plot/plan, \$100 fee.  
See Page 2 for Comments section and signature block.



**COMMENTS, ADDITIONAL INFORMATION, AND/OR LIST OF ATTACHMENTS**

**Documents forwarded by email: Drill plan with drilling fluids, Well Summary with casing/cement plans, Directional Plan & Plot, Plats**

**Lateral 2**

|                                                             |                        |                                                   |          |                                           |            |         |
|-------------------------------------------------------------|------------------------|---------------------------------------------------|----------|-------------------------------------------|------------|---------|
| KOP Lateral 2<br>Feet MD                                    | Azimuth Lateral 2<br>° | Estimated Total Depth Lateral 2<br>Feet MD        | Feet TVD | KOP Coordinates From Well Head<br>From WH |            | From WH |
| Formation Entry Point Coordinates From Well Head<br>From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH |          |                                           |            |         |
| KOP Footages From Nearest Section Line<br>F L               | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |
| Bottom Hole Footages From Nearest Section Line<br>F L       | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |

**Lateral 3**

|                                                             |                        |                                                   |          |                                           |            |         |
|-------------------------------------------------------------|------------------------|---------------------------------------------------|----------|-------------------------------------------|------------|---------|
| KOP Lateral 3<br>Feet MD                                    | Azimuth Lateral 3<br>° | Estimated Total Depth Lateral 3<br>Feet MD        | Feet TVD | KOP Coordinates From Well Head<br>From WH |            | From WH |
| Formation Entry Point Coordinates From Well Head<br>From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH |          |                                           |            |         |
| KOP Footages From Nearest Section Line<br>F L               | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |
| Bottom Hole Footages From Nearest Section Line<br>F L       | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |

**Lateral 4**

|                                                             |                        |                                                   |          |                                           |            |         |
|-------------------------------------------------------------|------------------------|---------------------------------------------------|----------|-------------------------------------------|------------|---------|
| KOP Lateral 4<br>Feet MD                                    | Azimuth Lateral 4<br>° | Estimated Total Depth Lateral 4<br>Feet MD        | Feet TVD | KOP Coordinates From Well Head<br>From WH |            | From WH |
| Formation Entry Point Coordinates From Well Head<br>From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH |          |                                           |            |         |
| KOP Footages From Nearest Section Line<br>F L               | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |
| Bottom Hole Footages From Nearest Section Line<br>F L       | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |

**Lateral 5**

|                                                             |                        |                                                   |          |                                           |            |         |
|-------------------------------------------------------------|------------------------|---------------------------------------------------|----------|-------------------------------------------|------------|---------|
| KOP Lateral 5<br>Feet MD                                    | Azimuth Lateral 5<br>° | Estimated Total Depth Lateral 5<br>Feet MD        | Feet TVD | KOP Coordinates From Well Head<br>From WH |            | From WH |
| Formation Entry Point Coordinates From Well Head<br>From WH |                        | Bottom Hole Coordinates From Well Head<br>From WH |          |                                           |            |         |
| KOP Footages From Nearest Section Line<br>F L               | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |
| Bottom Hole Footages From Nearest Section Line<br>F L       | F L                    | Qtr-Qtr                                           | Section  | Township<br>N                             | Range<br>W | County  |

I hereby swear or affirm the information provided is true, complete and correct as determined from all available records.

Date

**5 / 28 / 2014**

**ePermit**

Printed Name

**Heather McCowan**

Title

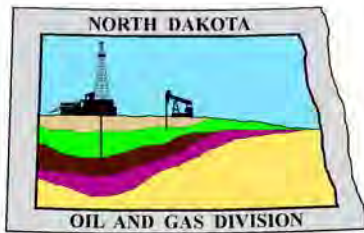
**Regulatory Assistant**

**FOR STATE USE ONLY**

|                                        |                                       |
|----------------------------------------|---------------------------------------|
| Permit and File Number<br><b>28600</b> | API Number<br><b>33 - 053 - 06011</b> |
| Field<br><b>BAKER</b>                  |                                       |
| Pool<br><b>BAKKEN</b>                  | Permit Type<br><b>DEVELOPMENT</b>     |

**FOR STATE USE ONLY**

|                                               |
|-----------------------------------------------|
| Date Approved<br><b>6 / 11 / 2014</b>         |
| By<br><b>Nathaniel Erbele</b>                 |
| Title<br><b>Petroleum Resource Specialist</b> |



# Oil and Gas Division

Lynn D. Helms - Director

Bruce E. Hicks - Assistant Director

## Department of Mineral Resources

Lynn D. Helms - Director

## North Dakota Industrial Commission

[www.oilgas.nd.gov](http://www.oilgas.nd.gov)

April 9, 2014

**RE: Filter Socks and Other Filter Media  
Leakproof Container Required  
Oil and Gas Wells**

Dear Operator,

North Dakota Administrative Code Section 43-02-03-19.2 states in part that all waste material associated with exploration or production of oil and gas must be properly disposed of in an authorized facility in accord with all applicable local, state, and federal laws and regulations.

Filtration systems are commonly used during oil and gas operations in North Dakota. The Commission is very concerned about the proper disposal of used filters (including filter socks) used by the oil and gas industry.

Effective June 1, 2014, a container must be maintained on each well drilled in North Dakota beginning when the well is spud and must remain on-site during clean-out, completion, and flow-back whenever filtration operations are conducted. The on-site container must be used to store filters until they can be properly disposed of in an authorized facility. Such containers must be:

- leakproof to prevent any fluids from escaping the container
- covered to prevent precipitation from entering the container
- placard to indicate only filters are to be placed in the container

If the operator will not utilize a filtration system, a waiver to the container requirement will be considered, but only upon the operator submitting a Sundry Notice (Form 4) justifying their request.

As previously stated in our March 13, 2014 letter, North Dakota Administrative Code Section 33-20-02.1-01 states in part that every person who transports solid waste (which includes oil and gas exploration and production wastes) is required to have a valid permit issued by the North Dakota Department of Health, Division of Waste Management. Please contact the Division of Waste Management at (701) 328-5166 with any questions on the solid waste program. Note oil and gas exploration and production wastes include produced water, drilling mud, invert mud, tank bottom sediment, pipe scale, filters, and fly ash.

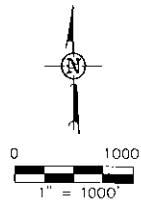
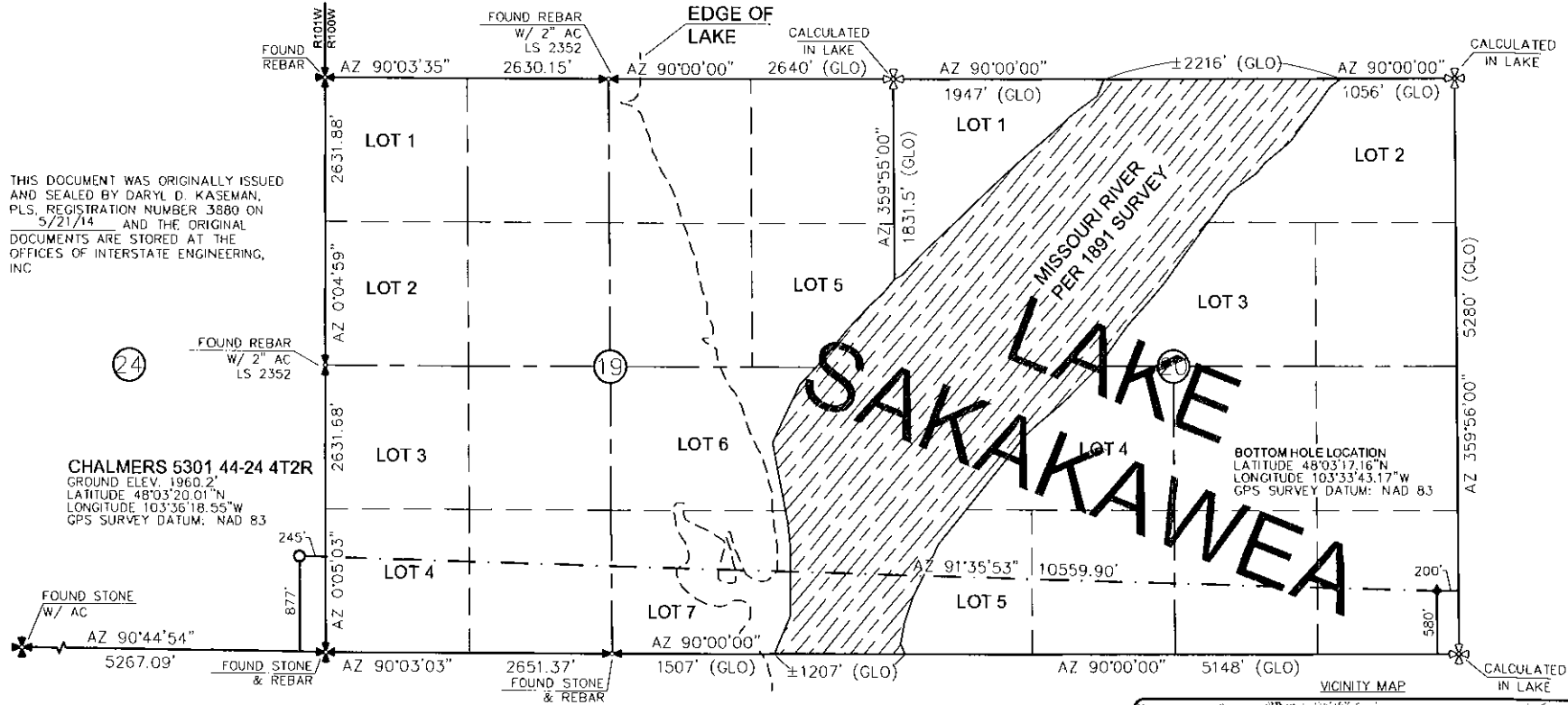
Thank you for your cooperation.

Sincerely,

*Bruce E. Hicks*

Assistant Director

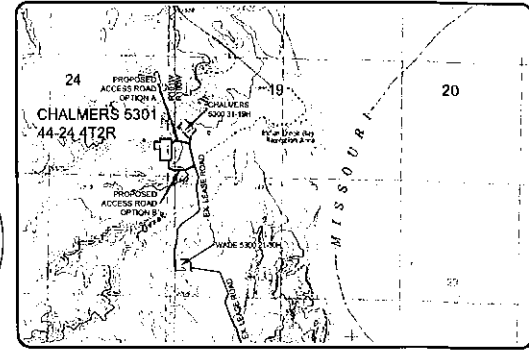
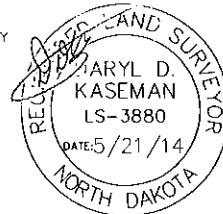
**WELL LOCATION PLAT**  
 OASIS PETROLEUM NORTH AMERICA, LLC  
 1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
 "CHALMERS 5301 44-24 4T2R"  
 877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
 SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



⊕ - MONUMENT - RECOVERED  
 ⊗ - MONUMENT - NOT RECOVERED

STAKED ON 9/3/13  
 VERTICAL CONTROL DATUM WAS BASED UPON  
 CONTROL POINT 16 WITH AN ELEVATION OF 2014.2'  
 THIS SURVEY AND PLAT IS BEING PROVIDED AT THE  
 REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY  
 THAT THIS PLAT CORRECTLY REPRESENTS WORK  
 PERFORMED BY ME OR UNDER MY SUPERVISION  
 AND IS TRUE AND CORRECT TO THE BEST OF MY  
 KNOWLEDGE AND BELIEF.

*Daryl D. Kaseman*  
 DARYL D. KASEMAN LS-3880



© 2014, INTERSTATE ENGINEERING, INC.

| Revision | By   | Description              |
|----------|------|--------------------------|
| REV 1    | JJ/S | CHANGED PAD LAYOUT       |
| REV 2    | JJ/S | CHANGED PAD LAYOUT       |
| REV 3    | JJ/S | CHANGED PAD LAYOUT       |
| REV 4    | JJ/S | ADDED WELL TO PAD        |
| REV 5    | JJ/S | CHANGED NAME, ADDED WELL |
| REV 6    | JJ/S | CHANGED NAME, ADDED WELL |

OASIS PETROLEUM NORTH AMERICA, LLC  
 WELL LOCATION PLAT  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA  
 Project No.: 21505-28-03  
 Drawn By: JJ/S  
 Checked By: JJ/S  
 Date: MAY 2014

Interstate Engineering, Inc.  
 P.O. Box 444  
 425 East Main Street  
 Sidney, Montana 59270  
 Tel: (406) 433-5017  
 Fax: (406) 433-5018  
 www.interstateeng.com  
 Other offices in Minnesota, South Dakota and Saskatchewan



1/8  
 SHEET NO.

| DRILLING PLAN                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                  |  |                                                                                      |                                          |          |                         |          |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------|--|--------------------------------------------------------------------------------------|------------------------------------------|----------|-------------------------|----------|----------------|
| OPERATOR                                                                                                                                                                                                                                                                                                                                                                                                         |  | Oasis Petroleum                  |  |                                                                                      | COUNTY/STATE                             |          | McKenzie Co., ND        |          |                |
| WELL NAME                                                                                                                                                                                                                                                                                                                                                                                                        |  | Chalmers 5301 44-24 4T2R         |  |                                                                                      | RIG                                      |          | Nabors B25              |          |                |
| WELL TYPE                                                                                                                                                                                                                                                                                                                                                                                                        |  | Horizontal Three Forks 2nd Bench |  |                                                                                      |                                          |          |                         |          |                |
| LOCATION                                                                                                                                                                                                                                                                                                                                                                                                         |  | SESE 24-153N-101W                |  |                                                                                      | Surface Location (survey plat): 877' FSL |          | 245' FEL                |          |                |
| EST. T.D.                                                                                                                                                                                                                                                                                                                                                                                                        |  | 21,106'                          |  |                                                                                      | GROUND ELEV:                             |          | 1943 Finished Pad Elev. |          | Sub Height: 25 |
| TOTAL LATERA                                                                                                                                                                                                                                                                                                                                                                                                     |  | 10,099'                          |  |                                                                                      | KB ELEV:                                 |          | 1968                    |          |                |
| PROGNOSIS:                                                                                                                                                                                                                                                                                                                                                                                                       |  |                                  |  | Based on 1,968' KB(est)                                                              |                                          |          |                         |          |                |
| LOGS:                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                  |  | Type Interval                                                                        |                                          |          |                         |          |                |
| OH Logs:                                                                                                                                                                                                                                                                                                                                                                                                         |  |                                  |  | Triple Combo KOP to Kibby (or min run of 1800' whichever is greater); GR/Res to BSC; |                                          |          |                         |          |                |
|                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                  |  | GR to surf; CND through the Dakota                                                   |                                          |          |                         |          |                |
| CBL/GR:                                                                                                                                                                                                                                                                                                                                                                                                          |  |                                  |  | Above top of cement/GR to base of casing                                             |                                          |          |                         |          |                |
| MWD GR:                                                                                                                                                                                                                                                                                                                                                                                                          |  |                                  |  | KOP to lateral TD                                                                    |                                          |          |                         |          |                |
| Logging Waiver                                                                                                                                                                                                                                                                                                                                                                                                   |  |                                  |  | Oasis Chalmers 5300 31-19H 894' to NE sec 19 153N 100W                               |                                          |          |                         |          |                |
| DEVIATION:                                                                                                                                                                                                                                                                                                                                                                                                       |  |                                  |  |                                                                                      |                                          |          |                         |          |                |
| Surf:                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                  |  | 3 deg. max., 1 deg / 100'; srvy every 500'                                           |                                          |          |                         |          |                |
| Prod:                                                                                                                                                                                                                                                                                                                                                                                                            |  |                                  |  | 5 deg. max., 1 deg / 100'; srvy every 100'                                           |                                          |          |                         |          |                |
| DST'S:                                                                                                                                                                                                                                                                                                                                                                                                           |  |                                  |  | None planned                                                                         |                                          |          |                         |          |                |
| CORES:                                                                                                                                                                                                                                                                                                                                                                                                           |  |                                  |  | None planned                                                                         |                                          |          |                         |          |                |
| MUDLOGGING:                                                                                                                                                                                                                                                                                                                                                                                                      |  |                                  |  |                                                                                      |                                          |          |                         |          |                |
| Two-Man:                                                                                                                                                                                                                                                                                                                                                                                                         |  |                                  |  | 8,233'                                                                               |                                          |          |                         |          |                |
|                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                  |  | ~200' above the Charles (Kibbey) to Casing point; Casing point to TD                 |                                          |          |                         |          |                |
|                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                                  |  | 30' samples at direction of wellsite geologist; 10' through target @ curve land      |                                          |          |                         |          |                |
| BOP:                                                                                                                                                                                                                                                                                                                                                                                                             |  |                                  |  | 11" 5000 psi blind, pipe & annular                                                   |                                          |          |                         |          |                |
| Dip Rate:                                                                                                                                                                                                                                                                                                                                                                                                        |  | -0.40° or .70' /100' down        |  |                                                                                      |                                          |          |                         |          |                |
| Max. Anticipated BHP:                                                                                                                                                                                                                                                                                                                                                                                            |  | 4642                             |  |                                                                                      |                                          |          |                         |          |                |
| Surface Formation:                                                                                                                                                                                                                                                                                                                                                                                               |  | Glacial till                     |  |                                                                                      |                                          |          |                         |          |                |
| MUD:                                                                                                                                                                                                                                                                                                                                                                                                             |  | Interval                         |  | Type                                                                                 |                                          | WT       |                         | Vis      |                |
| Surface:                                                                                                                                                                                                                                                                                                                                                                                                         |  | 0' - 2,018'                      |  | FW                                                                                   |                                          | 8.4-9.0  |                         | 28-32    |                |
| Intermediate:                                                                                                                                                                                                                                                                                                                                                                                                    |  | 2,018' - 11,007'                 |  | Invert                                                                               |                                          | 9.5-10.4 |                         | 40-50    |                |
| Lateral:                                                                                                                                                                                                                                                                                                                                                                                                         |  | 11,007' - 21,106'                |  | Salt Water                                                                           |                                          | 9.8-10.2 |                         | 28-32    |                |
| CASING:                                                                                                                                                                                                                                                                                                                                                                                                          |  | Size                             |  | Hole                                                                                 |                                          | Depth    |                         | Cement   |                |
| Surface:                                                                                                                                                                                                                                                                                                                                                                                                         |  | 13-3/8"                          |  | 54.5#                                                                                |                                          | 17-1/2"  |                         | 2,018'   |                |
| Intermediate (Dakota):                                                                                                                                                                                                                                                                                                                                                                                           |  | 9-5/8"                           |  | 40#                                                                                  |                                          | 12-1/4"  |                         | 6,000'   |                |
| Intermediate:                                                                                                                                                                                                                                                                                                                                                                                                    |  | 7"                               |  | 32#                                                                                  |                                          | 8-3/4"   |                         | 11,007'  |                |
| Production Liner:                                                                                                                                                                                                                                                                                                                                                                                                |  | 4.5"                             |  | 13.5#                                                                                |                                          | 6"       |                         | 21,106'  |                |
| TOL @ 10,209'                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                  |  |                                                                                      |                                          |          |                         |          |                |
| PROBABLE PLUGS, IF REQ'D:                                                                                                                                                                                                                                                                                                                                                                                        |  |                                  |  |                                                                                      |                                          |          |                         |          |                |
| OTHER:                                                                                                                                                                                                                                                                                                                                                                                                           |  | MD                               |  | TVD                                                                                  |                                          | FNL/FSL  |                         | FEL/FWL  |                |
| Surface:                                                                                                                                                                                                                                                                                                                                                                                                         |  | 2,018                            |  | 2,018                                                                                |                                          | 877' FSL |                         | 245' FEL |                |
| KOP:                                                                                                                                                                                                                                                                                                                                                                                                             |  | 10,259'                          |  | 10,259'                                                                              |                                          | 877' FSL |                         | 245' FEL |                |
| EOC:                                                                                                                                                                                                                                                                                                                                                                                                             |  | 11,007'                          |  | 10,737'                                                                              |                                          | 727' FSL |                         | 204' FWL |                |
| Casing Point:                                                                                                                                                                                                                                                                                                                                                                                                    |  | 11,007'                          |  | 10,737'                                                                              |                                          | 727' FSL |                         | 204' FWL |                |
| Middle Bakken Lateral TD:                                                                                                                                                                                                                                                                                                                                                                                        |  | 21,106'                          |  | 10,807'                                                                              |                                          | 580' FSL |                         | 200' FEL |                |
| S-T-R                                                                                                                                                                                                                                                                                                                                                                                                            |  | AZI                              |  |                                                                                      |                                          |          |                         |          |                |
| Sec. 24 153N R101W                                                                                                                                                                                                                                                                                                                                                                                               |  | Survey Company:                  |  |                                                                                      |                                          |          |                         |          |                |
| Sec. 24 153N R101W                                                                                                                                                                                                                                                                                                                                                                                               |  | Build Rate:                      |  | 12 deg /100'                                                                         |                                          |          |                         |          |                |
| Sec. 19 153N R100W                                                                                                                                                                                                                                                                                                                                                                                               |  | Turn Rate:                       |  | 2 deg /100'                                                                          |                                          |          |                         |          |                |
| Sec. 19 153N R100W                                                                                                                                                                                                                                                                                                                                                                                               |  | 108.0                            |  |                                                                                      |                                          |          |                         |          |                |
| Sec. 20 153N R100W                                                                                                                                                                                                                                                                                                                                                                                               |  | 108.0                            |  |                                                                                      |                                          |          |                         |          |                |
|                                                                                                                                                                                                                                                                                                                                                                                                                  |  | 90.0                             |  |                                                                                      |                                          |          |                         |          |                |
| <p><b>Comments:</b></p> <p><b>Request a Sundry for an Open Hole Log Waiver</b></p> <p><b>Exception well:</b> Oasis Petroleum's J O Anderson 5200 31-28T (33053039510000); 152N/100W/S28; located 0.30 miles S of the proposed location</p> <p>No frac string planned</p> <p>35 pkrs 2 slvs</p> <p><b>***Note Hard Lines</b></p> <p>Pit location as depicted on plats. Constructed on native ground, not fill</p> |  |                                  |  |                                                                                      |                                          |          |                         |          |                |
|                                                                                                                                                                                                                                                                                                                               |  |                                  |  |                                                                                      |                                          |          |                         |          |                |
| Geology: C. Hargett 9/26/2013                                                                                                                                                                                                                                                                                                                                                                                    |  |                                  |  |                                                                                      | Engineering: M. Brown 5-27-14            |          |                         |          |                |

**Oasis Petroleum  
Well Summary  
Chalmers 5301 44-24 4T2R  
Sec. 24 T153N R101W  
McKenzie County, North Dakota**

**SURFACE CASING AND CEMENT DESIGN**

|             |                 |               |              |                 |             |              | <b>Make-up Torque (ft-lbs)</b> |                |            |
|-------------|-----------------|---------------|--------------|-----------------|-------------|--------------|--------------------------------|----------------|------------|
| <b>Size</b> | <b>Interval</b> | <b>Weight</b> | <b>Grade</b> | <b>Coupling</b> | <b>I.D.</b> | <b>Drift</b> | <b>Minimum</b>                 | <b>Optimum</b> | <b>Max</b> |
| 13-3/8"     | 0' to 2,018'    | 54.5          | J-55         | STC             | 12.615"     | 12.459"      | 4,100                          | 5,470          | 6,840      |

| <b>Interval</b> | <b>Description</b>             | <b>Collapse</b> | <b>Burst</b> | <b>Tension</b> | <b>Cost per ft</b> |
|-----------------|--------------------------------|-----------------|--------------|----------------|--------------------|
|                 |                                | (psi) a         | (psi) b      | (1000 lbs) c   |                    |
| 0' to 2,018'    | 13-3/8", 54.5#, J-55, STC, 8rd | 1130 / 2.13     | 2730 / 2.89  | 514 / 2.63     |                    |

**API Rating & Safety Factor**

- a) Based on full casing evacuation with 9 ppg fluid on backside (2018' setting depth).
- b) Burst pressure based on 9 ppg fluid with no fluid on backside (2018' setting depth).
- c) Based on string weight in 9 ppg fluid at 2018' TVD plus 100k# overpull. (Buoyed weight equals 95k lbs.)

Cement volumes are based on 13-3/8" casing set in 17-1/2" hole with 50% excess to circulate cement back to surface. Mix and pump the following slurry.

**Pre-flush (Spacer):**      **20 bbls** fresh water

**Lead Slurry:**            **593 sks** (307 bbls) 2.9 yield conventional system with 94 lb/sk cement, .25 lb/sk D130 Lost Circulation Control Agent, 2% CaCL<sub>2</sub>, 4% D079 Extender, and 2% D053 Expanding Agent.

**Tail Slurry:**              **374 sks** (77 bbls) 1.16 yield conventional system with 94 lb/sk cement, .25 lb/sk Lost Circulation Control Agent, and .25% CaCL<sub>2</sub>.



**Oasis Petroleum  
Well Summary  
Chalmers 5301 44-24 4T2R  
Sec. 24 T153N R101W  
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

| Size   | Interval   | Weight | Grade  | Coupling | I.D.   | Drift    | Make-up Torque (ft-lbs) |         |       |
|--------|------------|--------|--------|----------|--------|----------|-------------------------|---------|-------|
|        |            |        |        |          |        |          | Minimum                 | Optimum | Max   |
| 9-5/8" | 0' - 6000' | 40     | HCL-80 | LTC      | 8.835" | 8.75"*** | 5,450                   | 7,270   | 9,090 |

\*\*\*Special Drift

| Interval   | Description                   | Collapse     | Burst       | Tension      |
|------------|-------------------------------|--------------|-------------|--------------|
|            |                               | (psi) a      | (psi) b     | (1000 lbs) c |
| 0' - 6000' | 9-5/8", 40#, HCL-80, LTC, 8rd | 3090 / 3.96* | 5750 / 1.23 | 837 / 2.75   |

API Rating & Safety Factor

- a) Burst pressure calculated from a gas kick coming from the production zone (Bakken Pool) at 9,000psi and a subsequent breakdown at the 9-5/8" shoe, based on a 13.5#/ft fracture gradient. Backup of 9 ppg fluid.
- b) Collapse pressure based on 11.5ppg fluid on backside and 9ppg fluid inside of casing.
- c) Yield based on string weight in 10 ppg fluid, (217k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 9-5/8" casing set in an 12-1/4" hole with **10%** excess in OH and **0%** excess inside surface casing. TOC at surface.

**Pre-flush (Spacer):**            **20 bbls** Chem wash

**Lead Slurry:**                    **540 sks** (280 bbls) Conventional system with 75 lb/sk cement, 0.5lb/sk lost circulation, 10% expanding agent, 2% extender, 2% CaCl<sub>2</sub>, 0.2% anti foam, and 0.4% fluid loss

**Tail Slurry:**                      **373 sks** (77 bbls) Conventional system with 94 lb/sk cement, 0.3% anti-settling agent, 0.3% fluid loss agent, 0.3 lb/sk lost circulation control agent, 0.2% anti foam, and 0.1% retarder

**Oasis Petroleum  
Well Summary  
Chalmers 5301 44-24 4T2R  
Sec. 24 T153N R101W  
McKenzie County, North Dakota**

INTERMEDIATE CASING AND CEMENT DESIGN

Intermediate Casing Design

| Size | Interval    | Weight | Grade   | Coupling | I.D.   | Drift  | Make-up Torque (ft-lbs) |         |       |
|------|-------------|--------|---------|----------|--------|--------|-------------------------|---------|-------|
|      |             |        |         |          |        |        | Minimum                 | Optimum | Max   |
| 7"   | 0' - 11540' | 32     | HCP-110 | LTC      | 6.094" | 6.000" | 6,730                   | 8,970   | 11210 |

\*\*Special Drift

| Interval      | Description                | Collapse<br>(psi) a | Burst<br>(psi) b | Tension<br>(1000 lbs) c |
|---------------|----------------------------|---------------------|------------------|-------------------------|
| 0' - 11540'   | 7", 32#, HCP-110, LTC, 8rd | 11890 / 2.13*       | 12450 / 1.28     | 897 / 2.17              |
| 6777' - 9095' | 7", 32#, HCP-110, LTC, 8rd | 11890 / 1.88**      | 12450 / 1.30     |                         |

API Rating & Safety Factor

- a. \*Assume full casing evacuation with 10 ppg fluid on backside. \*\*Assume full casing evacuation with 1.2 psi/ft equivalent fluid gradient across salt intervals.
- b. Burst pressure based on 9000 psig max press for stimulation plus 10.2 ppg fluid in casing and 9 ppg fluid on backside-to 10,630' TVD.
- c. Based on string weight in 10 ppg fluid, (295k lbs buoyed weight) plus 100k lbs overpull.

Cement volumes are estimates based on 7" casing set in an 8-3/4" hole with 30% excess.

**Pre-flush (Spacer):**      **50 bbls** Saltwater  
                                      **40 bbls** Weighted MudPush Express

**Lead Slurry:**                **219 sks** (86 bbls) 2.21 yield conventional system with 47 lb/sk cement, 37 lb/sk D035 Extender, 3.0% KCl, 3.0% D154 Extender, 0.3% D208 Viscosifier, 0.07% Retarder, 0.2% Anti Foam, 0.5lb/sk D130 LCM

**Tail Slurry:**                 **679 sks** (186 bbls) 1.54 yield conventional system with 94 lb/sk cement, 3.0% KCl, 35.0% Silica, 0.5% Retarder, 0.2% Fluid Loss, 0.2% Anti Foam, 0.5 lb/sk LCM

**Oasis Petroleum  
Well Summary  
Chalmers 5301 44-24 4T2R  
Sec. 24 T153N R101W  
McKenzie County, North Dakota**

**PRODUCTION LINER**

| <b>Size</b> | <b>Interval</b> | <b>Weight</b> | <b>Grade</b> | <b>Coupling</b> | <b>I.D.</b> | <b>Drift</b> | <b>Torque</b> |
|-------------|-----------------|---------------|--------------|-----------------|-------------|--------------|---------------|
| 4-1/2"      | 10175' - 21226' | 13.5          | P-110        | BTC             | 3.92"       | 3.795"       | 4,500         |

| <b>Interval</b> | <b>Length</b> | <b>Description</b>               | <b>Collapse</b> | <b>Burst</b>   | <b>Tension</b>      |
|-----------------|---------------|----------------------------------|-----------------|----------------|---------------------|
|                 |               |                                  | (psi) <b>a</b>  | (psi) <b>b</b> | (1000 lbs) <b>c</b> |
| 10175' - 21226' | 11,051        | 4-1/2", 13.5 lb, P-110, BTC, 8rd | 10680 / 2.00    | 12410 / 1.28   | 422 / 1.85          |

**API Rating & Safety Factor**

- a) Based on full casing evacuation with 9.5 ppg fluid on backside @ 10784' TVD.
- b) Burst pressure based on 9000 psi treating pressure with 10.2 ppg internal fluid gradient and 9 ppg external fluid gradient @ 10784' TVD.
- c) Based on string weight in 9.5 ppg fluid (Buoyed weight: 128k lbs.) plus 100k lbs overpull.

Oasis Petroleum does not use Diesel Fuel, as defined by the US EPA in the list below, in our hydraulic fracture operations.

**68334-30-5 (Primary Name: Fuels, diesel)**  
**68476-34-6 (Primary Name: Fuels, diesel, No. 2)**  
**68476-30-2 (Primary Name: Fuel oil No. 2)**  
**68476-31-3 (Primary Name: Fuel oil, No. 4)**  
**8008-20-6 (Primary Name: Kerosene)**

# **Oasis**

**Indian Hills**

**153N-100W-19/20\_Altered**

**Chalmers 5300 44-24 4T2R**

**Chalmers 5300 44-24 4T2R**

**Plan: Design #1**

## **Standard Planning Report**

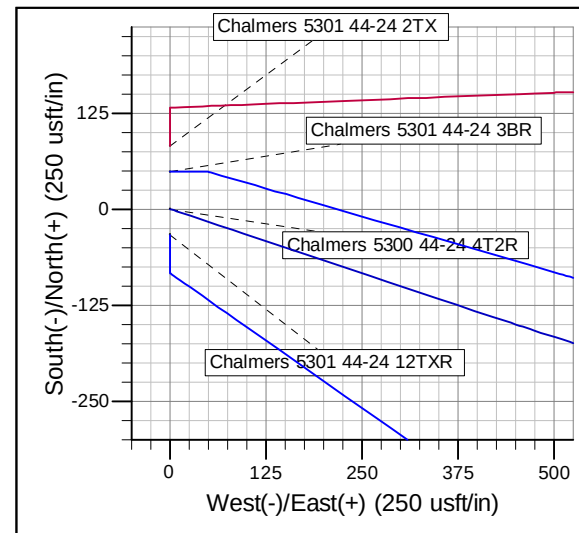
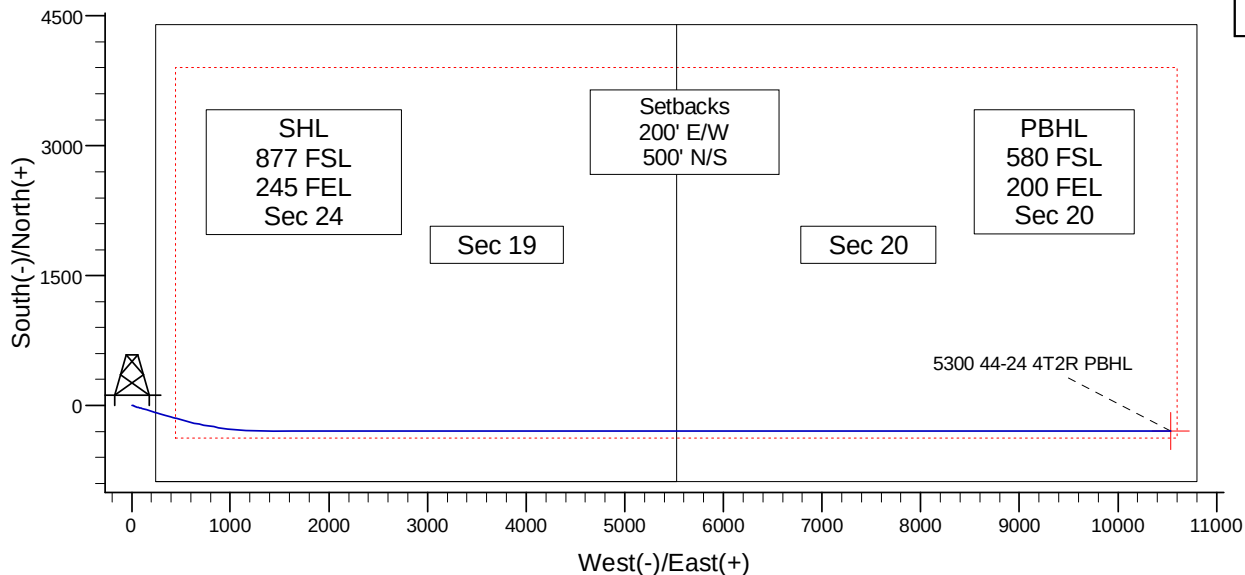
**22 May, 2014**



Project: Indian Hills  
Site: 153N-100W-19/20\_Alt  
Well: Chalmers 5300 44-24 4T2R  
Wellbore: Chalmers 5300 44-24 4T2R  
Design: Design #1

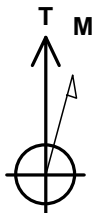
WELL DETAILS: Chalmers 5300 44-24 4T2R

|               |           |         |            |          |                 |           |                   |
|---------------|-----------|---------|------------|----------|-----------------|-----------|-------------------|
| Ground Level: | 1942.0    |         |            |          |                 |           |                   |
| Northing      | 400341.53 | Easting | 1209326.13 | Latitude | 48° 3' 20.010 N | Longitude | 103° 36' 18.550 W |



CASING DETAILS

|         |        |      |      |
|---------|--------|------|------|
| TVD     | MD     | Name | Size |
| 10737.0 | 1007.0 | 7"   | 7"   |

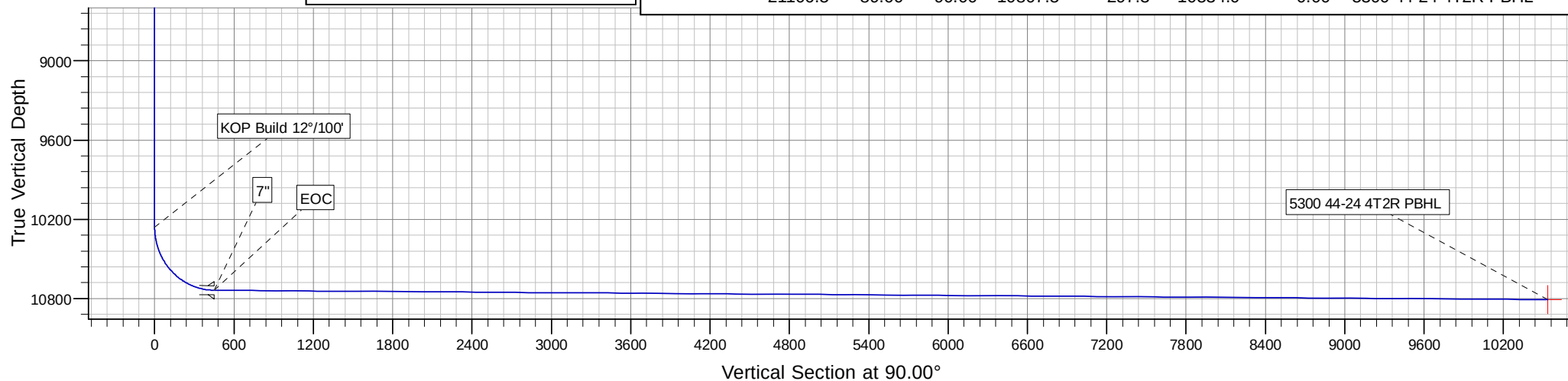


Azimuths to True North  
Magnetic North: 8.27°

Magnetic Field  
Strength: 56433.3snT  
Dip Angle: 72.98°  
Date: 5/22/2014  
Model: IGRF2010

SECTION DETAILS

| MD      | Inc   | Azi    | TVD     | +N/-S  | +E/-W   | Dleg  | Target               |
|---------|-------|--------|---------|--------|---------|-------|----------------------|
| 0.0     | 0.00  | 0.00   | 0.0     | 0.0    | 0.0     | 0.00  |                      |
| 10259.5 | 0.00  | 0.00   | 10259.5 | 0.0    | 0.0     | 0.00  |                      |
| 11006.2 | 89.60 | 108.45 | 10737.0 | -150.1 | 449.8   | 12.00 |                      |
| 11928.6 | 89.60 | 90.00  | 10743.4 | -297.3 | 1356.4  | 2.00  |                      |
| 21106.5 | 89.60 | 90.00  | 10807.5 | -297.3 | 10534.0 | 0.00  | 5300 44-24 4T2R PBHL |





# Ryan Directional Services

## Planning Report

|                  |                           |                                     |                               |
|------------------|---------------------------|-------------------------------------|-------------------------------|
| <b>Database:</b> | EDM 5000.1 Single User Db | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Company:</b>  | Oasis                     | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Project:</b>  | Indian Hills              | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site:</b>     | 153N-100W-19/20_Altered   | <b>North Reference:</b>             | True                          |
| <b>Well:</b>     | Chalmers 5300 44-24 4T2R  | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Wellbore:</b> | Chalmers 5300 44-24 4T2R  |                                     |                               |
| <b>Design:</b>   | Design #1                 |                                     |                               |

|                    |                            |                      |                |
|--------------------|----------------------------|----------------------|----------------|
| <b>Project</b>     | Indian Hills               |                      |                |
| <b>Map System:</b> | US State Plane 1983        | <b>System Datum:</b> | Mean Sea Level |
| <b>Geo Datum:</b>  | North American Datum 1983  |                      |                |
| <b>Map Zone:</b>   | North Dakota Northern Zone |                      |                |

|                       |          |                         |                   |                   |                   |
|-----------------------|----------|-------------------------|-------------------|-------------------|-------------------|
| Site                  |          | 153N-100W-19/20_Altered |                   |                   |                   |
| Site Position:        |          | Northing:               | 400,357.73 usft   | Latitude:         | 48° 3' 20.170 N   |
| From:                 | Lat/Long | Easting:                | 1,209,326.78 usft | Longitude:        | 103° 36' 18.550 W |
| Position Uncertainty: | 0.0 usft | Slot Radius:            | 13-3/16 "         | Grid Convergence: | -2.31 °           |

| Well                 | Chalmers 5300 44-24 4T2R |            |                     |                   |               |                   |
|----------------------|--------------------------|------------|---------------------|-------------------|---------------|-------------------|
| Well Position        | +N/-S                    | -16.2 usft | Northing:           | 400,341.53 usft   | Latitude:     | 48° 3' 20.010 N   |
|                      | +E/-W                    | 0.0 usft   | Easting:            | 1,209,326.13 usft | Longitude:    | 103° 36' 18.550 W |
| Position Uncertainty |                          | 2.0 usft   | Wellhead Elevation: | 0.0 usft          | Ground Level: | 1,942.0 usft      |

|                  |                          |                    |                        |                      |                            |
|------------------|--------------------------|--------------------|------------------------|----------------------|----------------------------|
| <b>Wellbore</b>  | Chalmers 5300 44-24 4T2R |                    |                        |                      |                            |
| <b>Magnetics</b> | <b>Model Name</b>        | <b>Sample Date</b> | <b>Declination (°)</b> | <b>Dip Angle (°)</b> | <b>Field Strength (nT)</b> |
|                  | IGRF2010                 | 5/22/2014          | 8.27                   | 72.98                | 56,433                     |

|                          |                                |                     |                      |                      |
|--------------------------|--------------------------------|---------------------|----------------------|----------------------|
| <b>Design</b>            | Design #1                      |                     |                      |                      |
| <b>Audit Notes:</b>      |                                |                     |                      |                      |
| <b>Version:</b>          | <b>Phase:</b>                  | PROTOTYPE           | <b>Tie On Depth:</b> | 0.0                  |
| <b>Vertical Section:</b> | <b>Depth From (TVD) (usft)</b> | <b>+N/-S (usft)</b> | <b>+E/-W (usft)</b>  | <b>Direction (°)</b> |
|                          | 0.0                            | 0.0                 | 0.0                  | 90.00                |

| <b>Plan Sections</b>  |                 |             |                       |              |              |                         |                        |                       |         |                     |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|------------------------|-----------------------|---------|---------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) | TFO (°) | Target              |
| 0.0                   | 0.00            | 0.00        | 0.0                   | 0.0          | 0.0          | 0.00                    | 0.00                   | 0.00                  | 0.00    |                     |
| 10,259.5              | 0.00            | 0.00        | 10,259.5              | 0.0          | 0.0          | 0.00                    | 0.00                   | 0.00                  | 0.00    |                     |
| 11,006.2              | 89.60           | 108.45      | 10,737.0              | -150.1       | 449.8        | 12.00                   | 12.00                  | 0.00                  | 108.45  |                     |
| 11,928.6              | 89.60           | 90.00       | 10,743.4              | -297.3       | 1,356.4      | 2.00                    | 0.00                   | -2.00                 | -90.06  |                     |
| 21,106.5              | 89.60           | 90.00       | 10,807.5              | -297.3       | 10,534.0     | 0.00                    | 0.00                   | 0.00                  | 0.00    | 5300 44-24 4T2R PBI |

# Ryan Directional Services

## Planning Report

|                  |                           |                                     |                               |
|------------------|---------------------------|-------------------------------------|-------------------------------|
| <b>Database:</b> | EDM 5000.1 Single User Db | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Company:</b>  | Oasis                     | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Project:</b>  | Indian Hills              | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site:</b>     | 153N-100W-19/20_Altered   | <b>North Reference:</b>             | True                          |
| <b>Well:</b>     | Chalmers 5300 44-24 4T2R  | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Wellbore:</b> | Chalmers 5300 44-24 4T2R  |                                     |                               |
| <b>Design:</b>   | Design #1                 |                                     |                               |

| Planned Survey                         |                 |             |                       |              |              |                         |                         |                        |                       |  |
|----------------------------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|--|
| Measured Depth (usft)                  | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |  |
| 10,259.5                               | 0.00            | 0.00        | 10,259.5              | 0.0          | 0.0          | 0.0                     | 0.00                    | 0.00                   | 0.00                  |  |
| <b>KOP Build 12°/100'</b>              |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,275.0                               | 1.86            | 108.45      | 10,275.0              | -0.1         | 0.2          | 0.2                     | 12.00                   | 12.00                  | 0.00                  |  |
| 10,300.0                               | 4.86            | 108.45      | 10,300.0              | -0.5         | 1.6          | 1.6                     | 12.00                   | 12.00                  | 0.00                  |  |
| 10,325.0                               | 7.86            | 108.45      | 10,324.8              | -1.4         | 4.3          | 4.3                     | 12.00                   | 12.00                  | 0.00                  |  |
| 10,350.0                               | 10.86           | 108.45      | 10,349.5              | -2.7         | 8.1          | 8.1                     | 12.00                   | 12.00                  | 0.00                  |  |
| 10,375.0                               | 13.86           | 108.45      | 10,373.9              | -4.4         | 13.2         | 13.2                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,400.0                               | 16.86           | 108.45      | 10,398.0              | -6.5         | 19.5         | 19.5                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,425.0                               | 19.86           | 108.45      | 10,421.7              | -9.0         | 26.9         | 26.9                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,450.0                               | 22.86           | 108.45      | 10,445.0              | -11.9        | 35.6         | 35.6                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,475.0                               | 25.86           | 108.45      | 10,467.8              | -15.1        | 45.4         | 45.4                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,500.0                               | 28.86           | 108.45      | 10,490.0              | -18.8        | 56.3         | 56.3                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,525.0                               | 31.86           | 108.45      | 10,511.5              | -22.8        | 68.2         | 68.2                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,550.0                               | 34.86           | 108.45      | 10,532.4              | -27.1        | 81.3         | 81.3                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,575.0                               | 37.86           | 108.45      | 10,552.5              | -31.8        | 95.3         | 95.3                    | 12.00                   | 12.00                  | 0.00                  |  |
| 10,600.0                               | 40.86           | 108.45      | 10,571.9              | -36.8        | 110.4        | 110.4                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,621.8                               | 43.47           | 108.45      | 10,588.0              | -41.4        | 124.2        | 124.2                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>False Bakken</b>                    |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,625.0                               | 43.86           | 108.45      | 10,590.3              | -42.2        | 126.3        | 126.3                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,637.2                               | 45.32           | 108.45      | 10,599.0              | -44.9        | 134.5        | 134.5                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>Upper Bakken</b>                    |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,650.0                               | 46.86           | 108.45      | 10,607.9              | -47.8        | 143.2        | 143.2                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,662.0                               | 48.30           | 108.45      | 10,616.0              | -50.6        | 151.6        | 151.6                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>Middle Bakken</b>                   |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,675.0                               | 49.86           | 108.45      | 10,624.5              | -53.7        | 160.9        | 160.9                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,700.0                               | 52.86           | 108.45      | 10,640.1              | -59.9        | 179.5        | 179.5                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,713.3                               | 54.46           | 108.45      | 10,648.0              | -63.3        | 189.6        | 189.6                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>Lower Bakken</b>                    |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,725.0                               | 55.86           | 108.45      | 10,654.7              | -66.3        | 198.7        | 198.7                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,732.8                               | 56.79           | 108.45      | 10,659.0              | -68.4        | 204.9        | 204.9                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>Pronghorn</b>                       |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,750.0                               | 58.86           | 108.45      | 10,668.2              | -73.0        | 218.7        | 218.7                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,765.6                               | 60.73           | 108.45      | 10,676.0              | -77.2        | 231.5        | 231.5                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>Three Forks 1st Bench</b>           |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,775.0                               | 61.86           | 108.45      | 10,680.5              | -79.8        | 239.3        | 239.3                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,800.0                               | 64.86           | 108.45      | 10,691.7              | -86.9        | 260.5        | 260.5                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,825.0                               | 67.86           | 108.45      | 10,701.8              | -94.2        | 282.2        | 282.2                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,825.6                               | 67.94           | 108.45      | 10,702.0              | -94.3        | 282.8        | 282.8                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>Three Forks 1st Bench Claystone</b> |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,850.0                               | 70.86           | 108.45      | 10,710.6              | -101.6       | 304.4        | 304.4                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,857.6                               | 71.77           | 108.45      | 10,713.0              | -103.8       | 311.2        | 311.2                   | 12.00                   | 12.00                  | 0.00                  |  |
| <b>Three Forks 2nd Bench</b>           |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 10,875.0                               | 73.86           | 108.45      | 10,718.1              | -109.1       | 327.0        | 327.0                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,900.0                               | 76.86           | 108.45      | 10,724.5              | -116.8       | 350.0        | 350.0                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,925.0                               | 79.86           | 108.45      | 10,729.5              | -124.5       | 373.2        | 373.2                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,950.0                               | 82.86           | 108.45      | 10,733.3              | -132.3       | 396.6        | 396.6                   | 12.00                   | 12.00                  | 0.00                  |  |
| 10,975.0                               | 85.86           | 108.45      | 10,735.7              | -140.2       | 420.2        | 420.2                   | 12.00                   | 12.00                  | 0.00                  |  |
| 11,000.0                               | 88.86           | 108.45      | 10,736.9              | -148.1       | 443.9        | 443.9                   | 12.00                   | 12.00                  | 0.00                  |  |
| 11,006.2                               | 89.60           | 108.45      | 10,737.0              | -150.1       | 449.8        | 449.8                   | 11.94                   | 11.94                  | 0.00                  |  |
| <b>EOC</b>                             |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 11,007.0                               | 89.60           | 108.43      | 10,737.0              | -150.3       | 450.6        | 450.6                   | 2.08                    | 0.00                   | -2.08                 |  |
| <b>7"</b>                              |                 |             |                       |              |              |                         |                         |                        |                       |  |
| 11,100.0                               | 89.60           | 106.57      | 10,737.6              | -178.3       | 539.2        | 539.2                   | 2.00                    | 0.00                   | -2.00                 |  |

# Ryan Directional Services

## Planning Report

|                  |                           |                                     |                               |
|------------------|---------------------------|-------------------------------------|-------------------------------|
| <b>Database:</b> | EDM 5000.1 Single User Db | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Company:</b>  | Oasis                     | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Project:</b>  | Indian Hills              | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site:</b>     | 153N-100W-19/20_Altered   | <b>North Reference:</b>             | True                          |
| <b>Well:</b>     | Chalmers 5300 44-24 4T2R  | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Wellbore:</b> | Chalmers 5300 44-24 4T2R  |                                     |                               |
| <b>Design:</b>   | Design #1                 |                                     |                               |

| Planned Survey                  |                 |             |                       |              |              |                         |                         |                        |                       |
|---------------------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft)           | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 11,200.0                        | 89.60           | 104.57      | 10,738.3              | -205.1       | 635.6        | 635.6                   | 2.00                    | 0.00                   | -2.00                 |
| 11,300.0                        | 89.60           | 102.57      | 10,739.0              | -228.6       | 732.8        | 732.8                   | 2.00                    | 0.00                   | -2.00                 |
| 11,400.0                        | 89.59           | 100.57      | 10,739.7              | -248.7       | 830.7        | 830.7                   | 2.00                    | 0.00                   | -2.00                 |
| 11,500.0                        | 89.59           | 98.57       | 10,740.4              | -265.3       | 929.3        | 929.3                   | 2.00                    | 0.00                   | -2.00                 |
| 11,600.0                        | 89.60           | 96.57       | 10,741.1              | -278.5       | 1,028.4      | 1,028.4                 | 2.00                    | 0.00                   | -2.00                 |
| 11,700.0                        | 89.60           | 94.57       | 10,741.8              | -288.2       | 1,128.0      | 1,128.0                 | 2.00                    | 0.00                   | -2.00                 |
| 11,722.1                        | 89.60           | 94.13       | 10,742.0              | -289.8       | 1,149.9      | 1,149.9                 | 2.00                    | 0.00                   | -2.00                 |
| Three Forks 2nd Bench Claystone |                 |             |                       |              |              |                         |                         |                        |                       |
| 11,800.0                        | 89.60           | 92.57       | 10,742.5              | -294.4       | 1,227.8      | 1,227.8                 | 2.00                    | 0.00                   | -2.00                 |
| 11,900.0                        | 89.60           | 90.57       | 10,743.2              | -297.2       | 1,327.7      | 1,327.7                 | 2.00                    | 0.00                   | -2.00                 |
| 11,928.6                        | 89.60           | 90.00       | 10,743.4              | -297.3       | 1,356.4      | 1,356.4                 | 2.00                    | 0.00                   | -2.00                 |
| 12,000.0                        | 89.60           | 90.00       | 10,743.9              | -297.3       | 1,427.7      | 1,427.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,100.0                        | 89.60           | 90.00       | 10,744.6              | -297.3       | 1,527.7      | 1,527.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,200.0                        | 89.60           | 90.00       | 10,745.3              | -297.3       | 1,627.7      | 1,627.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,300.0                        | 89.60           | 90.00       | 10,746.0              | -297.3       | 1,727.7      | 1,727.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,400.0                        | 89.60           | 90.00       | 10,746.7              | -297.3       | 1,827.7      | 1,827.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,500.0                        | 89.60           | 90.00       | 10,747.4              | -297.3       | 1,927.7      | 1,927.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,600.0                        | 89.60           | 90.00       | 10,748.1              | -297.3       | 2,027.7      | 2,027.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,700.0                        | 89.60           | 90.00       | 10,748.8              | -297.3       | 2,127.7      | 2,127.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,800.0                        | 89.60           | 90.00       | 10,749.5              | -297.3       | 2,227.7      | 2,227.7                 | 0.00                    | 0.00                   | 0.00                  |
| 12,900.0                        | 89.60           | 90.00       | 10,750.2              | -297.3       | 2,327.7      | 2,327.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,000.0                        | 89.60           | 90.00       | 10,750.9              | -297.3       | 2,427.7      | 2,427.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,100.0                        | 89.60           | 90.00       | 10,751.6              | -297.3       | 2,527.7      | 2,527.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,200.0                        | 89.60           | 90.00       | 10,752.3              | -297.3       | 2,627.7      | 2,627.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,300.0                        | 89.60           | 90.00       | 10,753.0              | -297.3       | 2,727.7      | 2,727.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,400.0                        | 89.60           | 90.00       | 10,753.7              | -297.3       | 2,827.7      | 2,827.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,500.0                        | 89.60           | 90.00       | 10,754.4              | -297.3       | 2,927.7      | 2,927.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,600.0                        | 89.60           | 90.00       | 10,755.1              | -297.3       | 3,027.7      | 3,027.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,700.0                        | 89.60           | 90.00       | 10,755.8              | -297.3       | 3,127.7      | 3,127.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,800.0                        | 89.60           | 90.00       | 10,756.5              | -297.3       | 3,227.7      | 3,227.7                 | 0.00                    | 0.00                   | 0.00                  |
| 13,900.0                        | 89.60           | 90.00       | 10,757.2              | -297.3       | 3,327.7      | 3,327.7                 | 0.00                    | 0.00                   | 0.00                  |
| 14,000.0                        | 89.60           | 90.00       | 10,757.9              | -297.3       | 3,427.7      | 3,427.7                 | 0.00                    | 0.00                   | 0.00                  |
| 14,100.0                        | 89.60           | 90.00       | 10,758.6              | -297.3       | 3,527.7      | 3,527.7                 | 0.00                    | 0.00                   | 0.00                  |
| 14,200.0                        | 89.60           | 90.00       | 10,759.3              | -297.3       | 3,627.7      | 3,627.7                 | 0.00                    | 0.00                   | 0.00                  |
| 14,300.0                        | 89.60           | 90.00       | 10,760.0              | -297.3       | 3,727.7      | 3,727.7                 | 0.00                    | 0.00                   | 0.00                  |
| 14,400.0                        | 89.60           | 90.00       | 10,760.7              | -297.3       | 3,827.7      | 3,827.7                 | 0.00                    | 0.00                   | 0.00                  |
| 14,442.6                        | 89.60           | 90.00       | 10,761.0              | -297.3       | 3,870.2      | 3,870.2                 | 0.00                    | 0.00                   | 0.00                  |
| Three Forks 3rd Bench           |                 |             |                       |              |              |                         |                         |                        |                       |
| 14,500.0                        | 89.60           | 90.00       | 10,761.4              | -297.3       | 3,927.6      | 3,927.6                 | 0.00                    | 0.00                   | 0.00                  |
| 14,600.0                        | 89.60           | 90.00       | 10,762.1              | -297.3       | 4,027.6      | 4,027.6                 | 0.00                    | 0.00                   | 0.00                  |
| 14,700.0                        | 89.60           | 90.00       | 10,762.8              | -297.3       | 4,127.6      | 4,127.6                 | 0.00                    | 0.00                   | 0.00                  |
| 14,800.0                        | 89.60           | 90.00       | 10,763.5              | -297.3       | 4,227.6      | 4,227.6                 | 0.00                    | 0.00                   | 0.00                  |
| 14,900.0                        | 89.60           | 90.00       | 10,764.2              | -297.3       | 4,327.6      | 4,327.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,000.0                        | 89.60           | 90.00       | 10,764.9              | -297.3       | 4,427.6      | 4,427.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,100.0                        | 89.60           | 90.00       | 10,765.6              | -297.3       | 4,527.6      | 4,527.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,200.0                        | 89.60           | 90.00       | 10,766.3              | -297.3       | 4,627.6      | 4,627.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,300.0                        | 89.60           | 90.00       | 10,767.0              | -297.3       | 4,727.6      | 4,727.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,400.0                        | 89.60           | 90.00       | 10,767.7              | -297.3       | 4,827.6      | 4,827.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,500.0                        | 89.60           | 90.00       | 10,768.4              | -297.3       | 4,927.6      | 4,927.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,600.0                        | 89.60           | 90.00       | 10,769.1              | -297.3       | 5,027.6      | 5,027.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,700.0                        | 89.60           | 90.00       | 10,769.8              | -297.3       | 5,127.6      | 5,127.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,800.0                        | 89.60           | 90.00       | 10,770.5              | -297.3       | 5,227.6      | 5,227.6                 | 0.00                    | 0.00                   | 0.00                  |
| 15,900.0                        | 89.60           | 90.00       | 10,771.2              | -297.3       | 5,327.6      | 5,327.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,000.0                        | 89.60           | 90.00       | 10,771.9              | -297.3       | 5,427.6      | 5,427.6                 | 0.00                    | 0.00                   | 0.00                  |

# Ryan Directional Services

## Planning Report

|                  |                           |                                     |                               |
|------------------|---------------------------|-------------------------------------|-------------------------------|
| <b>Database:</b> | EDM 5000.1 Single User Db | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Company:</b>  | Oasis                     | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Project:</b>  | Indian Hills              | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site:</b>     | 153N-100W-19/20_Altered   | <b>North Reference:</b>             | True                          |
| <b>Well:</b>     | Chalmers 5300 44-24 4T2R  | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Wellbore:</b> | Chalmers 5300 44-24 4T2R  |                                     |                               |
| <b>Design:</b>   | Design #1                 |                                     |                               |

| Planned Survey        |                 |             |                       |              |              |                         |                         |                        |                       |
|-----------------------|-----------------|-------------|-----------------------|--------------|--------------|-------------------------|-------------------------|------------------------|-----------------------|
| Measured Depth (usft) | Inclination (°) | Azimuth (°) | Vertical Depth (usft) | +N/-S (usft) | +E/-W (usft) | Vertical Section (usft) | Dogleg Rate (°/100usft) | Build Rate (°/100usft) | Turn Rate (°/100usft) |
| 16,100.0              | 89.60           | 90.00       | 10,772.6              | -297.3       | 5,527.6      | 5,527.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,200.0              | 89.60           | 90.00       | 10,773.3              | -297.3       | 5,627.6      | 5,627.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,300.0              | 89.60           | 90.00       | 10,774.0              | -297.3       | 5,727.6      | 5,727.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,400.0              | 89.60           | 90.00       | 10,774.7              | -297.3       | 5,827.6      | 5,827.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,500.0              | 89.60           | 90.00       | 10,775.4              | -297.3       | 5,927.6      | 5,927.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,600.0              | 89.60           | 90.00       | 10,776.1              | -297.3       | 6,027.6      | 6,027.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,700.0              | 89.60           | 90.00       | 10,776.8              | -297.3       | 6,127.6      | 6,127.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,800.0              | 89.60           | 90.00       | 10,777.5              | -297.3       | 6,227.6      | 6,227.6                 | 0.00                    | 0.00                   | 0.00                  |
| 16,900.0              | 89.60           | 90.00       | 10,778.2              | -297.3       | 6,327.6      | 6,327.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,000.0              | 89.60           | 90.00       | 10,778.9              | -297.3       | 6,427.6      | 6,427.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,100.0              | 89.60           | 90.00       | 10,779.6              | -297.3       | 6,527.6      | 6,527.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,200.0              | 89.60           | 90.00       | 10,780.3              | -297.3       | 6,627.6      | 6,627.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,300.0              | 89.60           | 90.00       | 10,780.9              | -297.3       | 6,727.6      | 6,727.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,400.0              | 89.60           | 90.00       | 10,781.6              | -297.3       | 6,827.6      | 6,827.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,500.0              | 89.60           | 90.00       | 10,782.3              | -297.3       | 6,927.6      | 6,927.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,600.0              | 89.60           | 90.00       | 10,783.0              | -297.3       | 7,027.6      | 7,027.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,700.0              | 89.60           | 90.00       | 10,783.7              | -297.3       | 7,127.6      | 7,127.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,800.0              | 89.60           | 90.00       | 10,784.4              | -297.3       | 7,227.6      | 7,227.6                 | 0.00                    | 0.00                   | 0.00                  |
| 17,900.0              | 89.60           | 90.00       | 10,785.1              | -297.3       | 7,327.6      | 7,327.6                 | 0.00                    | 0.00                   | 0.00                  |
| 18,000.0              | 89.60           | 90.00       | 10,785.8              | -297.3       | 7,427.6      | 7,427.6                 | 0.00                    | 0.00                   | 0.00                  |
| 18,100.0              | 89.60           | 90.00       | 10,786.5              | -297.3       | 7,527.6      | 7,527.6                 | 0.00                    | 0.00                   | 0.00                  |
| 18,200.0              | 89.60           | 90.00       | 10,787.2              | -297.3       | 7,627.6      | 7,627.6                 | 0.00                    | 0.00                   | 0.00                  |
| 18,300.0              | 89.60           | 90.00       | 10,787.9              | -297.3       | 7,727.6      | 7,727.6                 | 0.00                    | 0.00                   | 0.00                  |
| 18,400.0              | 89.60           | 90.00       | 10,788.6              | -297.3       | 7,827.6      | 7,827.6                 | 0.00                    | 0.00                   | 0.00                  |
| 18,500.0              | 89.60           | 90.00       | 10,789.3              | -297.3       | 7,927.6      | 7,927.6                 | 0.00                    | 0.00                   | 0.00                  |
| 18,600.0              | 89.60           | 90.00       | 10,790.0              | -297.3       | 8,027.5      | 8,027.5                 | 0.00                    | 0.00                   | 0.00                  |
| 18,700.0              | 89.60           | 90.00       | 10,790.7              | -297.3       | 8,127.5      | 8,127.5                 | 0.00                    | 0.00                   | 0.00                  |
| 18,800.0              | 89.60           | 90.00       | 10,791.4              | -297.3       | 8,227.5      | 8,227.5                 | 0.00                    | 0.00                   | 0.00                  |
| 18,900.0              | 89.60           | 90.00       | 10,792.1              | -297.3       | 8,327.5      | 8,327.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,000.0              | 89.60           | 90.00       | 10,792.8              | -297.3       | 8,427.5      | 8,427.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,100.0              | 89.60           | 90.00       | 10,793.5              | -297.3       | 8,527.5      | 8,527.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,200.0              | 89.60           | 90.00       | 10,794.2              | -297.3       | 8,627.5      | 8,627.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,300.0              | 89.60           | 90.00       | 10,794.9              | -297.3       | 8,727.5      | 8,727.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,400.0              | 89.60           | 90.00       | 10,795.6              | -297.3       | 8,827.5      | 8,827.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,500.0              | 89.60           | 90.00       | 10,796.3              | -297.3       | 8,927.5      | 8,927.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,600.0              | 89.60           | 90.00       | 10,797.0              | -297.3       | 9,027.5      | 9,027.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,700.0              | 89.60           | 90.00       | 10,797.7              | -297.3       | 9,127.5      | 9,127.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,800.0              | 89.60           | 90.00       | 10,798.4              | -297.3       | 9,227.5      | 9,227.5                 | 0.00                    | 0.00                   | 0.00                  |
| 19,900.0              | 89.60           | 90.00       | 10,799.1              | -297.3       | 9,327.5      | 9,327.5                 | 0.00                    | 0.00                   | 0.00                  |
| 20,000.0              | 89.60           | 90.00       | 10,799.8              | -297.3       | 9,427.5      | 9,427.5                 | 0.00                    | 0.00                   | 0.00                  |
| 20,100.0              | 89.60           | 90.00       | 10,800.5              | -297.3       | 9,527.5      | 9,527.5                 | 0.00                    | 0.00                   | 0.00                  |
| 20,200.0              | 89.60           | 90.00       | 10,801.2              | -297.3       | 9,627.5      | 9,627.5                 | 0.00                    | 0.00                   | 0.00                  |
| 20,300.0              | 89.60           | 90.00       | 10,801.9              | -297.3       | 9,727.5      | 9,727.5                 | 0.00                    | 0.00                   | 0.00                  |
| 20,400.0              | 89.60           | 90.00       | 10,802.6              | -297.3       | 9,827.5      | 9,827.5                 | 0.00                    | 0.00                   | 0.00                  |
| 20,500.0              | 89.60           | 90.00       | 10,803.3              | -297.3       | 9,927.5      | 9,927.5                 | 0.00                    | 0.00                   | 0.00                  |
| 20,600.0              | 89.60           | 90.00       | 10,804.0              | -297.3       | 10,027.5     | 10,027.5                | 0.00                    | 0.00                   | 0.00                  |
| 20,700.0              | 89.60           | 90.00       | 10,804.7              | -297.3       | 10,127.5     | 10,127.5                | 0.00                    | 0.00                   | 0.00                  |
| 20,800.0              | 89.60           | 90.00       | 10,805.4              | -297.3       | 10,227.5     | 10,227.5                | 0.00                    | 0.00                   | 0.00                  |
| 20,900.0              | 89.60           | 90.00       | 10,806.1              | -297.3       | 10,327.5     | 10,327.5                | 0.00                    | 0.00                   | 0.00                  |
| 21,000.0              | 89.60           | 90.00       | 10,806.8              | -297.3       | 10,427.5     | 10,427.5                | 0.00                    | 0.00                   | 0.00                  |
| 21,106.5              | 89.60           | 90.00       | 10,807.5              | -297.3       | 10,534.0     | 10,534.0                | 0.00                    | 0.00                   | 0.00                  |
| 5300 44-24 4T2R PBHL  |                 |             |                       |              |              |                         |                         |                        |                       |

# Ryan Directional Services

## Planning Report

|                  |                           |                                     |                               |
|------------------|---------------------------|-------------------------------------|-------------------------------|
| <b>Database:</b> | EDM 5000.1 Single User Db | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Company:</b>  | Oasis                     | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Project:</b>  | Indian Hills              | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site:</b>     | 153N-100W-19/20_Altered   | <b>North Reference:</b>             | True                          |
| <b>Well:</b>     | Chalmers 5300 44-24 4T2R  | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Wellbore:</b> | Chalmers 5300 44-24 4T2R  |                                     |                               |
| <b>Design:</b>   | Design #1                 |                                     |                               |

| Design Targets                                                                              |           |          |          |        |          |            |              |                 |                   |
|---------------------------------------------------------------------------------------------|-----------|----------|----------|--------|----------|------------|--------------|-----------------|-------------------|
| Target Name                                                                                 |           |          |          |        |          |            |              |                 |                   |
| - hit/miss target                                                                           | Dip Angle | Dip Dir. | TVD      | +N/-S  | +E/-W    | Northing   | Easting      | Latitude        | Longitude         |
| - Shape                                                                                     | (°)       | (°)      | (usft)   | (usft) | (usft)   | (usft)     | (usft)       |                 |                   |
| 5300 44-24 4T2R PBHL                                                                        | 0.00      | 0.00     | 10,807.5 | -297.0 | 10,534.0 | 399,620.07 | 1,219,839.59 | 48° 3' 17.050 N | 103° 33' 43.485 W |
| - plan misses target center by 0.3usft at 21106.5usft MD (10807.5 TVD, -297.3 N, 10534.0 E) |           |          |          |        |          |            |              |                 |                   |
| - Point                                                                                     |           |          |          |        |          |            |              |                 |                   |

| Casing Points |                |                |      |                 |               |
|---------------|----------------|----------------|------|-----------------|---------------|
|               | Measured Depth | Vertical Depth |      | Casing Diameter | Hole Diameter |
|               | (usft)         | (usft)         | Name | (")             | (")           |
|               | 11,007.0       | 10,737.0       | 7"   | 7               | 8-3/4         |



# Ryan Directional Services

## Planning Report

|                  |                           |                                     |                               |
|------------------|---------------------------|-------------------------------------|-------------------------------|
| <b>Database:</b> | EDM 5000.1 Single User Db | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Company:</b>  | Oasis                     | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Project:</b>  | Indian Hills              | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site:</b>     | 153N-100W-19/20_Altered   | <b>North Reference:</b>             | True                          |
| <b>Well:</b>     | Chalmers 5300 44-24 4T2R  | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Wellbore:</b> | Chalmers 5300 44-24 4T2R  |                                     |                               |
| <b>Design:</b>   | Design #1                 |                                     |                               |

| Formations            |                       |                                    |           |         |                   |  |
|-----------------------|-----------------------|------------------------------------|-----------|---------|-------------------|--|
| Measured Depth (usft) | Vertical Depth (usft) | Name                               | Lithology | Dip (°) | Dip Direction (°) |  |
| 1,918.0               | 1,918.0               | Pierre                             |           |         |                   |  |
| 4,510.0               | 4,510.0               | Greenhorn                          |           |         |                   |  |
| 4,914.0               | 4,914.0               | Mowry                              |           |         |                   |  |
| 5,341.0               | 5,341.0               | Dakota                             |           |         |                   |  |
| 6,356.0               | 6,356.0               | Rierdon                            |           |         |                   |  |
| 6,683.0               | 6,683.0               | Dunham Salt                        |           |         |                   |  |
| 6,800.0               | 6,800.0               | Dunham Salt Base                   |           |         |                   |  |
| 6,887.0               | 6,887.0               | Spearfish                          |           |         |                   |  |
| 7,135.0               | 7,135.0               | Pine Salt                          |           |         |                   |  |
| 7,168.0               | 7,168.0               | Pine Salt Base                     |           |         |                   |  |
| 7,235.0               | 7,235.0               | Opeche Salt                        |           |         |                   |  |
| 7,269.0               | 7,269.0               | Opeche Salt Base                   |           |         |                   |  |
| 7,469.0               | 7,469.0               | Broom Creek (Top of Minnelusa Gp.) |           |         |                   |  |
| 7,553.0               | 7,553.0               | Amsden                             |           |         |                   |  |
| 7,720.0               | 7,720.0               | Tyler                              |           |         |                   |  |
| 7,920.0               | 7,920.0               | Otter (Base of Minnelusa Gp.)      |           |         |                   |  |
| 8,269.0               | 8,269.0               | Kibbey Lime                        |           |         |                   |  |
| 8,433.0               | 8,433.0               | Charles Salt                       |           |         |                   |  |
| 9,010.0               | 9,010.0               | UB                                 |           |         |                   |  |
| 9,091.0               | 9,091.0               | Base Last Salt                     |           |         |                   |  |
| 9,141.0               | 9,141.0               | Ratcliffe                          |           |         |                   |  |
| 9,311.0               | 9,311.0               | Mission Canyon                     |           |         |                   |  |
| 9,876.0               | 9,876.0               | Lodgepole                          |           |         |                   |  |
| 10,109.0              | 10,109.0              | Lodgepole Fracture Zone            |           |         |                   |  |
| 10,621.8              | 10,588.0              | False Bakken                       |           |         |                   |  |
| 10,637.2              | 10,599.0              | Upper Bakken                       |           |         |                   |  |
| 10,662.0              | 10,616.0              | Middle Bakken                      |           |         |                   |  |
| 10,713.3              | 10,648.0              | Lower Bakken                       |           |         |                   |  |
| 10,732.8              | 10,659.0              | Pronghorn                          |           |         |                   |  |
| 10,765.6              | 10,676.0              | Three Forks 1st Bench              |           |         |                   |  |
| 10,825.6              | 10,702.0              | Three Forks 1st Bench Claystone    |           |         |                   |  |
| 10,857.6              | 10,713.0              | Three Forks 2nd Bench              |           |         |                   |  |
| 11,722.1              | 10,742.0              | Three Forks 2nd Bench Claystone    |           |         |                   |  |
| 14,442.6              | 10,761.0              | Three Forks 3rd Bench              |           |         |                   |  |

| Plan Annotations      |                       |                   |              |                    |
|-----------------------|-----------------------|-------------------|--------------|--------------------|
| Measured Depth (usft) | Vertical Depth (usft) | Local Coordinates |              | Comment            |
|                       |                       | +N/-S (usft)      | +E/-W (usft) |                    |
| 10,259.5              | 10,259.5              | 0.0               | 0.0          | KOP Build 12°/100' |
| 11,006.2              | 10,737.0              | -150.1            | 449.8        | EOC                |

# **Oasis**

**Indian Hills**

**153N-100W-19/20\_Altered**

**Chalmers 5300 44-24 4T2R**

**Chalmers 5300 44-24 4T2R**

**Design #1**

## **Anticollision Report**

**22 May, 2014**

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Reference                    | Design #1                                                           |                |                     |
|------------------------------|---------------------------------------------------------------------|----------------|---------------------|
| Filter type:                 | NO GLOBAL FILTER: Using user defined selection & filtering criteria |                |                     |
| Interpolation Method:        | Stations                                                            | Error Model:   | ISCWSA              |
| Depth Range:                 | Unlimited                                                           | Scan Method:   | Closest Approach 3D |
| Results Limited by:          | Maximum center-center distance of 10,000.0 usft                     | Error Surface: | Elliptical Conic    |
| Warning Levels Evaluated at: | 2.00 Sigma                                                          | Casing Method: | Not applied         |

| Survey Tool Program |              | Date                                 | 5/22/2014 |                |  |
|---------------------|--------------|--------------------------------------|-----------|----------------|--|
| From<br>(usft)      | To<br>(usft) | Survey (Wellbore)                    | Tool Name | Description    |  |
| 0.0                 | 21,106.5     | Design #1 (Chalmers 5300 44-24 4T2R) | MWD       | MWD - Standard |  |

| Summary                                             |  |                    |                 |                |                 |            |                 |
|-----------------------------------------------------|--|--------------------|-----------------|----------------|-----------------|------------|-----------------|
|                                                     |  | Reference Measured | Offset Measured | Distance       |                 |            |                 |
| Site Name                                           |  | Depth              | Depth           | Between        | Between         | Separation | Warning         |
| Offset Well - Wellbore - Design                     |  | Depth (usft)       | Depth (usft)    | Centres (usft) | Ellipses (usft) | Factor     |                 |
| 153N-100W-19/20_Altered                             |  |                    |                 |                |                 |            |                 |
| Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12T |  | 2,200.0            | 2,200.0         | 33.4           | 23.0            | 3.213      | CC              |
| Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12T |  | 21,106.5           | 21,226.5        | 579.9          | -34.3           | 0.944      | Level 1, ES, SF |
| Chalmers 5301 44-24 2TX - Chalmers #2T - Plan #1    |  | 2,166.7            | 2,166.7         | 82.2           | 72.3            | 8.334      | CC              |
| Chalmers 5301 44-24 2TX - Chalmers #2T - Plan #1    |  | 21,106.5           | 21,028.7        | 452.1          | -161.8          | 0.736      | Level 1, ES, SF |
| Chalmers 5301 44-24 3BR - Chalmers 5301 44-24 3BR - |  | 2,200.0            | 2,200.0         | 48.6           | 38.6            | 4.860      | CC              |
| Chalmers 5301 44-24 3BR - Chalmers 5301 44-24 3BR - |  | 21,106.5           | 20,938.2        | 147.4          | -285.9          | 0.340      | Level 1, ES, SF |

| Offset Design         |                       | 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1 |                       |                  |               |                       |                                     |              |                        |                         |                           | Offset Site Error: |         | 0.0 usft |
|-----------------------|-----------------------|---------------------------------------------------------------------------------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|--------------------|---------|----------|
| Survey Program:       |                       | 0-MWD                                                                                       |                       |                  |               |                       |                                     |              |                        |                         |                           | Offset Well Error: |         | 2.0 usft |
| Reference             |                       | Offset                                                                                      |                       | Semi Major Axis  |               |                       | Distance                            |              |                        |                         |                           |                    |         |          |
| Measured Depth (usft) | Vertical Depth (usft) | Measured Depth (usft)                                                                       | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor  | Warning |          |
| 0.0                   | 0.0                   | 0.0                                                                                         | 0.0                   | 2.0              | 2.0           | -180.00               | -33.4                               | 0.0          | 33.4                   |                         |                           |                    |         |          |
| 100.0                 | 100.0                 | 100.0                                                                                       | 100.0                 | 2.0              | 2.0           | -180.00               | -33.4                               | 0.0          | 33.4                   | 29.4                    | 4.00                      | 8.353              |         |          |
| 200.0                 | 200.0                 | 200.0                                                                                       | 200.0                 | 2.0              | 2.0           | -180.00               | -33.4                               | 0.0          | 33.4                   | 29.4                    | 4.05                      | 8.262              |         |          |
| 300.0                 | 300.0                 | 300.0                                                                                       | 300.0                 | 2.1              | 2.1           | -180.00               | -33.4                               | 0.0          | 33.4                   | 29.3                    | 4.14                      | 8.078              |         |          |
| 400.0                 | 400.0                 | 400.0                                                                                       | 400.0                 | 2.1              | 2.1           | -180.00               | -33.4                               | 0.0          | 33.4                   | 29.2                    | 4.28                      | 7.817              |         |          |
| 500.0                 | 500.0                 | 500.0                                                                                       | 500.0                 | 2.2              | 2.2           | -180.00               | -33.4                               | 0.0          | 33.4                   | 29.0                    | 4.46                      | 7.503              |         |          |
| 600.0                 | 600.0                 | 600.0                                                                                       | 600.0                 | 2.3              | 2.3           | -180.00               | -33.4                               | 0.0          | 33.4                   | 28.8                    | 4.67                      | 7.156              |         |          |
| 700.0                 | 700.0                 | 700.0                                                                                       | 700.0                 | 2.5              | 2.5           | -180.00               | -33.4                               | 0.0          | 33.4                   | 28.5                    | 4.92                      | 6.796              |         |          |
| 800.0                 | 800.0                 | 800.0                                                                                       | 800.0                 | 2.6              | 2.6           | -180.00               | -33.4                               | 0.0          | 33.4                   | 28.2                    | 5.20                      | 6.437              |         |          |
| 900.0                 | 900.0                 | 900.0                                                                                       | 900.0                 | 2.7              | 2.7           | -180.00               | -33.4                               | 0.0          | 33.4                   | 27.9                    | 5.49                      | 6.088              |         |          |
| 1,000.0               | 1,000.0               | 1,000.0                                                                                     | 1,000.0               | 2.9              | 2.9           | -180.00               | -33.4                               | 0.0          | 33.4                   | 27.6                    | 5.81                      | 5.755              |         |          |
| 1,100.0               | 1,100.0               | 1,100.0                                                                                     | 1,100.0               | 3.1              | 3.1           | -180.00               | -33.4                               | 0.0          | 33.4                   | 27.3                    | 6.14                      | 5.443              |         |          |
| 1,200.0               | 1,200.0               | 1,200.0                                                                                     | 1,200.0               | 3.2              | 3.2           | -180.00               | -33.4                               | 0.0          | 33.4                   | 26.9                    | 6.49                      | 5.151              |         |          |
| 1,300.0               | 1,300.0               | 1,300.0                                                                                     | 1,300.0               | 3.4              | 3.4           | -180.00               | -33.4                               | 0.0          | 33.4                   | 26.6                    | 6.85                      | 4.881              |         |          |
| 1,400.0               | 1,400.0               | 1,400.0                                                                                     | 1,400.0               | 3.6              | 3.6           | -180.00               | -33.4                               | 0.0          | 33.4                   | 26.2                    | 7.22                      | 4.631              |         |          |
| 1,500.0               | 1,500.0               | 1,500.0                                                                                     | 1,500.0               | 3.8              | 3.8           | -180.00               | -33.4                               | 0.0          | 33.4                   | 25.8                    | 7.60                      | 4.400              |         |          |
| 1,600.0               | 1,600.0               | 1,600.0                                                                                     | 1,600.0               | 4.0              | 4.0           | -180.00               | -33.4                               | 0.0          | 33.4                   | 25.5                    | 7.99                      | 4.188              |         |          |
| 1,700.0               | 1,700.0               | 1,700.0                                                                                     | 1,700.0               | 4.2              | 4.2           | -180.00               | -33.4                               | 0.0          | 33.4                   | 25.1                    | 8.38                      | 3.992              |         |          |
| 1,800.0               | 1,800.0               | 1,800.0                                                                                     | 1,800.0               | 4.4              | 4.4           | -180.00               | -33.4                               | 0.0          | 33.4                   | 24.7                    | 8.78                      | 3.811              |         |          |
| 1,900.0               | 1,900.0               | 1,900.0                                                                                     | 1,900.0               | 4.6              | 4.6           | -180.00               | -33.4                               | 0.0          | 33.4                   | 24.3                    | 9.18                      | 3.644              |         |          |
| 2,000.0               | 2,000.0               | 2,000.0                                                                                     | 2,000.0               | 4.8              | 4.8           | -180.00               | -33.4                               | 0.0          | 33.4                   | 23.9                    | 9.58                      | 3.489              |         |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: 0.0 usft |         |
|-----------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|-----------------------------|---------|
| Survey Program: 0-MWD                                                                                     |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: 2.0 usft |         |
| Reference                                                                                                 |                       | Offset                |                       | Semi Major Axis  |               |                       | Distance                            |              |                        |                         |                           |                   |                             | Warning |
| Measured Depth (usft)                                                                                     | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                             |         |
|                                                                                                           |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   |                             |         |
| 2,100.0                                                                                                   | 2,100.0               | 2,100.0               | 2,100.0               | 5.0              | 5.0           | -180.00               | -33.4                               | 0.0          | 33.4                   | 23.4                    | 9.99                      | 3.346             | CC                          |         |
| 2,200.0                                                                                                   | 2,200.0               | 2,200.0               | 2,200.0               | 5.2              | 5.2           | -180.00               | -33.4                               | 0.0          | 33.4                   | 23.0                    | 10.41                     | 3.213             |                             |         |
| 2,300.0                                                                                                   | 2,300.0               | 2,299.7               | 2,299.7               | 5.4              | 5.4           | -180.00               | -34.2                               | 0.0          | 34.2                   | 23.5                    | 10.77                     | 3.178             |                             |         |
| 2,400.0                                                                                                   | 2,400.0               | 2,399.7               | 2,399.7               | 5.6              | 5.5           | -180.00               | -35.1                               | 0.0          | 35.1                   | 24.0                    | 11.14                     | 3.151             |                             |         |
| 2,500.0                                                                                                   | 2,500.0               | 2,499.7               | 2,499.7               | 5.8              | 5.7           | -180.00               | -36.0                               | 0.0          | 36.0                   | 24.5                    | 11.52                     | 3.124             |                             |         |
| 2,600.0                                                                                                   | 2,600.0               | 2,599.7               | 2,599.7               | 6.0              | 5.9           | -180.00               | -36.9                               | 0.0          | 36.9                   | 25.0                    | 11.90                     | 3.098             |                             |         |
| 2,700.0                                                                                                   | 2,700.0               | 2,699.7               | 2,699.7               | 6.3              | 6.0           | -180.00               | -37.7                               | 0.0          | 37.7                   | 25.4                    | 12.28                     | 3.072             |                             |         |
| 2,800.0                                                                                                   | 2,800.0               | 2,799.7               | 2,799.7               | 6.5              | 6.2           | -180.00               | -38.6                               | 0.0          | 38.6                   | 25.9                    | 12.67                     | 3.047             |                             |         |
| 2,900.0                                                                                                   | 2,900.0               | 2,899.7               | 2,899.7               | 6.7              | 6.4           | -180.00               | -39.5                               | 0.0          | 39.5                   | 26.4                    | 13.06                     | 3.022             |                             |         |
| 3,000.0                                                                                                   | 3,000.0               | 2,999.7               | 2,999.6               | 6.9              | 6.6           | -180.00               | -40.3                               | 0.0          | 40.3                   | 26.9                    | 13.46                     | 2.998             |                             |         |
| 3,100.0                                                                                                   | 3,100.0               | 3,099.7               | 3,099.6               | 7.1              | 6.7           | -180.00               | -41.2                               | 0.0          | 41.2                   | 27.4                    | 13.86                     | 2.974             |                             |         |
| 3,200.0                                                                                                   | 3,200.0               | 3,199.7               | 3,199.6               | 7.3              | 6.9           | -180.00               | -42.1                               | 0.0          | 42.1                   | 27.8                    | 14.26                     | 2.952             |                             |         |
| 3,300.0                                                                                                   | 3,300.0               | 3,299.7               | 3,299.6               | 7.5              | 7.1           | -180.00               | -43.0                               | 0.0          | 43.0                   | 28.3                    | 14.67                     | 2.929             |                             |         |
| 3,400.0                                                                                                   | 3,400.0               | 3,399.7               | 3,399.6               | 7.8              | 7.3           | -180.00               | -43.8                               | 0.0          | 43.8                   | 28.8                    | 15.08                     | 2.908             |                             |         |
| 3,500.0                                                                                                   | 3,500.0               | 3,499.7               | 3,499.6               | 8.0              | 7.5           | -180.00               | -44.7                               | 0.0          | 44.7                   | 29.2                    | 15.49                     | 2.887             |                             |         |
| 3,600.0                                                                                                   | 3,600.0               | 3,599.7               | 3,599.6               | 8.2              | 7.7           | -180.00               | -45.6                               | 0.0          | 45.6                   | 29.7                    | 15.90                     | 2.867             |                             |         |
| 3,700.0                                                                                                   | 3,700.0               | 3,699.7               | 3,699.6               | 8.4              | 7.9           | -180.00               | -46.5                               | 0.0          | 46.5                   | 30.1                    | 16.31                     | 2.848             |                             |         |
| 3,800.0                                                                                                   | 3,800.0               | 3,799.6               | 3,799.6               | 8.6              | 8.1           | -180.00               | -47.3                               | 0.0          | 47.3                   | 30.6                    | 16.73                     | 2.829             |                             |         |
| 3,900.0                                                                                                   | 3,900.0               | 3,899.6               | 3,899.6               | 8.9              | 8.3           | -180.00               | -48.2                               | 0.0          | 48.2                   | 31.1                    | 17.15                     | 2.811             |                             |         |
| 4,000.0                                                                                                   | 4,000.0               | 3,999.6               | 3,999.6               | 9.1              | 8.5           | -180.00               | -49.1                               | 0.0          | 49.1                   | 31.5                    | 17.57                     | 2.794             |                             |         |
| 4,100.0                                                                                                   | 4,100.0               | 4,099.6               | 4,099.6               | 9.3              | 8.7           | -180.00               | -49.9                               | 0.0          | 49.9                   | 32.0                    | 17.99                     | 2.777             |                             |         |
| 4,200.0                                                                                                   | 4,200.0               | 4,199.6               | 4,199.6               | 9.5              | 8.9           | -180.00               | -50.8                               | 0.0          | 50.8                   | 32.4                    | 18.41                     | 2.760             |                             |         |
| 4,300.0                                                                                                   | 4,300.0               | 4,299.6               | 4,299.5               | 9.7              | 9.1           | -180.00               | -51.7                               | 0.0          | 51.7                   | 32.9                    | 18.84                     | 2.744             |                             |         |
| 4,400.0                                                                                                   | 4,400.0               | 4,399.6               | 4,399.5               | 10.0             | 9.3           | -180.00               | -52.6                               | 0.0          | 52.6                   | 33.3                    | 19.26                     | 2.729             |                             |         |
| 4,500.0                                                                                                   | 4,500.0               | 4,499.6               | 4,499.5               | 10.2             | 9.5           | -180.00               | -53.4                               | 0.0          | 53.4                   | 33.8                    | 19.69                     | 2.714             |                             |         |
| 4,600.0                                                                                                   | 4,600.0               | 4,599.6               | 4,599.5               | 10.4             | 9.7           | -180.00               | -54.3                               | 0.0          | 54.3                   | 34.2                    | 20.12                     | 2.700             |                             |         |
| 4,700.0                                                                                                   | 4,700.0               | 4,699.6               | 4,699.5               | 10.6             | 9.9           | -180.00               | -55.2                               | 0.0          | 55.2                   | 34.6                    | 20.54                     | 2.686             |                             |         |
| 4,800.0                                                                                                   | 4,800.0               | 4,799.6               | 4,799.5               | 10.8             | 10.1          | -180.00               | -56.1                               | 0.0          | 56.1                   | 35.1                    | 20.97                     | 2.673             |                             |         |
| 4,900.0                                                                                                   | 4,900.0               | 4,899.6               | 4,899.5               | 11.1             | 10.3          | -180.00               | -56.9                               | 0.0          | 56.9                   | 35.5                    | 21.40                     | 2.660             |                             |         |
| 5,000.0                                                                                                   | 5,000.0               | 4,999.6               | 4,999.5               | 11.3             | 10.6          | -180.00               | -57.8                               | 0.0          | 57.8                   | 36.0                    | 21.84                     | 2.647             |                             |         |
| 5,100.0                                                                                                   | 5,100.0               | 5,099.6               | 5,099.5               | 11.5             | 10.8          | -180.00               | -58.7                               | 0.0          | 58.7                   | 36.4                    | 22.27                     | 2.635             |                             |         |
| 5,200.0                                                                                                   | 5,200.0               | 5,199.6               | 5,199.5               | 11.7             | 11.0          | -180.00               | -59.5                               | 0.0          | 59.5                   | 36.8                    | 22.70                     | 2.623             |                             |         |
| 5,300.0                                                                                                   | 5,300.0               | 5,299.6               | 5,299.5               | 11.9             | 11.2          | -180.00               | -60.4                               | 0.0          | 60.4                   | 37.3                    | 23.13                     | 2.612             |                             |         |
| 5,400.0                                                                                                   | 5,400.0               | 5,399.6               | 5,399.5               | 12.2             | 11.4          | -180.00               | -61.3                               | 0.0          | 61.3                   | 37.7                    | 23.57                     | 2.601             |                             |         |
| 5,500.0                                                                                                   | 5,500.0               | 5,499.6               | 5,499.5               | 12.4             | 11.6          | -180.00               | -62.2                               | 0.0          | 62.2                   | 38.2                    | 24.00                     | 2.590             |                             |         |
| 5,600.0                                                                                                   | 5,600.0               | 5,599.6               | 5,599.4               | 12.6             | 11.8          | -180.00               | -63.0                               | 0.0          | 63.0                   | 38.6                    | 24.44                     | 2.580             |                             |         |
| 5,700.0                                                                                                   | 5,700.0               | 5,699.6               | 5,699.4               | 12.8             | 12.0          | -180.00               | -63.9                               | 0.0          | 63.9                   | 39.0                    | 24.87                     | 2.569             |                             |         |
| 5,800.0                                                                                                   | 5,800.0               | 5,799.6               | 5,799.4               | 13.1             | 12.3          | -180.00               | -64.8                               | 0.0          | 64.8                   | 39.5                    | 25.31                     | 2.560             |                             |         |
| 5,900.0                                                                                                   | 5,900.0               | 5,899.6               | 5,899.4               | 13.3             | 12.5          | -180.00               | -65.7                               | 0.0          | 65.7                   | 39.9                    | 25.75                     | 2.550             |                             |         |
| 6,000.0                                                                                                   | 6,000.0               | 5,999.6               | 5,999.4               | 13.5             | 12.7          | -180.00               | -66.5                               | 0.0          | 66.5                   | 40.3                    | 26.18                     | 2.541             |                             |         |
| 6,100.0                                                                                                   | 6,100.0               | 6,099.6               | 6,099.4               | 13.7             | 12.9          | -180.00               | -67.4                               | 0.0          | 67.4                   | 40.8                    | 26.62                     | 2.532             |                             |         |
| 6,200.0                                                                                                   | 6,200.0               | 6,199.6               | 6,199.4               | 13.9             | 13.1          | -180.00               | -68.3                               | 0.0          | 68.3                   | 41.2                    | 27.06                     | 2.523             |                             |         |
| 6,300.0                                                                                                   | 6,300.0               | 6,299.6               | 6,299.4               | 14.2             | 13.3          | -180.00               | -69.1                               | 0.0          | 69.1                   | 41.6                    | 27.50                     | 2.514             |                             |         |
| 6,400.0                                                                                                   | 6,400.0               | 6,399.5               | 6,399.4               | 14.4             | 13.6          | -180.00               | -70.0                               | 0.0          | 70.0                   | 42.1                    | 27.94                     | 2.506             |                             |         |
| 6,500.0                                                                                                   | 6,500.0               | 6,499.5               | 6,499.4               | 14.6             | 13.8          | -180.00               | -70.9                               | 0.0          | 70.9                   | 42.5                    | 28.38                     | 2.498             |                             |         |
| 6,600.0                                                                                                   | 6,600.0               | 6,599.5               | 6,599.4               | 14.8             | 14.0          | -180.00               | -71.8                               | 0.0          | 71.8                   | 42.9                    | 28.82                     | 2.490             |                             |         |
| 6,700.0                                                                                                   | 6,700.0               | 6,699.5               | 6,699.4               | 15.1             | 14.2          | -180.00               | -72.6                               | 0.0          | 72.6                   | 43.4                    | 29.26                     | 2.483             |                             |         |
| 6,800.0                                                                                                   | 6,800.0               | 6,799.5               | 6,799.4               | 15.3             | 14.4          | -180.00               | -73.5                               | 0.0          | 73.5                   | 43.8                    | 29.70                     | 2.475             |                             |         |
| 6,900.0                                                                                                   | 6,900.0               | 6,899.5               | 6,899.4               | 15.5             | 14.6          | -180.00               | -74.4                               | 0.0          | 74.4                   | 44.2                    | 30.14                     | 2.468             |                             |         |
| 7,000.0                                                                                                   | 7,000.0               | 6,999.5               | 6,999.3               | 15.7             | 14.9          | -180.00               | -75.3                               | 0.0          | 75.3                   | 44.7                    | 30.58                     | 2.461             |                             |         |
| 7,100.0                                                                                                   | 7,100.0               | 7,099.5               | 7,099.3               | 15.9             | 15.1          | -180.00               | -76.1                               | 0.0          | 76.1                   | 45.1                    | 31.02                     | 2.454             |                             |         |
| 7,200.0                                                                                                   | 7,200.0               | 7,199.5               | 7,199.3               | 16.2             | 15.3          | -180.00               | -77.0                               | 0.0          | 77.0                   | 45.5                    | 31.46                     | 2.447             |                             |         |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-----------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                     |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 2.0 usft |
| Reference                                                                                                 |                       | Offset                |                       | Semi Major Axis  |               | Distance              |                                     |              |                        |                         |                           |                   |                    | Warning  |
| Measured Depth (usft)                                                                                     | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 7,300.0                                                                                                   | 7,300.0               | 7,299.5               | 7,299.3               | 16.4             | 15.5          | -180.00               | -77.9                               | 0.0          | 77.9                   | 46.0                    | 31.91                     | 2.441             |                    |          |
| 7,400.0                                                                                                   | 7,400.0               | 7,399.5               | 7,399.3               | 16.6             | 15.7          | -180.00               | -78.7                               | 0.0          | 78.7                   | 46.4                    | 32.35                     | 2.434             |                    |          |
| 7,500.0                                                                                                   | 7,500.0               | 7,499.5               | 7,499.3               | 16.8             | 16.0          | -180.00               | -79.6                               | 0.0          | 79.6                   | 46.8                    | 32.79                     | 2.428             |                    |          |
| 7,600.0                                                                                                   | 7,600.0               | 7,599.5               | 7,599.3               | 17.1             | 16.2          | -180.00               | -80.5                               | 0.0          | 80.5                   | 47.3                    | 33.23                     | 2.422             |                    |          |
| 7,700.0                                                                                                   | 7,700.0               | 7,699.5               | 7,699.3               | 17.3             | 16.4          | -180.00               | -81.4                               | 0.0          | 81.4                   | 47.7                    | 33.68                     | 2.416             |                    |          |
| 7,800.0                                                                                                   | 7,800.0               | 7,799.5               | 7,799.3               | 17.5             | 16.6          | -180.00               | -82.2                               | 0.0          | 82.2                   | 48.1                    | 34.12                     | 2.410             |                    |          |
| 7,900.0                                                                                                   | 7,900.0               | 7,899.5               | 7,899.3               | 17.7             | 16.8          | -180.00               | -83.1                               | 0.0          | 83.1                   | 48.5                    | 34.56                     | 2.405             |                    |          |
| 8,000.0                                                                                                   | 8,000.0               | 8,000.2               | 8,000.0               | 18.0             | 17.1          | -180.00               | -83.4                               | 0.0          | 83.4                   | 48.4                    | 35.00                     | 2.383             |                    |          |
| 8,100.0                                                                                                   | 8,100.0               | 8,100.2               | 8,100.0               | 18.2             | 17.2          | -180.00               | -83.4                               | 0.0          | 83.4                   | 48.0                    | 35.41                     | 2.356             |                    |          |
| 8,200.0                                                                                                   | 8,200.0               | 8,200.2               | 8,200.0               | 18.4             | 17.4          | -180.00               | -83.4                               | 0.0          | 83.4                   | 47.6                    | 35.83                     | 2.328             |                    |          |
| 8,300.0                                                                                                   | 8,300.0               | 8,300.2               | 8,300.0               | 18.6             | 17.6          | -180.00               | -83.4                               | 0.0          | 83.4                   | 47.2                    | 36.24                     | 2.302             |                    |          |
| 8,400.0                                                                                                   | 8,400.0               | 8,400.2               | 8,400.0               | 18.8             | 17.8          | -180.00               | -83.4                               | 0.0          | 83.4                   | 46.8                    | 36.66                     | 2.276             |                    |          |
| 8,500.0                                                                                                   | 8,500.0               | 8,500.2               | 8,500.0               | 19.1             | 18.0          | -180.00               | -83.4                               | 0.0          | 83.4                   | 46.3                    | 37.07                     | 2.250             |                    |          |
| 8,600.0                                                                                                   | 8,600.0               | 8,600.2               | 8,600.0               | 19.3             | 18.2          | -180.00               | -83.4                               | 0.0          | 83.4                   | 45.9                    | 37.49                     | 2.225             |                    |          |
| 8,700.0                                                                                                   | 8,700.0               | 8,700.2               | 8,700.0               | 19.5             | 18.4          | -180.00               | -83.4                               | 0.0          | 83.4                   | 45.5                    | 37.91                     | 2.200             |                    |          |
| 8,800.0                                                                                                   | 8,800.0               | 8,800.2               | 8,800.0               | 19.7             | 18.6          | -180.00               | -83.4                               | 0.0          | 83.4                   | 45.1                    | 38.33                     | 2.176             |                    |          |
| 8,900.0                                                                                                   | 8,900.0               | 8,900.2               | 8,900.0               | 20.0             | 18.8          | -180.00               | -83.4                               | 0.0          | 83.4                   | 44.7                    | 38.75                     | 2.153             |                    |          |
| 9,000.0                                                                                                   | 9,000.0               | 9,000.2               | 9,000.0               | 20.2             | 19.0          | -180.00               | -83.4                               | 0.0          | 83.4                   | 44.2                    | 39.17                     | 2.130             |                    |          |
| 9,100.0                                                                                                   | 9,100.0               | 9,100.2               | 9,100.0               | 20.4             | 19.2          | -180.00               | -83.4                               | 0.0          | 83.4                   | 43.8                    | 39.59                     | 2.107             |                    |          |
| 9,200.0                                                                                                   | 9,200.0               | 9,200.2               | 9,200.0               | 20.6             | 19.4          | -180.00               | -83.4                               | 0.0          | 83.4                   | 43.4                    | 40.01                     | 2.085             |                    |          |
| 9,300.0                                                                                                   | 9,300.0               | 9,300.2               | 9,300.0               | 20.9             | 19.6          | -180.00               | -83.4                               | 0.0          | 83.4                   | 43.0                    | 40.43                     | 2.063             |                    |          |
| 9,400.0                                                                                                   | 9,400.0               | 9,400.2               | 9,400.0               | 21.1             | 19.8          | -180.00               | -83.4                               | 0.0          | 83.4                   | 42.6                    | 40.86                     | 2.042             |                    |          |
| 9,500.0                                                                                                   | 9,500.0               | 9,500.2               | 9,500.0               | 21.3             | 20.0          | -180.00               | -83.4                               | 0.0          | 83.4                   | 42.1                    | 41.28                     | 2.021             |                    |          |
| 9,600.0                                                                                                   | 9,600.0               | 9,600.2               | 9,600.0               | 21.5             | 20.2          | -180.00               | -83.4                               | 0.0          | 83.4                   | 41.7                    | 41.70                     | 2.000             |                    |          |
| 9,700.0                                                                                                   | 9,700.0               | 9,700.2               | 9,700.0               | 21.8             | 20.4          | -180.00               | -83.4                               | 0.0          | 83.4                   | 41.3                    | 42.13                     | 1.980             |                    |          |
| 9,800.0                                                                                                   | 9,800.0               | 9,800.2               | 9,800.0               | 22.0             | 20.6          | -180.00               | -83.4                               | 0.0          | 83.4                   | 40.9                    | 42.55                     | 1.960             |                    |          |
| 9,900.0                                                                                                   | 9,900.0               | 9,900.2               | 9,900.0               | 22.2             | 20.8          | -180.00               | -83.4                               | 0.0          | 83.4                   | 40.4                    | 42.98                     | 1.941             |                    |          |
| 10,000.0                                                                                                  | 10,000.0              | 10,000.2              | 10,000.0              | 22.4             | 21.0          | -180.00               | -83.4                               | 0.0          | 83.4                   | 40.0                    | 43.40                     | 1.922             |                    |          |
| 10,100.0                                                                                                  | 10,100.0              | 10,100.2              | 10,100.0              | 22.6             | 21.2          | -180.00               | -83.4                               | 0.0          | 83.4                   | 39.6                    | 43.83                     | 1.903             |                    |          |
| 10,200.0                                                                                                  | 10,200.0              | 10,200.2              | 10,200.0              | 22.9             | 21.4          | -180.00               | -83.4                               | 0.0          | 83.4                   | 39.2                    | 44.26                     | 1.885             |                    |          |
| 10,213.8                                                                                                  | 10,213.8              | 10,214.0              | 10,213.8              | 22.9             | 21.4          | -180.00               | -83.4                               | 0.0          | 83.4                   | 39.1                    | 44.32                     | 1.882             |                    |          |
| 10,259.5                                                                                                  | 10,259.5              | 10,256.6              | 10,256.3              | 23.0             | 21.5          | 179.45                | -84.0                               | 0.8          | 84.0                   | 39.5                    | 44.51                     | 1.888             |                    |          |
| 10,275.0                                                                                                  | 10,275.0              | 10,270.6              | 10,270.3              | 23.0             | 21.5          | 70.46                 | -84.6                               | 1.7          | 84.7                   | 40.1                    | 44.54                     | 1.901             |                    |          |
| 10,300.0                                                                                                  | 10,300.0              | 10,293.1              | 10,292.7              | 23.1             | 21.6          | 69.72                 | -86.1                               | 3.9          | 85.9                   | 41.3                    | 44.63                     | 1.926             |                    |          |
| 10,325.0                                                                                                  | 10,324.8              | 10,315.6              | 10,314.9              | 23.1             | 21.6          | 69.15                 | -88.3                               | 6.9          | 87.4                   | 42.7                    | 44.70                     | 1.956             |                    |          |
| 10,350.0                                                                                                  | 10,349.5              | 10,338.0              | 10,336.8              | 23.2             | 21.7          | 68.73                 | -91.0                               | 10.7         | 89.2                   | 44.4                    | 44.76                     | 1.993             |                    |          |
| 10,375.0                                                                                                  | 10,373.9              | 10,360.4              | 10,358.4              | 23.2             | 21.7          | 68.46                 | -94.2                               | 15.4         | 91.2                   | 46.4                    | 44.81                     | 2.035             |                    |          |
| 10,400.0                                                                                                  | 10,398.0              | 10,382.7              | 10,379.6              | 23.3             | 21.8          | 68.32                 | -98.1                               | 20.9         | 93.4                   | 48.6                    | 44.85                     | 2.083             |                    |          |
| 10,425.0                                                                                                  | 10,421.7              | 10,404.9              | 10,400.5              | 23.3             | 21.8          | 68.32                 | -102.5                              | 27.1         | 95.9                   | 51.0                    | 44.89                     | 2.136             |                    |          |
| 10,450.0                                                                                                  | 10,445.0              | 10,427.0              | 10,420.9              | 23.4             | 21.9          | 68.43                 | -107.4                              | 34.2         | 98.6                   | 53.6                    | 44.92                     | 2.194             |                    |          |
| 10,475.0                                                                                                  | 10,467.8              | 10,450.0              | 10,441.6              | 23.5             | 22.0          | 68.67                 | -113.1                              | 42.3         | 101.5                  | 56.5                    | 44.95                     | 2.257             |                    |          |
| 10,500.0                                                                                                  | 10,490.0              | 10,471.1              | 10,460.2              | 23.5             | 22.0          | 68.95                 | -118.8                              | 50.4         | 104.6                  | 59.6                    | 44.99                     | 2.324             |                    |          |
| 10,525.0                                                                                                  | 10,511.5              | 10,493.0              | 10,479.0              | 23.6             | 22.1          | 69.32                 | -125.3                              | 59.6         | 107.9                  | 62.9                    | 45.03                     | 2.396             |                    |          |
| 10,550.0                                                                                                  | 10,532.4              | 10,514.9              | 10,497.3              | 23.7             | 22.1          | 69.76                 | -132.2                              | 69.4         | 111.4                  | 66.3                    | 45.09                     | 2.471             |                    |          |
| 10,575.0                                                                                                  | 10,552.5              | 10,536.7              | 10,514.9              | 23.7             | 22.2          | 70.25                 | -139.6                              | 79.9         | 115.2                  | 70.0                    | 45.17                     | 2.550             |                    |          |
| 10,600.0                                                                                                  | 10,571.9              | 10,558.4              | 10,531.9              | 23.8             | 22.3          | 70.78                 | -147.4                              | 91.0         | 119.1                  | 73.9                    | 45.27                     | 2.632             |                    |          |
| 10,625.0                                                                                                  | 10,590.3              | 10,580.1              | 10,548.3              | 23.9             | 22.4          | 71.34                 | -155.6                              | 102.7        | 123.3                  | 77.9                    | 45.39                     | 2.716             |                    |          |
| 10,650.0                                                                                                  | 10,607.9              | 10,601.8              | 10,563.9              | 24.0             | 22.5          | 71.91                 | -164.2                              | 115.0        | 127.6                  | 82.1                    | 45.53                     | 2.803             |                    |          |
| 10,675.0                                                                                                  | 10,624.5              | 10,623.4              | 10,578.8              | 24.2             | 22.6          | 72.50                 | -173.2                              | 127.8        | 132.2                  | 86.5                    | 45.71                     | 2.892             |                    |          |
| 10,700.0                                                                                                  | 10,640.1              | 10,645.0              | 10,592.9              | 24.3             | 22.7          | 73.09                 | -182.6                              | 141.1        | 137.0                  | 91.0                    | 45.92                     | 2.983             |                    |          |
| 10,725.0                                                                                                  | 10,654.7              | 10,666.5              | 10,606.3              | 24.4             | 22.8          | 73.67                 | -192.3                              | 154.9        | 141.9                  | 95.7                    | 46.15                     | 3.074             |                    |          |
| 10,750.0                                                                                                  | 10,668.2              | 10,688.0              | 10,618.9              | 24.6             | 22.9          | 74.25                 | -202.3                              | 169.2        | 147.0                  | 100.6                   | 46.43                     | 3.166             |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-----------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                     |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 2.0 usft |
| Reference                                                                                                 |                       | Offset                |                       | Semi Major Axis  |               | Distance              |                                     |              |                        |                         |                           |                   |                    | Warning  |
| Measured Depth (usft)                                                                                     | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 10,775.0                                                                                                  | 10,680.5              | 10,709.6              | 10,630.7              | 24.8             | 23.1          | 74.81                 | -212.7                              | 183.9        | 152.3                  | 105.6                   | 46.73                     | 3.259             |                    |          |
| 10,800.0                                                                                                  | 10,691.7              | 10,731.1              | 10,641.7              | 25.0             | 23.2          | 75.36                 | -223.3                              | 199.1        | 157.8                  | 110.7                   | 47.08                     | 3.351             |                    |          |
| 10,825.0                                                                                                  | 10,701.8              | 10,752.6              | 10,651.8              | 25.2             | 23.4          | 75.89                 | -234.2                              | 214.6        | 163.4                  | 115.9                   | 47.45                     | 3.443             |                    |          |
| 10,850.0                                                                                                  | 10,710.6              | 10,775.0              | 10,661.4              | 25.4             | 23.6          | 76.44                 | -245.9                              | 231.2        | 169.1                  | 121.3                   | 47.87                     | 3.533             |                    |          |
| 10,875.0                                                                                                  | 10,718.1              | 10,795.7              | 10,669.5              | 25.7             | 23.7          | 76.88                 | -256.8                              | 246.8        | 175.0                  | 126.7                   | 48.31                     | 3.623             |                    |          |
| 10,900.0                                                                                                  | 10,724.5              | 10,817.3              | 10,677.0              | 25.9             | 23.9          | 77.34                 | -268.5                              | 263.3        | 181.1                  | 132.3                   | 48.78                     | 3.711             |                    |          |
| 10,925.0                                                                                                  | 10,729.5              | 10,839.0              | 10,683.6              | 26.2             | 24.2          | 77.78                 | -280.3                              | 280.2        | 187.2                  | 137.9                   | 49.29                     | 3.798             |                    |          |
| 10,950.0                                                                                                  | 10,733.3              | 10,860.7              | 10,689.2              | 26.5             | 24.4          | 78.20                 | -292.4                              | 297.4        | 193.4                  | 143.6                   | 49.83                     | 3.882             |                    |          |
| 10,975.0                                                                                                  | 10,735.7              | 10,882.5              | 10,694.0              | 26.8             | 24.6          | 78.60                 | -304.6                              | 314.8        | 199.8                  | 149.4                   | 50.40                     | 3.964             |                    |          |
| 11,000.0                                                                                                  | 10,736.9              | 11,004.4              | 10,697.7              | 27.1             | 24.9          | 78.97                 | -317.0                              | 332.4        | 206.2                  | 155.2                   | 50.99                     | 4.043             |                    |          |
| 11,006.2                                                                                                  | 10,737.0              | 10,909.8              | 10,698.5              | 27.2             | 24.9          | 79.06                 | -320.1                              | 336.8        | 207.8                  | 156.6                   | 51.14                     | 4.063             |                    |          |
| 11,100.0                                                                                                  | 10,737.6              | 10,996.3              | 10,703.2              | 28.5             | 26.0          | 81.28                 | -369.7                              | 407.5        | 234.9                  | 181.3                   | 53.63                     | 4.380             |                    |          |
| 11,200.0                                                                                                  | 10,738.3              | 11,100.4              | 10,704.0              | 30.2             | 27.6          | 82.34                 | -427.2                              | 494.2        | 265.5                  | 208.9                   | 56.62                     | 4.689             |                    |          |
| 11,300.0                                                                                                  | 10,739.0              | 11,205.8              | 10,704.9              | 32.0             | 29.3          | 83.18                 | -482.3                              | 584.1        | 296.0                  | 236.1                   | 59.94                     | 4.939             |                    |          |
| 11,400.0                                                                                                  | 10,739.7              | 11,312.6              | 10,705.8              | 33.9             | 31.3          | 83.86                 | -534.6                              | 677.1        | 326.4                  | 262.8                   | 63.54                     | 5.137             |                    |          |
| 11,500.0                                                                                                  | 10,740.4              | 11,420.7              | 10,706.6              | 36.0             | 33.5          | 84.42                 | -584.0                              | 773.3        | 356.5                  | 289.1                   | 67.36                     | 5.292             |                    |          |
| 11,600.0                                                                                                  | 10,741.1              | 11,530.2              | 10,707.5              | 38.2             | 35.9          | 84.90                 | -630.4                              | 872.5        | 386.4                  | 315.0                   | 71.36                     | 5.414             |                    |          |
| 11,700.0                                                                                                  | 10,741.8              | 11,641.2              | 10,708.4              | 40.5             | 38.4          | 85.30                 | -673.4                              | 974.8        | 415.9                  | 340.4                   | 75.50                     | 5.509             |                    |          |
| 11,800.0                                                                                                  | 10,742.5              | 11,753.7              | 10,709.4              | 42.9             | 41.0          | 85.65                 | -713.0                              | 1,080.1      | 445.1                  | 365.4                   | 79.74                     | 5.582             |                    |          |
| 11,900.0                                                                                                  | 10,743.2              | 11,867.7              | 10,710.3              | 45.3             | 43.7          | 85.96                 | -748.8                              | 1,188.3      | 473.8                  | 389.8                   | 84.05                     | 5.637             |                    |          |
| 11,928.6                                                                                                  | 10,743.4              | 11,900.6              | 10,710.6              | 45.9             | 44.5          | 86.04                 | -758.4                              | 1,219.8      | 482.0                  | 396.7                   | 85.29                     | 5.651             |                    |          |
| 12,000.0                                                                                                  | 10,743.9              | 11,983.5              | 10,711.3              | 47.7             | 46.6          | 86.24                 | -780.8                              | 1,299.6      | 501.3                  | 412.2                   | 89.03                     | 5.630             |                    |          |
| 12,100.0                                                                                                  | 10,744.6              | 12,101.8              | 10,712.2              | 50.2             | 49.5          | 86.46                 | -808.8                              | 1,414.6      | 524.9                  | 430.4                   | 94.44                     | 5.558             |                    |          |
| 12,200.0                                                                                                  | 10,745.3              | 12,222.4              | 10,713.2              | 52.7             | 52.6          | 86.63                 | -832.4                              | 1,532.7      | 544.4                  | 444.4                   | 100.05                    | 5.441             |                    |          |
| 12,300.0                                                                                                  | 10,746.0              | 12,344.7              | 10,714.2              | 55.3             | 55.7          | 86.76                 | -851.3                              | 1,653.6      | 559.8                  | 454.0                   | 105.82                    | 5.290             |                    |          |
| 12,400.0                                                                                                  | 10,746.7              | 12,468.5              | 10,715.2              | 57.9             | 58.8          | 86.86                 | -865.1                              | 1,776.6      | 571.0                  | 459.2                   | 111.72                    | 5.111             |                    |          |
| 12,500.0                                                                                                  | 10,747.4              | 12,593.2              | 10,716.2              | 60.5             | 62.0          | 86.92                 | -873.6                              | 1,901.0      | 577.8                  | 460.1                   | 117.71                    | 4.908             |                    |          |
| 12,600.0                                                                                                  | 10,748.1              | 12,718.5              | 10,717.2              | 63.2             | 65.2          | 86.94                 | -876.7                              | 2,026.2      | 580.3                  | 456.5                   | 123.78                    | 4.688             |                    |          |
| 12,700.0                                                                                                  | 10,748.8              | 12,820.2              | 10,718.0              | 65.9             | 67.8          | 86.95                 | -876.7                              | 2,127.9      | 580.3                  | 451.0                   | 129.25                    | 4.489             |                    |          |
| 12,800.0                                                                                                  | 10,749.5              | 12,920.2              | 10,718.8              | 68.6             | 70.4          | 86.96                 | -876.7                              | 2,227.9      | 580.3                  | 445.5                   | 134.71                    | 4.307             |                    |          |
| 12,900.0                                                                                                  | 10,750.2              | 13,020.2              | 10,719.6              | 71.3             | 73.0          | 86.97                 | -876.7                              | 2,327.9      | 580.2                  | 440.0                   | 140.20                    | 4.139             |                    |          |
| 13,000.0                                                                                                  | 10,750.9              | 13,120.2              | 10,720.4              | 74.1             | 75.7          | 86.98                 | -876.7                              | 2,427.9      | 580.2                  | 434.5                   | 145.73                    | 3.982             |                    |          |
| 13,100.0                                                                                                  | 10,751.6              | 13,220.2              | 10,721.1              | 76.8             | 78.3          | 86.99                 | -876.7                              | 2,527.9      | 580.2                  | 429.0                   | 151.27                    | 3.836             |                    |          |
| 13,200.0                                                                                                  | 10,752.3              | 13,320.2              | 10,721.9              | 79.6             | 81.0          | 87.00                 | -876.7                              | 2,627.9      | 580.2                  | 423.4                   | 156.84                    | 3.700             |                    |          |
| 13,300.0                                                                                                  | 10,753.0              | 13,420.2              | 10,722.7              | 82.4             | 83.7          | 87.00                 | -876.7                              | 2,727.9      | 580.2                  | 417.8                   | 162.43                    | 3.572             |                    |          |
| 13,400.0                                                                                                  | 10,753.7              | 13,520.2              | 10,723.5              | 85.2             | 86.4          | 87.01                 | -876.7                              | 2,827.9      | 580.2                  | 412.2                   | 168.04                    | 3.453             |                    |          |
| 13,500.0                                                                                                  | 10,754.4              | 13,620.2              | 10,724.3              | 88.0             | 89.2          | 87.02                 | -876.7                              | 2,927.9      | 580.2                  | 406.6                   | 173.66                    | 3.341             |                    |          |
| 13,600.0                                                                                                  | 10,755.1              | 13,720.2              | 10,725.1              | 90.8             | 91.9          | 87.03                 | -876.7                              | 3,027.9      | 580.2                  | 400.9                   | 179.30                    | 3.236             |                    |          |
| 13,700.0                                                                                                  | 10,755.8              | 13,820.2              | 10,725.8              | 93.6             | 94.6          | 87.04                 | -876.7                              | 3,127.9      | 580.2                  | 395.3                   | 184.95                    | 3.137             |                    |          |
| 13,800.0                                                                                                  | 10,756.5              | 13,920.2              | 10,726.6              | 96.4             | 97.4          | 87.05                 | -876.7                              | 3,227.9      | 580.2                  | 389.6                   | 190.62                    | 3.044             |                    |          |
| 13,900.0                                                                                                  | 10,757.2              | 14,020.2              | 10,727.4              | 99.2             | 100.2         | 87.06                 | -876.7                              | 3,327.9      | 580.2                  | 383.9                   | 196.29                    | 2.956             |                    |          |
| 14,000.0                                                                                                  | 10,757.9              | 14,120.2              | 10,728.2              | 102.1            | 103.0         | 87.07                 | -876.7                              | 3,427.9      | 580.2                  | 378.2                   | 201.98                    | 2.873             |                    |          |
| 14,100.0                                                                                                  | 10,758.6              | 14,220.2              | 10,729.0              | 104.9            | 105.7         | 87.07                 | -876.7                              | 3,527.9      | 580.2                  | 372.5                   | 207.67                    | 2.794             |                    |          |
| 14,200.0                                                                                                  | 10,759.3              | 14,320.2              | 10,729.8              | 107.7            | 108.5         | 87.08                 | -876.7                              | 3,627.9      | 580.2                  | 366.8                   | 213.38                    | 2.719             |                    |          |
| 14,300.0                                                                                                  | 10,760.0              | 14,420.2              | 10,730.6              | 110.6            | 111.3         | 87.09                 | -876.7                              | 3,727.9      | 580.2                  | 361.1                   | 219.09                    | 2.648             |                    |          |
| 14,400.0                                                                                                  | 10,760.7              | 14,520.2              | 10,731.3              | 113.4            | 114.1         | 87.10                 | -876.7                              | 3,827.9      | 580.2                  | 355.4                   | 224.81                    | 2.581             |                    |          |
| 14,500.0                                                                                                  | 10,761.4              | 14,620.2              | 10,732.1              | 116.3            | 117.0         | 87.11                 | -876.7                              | 3,927.9      | 580.2                  | 349.6                   | 230.54                    | 2.517             |                    |          |
| 14,600.0                                                                                                  | 10,762.1              | 14,720.2              | 10,732.9              | 119.2            | 119.8         | 87.12                 | -876.7                              | 4,027.9      | 580.2                  | 343.9                   | 236.27                    | 2.456             |                    |          |
| 14,700.0                                                                                                  | 10,762.8              | 14,820.2              | 10,733.7              | 122.0            | 122.6         | 87.13                 | -876.7                              | 4,127.9      | 580.2                  | 338.2                   | 242.01                    | 2.397             |                    |          |
| 14,800.0                                                                                                  | 10,763.5              | 14,920.2              | 10,734.5              | 124.9            | 125.4         | 87.13                 | -876.7                              | 4,227.9      | 580.2                  | 332.4                   | 247.76                    | 2.342             |                    |          |
| 14,900.0                                                                                                  | 10,764.2              | 15,020.2              | 10,735.3              | 127.8            | 128.3         | 87.14                 | -876.7                              | 4,327.9      | 580.2                  | 326.7                   | 253.50                    | 2.289             |                    |          |
| 15,000.0                                                                                                  | 10,764.9              | 15,120.2              | 10,736.1              | 130.6            | 131.1         | 87.15                 | -876.7                              | 4,427.9      | 580.2                  | 320.9                   | 259.26                    | 2.238             |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-----------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                     |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 2.0 usft |
| Reference                                                                                                 |                       | Offset                |                       | Semi Major Axis  |               | Distance              |                                     |              |                        |                         |                           |                   |                    | Warning  |
| Measured Depth (usft)                                                                                     | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 15,100.0                                                                                                  | 10,765.6              | 15,220.2              | 10,736.8              | 133.5            | 134.0         | 87.16                 | -876.7                              | 4,527.9      | 580.2                  | 315.1                   | 265.02                    | 2.189             |                    |          |
| 15,200.0                                                                                                  | 10,766.3              | 15,320.2              | 10,737.6              | 136.4            | 136.8         | 87.17                 | -876.7                              | 4,627.9      | 580.1                  | 309.4                   | 270.78                    | 2.142             |                    |          |
| 15,300.0                                                                                                  | 10,767.0              | 15,420.2              | 10,738.4              | 139.3            | 139.7         | 87.18                 | -876.7                              | 4,727.9      | 580.1                  | 303.6                   | 276.55                    | 2.098             |                    |          |
| 15,400.0                                                                                                  | 10,767.7              | 15,520.2              | 10,739.2              | 142.1            | 142.5         | 87.19                 | -876.7                              | 4,827.9      | 580.1                  | 297.8                   | 282.32                    | 2.055             |                    |          |
| 15,500.0                                                                                                  | 10,768.4              | 15,620.2              | 10,740.0              | 145.0            | 145.4         | 87.19                 | -876.7                              | 4,927.8      | 580.1                  | 292.0                   | 288.10                    | 2.014             |                    |          |
| 15,600.0                                                                                                  | 10,769.1              | 15,720.2              | 10,740.8              | 147.9            | 148.2         | 87.20                 | -876.7                              | 5,027.8      | 580.1                  | 286.3                   | 293.87                    | 1.974             |                    |          |
| 15,700.0                                                                                                  | 10,769.8              | 15,820.2              | 10,741.6              | 150.8            | 151.1         | 87.21                 | -876.7                              | 5,127.8      | 580.1                  | 280.5                   | 299.65                    | 1.936             |                    |          |
| 15,800.0                                                                                                  | 10,770.5              | 15,920.2              | 10,742.3              | 153.7            | 153.9         | 87.22                 | -876.7                              | 5,227.8      | 580.1                  | 274.7                   | 305.44                    | 1.899             |                    |          |
| 15,900.0                                                                                                  | 10,771.2              | 16,020.2              | 10,743.1              | 156.6            | 156.8         | 87.23                 | -876.7                              | 5,327.8      | 580.1                  | 268.9                   | 311.23                    | 1.864             |                    |          |
| 16,000.0                                                                                                  | 10,771.9              | 16,120.2              | 10,743.9              | 159.5            | 159.7         | 87.24                 | -876.7                              | 5,427.8      | 580.1                  | 263.1                   | 317.01                    | 1.830             |                    |          |
| 16,100.0                                                                                                  | 10,772.6              | 16,220.2              | 10,744.7              | 162.4            | 162.6         | 87.25                 | -876.7                              | 5,527.8      | 580.1                  | 257.3                   | 322.81                    | 1.797             |                    |          |
| 16,200.0                                                                                                  | 10,773.3              | 16,320.2              | 10,745.5              | 165.2            | 165.4         | 87.25                 | -876.7                              | 5,627.8      | 580.1                  | 251.5                   | 328.60                    | 1.765             |                    |          |
| 16,300.0                                                                                                  | 10,774.0              | 16,420.2              | 10,746.3              | 168.1            | 168.3         | 87.26                 | -876.7                              | 5,727.8      | 580.1                  | 245.7                   | 334.40                    | 1.735             |                    |          |
| 16,400.0                                                                                                  | 10,774.7              | 16,520.2              | 10,747.1              | 171.0            | 171.2         | 87.27                 | -876.7                              | 5,827.8      | 580.1                  | 239.9                   | 340.20                    | 1.705             |                    |          |
| 16,500.0                                                                                                  | 10,775.4              | 16,620.2              | 10,747.8              | 173.9            | 174.1         | 87.28                 | -876.7                              | 5,927.8      | 580.1                  | 234.1                   | 346.00                    | 1.677             |                    |          |
| 16,600.0                                                                                                  | 10,776.1              | 16,720.2              | 10,748.6              | 176.8            | 176.9         | 87.29                 | -876.7                              | 6,027.8      | 580.1                  | 228.3                   | 351.80                    | 1.649             |                    |          |
| 16,700.0                                                                                                  | 10,776.8              | 16,820.2              | 10,749.4              | 179.7            | 179.8         | 87.30                 | -876.7                              | 6,127.8      | 580.1                  | 222.5                   | 357.60                    | 1.622             |                    |          |
| 16,800.0                                                                                                  | 10,777.5              | 16,920.2              | 10,750.2              | 182.6            | 182.7         | 87.31                 | -876.7                              | 6,227.8      | 580.1                  | 216.7                   | 363.41                    | 1.596             |                    |          |
| 16,900.0                                                                                                  | 10,778.2              | 17,020.2              | 10,751.0              | 185.5            | 185.6         | 87.31                 | -876.7                              | 6,327.8      | 580.1                  | 210.9                   | 369.22                    | 1.571             |                    |          |
| 17,000.0                                                                                                  | 10,778.9              | 17,120.2              | 10,751.8              | 188.4            | 188.5         | 87.32                 | -876.7                              | 6,427.8      | 580.1                  | 205.0                   | 375.03                    | 1.547             |                    |          |
| 17,100.0                                                                                                  | 10,779.6              | 17,220.2              | 10,752.6              | 191.3            | 191.4         | 87.33                 | -876.7                              | 6,527.8      | 580.1                  | 199.2                   | 380.84                    | 1.523             |                    |          |
| 17,200.0                                                                                                  | 10,780.3              | 17,320.2              | 10,753.3              | 194.3            | 194.3         | 87.34                 | -876.7                              | 6,627.8      | 580.1                  | 193.4                   | 386.65                    | 1.500             |                    |          |
| 17,300.0                                                                                                  | 10,780.9              | 17,420.2              | 10,754.1              | 197.2            | 197.2         | 87.35                 | -876.7                              | 6,727.8      | 580.1                  | 187.6                   | 392.47                    | 1.478 Level 3     | Level 3            |          |
| 17,400.0                                                                                                  | 10,781.6              | 17,520.2              | 10,754.9              | 200.1            | 200.0         | 87.36                 | -876.7                              | 6,827.8      | 580.1                  | 181.8                   | 398.28                    | 1.456 Level 3     | Level 3            |          |
| 17,500.0                                                                                                  | 10,782.3              | 17,620.2              | 10,755.7              | 203.0            | 202.9         | 87.37                 | -876.7                              | 6,927.8      | 580.1                  | 176.0                   | 404.10                    | 1.435 Level 3     | Level 3            |          |
| 17,600.0                                                                                                  | 10,783.0              | 17,720.2              | 10,756.5              | 205.9            | 205.8         | 87.38                 | -876.7                              | 7,027.8      | 580.0                  | 170.1                   | 409.92                    | 1.415 Level 3     | Level 3            |          |
| 17,700.0                                                                                                  | 10,783.7              | 17,820.2              | 10,757.3              | 208.8            | 208.7         | 87.38                 | -876.7                              | 7,127.8      | 580.0                  | 164.3                   | 415.74                    | 1.395 Level 3     | Level 3            |          |
| 17,800.0                                                                                                  | 10,784.4              | 17,920.2              | 10,758.1              | 211.7            | 211.6         | 87.39                 | -876.7                              | 7,227.8      | 580.0                  | 158.5                   | 421.56                    | 1.376 Level 3     | Level 3            |          |
| 17,900.0                                                                                                  | 10,785.1              | 18,020.2              | 10,758.8              | 214.6            | 214.5         | 87.40                 | -876.7                              | 7,327.8      | 580.0                  | 152.7                   | 427.38                    | 1.357 Level 3     | Level 3            |          |
| 18,000.0                                                                                                  | 10,785.8              | 18,120.2              | 10,759.6              | 217.5            | 217.4         | 87.41                 | -876.7                              | 7,427.8      | 580.0                  | 146.8                   | 433.20                    | 1.339 Level 3     | Level 3            |          |
| 18,100.0                                                                                                  | 10,786.5              | 18,220.2              | 10,760.4              | 220.4            | 220.3         | 87.42                 | -876.7                              | 7,527.8      | 580.0                  | 141.0                   | 439.02                    | 1.321 Level 3     | Level 3            |          |
| 18,200.0                                                                                                  | 10,787.2              | 18,320.2              | 10,761.2              | 223.3            | 223.2         | 87.43                 | -876.7                              | 7,627.8      | 580.0                  | 135.2                   | 444.85                    | 1.304 Level 3     | Level 3            |          |
| 18,300.0                                                                                                  | 10,787.9              | 18,420.2              | 10,762.0              | 226.2            | 226.1         | 87.44                 | -876.7                              | 7,727.8      | 580.0                  | 129.3                   | 450.67                    | 1.287 Level 3     | Level 3            |          |
| 18,400.0                                                                                                  | 10,788.6              | 18,520.2              | 10,762.8              | 229.2            | 229.0         | 87.44                 | -876.7                              | 7,827.8      | 580.0                  | 123.5                   | 456.50                    | 1.271 Level 3     | Level 3            |          |
| 18,500.0                                                                                                  | 10,789.3              | 18,620.2              | 10,763.5              | 232.1            | 231.9         | 87.45                 | -876.7                              | 7,927.8      | 580.0                  | 117.7                   | 462.33                    | 1.255 Level 3     | Level 3            |          |
| 18,600.0                                                                                                  | 10,790.0              | 18,720.2              | 10,764.3              | 235.0            | 234.8         | 87.46                 | -876.7                              | 8,027.8      | 580.0                  | 111.9                   | 468.16                    | 1.239 Level 2     |                    |          |
| 18,700.0                                                                                                  | 10,790.7              | 18,820.2              | 10,765.1              | 237.9            | 237.7         | 87.47                 | -876.7                              | 8,127.7      | 580.0                  | 106.0                   | 473.98                    | 1.224 Level 2     |                    |          |
| 18,800.0                                                                                                  | 10,791.4              | 18,920.2              | 10,765.9              | 240.8            | 240.6         | 87.48                 | -876.7                              | 8,227.7      | 580.0                  | 100.2                   | 479.81                    | 1.209 Level 2     |                    |          |
| 18,900.0                                                                                                  | 10,792.1              | 19,020.2              | 10,766.7              | 243.7            | 243.5         | 87.49                 | -876.7                              | 8,327.7      | 580.0                  | 94.4                    | 485.64                    | 1.194 Level 2     |                    |          |
| 19,000.0                                                                                                  | 10,792.8              | 19,120.2              | 10,767.5              | 246.6            | 246.4         | 87.50                 | -876.7                              | 8,427.7      | 580.0                  | 88.5                    | 491.48                    | 1.180 Level 2     |                    |          |
| 19,100.0                                                                                                  | 10,793.5              | 19,220.2              | 10,768.3              | 249.6            | 249.4         | 87.50                 | -876.7                              | 8,527.7      | 580.0                  | 82.7                    | 497.31                    | 1.166 Level 2     |                    |          |
| 19,200.0                                                                                                  | 10,794.2              | 19,320.2              | 10,769.0              | 252.5            | 252.3         | 87.51                 | -876.7                              | 8,627.7      | 580.0                  | 76.8                    | 503.14                    | 1.153 Level 2     |                    |          |
| 19,300.0                                                                                                  | 10,794.9              | 19,420.2              | 10,769.8              | 255.4            | 255.2         | 87.52                 | -876.7                              | 8,727.7      | 580.0                  | 71.0                    | 508.97                    | 1.140 Level 2     |                    |          |
| 19,400.0                                                                                                  | 10,795.6              | 19,520.2              | 10,770.6              | 258.3            | 258.1         | 87.53                 | -876.7                              | 8,827.7      | 580.0                  | 65.2                    | 514.81                    | 1.127 Level 2     |                    |          |
| 19,500.0                                                                                                  | 10,796.3              | 19,620.2              | 10,771.4              | 261.2            | 261.0         | 87.54                 | -876.7                              | 8,927.7      | 580.0                  | 59.3                    | 520.64                    | 1.114 Level 2     |                    |          |
| 19,600.0                                                                                                  | 10,797.0              | 19,720.2              | 10,772.2              | 264.1            | 263.9         | 87.55                 | -876.7                              | 9,027.7      | 580.0                  | 53.5                    | 526.48                    | 1.102 Level 2     |                    |          |
| 19,700.0                                                                                                  | 10,797.7              | 19,820.2              | 10,773.0              | 267.1            | 266.8         | 87.56                 | -876.7                              | 9,127.7      | 580.0                  | 47.7                    | 532.31                    | 1.090 Level 2     |                    |          |
| 19,800.0                                                                                                  | 10,798.4              | 19,920.2              | 10,773.8              | 270.0            | 269.7         | 87.56                 | -876.7                              | 9,227.7      | 580.0                  | 41.8                    | 538.15                    | 1.078 Level 2     |                    |          |
| 19,900.0                                                                                                  | 10,799.1              | 20,020.2              | 10,774.5              | 272.9            | 272.6         | 87.57                 | -876.7                              | 9,327.7      | 580.0                  | 36.0                    | 543.99                    | 1.066 Level 2     |                    |          |
| 20,000.0                                                                                                  | 10,799.8              | 20,120.2              | 10,775.3              | 275.8            | 275.5         | 87.58                 | -876.7                              | 9,427.7      | 580.0                  | 30.1                    | 549.82                    | 1.055 Level 2     |                    |          |
| 20,100.0                                                                                                  | 10,800.5              | 20,220.2              | 10,776.1              | 278.7            | 278.4         | 87.59                 | -876.7                              | 9,527.7      | 580.0                  | 24.3                    | 555.66                    | 1.044 Level 2     |                    |          |
| 20,200.0                                                                                                  | 10,801.2              | 20,320.2              | 10,776.9              | 281.6            | 281.4         | 87.60                 | -876.7                              | 9,627.7      | 579.9                  | 18.4                    | 561.50                    | 1.033 Level 2     |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| <b>Offset Design</b> 153N-100W-19/20_Altered - Chalmers 5301 44-24 12TXR - Chalmers 5301 44-24 12TXR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           | <b>Offset Site Error:</b> | 0.0 usft        |
|------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|---------------------------|-----------------|
| Survey Program: 0-MWD                                                                                            |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           | <b>Offset Well Error:</b> | 2.0 usft        |
| Reference                                                                                                        |                       | Offset                |                       | Semi Major Axis  |               |                       | Distance                            |              |                        |                         |                           |                           | Warning         |
| Measured Depth (usft)                                                                                            | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor         |                 |
| 20,300.0                                                                                                         | 10,801.9              | 20,420.2              | 10,777.7              | 284.6            | 284.3         | 87.61                 | -876.7                              | 9,727.7      | 579.9                  | 12.6                    | 567.34                    | 1.022                     | Level 2         |
| 20,400.0                                                                                                         | 10,802.6              | 20,520.2              | 10,778.5              | 287.5            | 287.2         | 87.62                 | -876.7                              | 9,827.7      | 579.9                  | 6.8                     | 573.18                    | 1.012                     | Level 2         |
| 20,500.0                                                                                                         | 10,803.3              | 20,620.2              | 10,779.3              | 290.4            | 290.1         | 87.62                 | -876.7                              | 9,927.7      | 579.9                  | 0.9                     | 579.02                    | 1.002                     | Level 2         |
| 20,600.0                                                                                                         | 10,804.0              | 20,720.2              | 10,780.0              | 293.3            | 293.0         | 87.63                 | -876.7                              | 10,027.7     | 579.9                  | -4.9                    | 584.86                    | 0.992                     | Level 1         |
| 20,700.0                                                                                                         | 10,804.7              | 20,820.2              | 10,780.8              | 296.2            | 295.9         | 87.64                 | -876.7                              | 10,127.7     | 579.9                  | -10.8                   | 590.70                    | 0.982                     | Level 1         |
| 20,800.0                                                                                                         | 10,805.4              | 20,920.2              | 10,781.6              | 299.2            | 298.8         | 87.65                 | -876.7                              | 10,227.7     | 579.9                  | -16.6                   | 596.54                    | 0.972                     | Level 1         |
| 20,900.0                                                                                                         | 10,806.1              | 21,020.2              | 10,782.4              | 302.1            | 301.8         | 87.66                 | -876.7                              | 10,327.7     | 579.9                  | -22.5                   | 602.38                    | 0.963                     | Level 1         |
| 21,000.0                                                                                                         | 10,806.8              | 21,120.2              | 10,783.2              | 305.0            | 304.7         | 87.67                 | -876.7                              | 10,427.7     | 579.9                  | -28.3                   | 608.23                    | 0.953                     | Level 1         |
| 21,076.5                                                                                                         | 10,807.3              | 21,196.7              | 10,783.8              | 307.2            | 306.9         | 87.67                 | -876.7                              | 10,504.1     | 579.9                  | -32.8                   | 612.69                    | 0.947                     | Level 1         |
| 21,106.5                                                                                                         | 10,807.5              | 21,226.5              | 10,784.0              | 308.1            | 307.5         | 87.68                 | -876.7                              | 10,534.0     | 579.9                  | -34.3                   | 614.18                    | 0.944                     | Level 1, ES, SF |

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 2TX - Chalmers #2 T - Plan #1 |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|---------------------|------------------|-----------------------------|-------------------------------------------|-----------------|------------------------------|-------------------------------|---------------------------------|----------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                     |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Well Error: | 0.0 usft |
| Reference                                                                                 |                             | Offset                      |                             | Semi Major Axis     |                  | Highside<br>Toolface<br>(°) | Distance                                  |                 | Between<br>Centres<br>(usft) | Between<br>Ellipses<br>(usft) | Minimum<br>Separation<br>(usft) | Separation<br>Factor | Warning            |          |
| Measured<br>Depth<br>(usft)                                                               | Vertical<br>Depth<br>(usft) | Measured<br>Depth<br>(usft) | Vertical<br>Depth<br>(usft) | Reference<br>(usft) | Offset<br>(usft) |                             | Offset Wellbore Centre<br>+N/-S<br>(usft) | +E/-W<br>(usft) |                              |                               |                                 |                      |                    |          |
| 0.0                                                                                       | 0.0                         | 0.0                         | 0.0                         | 2.0                 | 0.0              | 0.00                        | 82.2                                      | 0.0             | 82.2                         |                               |                                 |                      |                    |          |
| 100.0                                                                                     | 100.0                       | 100.0                       | 100.0                       | 2.0                 | 0.1              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 80.1                          | 2.09                            | 39.411               |                    |          |
| 200.0                                                                                     | 200.0                       | 200.0                       | 200.0                       | 2.0                 | 0.3              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 79.9                          | 2.33                            | 35.243               |                    |          |
| 300.0                                                                                     | 300.0                       | 300.0                       | 300.0                       | 2.1                 | 0.5              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 79.6                          | 2.60                            | 31.574               |                    |          |
| 400.0                                                                                     | 400.0                       | 400.0                       | 400.0                       | 2.1                 | 0.8              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 79.3                          | 2.90                            | 28.373               |                    |          |
| 500.0                                                                                     | 500.0                       | 500.0                       | 500.0                       | 2.2                 | 1.0              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 79.0                          | 3.21                            | 25.596               |                    |          |
| 600.0                                                                                     | 600.0                       | 600.0                       | 600.0                       | 2.3                 | 1.2              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 78.7                          | 3.54                            | 23.194               |                    |          |
| 700.0                                                                                     | 700.0                       | 700.0                       | 700.0                       | 2.5                 | 1.4              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 78.3                          | 3.89                            | 21.117               |                    |          |
| 800.0                                                                                     | 800.0                       | 800.0                       | 800.0                       | 2.6                 | 1.7              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 78.0                          | 4.26                            | 19.320               |                    |          |
| 900.0                                                                                     | 900.0                       | 900.0                       | 900.0                       | 2.7                 | 1.9              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 77.6                          | 4.63                            | 17.761               |                    |          |
| 1,000.0                                                                                   | 1,000.0                     | 1,000.0                     | 1,000.0                     | 2.9                 | 2.1              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 77.2                          | 5.01                            | 16.402               |                    |          |
| 1,100.0                                                                                   | 1,100.0                     | 1,100.0                     | 1,100.0                     | 3.1                 | 2.3              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 76.8                          | 5.40                            | 15.213               |                    |          |
| 1,200.0                                                                                   | 1,200.0                     | 1,200.0                     | 1,200.0                     | 3.2                 | 2.6              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 76.4                          | 5.80                            | 14.168               |                    |          |
| 1,300.0                                                                                   | 1,300.0                     | 1,300.0                     | 1,300.0                     | 3.4                 | 2.8              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 76.0                          | 6.21                            | 13.245               |                    |          |
| 1,400.0                                                                                   | 1,400.0                     | 1,400.0                     | 1,400.0                     | 3.6                 | 3.0              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 75.6                          | 6.62                            | 12.425               |                    |          |
| 1,500.0                                                                                   | 1,500.0                     | 1,500.0                     | 1,500.0                     | 3.8                 | 3.2              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 75.2                          | 7.03                            | 11.693               |                    |          |
| 1,600.0                                                                                   | 1,600.0                     | 1,600.0                     | 1,600.0                     | 4.0                 | 3.5              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 74.8                          | 7.45                            | 11.038               |                    |          |
| 1,700.0                                                                                   | 1,700.0                     | 1,700.0                     | 1,700.0                     | 4.2                 | 3.7              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 74.3                          | 7.87                            | 10.447               |                    |          |
| 1,800.0                                                                                   | 1,800.0                     | 1,800.0                     | 1,800.0                     | 4.4                 | 3.9              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 73.9                          | 8.29                            | 9.914                |                    |          |
| 1,900.0                                                                                   | 1,900.0                     | 1,900.0                     | 1,900.0                     | 4.6                 | 4.1              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 73.5                          | 8.72                            | 9.429                |                    |          |
| 2,000.0                                                                                   | 2,000.0                     | 2,000.0                     | 2,000.0                     | 4.8                 | 4.4              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 73.1                          | 9.15                            | 8.988                |                    |          |
| 2,100.0                                                                                   | 2,100.0                     | 2,100.0                     | 2,100.0                     | 5.0                 | 4.6              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 72.6                          | 9.58                            | 8.585                |                    |          |
| 2,166.7                                                                                   | 2,166.7                     | 2,166.7                     | 2,166.7                     | 5.1                 | 4.7              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 72.3                          | 9.86                            | 8.334 CC             |                    |          |
| 2,200.0                                                                                   | 2,200.0                     | 2,200.0                     | 2,200.0                     | 5.2                 | 4.8              | 0.00                        | 82.2                                      | 0.0             | 82.2                         | 72.2                          | 10.01                           | 8.215                |                    |          |
| 2,300.0                                                                                   | 2,300.0                     | 2,299.3                     | 2,299.3                     | 5.4                 | 5.0              | 0.00                        | 83.0                                      | 0.0             | 83.0                         | 72.6                          | 10.44                           | 7.955                |                    |          |
| 2,400.0                                                                                   | 2,400.0                     | 2,399.3                     | 2,399.3                     | 5.6                 | 5.3              | 0.00                        | 83.9                                      | 0.0             | 83.9                         | 73.0                          | 10.87                           | 7.718                |                    |          |
| 2,500.0                                                                                   | 2,500.0                     | 2,499.3                     | 2,499.3                     | 5.8                 | 5.5              | 0.00                        | 84.8                                      | 0.0             | 84.8                         | 73.5                          | 11.31                           | 7.498                |                    |          |
| 2,600.0                                                                                   | 2,600.0                     | 2,599.3                     | 2,599.2                     | 6.0                 | 5.7              | 0.00                        | 85.7                                      | 0.0             | 85.7                         | 73.9                          | 11.74                           | 7.294                |                    |          |
| 2,700.0                                                                                   | 2,700.0                     | 2,699.3                     | 2,699.2                     | 6.3                 | 5.9              | 0.00                        | 86.5                                      | 0.0             | 86.5                         | 74.4                          | 12.18                           | 7.104                |                    |          |
| 2,800.0                                                                                   | 2,800.0                     | 2,799.3                     | 2,799.2                     | 6.5                 | 6.2              | 0.00                        | 87.4                                      | 0.0             | 87.4                         | 74.8                          | 12.62                           | 6.927                |                    |          |
| 2,900.0                                                                                   | 2,900.0                     | 2,899.3                     | 2,899.2                     | 6.7                 | 6.4              | 0.00                        | 88.3                                      | 0.0             | 88.3                         | 75.2                          | 13.06                           | 6.761                |                    |          |
| 3,000.0                                                                                   | 3,000.0                     | 2,999.2                     | 2,999.2                     | 6.9                 | 6.6              | 0.00                        | 89.1                                      | 0.0             | 89.1                         | 75.7                          | 13.50                           | 6.606                |                    |          |
| 3,100.0                                                                                   | 3,100.0                     | 3,099.2                     | 3,099.2                     | 7.1                 | 6.8              | 0.00                        | 90.0                                      | 0.0             | 90.0                         | 76.1                          | 13.94                           | 6.460                |                    |          |
| 3,200.0                                                                                   | 3,200.0                     | 3,199.2                     | 3,199.2                     | 7.3                 | 7.1              | 0.00                        | 90.9                                      | 0.0             | 90.9                         | 76.5                          | 14.38                           | 6.323                |                    |          |
| 3,300.0                                                                                   | 3,300.0                     | 3,299.2                     | 3,299.2                     | 7.5                 | 7.3              | 0.00                        | 91.8                                      | 0.0             | 91.8                         | 77.0                          | 14.82                           | 6.194                |                    |          |
| 3,400.0                                                                                   | 3,400.0                     | 3,399.2                     | 3,399.2                     | 7.8                 | 7.5              | 0.00                        | 92.6                                      | 0.0             | 92.6                         | 77.4                          | 15.26                           | 6.072                |                    |          |
| 3,500.0                                                                                   | 3,500.0                     | 3,499.2                     | 3,499.2                     | 8.0                 | 7.7              | 0.00                        | 93.5                                      | 0.0             | 93.5                         | 77.8                          | 15.70                           | 5.956                |                    |          |
| 3,600.0                                                                                   | 3,600.0                     | 3,599.2                     | 3,599.2                     | 8.2                 | 7.9              | 0.00                        | 94.4                                      | 0.0             | 94.4                         | 78.2                          | 16.14                           | 5.847                |                    |          |
| 3,700.0                                                                                   | 3,700.0                     | 3,699.2                     | 3,699.2                     | 8.4                 | 8.2              | 0.00                        | 95.3                                      | 0.0             | 95.3                         | 78.7                          | 16.58                           | 5.744                |                    |          |
| 3,800.0                                                                                   | 3,800.0                     | 3,799.2                     | 3,799.2                     | 8.6                 | 8.4              | 0.00                        | 96.1                                      | 0.0             | 96.1                         | 79.1                          | 17.03                           | 5.646                |                    |          |
| 3,900.0                                                                                   | 3,900.0                     | 3,899.2                     | 3,899.1                     | 8.9                 | 8.6              | 0.00                        | 97.0                                      | 0.0             | 97.0                         | 79.5                          | 17.47                           | 5.552                |                    |          |
| 4,000.0                                                                                   | 4,000.0                     | 3,999.2                     | 3,999.1                     | 9.1                 | 8.8              | 0.00                        | 97.9                                      | 0.0             | 97.9                         | 80.0                          | 17.91                           | 5.464                |                    |          |
| 4,100.0                                                                                   | 4,100.0                     | 4,099.2                     | 4,099.1                     | 9.3                 | 9.1              | 0.00                        | 98.7                                      | 0.0             | 98.7                         | 80.4                          | 18.36                           | 5.379                |                    |          |
| 4,200.0                                                                                   | 4,200.0                     | 4,199.2                     | 4,199.1                     | 9.5                 | 9.3              | 0.00                        | 99.6                                      | 0.0             | 99.6                         | 80.8                          | 18.80                           | 5.298                |                    |          |
| 4,300.0                                                                                   | 4,300.0                     | 4,299.2                     | 4,299.1                     | 9.7                 | 9.5              | 0.00                        | 100.5                                     | 0.0             | 100.5                        | 81.2                          | 19.25                           | 5.221                |                    |          |
| 4,400.0                                                                                   | 4,400.0                     | 4,399.2                     | 4,399.1                     | 10.0                | 9.7              | 0.00                        | 101.4                                     | 0.0             | 101.4                        | 81.7                          | 19.69                           | 5.148                |                    |          |
| 4,500.0                                                                                   | 4,500.0                     | 4,499.2                     | 4,499.1                     | 10.2                | 10.0             | 0.00                        | 102.2                                     | 0.0             | 102.2                        | 82.1                          | 20.14                           | 5.077                |                    |          |
| 4,600.0                                                                                   | 4,600.0                     | 4,599.2                     | 4,599.1                     | 10.4                | 10.2             | 0.00                        | 103.1                                     | 0.0             | 103.1                        | 82.5                          | 20.58                           | 5.010                |                    |          |
| 4,700.0                                                                                   | 4,700.0                     | 4,699.2                     | 4,699.1                     | 10.6                | 10.4             | 0.00                        | 104.0                                     | 0.0             | 104.0                        | 83.0                          | 21.03                           | 4.945                |                    |          |
| 4,800.0                                                                                   | 4,800.0                     | 4,799.2                     | 4,799.1                     | 10.8                | 10.6             | 0.00                        | 104.9                                     | 0.0             | 104.9                        | 83.4                          | 21.47                           | 4.883                |                    |          |
| 4,900.0                                                                                   | 4,900.0                     | 4,899.2                     | 4,899.1                     | 11.1                | 10.9             | 0.00                        | 105.7                                     | 0.0             | 105.7                        | 83.8                          | 21.92                           | 4.824                |                    |          |
| 5,000.0                                                                                   | 5,000.0                     | 4,999.2                     | 4,999.1                     | 11.3                | 11.1             | 0.00                        | 106.6                                     | 0.0             | 106.6                        | 84.2                          | 22.36                           | 4.767                |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 2TX - Chalmers #2 T - Plan #1 |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|---------------------|------------------|-----------------------------|-------------------------------------------|-----------------|------------------------------|-------------------------------|---------------------------------|----------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                     |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Well Error: | 0.0 usft |
| Reference                                                                                 |                             | Offset                      |                             | Semi Major Axis     |                  | Highside<br>Toolface<br>(°) | Distance                                  |                 | Between<br>Centres<br>(usft) | Between<br>Ellipses<br>(usft) | Minimum<br>Separation<br>(usft) | Separation<br>Factor | Warning            |          |
| Measured<br>Depth<br>(usft)                                                               | Vertical<br>Depth<br>(usft) | Measured<br>Depth<br>(usft) | Vertical<br>Depth<br>(usft) | Reference<br>(usft) | Offset<br>(usft) |                             | Offset Wellbore Centre<br>+N/-S<br>(usft) | +E/-W<br>(usft) |                              |                               |                                 |                      |                    |          |
| 5,100.0                                                                                   | 5,100.0                     | 5,099.2                     | 5,099.1                     | 11.5                | 11.3             | 0.00                        | 107.5                                     | 0.0             | 107.5                        | 84.7                          | 22.81                           | 4.712                |                    |          |
| 5,200.0                                                                                   | 5,200.0                     | 5,199.2                     | 5,199.1                     | 11.7                | 11.5             | 0.00                        | 108.3                                     | 0.0             | 108.3                        | 85.1                          | 23.25                           | 4.659                |                    |          |
| 5,300.0                                                                                   | 5,300.0                     | 5,299.2                     | 5,299.0                     | 11.9                | 11.8             | 0.00                        | 109.2                                     | 0.0             | 109.2                        | 85.5                          | 23.70                           | 4.608                |                    |          |
| 5,400.0                                                                                   | 5,400.0                     | 5,399.2                     | 5,399.0                     | 12.2                | 12.0             | 0.00                        | 110.1                                     | 0.0             | 110.1                        | 85.9                          | 24.15                           | 4.559                |                    |          |
| 5,500.0                                                                                   | 5,500.0                     | 5,499.2                     | 5,499.0                     | 12.4                | 12.2             | 0.00                        | 111.0                                     | 0.0             | 111.0                        | 86.4                          | 24.59                           | 4.512                |                    |          |
| 5,600.0                                                                                   | 5,600.0                     | 5,599.1                     | 5,599.0                     | 12.6                | 12.4             | 0.00                        | 111.8                                     | 0.0             | 111.8                        | 86.8                          | 25.04                           | 4.466                |                    |          |
| 5,700.0                                                                                   | 5,700.0                     | 5,699.1                     | 5,699.0                     | 12.8                | 12.7             | 0.00                        | 112.7                                     | 0.0             | 112.7                        | 87.2                          | 25.49                           | 4.422                |                    |          |
| 5,800.0                                                                                   | 5,800.0                     | 5,799.1                     | 5,799.0                     | 13.1                | 12.9             | 0.00                        | 113.6                                     | 0.0             | 113.6                        | 87.6                          | 25.93                           | 4.380                |                    |          |
| 5,900.0                                                                                   | 5,900.0                     | 5,899.1                     | 5,899.0                     | 13.3                | 13.1             | 0.00                        | 114.5                                     | 0.0             | 114.5                        | 88.1                          | 26.38                           | 4.339                |                    |          |
| 6,000.0                                                                                   | 6,000.0                     | 5,999.1                     | 5,999.0                     | 13.5                | 13.3             | 0.00                        | 115.3                                     | 0.0             | 115.3                        | 88.5                          | 26.83                           | 4.299                |                    |          |
| 6,100.0                                                                                   | 6,100.0                     | 6,099.1                     | 6,099.0                     | 13.7                | 13.6             | 0.00                        | 116.2                                     | 0.0             | 116.2                        | 88.9                          | 27.27                           | 4.261                |                    |          |
| 6,200.0                                                                                   | 6,200.0                     | 6,199.1                     | 6,199.0                     | 13.9                | 13.8             | 0.00                        | 117.1                                     | 0.0             | 117.1                        | 89.4                          | 27.72                           | 4.223                |                    |          |
| 6,300.0                                                                                   | 6,300.0                     | 6,299.1                     | 6,299.0                     | 14.2                | 14.0             | 0.00                        | 117.9                                     | 0.0             | 117.9                        | 89.8                          | 28.17                           | 4.187                |                    |          |
| 6,400.0                                                                                   | 6,400.0                     | 6,399.1                     | 6,399.0                     | 14.4                | 14.2             | 0.00                        | 118.8                                     | 0.0             | 118.8                        | 90.2                          | 28.61                           | 4.152                |                    |          |
| 6,500.0                                                                                   | 6,500.0                     | 6,499.1                     | 6,499.0                     | 14.6                | 14.5             | 0.00                        | 119.7                                     | 0.0             | 119.7                        | 90.6                          | 29.06                           | 4.118                |                    |          |
| 6,600.0                                                                                   | 6,600.0                     | 6,599.1                     | 6,598.9                     | 14.8                | 14.7             | 0.00                        | 120.6                                     | 0.0             | 120.6                        | 91.1                          | 29.51                           | 4.086                |                    |          |
| 6,700.0                                                                                   | 6,700.0                     | 6,699.1                     | 6,698.9                     | 15.1                | 14.9             | 0.00                        | 121.4                                     | 0.0             | 121.4                        | 91.5                          | 29.96                           | 4.054                |                    |          |
| 6,800.0                                                                                   | 6,800.0                     | 6,799.1                     | 6,798.9                     | 15.3                | 15.1             | 0.00                        | 122.3                                     | 0.0             | 122.3                        | 91.9                          | 30.40                           | 4.023                |                    |          |
| 6,900.0                                                                                   | 6,900.0                     | 6,899.1                     | 6,898.9                     | 15.5                | 15.4             | 0.00                        | 123.2                                     | 0.0             | 123.2                        | 92.3                          | 30.85                           | 3.993                |                    |          |
| 7,000.0                                                                                   | 7,000.0                     | 6,999.1                     | 6,998.9                     | 15.7                | 15.6             | 0.00                        | 124.0                                     | 0.0             | 124.1                        | 92.8                          | 31.30                           | 3.964                |                    |          |
| 7,100.0                                                                                   | 7,100.0                     | 7,099.1                     | 7,098.9                     | 15.9                | 15.8             | 0.00                        | 124.9                                     | 0.0             | 124.9                        | 93.2                          | 31.75                           | 3.935                |                    |          |
| 7,200.0                                                                                   | 7,200.0                     | 7,199.1                     | 7,198.9                     | 16.2                | 16.0             | 0.00                        | 125.8                                     | 0.0             | 125.8                        | 93.6                          | 32.19                           | 3.907                |                    |          |
| 7,300.0                                                                                   | 7,300.0                     | 7,299.1                     | 7,298.9                     | 16.4                | 16.3             | 0.00                        | 126.7                                     | 0.0             | 126.7                        | 94.0                          | 32.64                           | 3.881                |                    |          |
| 7,400.0                                                                                   | 7,400.0                     | 7,399.1                     | 7,398.9                     | 16.6                | 16.5             | 0.00                        | 127.5                                     | 0.0             | 127.5                        | 94.5                          | 33.09                           | 3.855                |                    |          |
| 7,500.0                                                                                   | 7,500.0                     | 7,499.1                     | 7,498.9                     | 16.8                | 16.7             | 0.00                        | 128.4                                     | 0.0             | 128.4                        | 94.9                          | 33.54                           | 3.829                |                    |          |
| 7,600.0                                                                                   | 7,600.0                     | 7,599.1                     | 7,598.9                     | 17.1                | 16.9             | 0.00                        | 129.3                                     | 0.0             | 129.3                        | 95.3                          | 33.99                           | 3.804                |                    |          |
| 7,700.0                                                                                   | 7,700.0                     | 7,699.1                     | 7,698.9                     | 17.3                | 17.2             | 0.00                        | 130.2                                     | 0.0             | 130.2                        | 95.7                          | 34.43                           | 3.780                |                    |          |
| 7,800.0                                                                                   | 7,800.0                     | 7,799.1                     | 7,798.9                     | 17.5                | 17.4             | 0.00                        | 131.0                                     | 0.0             | 131.0                        | 96.2                          | 34.88                           | 3.757                |                    |          |
| 7,900.0                                                                                   | 7,900.0                     | 7,899.1                     | 7,898.8                     | 17.7                | 17.6             | 0.00                        | 131.9                                     | 0.0             | 131.9                        | 96.6                          | 35.33                           | 3.734                |                    |          |
| 8,000.0                                                                                   | 8,000.0                     | 8,000.2                     | 8,000.0                     | 18.0                | 17.8             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 96.5                          | 35.76                           | 3.697                |                    |          |
| 8,100.0                                                                                   | 8,100.0                     | 8,100.2                     | 8,100.0                     | 18.2                | 18.0             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 96.0                          | 36.21                           | 3.652                |                    |          |
| 8,200.0                                                                                   | 8,200.0                     | 8,200.2                     | 8,200.0                     | 18.4                | 18.3             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 95.6                          | 36.66                           | 3.607                |                    |          |
| 8,300.0                                                                                   | 8,300.0                     | 8,300.2                     | 8,300.0                     | 18.6                | 18.5             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 95.1                          | 37.10                           | 3.563                |                    |          |
| 8,400.0                                                                                   | 8,400.0                     | 8,400.2                     | 8,400.0                     | 18.8                | 18.7             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 94.7                          | 37.55                           | 3.521                |                    |          |
| 8,500.0                                                                                   | 8,500.0                     | 8,500.2                     | 8,500.0                     | 19.1                | 18.9             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 94.2                          | 38.00                           | 3.479                |                    |          |
| 8,600.0                                                                                   | 8,600.0                     | 8,600.2                     | 8,600.0                     | 19.3                | 19.2             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 93.8                          | 38.45                           | 3.439                |                    |          |
| 8,700.0                                                                                   | 8,700.0                     | 8,700.2                     | 8,700.0                     | 19.5                | 19.4             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 93.3                          | 38.89                           | 3.399                |                    |          |
| 8,800.0                                                                                   | 8,800.0                     | 8,800.2                     | 8,800.0                     | 19.7                | 19.6             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 92.9                          | 39.34                           | 3.361                |                    |          |
| 8,900.0                                                                                   | 8,900.0                     | 8,900.2                     | 8,900.0                     | 20.0                | 19.8             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 92.4                          | 39.79                           | 3.323                |                    |          |
| 9,000.0                                                                                   | 9,000.0                     | 9,000.2                     | 9,000.0                     | 20.2                | 20.1             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 92.0                          | 40.24                           | 3.286                |                    |          |
| 9,100.0                                                                                   | 9,100.0                     | 9,100.2                     | 9,100.0                     | 20.4                | 20.3             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 91.5                          | 40.69                           | 3.250                |                    |          |
| 9,200.0                                                                                   | 9,200.0                     | 9,200.2                     | 9,200.0                     | 20.6                | 20.5             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 91.1                          | 41.14                           | 3.214                |                    |          |
| 9,300.0                                                                                   | 9,300.0                     | 9,300.2                     | 9,300.0                     | 20.9                | 20.7             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 90.6                          | 41.58                           | 3.179                |                    |          |
| 9,400.0                                                                                   | 9,400.0                     | 9,400.2                     | 9,400.0                     | 21.1                | 21.0             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 90.2                          | 42.03                           | 3.146                |                    |          |
| 9,500.0                                                                                   | 9,500.0                     | 9,500.2                     | 9,500.0                     | 21.3                | 21.2             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 89.7                          | 42.48                           | 3.112                |                    |          |
| 9,600.0                                                                                   | 9,600.0                     | 9,600.2                     | 9,600.0                     | 21.5                | 21.4             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 89.3                          | 42.93                           | 3.080                |                    |          |
| 9,700.0                                                                                   | 9,700.0                     | 9,700.2                     | 9,700.0                     | 21.8                | 21.6             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 88.8                          | 43.38                           | 3.048                |                    |          |
| 9,800.0                                                                                   | 9,800.0                     | 9,800.2                     | 9,800.0                     | 22.0                | 21.9             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 88.4                          | 43.83                           | 3.017                |                    |          |
| 9,900.0                                                                                   | 9,900.0                     | 9,900.2                     | 9,900.0                     | 22.2                | 22.1             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 87.9                          | 44.27                           | 2.986                |                    |          |
| 10,000.0                                                                                  | 10,000.0                    | 10,000.2                    | 10,000.0                    | 22.4                | 22.3             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 87.5                          | 44.72                           | 2.956                |                    |          |
| 10,100.0                                                                                  | 10,100.0                    | 10,100.2                    | 10,100.0                    | 22.6                | 22.5             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 87.0                          | 45.17                           | 2.927                |                    |          |
| 10,200.0                                                                                  | 10,200.0                    | 10,200.2                    | 10,200.0                    | 22.9                | 22.8             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 86.6                          | 45.62                           | 2.898                |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 2TX - Chalmers #2 T - Plan #1 |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|---------------------|------------------|-----------------------------|-------------------------------------------|-----------------|------------------------------|-------------------------------|---------------------------------|----------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                     |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Well Error: | 0.0 usft |
| Reference                                                                                 |                             | Offset                      |                             | Semi Major Axis     |                  | Highside<br>Toolface<br>(°) | Distance                                  |                 | Between<br>Centres<br>(usft) | Between<br>Ellipses<br>(usft) | Minimum<br>Separation<br>(usft) | Separation<br>Factor | Warning            |          |
| Measured<br>Depth<br>(usft)                                                               | Vertical<br>Depth<br>(usft) | Measured<br>Depth<br>(usft) | Vertical<br>Depth<br>(usft) | Reference<br>(usft) | Offset<br>(usft) |                             | Offset Wellbore Centre<br>+N/-S<br>(usft) | +E/-W<br>(usft) |                              |                               |                                 |                      |                    |          |
| 10,213.4                                                                                  | 10,213.4                    | 10,213.6                    | 10,213.4                    | 22.9                | 22.8             | 0.00                        | 132.2                                     | 0.0             | 132.2                        | 86.5                          | 45.68                           | 2.894                |                    |          |
| 10,259.5                                                                                  | 10,259.5                    | 10,259.2                    | 10,259.0                    | 23.0                | 22.9             | 0.64                        | 132.3                                     | 1.5             | 132.3                        | 86.4                          | 45.87                           | 2.884                |                    |          |
| 10,275.0                                                                                  | 10,275.0                    | 10,274.4                    | 10,274.1                    | 23.0                | 22.9             | -107.27                     | 132.3                                     | 2.9             | 132.4                        | 86.5                          | 45.96                           | 2.882                |                    |          |
| 10,300.0                                                                                  | 10,300.0                    | 10,298.9                    | 10,298.3                    | 23.1                | 23.0             | -106.35                     | 132.5                                     | 6.2             | 133.1                        | 87.0                          | 46.05                           | 2.890                |                    |          |
| 10,325.0                                                                                  | 10,324.8                    | 10,323.2                    | 10,322.3                    | 23.1                | 23.0             | -105.38                     | 132.6                                     | 10.8            | 134.2                        | 88.1                          | 46.14                           | 2.909                |                    |          |
| 10,350.0                                                                                  | 10,349.5                    | 10,347.4                    | 10,345.8                    | 23.2                | 23.1             | -104.38                     | 132.9                                     | 16.5            | 135.9                        | 89.6                          | 46.23                           | 2.939                |                    |          |
| 10,375.0                                                                                  | 10,373.9                    | 10,371.5                    | 10,368.8                    | 23.2                | 23.1             | -103.34                     | 133.1                                     | 23.3            | 138.0                        | 91.7                          | 46.32                           | 2.979                |                    |          |
| 10,400.0                                                                                  | 10,398.0                    | 10,395.4                    | 10,391.4                    | 23.3                | 23.2             | -102.29                     | 133.4                                     | 31.3            | 140.6                        | 94.2                          | 46.41                           | 3.029                |                    |          |
| 10,425.0                                                                                  | 10,421.7                    | 10,419.2                    | 10,413.4                    | 23.3                | 23.2             | -101.24                     | 133.8                                     | 40.3            | 143.6                        | 97.1                          | 46.52                           | 3.088                |                    |          |
| 10,450.0                                                                                  | 10,445.0                    | 10,442.8                    | 10,434.8                    | 23.4                | 23.3             | -100.19                     | 134.2                                     | 50.3            | 147.1                        | 100.5                         | 46.63                           | 3.156                |                    |          |
| 10,475.0                                                                                  | 10,467.8                    | 10,466.3                    | 10,455.5                    | 23.5                | 23.4             | -99.14                      | 134.6                                     | 61.3            | 151.1                        | 104.3                         | 46.75                           | 3.232                |                    |          |
| 10,500.0                                                                                  | 10,490.0                    | 10,489.6                    | 10,475.5                    | 23.5                | 23.4             | -98.11                      | 135.1                                     | 73.2            | 155.4                        | 108.6                         | 46.88                           | 3.316                |                    |          |
| 10,525.0                                                                                  | 10,511.5                    | 10,512.7                    | 10,494.8                    | 23.6                | 23.5             | -97.10                      | 135.6                                     | 86.0            | 160.2                        | 113.2                         | 47.02                           | 3.407                |                    |          |
| 10,550.0                                                                                  | 10,532.4                    | 10,535.7                    | 10,513.3                    | 23.7                | 23.6             | -96.12                      | 136.1                                     | 99.6            | 165.3                        | 118.2                         | 47.17                           | 3.505                |                    |          |
| 10,575.0                                                                                  | 10,552.5                    | 10,558.6                    | 10,531.1                    | 23.7                | 23.7             | -95.15                      | 136.7                                     | 114.0           | 170.9                        | 123.5                         | 47.33                           | 3.610                |                    |          |
| 10,600.0                                                                                  | 10,571.9                    | 10,581.2                    | 10,548.0                    | 23.8                | 23.8             | -94.21                      | 137.2                                     | 129.1           | 176.7                        | 129.2                         | 47.51                           | 3.719                |                    |          |
| 10,625.0                                                                                  | 10,590.3                    | 10,603.8                    | 10,564.0                    | 23.9                | 23.9             | -93.30                      | 137.9                                     | 144.9           | 182.9                        | 135.2                         | 47.71                           | 3.833                |                    |          |
| 10,650.0                                                                                  | 10,607.9                    | 10,626.2                    | 10,579.2                    | 24.0                | 24.0             | -92.41                      | 138.5                                     | 161.3           | 189.3                        | 141.4                         | 47.92                           | 3.951                |                    |          |
| 10,675.0                                                                                  | 10,624.5                    | 10,648.4                    | 10,593.6                    | 24.2                | 24.1             | -91.55                      | 139.2                                     | 178.3           | 196.1                        | 148.0                         | 48.14                           | 4.073                |                    |          |
| 10,700.0                                                                                  | 10,640.1                    | 10,670.5                    | 10,607.0                    | 24.3                | 24.2             | -90.71                      | 139.9                                     | 195.8           | 203.1                        | 154.7                         | 48.39                           | 4.197                |                    |          |
| 10,725.0                                                                                  | 10,654.7                    | 10,692.5                    | 10,619.6                    | 24.4                | 24.3             | -89.89                      | 140.6                                     | 213.9           | 210.4                        | 161.7                         | 48.66                           | 4.323                |                    |          |
| 10,750.0                                                                                  | 10,668.2                    | 10,714.4                    | 10,631.2                    | 24.6                | 24.5             | -89.10                      | 141.3                                     | 232.4           | 217.8                        | 168.9                         | 48.94                           | 4.451                |                    |          |
| 10,775.0                                                                                  | 10,680.5                    | 10,736.2                    | 10,642.0                    | 24.8                | 24.7             | -88.33                      | 142.0                                     | 251.4           | 225.5                        | 176.3                         | 49.25                           | 4.579                |                    |          |
| 10,800.0                                                                                  | 10,691.7                    | 10,758.0                    | 10,651.8                    | 25.0                | 24.8             | -87.58                      | 142.8                                     | 270.7           | 233.3                        | 183.8                         | 49.57                           | 4.707                |                    |          |
| 10,825.0                                                                                  | 10,701.8                    | 10,779.6                    | 10,660.7                    | 25.2                | 25.0             | -86.85                      | 143.5                                     | 290.4           | 241.4                        | 191.4                         | 49.92                           | 4.835                |                    |          |
| 10,850.0                                                                                  | 10,710.6                    | 10,801.2                    | 10,668.7                    | 25.4                | 25.2             | -86.14                      | 144.3                                     | 310.4           | 249.5                        | 199.2                         | 50.29                           | 4.961                |                    |          |
| 10,875.0                                                                                  | 10,718.1                    | 10,822.7                    | 10,675.8                    | 25.7                | 25.4             | -85.45                      | 145.1                                     | 330.8           | 257.7                        | 207.1                         | 50.69                           | 5.085                |                    |          |
| 10,900.0                                                                                  | 10,724.5                    | 10,844.2                    | 10,681.9                    | 25.9                | 25.6             | -84.78                      | 145.9                                     | 351.3           | 266.1                        | 215.0                         | 51.10                           | 5.207                |                    |          |
| 10,925.0                                                                                  | 10,729.5                    | 10,865.7                    | 10,687.1                    | 26.2                | 25.9             | -84.12                      | 146.7                                     | 372.2           | 274.5                        | 223.0                         | 51.54                           | 5.326                |                    |          |
| 10,950.0                                                                                  | 10,733.3                    | 10,887.2                    | 10,691.4                    | 26.5                | 26.1             | -83.49                      | 147.5                                     | 393.2           | 283.0                        | 231.0                         | 52.01                           | 5.442                |                    |          |
| 10,975.0                                                                                  | 10,735.7                    | 10,908.6                    | 10,694.7                    | 26.8                | 26.3             | -82.88                      | 148.4                                     | 414.4           | 291.5                        | 239.0                         | 52.49                           | 5.554                |                    |          |
| 11,000.0                                                                                  | 10,736.9                    | 10,930.2                    | 10,697.0                    | 27.1                | 26.6             | -82.29                      | 149.2                                     | 435.8           | 300.1                        | 247.1                         | 53.00                           | 5.662                |                    |          |
| 11,006.2                                                                                  | 10,737.0                    | 10,935.5                    | 10,697.4                    | 27.2                | 26.7             | -82.15                      | 149.4                                     | 441.1           | 302.2                        | 249.1                         | 53.13                           | 5.688                |                    |          |
| 11,100.0                                                                                  | 10,737.6                    | 11,026.7                    | 10,699.2                    | 28.5                | 27.9             | -83.18                      | 152.7                                     | 532.2           | 333.3                        | 277.6                         | 55.70                           | 5.983                |                    |          |
| 11,200.0                                                                                  | 10,738.3                    | 11,130.3                    | 10,699.9                    | 30.2                | 29.6             | -83.79                      | 153.2                                     | 635.8           | 360.3                        | 301.4                         | 58.97                           | 6.111                |                    |          |
| 11,300.0                                                                                  | 10,739.0                    | 11,227.5                    | 10,700.6                    | 32.0                | 31.3             | -84.21                      | 153.2                                     | 733.0           | 383.7                        | 321.2                         | 62.51                           | 6.138                |                    |          |
| 11,400.0                                                                                  | 10,739.7                    | 11,325.5                    | 10,701.3                    | 33.9                | 33.2             | -84.52                      | 153.2                                     | 831.0           | 403.7                        | 337.3                         | 66.39                           | 6.080                |                    |          |
| 11,500.0                                                                                  | 10,740.4                    | 11,424.1                    | 10,702.0                    | 36.0                | 35.2             | -84.75                      | 153.2                                     | 929.6           | 420.2                        | 349.7                         | 70.53                           | 5.958                |                    |          |
| 11,600.0                                                                                  | 10,741.1                    | 11,523.2                    | 10,702.7                    | 38.2                | 37.3             | -84.92                      | 153.2                                     | 1,028.7         | 433.3                        | 358.5                         | 74.88                           | 5.787                |                    |          |
| 11,700.0                                                                                  | 10,741.8                    | 11,622.7                    | 10,703.4                    | 40.5                | 39.6             | -85.04                      | 153.2                                     | 1,128.2         | 443.0                        | 363.6                         | 79.38                           | 5.581                |                    |          |
| 11,800.0                                                                                  | 10,742.5                    | 11,722.5                    | 10,704.1                    | 42.9                | 41.9             | -85.10                      | 153.2                                     | 1,228.0         | 449.2                        | 365.2                         | 84.00                           | 5.348                |                    |          |
| 11,900.0                                                                                  | 10,743.2                    | 11,822.5                    | 10,704.8                    | 45.3                | 44.4             | -85.12                      | 153.2                                     | 1,328.0         | 452.0                        | 363.3                         | 88.67                           | 5.097                |                    |          |
| 11,928.6                                                                                  | 10,743.4                    | 11,851.1                    | 10,705.0                    | 45.9                | 45.1             | -85.12                      | 153.2                                     | 1,356.6         | 452.1                        | 362.1                         | 90.02                           | 5.022                |                    |          |
| 12,000.0                                                                                  | 10,743.9                    | 11,922.5                    | 10,705.5                    | 47.7                | 46.8             | -85.12                      | 153.2                                     | 1,428.0         | 452.1                        | 358.5                         | 93.55                           | 4.833                |                    |          |
| 12,100.0                                                                                  | 10,744.6                    | 12,022.5                    | 10,706.2                    | 50.2                | 49.4             | -85.12                      | 153.2                                     | 1,528.0         | 452.1                        | 353.5                         | 98.59                           | 4.586                |                    |          |
| 12,200.0                                                                                  | 10,745.3                    | 12,122.5                    | 10,706.9                    | 52.7                | 51.9             | -85.12                      | 153.2                                     | 1,628.0         | 452.1                        | 348.4                         | 103.71                          | 4.359                |                    |          |
| 12,300.0                                                                                  | 10,746.0                    | 12,222.5                    | 10,707.6                    | 55.3                | 54.6             | -85.12                      | 153.2                                     | 1,728.0         | 452.1                        | 343.2                         | 108.91                          | 4.151                |                    |          |
| 12,400.0                                                                                  | 10,746.7                    | 12,322.5                    | 10,708.3                    | 57.9                | 57.2             | -85.12                      | 153.2                                     | 1,828.0         | 452.1                        | 337.9                         | 114.17                          | 3.960                |                    |          |
| 12,500.0                                                                                  | 10,747.4                    | 12,422.5                    | 10,709.0                    | 60.5                | 59.9             | -85.12                      | 153.2                                     | 1,928.0         | 452.1                        | 332.6                         | 119.48                          | 3.784                |                    |          |
| 12,600.0                                                                                  | 10,748.1                    | 12,522.5                    | 10,709.7                    | 63.2                | 62.6             | -85.12                      | 153.2                                     | 2,028.0         | 452.1                        | 327.3                         | 124.84                          | 3.621                |                    |          |
| 12,700.0                                                                                  | 10,748.8                    | 12,622.5                    | 10,710.4                    | 65.9                | 65.3             | -85.12                      | 153.2                                     | 2,128.0         | 452.1                        | 321.9                         | 130.24                          | 3.471                |                    |          |
| 12,800.0                                                                                  | 10,749.5                    | 12,722.5                    | 10,711.1                    | 68.6                | 68.0             | -85.12                      | 153.2                                     | 2,228.0         | 452.1                        | 316.4                         | 135.68                          | 3.332                |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 2TX - Chalmers #2 T - Plan #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                     |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 0.0 usft |
| Reference                                                                                 |                       | Offset                |                       | Semi Major Axis  |               |                       | Distance                            |              |                        |                         |                           |                   |                    | Warning  |
| Measured Depth (usft)                                                                     | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 12,900.0                                                                                  | 10,750.2              | 12,822.5              | 10,711.8              | 71.3             | 70.7          | -85.12                | 153.2                               | 2,328.0      | 452.1                  | 310.9                   | 141.15                    | 3.203             |                    |          |
| 13,000.0                                                                                  | 10,750.9              | 12,922.5              | 10,712.5              | 74.1             | 73.5          | -85.12                | 153.2                               | 2,428.0      | 452.1                  | 305.4                   | 146.65                    | 3.083             |                    |          |
| 13,100.0                                                                                  | 10,751.6              | 13,022.5              | 10,713.2              | 76.8             | 76.3          | -85.13                | 153.2                               | 2,528.0      | 452.1                  | 299.9                   | 152.17                    | 2.971             |                    |          |
| 13,200.0                                                                                  | 10,752.3              | 13,122.5              | 10,713.9              | 79.6             | 79.0          | -85.13                | 153.2                               | 2,628.0      | 452.1                  | 294.4                   | 157.72                    | 2.866             |                    |          |
| 13,300.0                                                                                  | 10,753.0              | 13,222.5              | 10,714.6              | 82.4             | 81.8          | -85.13                | 153.2                               | 2,727.9      | 452.1                  | 288.8                   | 163.29                    | 2.769             |                    |          |
| 13,400.0                                                                                  | 10,753.7              | 13,322.5              | 10,715.3              | 85.2             | 84.6          | -85.13                | 153.2                               | 2,827.9      | 452.1                  | 283.2                   | 168.87                    | 2.677             |                    |          |
| 13,500.0                                                                                  | 10,754.4              | 13,422.5              | 10,716.0              | 88.0             | 87.5          | -85.13                | 153.2                               | 2,927.9      | 452.1                  | 277.6                   | 174.48                    | 2.591             |                    |          |
| 13,600.0                                                                                  | 10,755.1              | 13,522.5              | 10,716.7              | 90.8             | 90.3          | -85.13                | 153.2                               | 3,027.9      | 452.1                  | 272.0                   | 180.10                    | 2.510             |                    |          |
| 13,700.0                                                                                  | 10,755.8              | 13,622.5              | 10,717.4              | 93.6             | 93.1          | -85.13                | 153.2                               | 3,127.9      | 452.1                  | 266.4                   | 185.73                    | 2.434             |                    |          |
| 13,800.0                                                                                  | 10,756.5              | 13,722.5              | 10,718.1              | 96.4             | 95.9          | -85.13                | 153.2                               | 3,227.9      | 452.1                  | 260.7                   | 191.37                    | 2.362             |                    |          |
| 13,900.0                                                                                  | 10,757.2              | 13,822.5              | 10,718.8              | 99.2             | 98.8          | -85.13                | 153.2                               | 3,327.9      | 452.1                  | 255.1                   | 197.03                    | 2.295             |                    |          |
| 14,000.0                                                                                  | 10,757.9              | 13,922.5              | 10,719.5              | 102.1            | 101.6         | -85.13                | 153.2                               | 3,427.9      | 452.1                  | 249.4                   | 202.70                    | 2.230             |                    |          |
| 14,100.0                                                                                  | 10,758.6              | 14,022.5              | 10,720.2              | 104.9            | 104.5         | -85.13                | 153.2                               | 3,527.9      | 452.1                  | 243.7                   | 208.37                    | 2.170             |                    |          |
| 14,200.0                                                                                  | 10,759.3              | 14,122.5              | 10,720.9              | 107.7            | 107.3         | -85.13                | 153.2                               | 3,627.9      | 452.1                  | 238.0                   | 214.06                    | 2.112             |                    |          |
| 14,300.0                                                                                  | 10,760.0              | 14,222.5              | 10,721.6              | 110.6            | 110.2         | -85.13                | 153.2                               | 3,727.9      | 452.1                  | 232.3                   | 219.75                    | 2.057             |                    |          |
| 14,400.0                                                                                  | 10,760.7              | 14,322.5              | 10,722.3              | 113.4            | 113.0         | -85.13                | 153.2                               | 3,827.9      | 452.1                  | 226.6                   | 225.45                    | 2.005             |                    |          |
| 14,500.0                                                                                  | 10,761.4              | 14,422.5              | 10,723.0              | 116.3            | 115.9         | -85.13                | 153.2                               | 3,927.9      | 452.1                  | 220.9                   | 231.16                    | 1.956             |                    |          |
| 14,600.0                                                                                  | 10,762.1              | 14,522.5              | 10,723.7              | 119.2            | 118.8         | -85.13                | 153.2                               | 4,027.9      | 452.1                  | 215.2                   | 236.88                    | 1.909             |                    |          |
| 14,700.0                                                                                  | 10,762.8              | 14,622.5              | 10,724.4              | 122.0            | 121.6         | -85.13                | 153.2                               | 4,127.9      | 452.1                  | 209.5                   | 242.60                    | 1.864             |                    |          |
| 14,800.0                                                                                  | 10,763.5              | 14,722.5              | 10,725.1              | 124.9            | 124.5         | -85.13                | 153.2                               | 4,227.9      | 452.1                  | 203.8                   | 248.33                    | 1.821             |                    |          |
| 14,900.0                                                                                  | 10,764.2              | 14,822.5              | 10,725.8              | 127.8            | 127.4         | -85.13                | 153.2                               | 4,327.9      | 452.1                  | 198.0                   | 254.06                    | 1.779             |                    |          |
| 15,000.0                                                                                  | 10,764.9              | 14,922.5              | 10,726.5              | 130.6            | 130.2         | -85.13                | 153.2                               | 4,427.9      | 452.1                  | 192.3                   | 259.79                    | 1.740             |                    |          |
| 15,100.0                                                                                  | 10,765.6              | 15,022.5              | 10,727.2              | 133.5            | 133.1         | -85.13                | 153.2                               | 4,527.9      | 452.1                  | 186.6                   | 265.54                    | 1.703             |                    |          |
| 15,200.0                                                                                  | 10,766.3              | 15,122.5              | 10,727.9              | 136.4            | 136.0         | -85.13                | 153.2                               | 4,627.9      | 452.1                  | 180.8                   | 271.28                    | 1.666             |                    |          |
| 15,300.0                                                                                  | 10,767.0              | 15,222.5              | 10,728.6              | 139.3            | 138.9         | -85.14                | 153.2                               | 4,727.9      | 452.1                  | 175.1                   | 277.03                    | 1.632             |                    |          |
| 15,400.0                                                                                  | 10,767.7              | 15,322.5              | 10,729.3              | 142.1            | 141.8         | -85.14                | 153.2                               | 4,827.9      | 452.1                  | 169.3                   | 282.78                    | 1.599             |                    |          |
| 15,500.0                                                                                  | 10,768.4              | 15,422.5              | 10,730.0              | 145.0            | 144.7         | -85.14                | 153.2                               | 4,927.9      | 452.1                  | 163.5                   | 288.54                    | 1.567             |                    |          |
| 15,600.0                                                                                  | 10,769.1              | 15,522.5              | 10,730.7              | 147.9            | 147.6         | -85.14                | 153.2                               | 5,027.9      | 452.1                  | 157.8                   | 294.30                    | 1.536             |                    |          |
| 15,700.0                                                                                  | 10,769.8              | 15,622.5              | 10,731.5              | 150.8            | 150.4         | -85.14                | 153.2                               | 5,127.9      | 452.1                  | 152.0                   | 300.07                    | 1.507             |                    |          |
| 15,800.0                                                                                  | 10,770.5              | 15,722.5              | 10,732.2              | 153.7            | 153.3         | -85.14                | 153.2                               | 5,227.9      | 452.1                  | 146.3                   | 305.83                    | 1.478 Level 3     | Level 3            |          |
| 15,900.0                                                                                  | 10,771.2              | 15,822.5              | 10,732.9              | 156.6            | 156.2         | -85.14                | 153.2                               | 5,327.9      | 452.1                  | 140.5                   | 311.60                    | 1.451 Level 3     | Level 3            |          |
| 16,000.0                                                                                  | 10,771.9              | 15,922.5              | 10,733.6              | 159.5            | 159.1         | -85.14                | 153.2                               | 5,427.9      | 452.1                  | 134.7                   | 317.37                    | 1.424 Level 3     | Level 3            |          |
| 16,100.0                                                                                  | 10,772.6              | 16,022.5              | 10,734.3              | 162.4            | 162.0         | -85.14                | 153.2                               | 5,527.9      | 452.1                  | 128.9                   | 323.15                    | 1.399 Level 3     | Level 3            |          |
| 16,200.0                                                                                  | 10,773.3              | 16,122.5              | 10,735.0              | 165.2            | 164.9         | -85.14                | 153.2                               | 5,627.9      | 452.1                  | 123.2                   | 328.93                    | 1.374 Level 3     | Level 3            |          |
| 16,300.0                                                                                  | 10,774.0              | 16,222.5              | 10,735.7              | 168.1            | 167.8         | -85.14                | 153.2                               | 5,727.9      | 452.1                  | 117.4                   | 334.71                    | 1.351 Level 3     | Level 3            |          |
| 16,400.0                                                                                  | 10,774.7              | 16,322.5              | 10,736.4              | 171.0            | 170.7         | -85.14                | 153.2                               | 5,827.9      | 452.1                  | 111.6                   | 340.49                    | 1.328 Level 3     | Level 3            |          |
| 16,500.0                                                                                  | 10,775.4              | 16,422.5              | 10,737.1              | 173.9            | 173.6         | -85.14                | 153.2                               | 5,927.9      | 452.1                  | 105.8                   | 346.27                    | 1.306 Level 3     | Level 3            |          |
| 16,600.0                                                                                  | 10,776.1              | 16,522.5              | 10,737.8              | 176.8            | 176.5         | -85.14                | 153.2                               | 6,027.9      | 452.1                  | 100.0                   | 352.06                    | 1.284 Level 3     | Level 3            |          |
| 16,700.0                                                                                  | 10,776.8              | 16,622.5              | 10,738.5              | 179.7            | 179.4         | -85.14                | 153.2                               | 6,127.9      | 452.1                  | 94.2                    | 357.84                    | 1.263 Level 3     | Level 3            |          |
| 16,800.0                                                                                  | 10,777.5              | 16,722.5              | 10,739.2              | 182.6            | 182.3         | -85.14                | 153.2                               | 6,227.9      | 452.1                  | 88.4                    | 363.63                    | 1.243 Level 2     |                    |          |
| 16,900.0                                                                                  | 10,778.2              | 16,822.5              | 10,739.9              | 185.5            | 185.2         | -85.14                | 153.2                               | 6,327.9      | 452.1                  | 82.7                    | 369.42                    | 1.224 Level 2     |                    |          |
| 17,000.0                                                                                  | 10,778.9              | 16,922.5              | 10,740.6              | 188.4            | 188.1         | -85.14                | 153.2                               | 6,427.9      | 452.1                  | 76.9                    | 375.22                    | 1.205 Level 2     |                    |          |
| 17,100.0                                                                                  | 10,779.6              | 17,022.5              | 10,741.3              | 191.3            | 191.0         | -85.14                | 153.2                               | 6,527.9      | 452.1                  | 71.1                    | 381.01                    | 1.187 Level 2     |                    |          |
| 17,200.0                                                                                  | 10,780.3              | 17,122.5              | 10,742.0              | 194.3            | 193.9         | -85.14                | 153.2                               | 6,627.9      | 452.1                  | 65.3                    | 386.81                    | 1.169 Level 2     |                    |          |
| 17,300.0                                                                                  | 10,780.9              | 17,222.5              | 10,742.7              | 197.2            | 196.9         | -85.14                | 153.2                               | 6,727.8      | 452.1                  | 59.5                    | 392.60                    | 1.151 Level 2     |                    |          |
| 17,400.0                                                                                  | 10,781.6              | 17,322.5              | 10,743.4              | 200.1            | 199.8         | -85.14                | 153.2                               | 6,827.8      | 452.1                  | 53.7                    | 398.40                    | 1.135 Level 2     |                    |          |
| 17,500.0                                                                                  | 10,782.3              | 17,422.5              | 10,744.1              | 203.0            | 202.7         | -85.15                | 153.2                               | 6,927.8      | 452.1                  | 47.9                    | 404.20                    | 1.118 Level 2     |                    |          |
| 17,600.0                                                                                  | 10,783.0              | 17,522.5              | 10,744.8              | 205.9            | 205.6         | -85.15                | 153.2                               | 7,027.8      | 452.1                  | 42.1                    | 410.00                    | 1.103 Level 2     |                    |          |
| 17,700.0                                                                                  | 10,783.7              | 17,622.5              | 10,745.5              | 208.8            | 208.5         | -85.15                | 153.2                               | 7,127.8      | 452.1                  | 36.3                    | 415.80                    | 1.087 Level 2     |                    |          |
| 17,800.0                                                                                  | 10,784.4              | 17,722.5              | 10,746.2              | 211.7            | 211.4         | -85.15                | 153.2                               | 7,227.8      | 452.1                  | 30.5                    | 421.61                    | 1.072 Level 2     |                    |          |
| 17,900.0                                                                                  | 10,785.1              | 17,822.5              | 10,746.9              | 214.6            | 214.3         | -85.15                | 153.2                               | 7,327.8      | 452.1                  | 24.7                    | 427.41                    | 1.058 Level 2     |                    |          |
| 18,000.0                                                                                  | 10,785.8              | 17,922.5              | 10,747.6              | 217.5            | 217.2         | -85.15                | 153.2                               | 7,427.8      | 452.1                  | 18.9                    | 433.22                    | 1.044 Level 2     |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 2TX - Chalmers #2 T - Plan #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                     |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 0.0 usft |
| Reference                                                                                 |                       | Offset                |                       | Semi Major Axis  |               |                       | Distance                            |              |                        |                         |                           |                   |                    | Warning  |
| Measured Depth (usft)                                                                     | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 18,100.0                                                                                  | 10,786.5              | 18,022.5              | 10,748.3              | 220.4            | 220.1         | -85.15                | 153.2                               | 7,527.8      | 452.1                  | 13.1                    | 439.02                    | 1.030             | Level 2            |          |
| 18,200.0                                                                                  | 10,787.2              | 18,122.5              | 10,749.0              | 223.3            | 223.0         | -85.15                | 153.2                               | 7,627.8      | 452.1                  | 7.2                     | 444.83                    | 1.016             | Level 2            |          |
| 18,300.0                                                                                  | 10,787.9              | 18,222.5              | 10,749.7              | 226.2            | 226.0         | -85.15                | 153.2                               | 7,727.8      | 452.1                  | 1.4                     | 450.64                    | 1.003             | Level 2            |          |
| 18,400.0                                                                                  | 10,788.6              | 18,322.5              | 10,750.4              | 229.2            | 228.9         | -85.15                | 153.2                               | 7,827.8      | 452.1                  | -4.4                    | 456.45                    | 0.990             | Level 1            |          |
| 18,500.0                                                                                  | 10,789.3              | 18,422.5              | 10,751.1              | 232.1            | 231.8         | -85.15                | 153.2                               | 7,927.8      | 452.1                  | -10.2                   | 462.26                    | 0.978             | Level 1            |          |
| 18,600.0                                                                                  | 10,790.0              | 18,522.5              | 10,751.8              | 235.0            | 234.7         | -85.15                | 153.2                               | 8,027.8      | 452.1                  | -16.0                   | 468.07                    | 0.966             | Level 1            |          |
| 18,700.0                                                                                  | 10,790.7              | 18,622.5              | 10,752.5              | 237.9            | 237.6         | -85.15                | 153.2                               | 8,127.8      | 452.1                  | -21.8                   | 473.88                    | 0.954             | Level 1            |          |
| 18,800.0                                                                                  | 10,791.4              | 18,722.5              | 10,753.2              | 240.8            | 240.5         | -85.15                | 153.2                               | 8,227.8      | 452.1                  | -27.6                   | 479.69                    | 0.942             | Level 1            |          |
| 18,900.0                                                                                  | 10,792.1              | 18,822.5              | 10,753.9              | 243.7            | 243.4         | -85.15                | 153.2                               | 8,327.8      | 452.1                  | -33.4                   | 485.50                    | 0.931             | Level 1            |          |
| 19,000.0                                                                                  | 10,792.8              | 18,922.5              | 10,754.6              | 246.6            | 246.4         | -85.15                | 153.2                               | 8,427.8      | 452.1                  | -39.2                   | 491.32                    | 0.920             | Level 1            |          |
| 19,100.0                                                                                  | 10,793.5              | 19,022.5              | 10,755.3              | 249.6            | 249.3         | -85.15                | 153.2                               | 8,527.8      | 452.1                  | -45.1                   | 497.13                    | 0.909             | Level 1            |          |
| 19,200.0                                                                                  | 10,794.2              | 19,122.5              | 10,756.0              | 252.5            | 252.2         | -85.15                | 153.2                               | 8,627.8      | 452.1                  | -50.9                   | 502.95                    | 0.899             | Level 1            |          |
| 19,300.0                                                                                  | 10,794.9              | 19,222.5              | 10,756.7              | 255.4            | 255.1         | -85.15                | 153.2                               | 8,727.8      | 452.1                  | -56.7                   | 508.76                    | 0.889             | Level 1            |          |
| 19,400.0                                                                                  | 10,795.6              | 19,322.5              | 10,757.4              | 258.3            | 258.0         | -85.15                | 153.2                               | 8,827.8      | 452.1                  | -62.5                   | 514.58                    | 0.879             | Level 1            |          |
| 19,500.0                                                                                  | 10,796.3              | 19,422.5              | 10,758.1              | 261.2            | 260.9         | -85.15                | 153.2                               | 8,927.8      | 452.1                  | -68.3                   | 520.39                    | 0.869             | Level 1            |          |
| 19,600.0                                                                                  | 10,797.0              | 19,522.5              | 10,758.8              | 264.1            | 263.9         | -85.15                | 153.2                               | 9,027.8      | 452.1                  | -74.1                   | 526.21                    | 0.859             | Level 1            |          |
| 19,700.0                                                                                  | 10,797.7              | 19,622.5              | 10,759.5              | 267.1            | 266.8         | -85.16                | 153.2                               | 9,127.8      | 452.1                  | -80.0                   | 532.03                    | 0.850             | Level 1            |          |
| 19,800.0                                                                                  | 10,798.4              | 19,722.5              | 10,760.2              | 270.0            | 269.7         | -85.16                | 153.2                               | 9,227.8      | 452.1                  | -85.8                   | 537.85                    | 0.841             | Level 1            |          |
| 19,900.0                                                                                  | 10,799.1              | 19,822.5              | 10,760.9              | 272.9            | 272.6         | -85.16                | 153.2                               | 9,327.8      | 452.1                  | -91.6                   | 543.67                    | 0.832             | Level 1            |          |
| 20,000.0                                                                                  | 10,799.8              | 19,922.5              | 10,761.6              | 275.8            | 275.5         | -85.16                | 153.2                               | 9,427.8      | 452.1                  | -97.4                   | 549.49                    | 0.823             | Level 1            |          |
| 20,100.0                                                                                  | 10,800.5              | 20,022.5              | 10,762.3              | 278.7            | 278.4         | -85.16                | 153.2                               | 9,527.8      | 452.1                  | -103.2                  | 555.31                    | 0.814             | Level 1            |          |
| 20,200.0                                                                                  | 10,801.2              | 20,122.5              | 10,763.0              | 281.6            | 281.4         | -85.16                | 153.2                               | 9,627.8      | 452.1                  | -109.1                  | 561.13                    | 0.806             | Level 1            |          |
| 20,300.0                                                                                  | 10,801.9              | 20,222.5              | 10,763.7              | 284.6            | 284.3         | -85.16                | 153.2                               | 9,727.8      | 452.1                  | -114.9                  | 566.95                    | 0.797             | Level 1            |          |
| 20,400.0                                                                                  | 10,802.6              | 20,322.5              | 10,764.4              | 287.5            | 287.2         | -85.16                | 153.2                               | 9,827.8      | 452.1                  | -120.7                  | 572.77                    | 0.789             | Level 1            |          |
| 20,500.0                                                                                  | 10,803.3              | 20,422.5              | 10,765.1              | 290.4            | 290.1         | -85.16                | 153.2                               | 9,927.8      | 452.1                  | -126.5                  | 578.59                    | 0.781             | Level 1            |          |
| 20,600.0                                                                                  | 10,804.0              | 20,522.5              | 10,765.8              | 293.3            | 293.0         | -85.16                | 153.2                               | 10,027.8     | 452.1                  | -132.3                  | 584.41                    | 0.774             | Level 1            |          |
| 20,700.0                                                                                  | 10,804.7              | 20,622.5              | 10,766.5              | 296.2            | 296.0         | -85.16                | 153.2                               | 10,127.8     | 452.1                  | -138.2                  | 590.24                    | 0.766             | Level 1            |          |
| 20,800.0                                                                                  | 10,805.4              | 20,722.5              | 10,767.2              | 299.2            | 298.9         | -85.16                | 153.2                               | 10,227.8     | 452.1                  | -144.0                  | 596.06                    | 0.758             | Level 1            |          |
| 20,900.0                                                                                  | 10,806.1              | 20,822.5              | 10,767.9              | 302.1            | 301.8         | -85.16                | 153.2                               | 10,327.8     | 452.1                  | -149.8                  | 601.88                    | 0.751             | Level 1            |          |
| 21,000.0                                                                                  | 10,806.8              | 20,922.5              | 10,768.6              | 305.0            | 304.7         | -85.16                | 153.2                               | 10,427.8     | 452.1                  | -155.6                  | 607.71                    | 0.744             | Level 1            |          |
| 21,076.2                                                                                  | 10,807.3              | 20,998.7              | 10,769.2              | 307.2            | 306.9         | -85.16                | 153.2                               | 10,504.0     | 452.1                  | -160.1                  | 612.15                    | 0.739             | Level 1            |          |
| 21,106.5                                                                                  | 10,807.5              | 21,028.7              | 10,769.4              | 308.1            | 307.8         | -85.16                | 153.2                               | 10,534.0     | 452.1                  | -161.8                  | 613.90                    | 0.736             | Level 1, ES, SF    |          |

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 3BR - Chalmers 5301 44-24 3BR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 0.0 usft |
| Reference                                                                                             |                       | Offset                |                       | Semi Major Axis  |               | Distance              |                                     |              |                        |                         |                           |                   | Warning            |          |
| Measured Depth (usft)                                                                                 | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 0.0                                                                                                   | 0.0                   | 0.0                   | 0.0                   | 2.0              | 0.0           | 0.01                  | 48.6                                | 0.0          | 48.6                   |                         |                           |                   |                    |          |
| 100.0                                                                                                 | 100.0                 | 100.0                 | 100.0                 | 2.0              | 0.1           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 46.6                    | 2.09                      | 23.315            |                    |          |
| 200.0                                                                                                 | 200.0                 | 200.0                 | 200.0                 | 2.0              | 0.3           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 46.3                    | 2.33                      | 20.849            |                    |          |
| 300.0                                                                                                 | 300.0                 | 300.0                 | 300.0                 | 2.1              | 0.5           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 46.0                    | 2.60                      | 18.679            |                    |          |
| 400.0                                                                                                 | 400.0                 | 400.0                 | 400.0                 | 2.1              | 0.8           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 45.7                    | 2.90                      | 16.785            |                    |          |
| 500.0                                                                                                 | 500.0                 | 500.0                 | 500.0                 | 2.2              | 1.0           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 45.4                    | 3.21                      | 15.142            |                    |          |
| 600.0                                                                                                 | 600.0                 | 600.0                 | 600.0                 | 2.3              | 1.2           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 45.1                    | 3.54                      | 13.721            |                    |          |
| 700.0                                                                                                 | 700.0                 | 700.0                 | 700.0                 | 2.5              | 1.4           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 44.7                    | 3.89                      | 12.493            |                    |          |
| 800.0                                                                                                 | 800.0                 | 800.0                 | 800.0                 | 2.6              | 1.7           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 44.4                    | 4.26                      | 11.430            |                    |          |
| 900.0                                                                                                 | 900.0                 | 900.0                 | 900.0                 | 2.7              | 1.9           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 44.0                    | 4.63                      | 10.507            |                    |          |
| 1,000.0                                                                                               | 1,000.0               | 1,000.0               | 1,000.0               | 2.9              | 2.1           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 43.6                    | 5.01                      | 9.703             |                    |          |
| 1,100.0                                                                                               | 1,100.0               | 1,100.0               | 1,100.0               | 3.1              | 2.3           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 43.2                    | 5.40                      | 9.000             |                    |          |
| 1,200.0                                                                                               | 1,200.0               | 1,200.0               | 1,200.0               | 3.2              | 2.6           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 42.8                    | 5.80                      | 8.382             |                    |          |
| 1,300.0                                                                                               | 1,300.0               | 1,300.0               | 1,300.0               | 3.4              | 2.8           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 42.4                    | 6.21                      | 7.835             |                    |          |
| 1,400.0                                                                                               | 1,400.0               | 1,400.0               | 1,400.0               | 3.6              | 3.0           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 42.0                    | 6.62                      | 7.350             |                    |          |
| 1,500.0                                                                                               | 1,500.0               | 1,500.0               | 1,500.0               | 3.8              | 3.2           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 41.6                    | 7.03                      | 6.918             |                    |          |
| 1,600.0                                                                                               | 1,600.0               | 1,600.0               | 1,600.0               | 4.0              | 3.5           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 41.2                    | 7.45                      | 6.530             |                    |          |
| 1,700.0                                                                                               | 1,700.0               | 1,700.0               | 1,700.0               | 4.2              | 3.7           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 40.8                    | 7.87                      | 6.181             |                    |          |
| 1,800.0                                                                                               | 1,800.0               | 1,800.0               | 1,800.0               | 4.4              | 3.9           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 40.3                    | 8.29                      | 5.865             |                    |          |
| 1,900.0                                                                                               | 1,900.0               | 1,900.0               | 1,900.0               | 4.6              | 4.1           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 39.9                    | 8.72                      | 5.578             |                    |          |
| 2,000.0                                                                                               | 2,000.0               | 2,000.0               | 2,000.0               | 4.8              | 4.4           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 39.5                    | 9.15                      | 5.317             |                    |          |
| 2,100.0                                                                                               | 2,100.0               | 2,100.0               | 2,100.0               | 5.0              | 4.6           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 39.1                    | 9.58                      | 5.079             |                    |          |
| 2,200.0                                                                                               | 2,200.0               | 2,200.0               | 2,200.0               | 5.2              | 4.8           | 0.01                  | 48.6                                | 0.0          | 48.6                   | 38.6                    | 10.01                     | 4.860 CC          |                    |          |
| 2,300.0                                                                                               | 2,300.0               | 2,300.0               | 2,300.0               | 5.4              | 5.0           | 0.94                  | 48.6                                | 0.8          | 48.6                   | 38.2                    | 10.42                     | 4.670             |                    |          |
| 2,400.0                                                                                               | 2,400.0               | 2,400.0               | 2,400.0               | 5.6              | 5.2           | 1.97                  | 48.6                                | 1.7          | 48.7                   | 37.8                    | 10.82                     | 4.496             |                    |          |
| 2,500.0                                                                                               | 2,500.0               | 2,500.0               | 2,500.0               | 5.8              | 5.4           | 3.00                  | 48.6                                | 2.5          | 48.7                   | 37.5                    | 11.23                     | 4.335             |                    |          |
| 2,600.0                                                                                               | 2,600.0               | 2,600.0               | 2,600.0               | 6.0              | 5.6           | 4.02                  | 48.6                                | 3.4          | 48.8                   | 37.1                    | 11.65                     | 4.186             |                    |          |
| 2,700.0                                                                                               | 2,700.0               | 2,700.0               | 2,700.0               | 6.3              | 5.8           | 5.04                  | 48.6                                | 4.3          | 48.8                   | 36.8                    | 12.07                     | 4.047             |                    |          |
| 2,800.0                                                                                               | 2,800.0               | 2,800.0               | 2,800.0               | 6.5              | 6.0           | 6.06                  | 48.6                                | 5.2          | 48.9                   | 36.4                    | 12.48                     | 3.918             |                    |          |
| 2,900.0                                                                                               | 2,900.0               | 2,900.0               | 2,899.9               | 6.7              | 6.2           | 7.07                  | 48.6                                | 6.0          | 49.0                   | 36.1                    | 12.91                     | 3.798             |                    |          |
| 3,000.0                                                                                               | 3,000.0               | 3,000.0               | 2,999.9               | 6.9              | 6.4           | 8.08                  | 48.6                                | 6.9          | 49.1                   | 35.8                    | 13.33                     | 3.686             |                    |          |
| 3,100.0                                                                                               | 3,100.0               | 3,100.0               | 3,099.9               | 7.1              | 6.6           | 9.09                  | 48.6                                | 7.8          | 49.3                   | 35.5                    | 13.75                     | 3.581             |                    |          |
| 3,200.0                                                                                               | 3,200.0               | 3,200.0               | 3,199.9               | 7.3              | 6.9           | 10.09                 | 48.6                                | 8.7          | 49.4                   | 35.2                    | 14.18                     | 3.484             |                    |          |
| 3,300.0                                                                                               | 3,300.0               | 3,300.0               | 3,299.9               | 7.5              | 7.1           | 11.08                 | 48.6                                | 9.5          | 49.6                   | 35.0                    | 14.61                     | 3.393             |                    |          |
| 3,400.0                                                                                               | 3,400.0               | 3,400.0               | 3,399.9               | 7.8              | 7.3           | 12.07                 | 48.6                                | 10.4         | 49.7                   | 34.7                    | 15.04                     | 3.308             |                    |          |
| 3,500.0                                                                                               | 3,500.0               | 3,500.0               | 3,499.9               | 8.0              | 7.5           | 13.05                 | 48.6                                | 11.3         | 49.9                   | 34.5                    | 15.47                     | 3.228             |                    |          |
| 3,600.0                                                                                               | 3,600.0               | 3,599.9               | 3,599.9               | 8.2              | 7.7           | 14.02                 | 48.6                                | 12.1         | 50.1                   | 34.2                    | 15.90                     | 3.154             |                    |          |
| 3,700.0                                                                                               | 3,700.0               | 3,699.9               | 3,699.9               | 8.4              | 7.9           | 14.98                 | 48.6                                | 13.0         | 50.3                   | 34.0                    | 16.33                     | 3.083             |                    |          |
| 3,800.0                                                                                               | 3,800.0               | 3,799.9               | 3,799.9               | 8.6              | 8.1           | 15.94                 | 48.6                                | 13.9         | 50.6                   | 33.8                    | 16.76                     | 3.018             |                    |          |
| 3,900.0                                                                                               | 3,900.0               | 3,899.9               | 3,899.9               | 8.9              | 8.4           | 16.88                 | 48.6                                | 14.8         | 50.8                   | 33.6                    | 17.20                     | 2.956             |                    |          |
| 4,000.0                                                                                               | 4,000.0               | 3,999.9               | 3,999.9               | 9.1              | 8.6           | 17.82                 | 48.6                                | 15.6         | 51.1                   | 33.5                    | 17.63                     | 2.898             |                    |          |
| 4,100.0                                                                                               | 4,100.0               | 4,099.9               | 4,099.9               | 9.3              | 8.8           | 18.75                 | 48.6                                | 16.5         | 51.4                   | 33.3                    | 18.07                     | 2.843             |                    |          |
| 4,200.0                                                                                               | 4,200.0               | 4,199.9               | 4,199.8               | 9.5              | 9.0           | 19.66                 | 48.6                                | 17.4         | 51.6                   | 33.1                    | 18.50                     | 2.792             |                    |          |
| 4,300.0                                                                                               | 4,300.0               | 4,299.9               | 4,299.8               | 9.7              | 9.2           | 20.57                 | 48.6                                | 18.3         | 51.9                   | 33.0                    | 18.94                     | 2.743             |                    |          |
| 4,400.0                                                                                               | 4,400.0               | 4,399.9               | 4,399.8               | 10.0             | 9.4           | 21.47                 | 48.6                                | 19.1         | 52.3                   | 32.9                    | 19.38                     | 2.697             |                    |          |
| 4,500.0                                                                                               | 4,500.0               | 4,499.9               | 4,499.8               | 10.2             | 9.7           | 22.35                 | 48.6                                | 20.0         | 52.6                   | 32.8                    | 19.81                     | 2.654             |                    |          |
| 4,600.0                                                                                               | 4,600.0               | 4,599.9               | 4,599.8               | 10.4             | 9.9           | 23.22                 | 48.6                                | 20.9         | 52.9                   | 32.7                    | 20.25                     | 2.613             |                    |          |
| 4,700.0                                                                                               | 4,700.0               | 4,699.9               | 4,699.8               | 10.6             | 10.1          | 24.09                 | 48.6                                | 21.7         | 53.3                   | 32.6                    | 20.69                     | 2.575             |                    |          |
| 4,800.0                                                                                               | 4,800.0               | 4,799.9               | 4,799.8               | 10.8             | 10.3          | 24.94                 | 48.6                                | 22.6         | 53.6                   | 32.5                    | 21.13                     | 2.539             |                    |          |
| 4,900.0                                                                                               | 4,900.0               | 4,899.9               | 4,899.8               | 11.1             | 10.5          | 25.78                 | 48.6                                | 23.5         | 54.0                   | 32.4                    | 21.57                     | 2.504             |                    |          |
| 5,000.0                                                                                               | 5,000.0               | 4,999.9               | 4,999.8               | 11.3             | 10.8          | 26.60                 | 48.6                                | 24.4         | 54.4                   | 32.4                    | 22.01                     | 2.472             |                    |          |
| 5,100.0                                                                                               | 5,100.0               | 5,099.9               | 5,099.8               | 11.5             | 11.0          | 27.42                 | 48.6                                | 25.2         | 54.8                   | 32.3                    | 22.45                     | 2.441             |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 3BR - Chalmers 5301 44-24 3BR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 0.0 usft |
| Reference                                                                                             |                       | Offset                |                       | Semi Major Axis  |               | Distance              |                                     |              |                        |                         |                           |                   | Warning            |          |
| Measured Depth (usft)                                                                                 | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 5,200.0                                                                                               | 5,200.0               | 5,199.9               | 5,199.8               | 11.7             | 11.2          | 28.22                 | 48.6                                | 26.1         | 55.2                   | 32.3                    | 22.89                     | 2.412             |                    |          |
| 5,300.0                                                                                               | 5,300.0               | 5,299.9               | 5,299.8               | 11.9             | 11.4          | 29.02                 | 48.6                                | 27.0         | 55.6                   | 32.3                    | 23.33                     | 2.384             |                    |          |
| 5,400.0                                                                                               | 5,400.0               | 5,399.9               | 5,399.8               | 12.2             | 11.6          | 29.80                 | 48.6                                | 27.9         | 56.0                   | 32.3                    | 23.77                     | 2.358             |                    |          |
| 5,500.0                                                                                               | 5,500.0               | 5,499.9               | 5,499.7               | 12.4             | 11.9          | 30.57                 | 48.6                                | 28.7         | 56.5                   | 32.3                    | 24.21                     | 2.333             |                    |          |
| 5,600.0                                                                                               | 5,600.0               | 5,599.9               | 5,599.7               | 12.6             | 12.1          | 31.32                 | 48.6                                | 29.6         | 56.9                   | 32.3                    | 24.66                     | 2.309             |                    |          |
| 5,700.0                                                                                               | 5,700.0               | 5,699.9               | 5,699.7               | 12.8             | 12.3          | 32.07                 | 48.6                                | 30.5         | 57.4                   | 32.3                    | 25.10                     | 2.287             |                    |          |
| 5,800.0                                                                                               | 5,800.0               | 5,799.9               | 5,799.7               | 13.1             | 12.5          | 32.80                 | 48.6                                | 31.3         | 57.9                   | 32.3                    | 25.54                     | 2.266             |                    |          |
| 5,900.0                                                                                               | 5,900.0               | 5,899.9               | 5,899.7               | 13.3             | 12.7          | 33.52                 | 48.6                                | 32.2         | 58.3                   | 32.4                    | 25.98                     | 2.245             |                    |          |
| 6,000.0                                                                                               | 6,000.0               | 5,999.9               | 5,999.7               | 13.5             | 13.0          | 34.23                 | 48.6                                | 33.1         | 58.8                   | 32.4                    | 26.43                     | 2.226             |                    |          |
| 6,100.0                                                                                               | 6,100.0               | 6,099.9               | 6,099.7               | 13.7             | 13.2          | 34.92                 | 48.6                                | 34.0         | 59.3                   | 32.5                    | 26.87                     | 2.208             |                    |          |
| 6,200.0                                                                                               | 6,200.0               | 6,199.8               | 6,199.7               | 13.9             | 13.4          | 35.61                 | 48.6                                | 34.8         | 59.8                   | 32.5                    | 27.31                     | 2.190             |                    |          |
| 6,300.0                                                                                               | 6,300.0               | 6,299.8               | 6,299.7               | 14.2             | 13.6          | 36.28                 | 48.6                                | 35.7         | 60.3                   | 32.6                    | 27.76                     | 2.174             |                    |          |
| 6,400.0                                                                                               | 6,400.0               | 6,399.8               | 6,399.7               | 14.4             | 13.9          | 36.94                 | 48.6                                | 36.6         | 60.9                   | 32.7                    | 28.20                     | 2.158             |                    |          |
| 6,500.0                                                                                               | 6,500.0               | 6,499.8               | 6,499.7               | 14.6             | 14.1          | 37.60                 | 48.6                                | 37.4         | 61.4                   | 32.7                    | 28.64                     | 2.143             |                    |          |
| 6,600.0                                                                                               | 6,600.0               | 6,599.8               | 6,599.7               | 14.8             | 14.3          | 38.24                 | 48.6                                | 38.3         | 61.9                   | 32.8                    | 29.09                     | 2.129             |                    |          |
| 6,700.0                                                                                               | 6,700.0               | 6,699.8               | 6,699.7               | 15.1             | 14.5          | 38.86                 | 48.6                                | 39.2         | 62.5                   | 32.9                    | 29.53                     | 2.115             |                    |          |
| 6,800.0                                                                                               | 6,800.0               | 6,799.8               | 6,799.7               | 15.3             | 14.7          | 39.48                 | 48.6                                | 40.1         | 63.0                   | 33.0                    | 29.98                     | 2.102             |                    |          |
| 6,900.0                                                                                               | 6,900.0               | 6,899.8               | 6,899.6               | 15.5             | 15.0          | 40.09                 | 48.6                                | 40.9         | 63.6                   | 33.2                    | 30.42                     | 2.090             |                    |          |
| 7,000.0                                                                                               | 7,000.0               | 6,999.8               | 6,999.6               | 15.7             | 15.2          | 40.69                 | 48.6                                | 41.8         | 64.1                   | 33.3                    | 30.86                     | 2.078             |                    |          |
| 7,100.0                                                                                               | 7,100.0               | 7,099.8               | 7,099.6               | 15.9             | 15.4          | 41.27                 | 48.6                                | 42.7         | 64.7                   | 33.4                    | 31.31                     | 2.067             |                    |          |
| 7,200.0                                                                                               | 7,200.0               | 7,199.8               | 7,199.6               | 16.2             | 15.6          | 41.85                 | 48.6                                | 43.6         | 65.3                   | 33.5                    | 31.75                     | 2.056             |                    |          |
| 7,300.0                                                                                               | 7,300.0               | 7,299.8               | 7,299.6               | 16.4             | 15.9          | 42.41                 | 48.6                                | 44.4         | 65.9                   | 33.7                    | 32.20                     | 2.046             |                    |          |
| 7,400.0                                                                                               | 7,400.0               | 7,399.8               | 7,399.6               | 16.6             | 16.1          | 42.97                 | 48.6                                | 45.3         | 66.5                   | 33.8                    | 32.64                     | 2.036             |                    |          |
| 7,500.0                                                                                               | 7,500.0               | 7,499.8               | 7,499.6               | 16.8             | 16.3          | 43.51                 | 48.6                                | 46.2         | 67.1                   | 34.0                    | 33.09                     | 2.027             |                    |          |
| 7,600.0                                                                                               | 7,600.0               | 7,599.8               | 7,599.6               | 17.1             | 16.5          | 44.05                 | 48.6                                | 47.0         | 67.7                   | 34.1                    | 33.54                     | 2.018             |                    |          |
| 7,700.0                                                                                               | 7,700.0               | 7,699.8               | 7,699.6               | 17.3             | 16.7          | 44.58                 | 48.6                                | 47.9         | 68.3                   | 34.3                    | 33.98                     | 2.009             |                    |          |
| 7,800.0                                                                                               | 7,800.0               | 7,799.8               | 7,799.6               | 17.5             | 17.0          | 45.09                 | 48.6                                | 48.8         | 68.9                   | 34.5                    | 34.43                     | 2.001             |                    |          |
| 7,900.0                                                                                               | 7,900.0               | 7,899.8               | 7,899.6               | 17.7             | 17.2          | 45.60                 | 48.6                                | 49.7         | 69.5                   | 34.6                    | 34.87                     | 1.993             |                    |          |
| 8,000.0                                                                                               | 8,000.0               | 8,000.2               | 8,000.0               | 18.0             | 17.4          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 34.4                    | 35.30                     | 1.975             |                    |          |
| 8,100.0                                                                                               | 8,100.0               | 8,100.2               | 8,100.0               | 18.2             | 17.6          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 34.0                    | 35.74                     | 1.951             |                    |          |
| 8,200.0                                                                                               | 8,200.0               | 8,200.2               | 8,200.0               | 18.4             | 17.8          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 33.6                    | 36.17                     | 1.928             |                    |          |
| 8,300.0                                                                                               | 8,300.0               | 8,300.2               | 8,300.0               | 18.6             | 18.0          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 33.1                    | 36.60                     | 1.905             |                    |          |
| 8,400.0                                                                                               | 8,400.0               | 8,400.2               | 8,400.0               | 18.8             | 18.2          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 32.7                    | 37.04                     | 1.883             |                    |          |
| 8,500.0                                                                                               | 8,500.0               | 8,500.2               | 8,500.0               | 19.1             | 18.5          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 32.3                    | 37.47                     | 1.861             |                    |          |
| 8,600.0                                                                                               | 8,600.0               | 8,600.2               | 8,600.0               | 19.3             | 18.7          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 31.8                    | 37.91                     | 1.840             |                    |          |
| 8,700.0                                                                                               | 8,700.0               | 8,700.2               | 8,700.0               | 19.5             | 18.9          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 31.4                    | 38.34                     | 1.819             |                    |          |
| 8,800.0                                                                                               | 8,800.0               | 8,800.2               | 8,800.0               | 19.7             | 19.1          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 31.0                    | 38.78                     | 1.798             |                    |          |
| 8,900.0                                                                                               | 8,900.0               | 8,900.2               | 8,900.0               | 20.0             | 19.3          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 30.5                    | 39.21                     | 1.778             |                    |          |
| 9,000.0                                                                                               | 9,000.0               | 9,000.2               | 9,000.0               | 20.2             | 19.5          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 30.1                    | 39.65                     | 1.759             |                    |          |
| 9,100.0                                                                                               | 9,100.0               | 9,100.2               | 9,100.0               | 20.4             | 19.7          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 29.7                    | 40.08                     | 1.740             |                    |          |
| 9,200.0                                                                                               | 9,200.0               | 9,200.2               | 9,200.0               | 20.6             | 19.9          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 29.2                    | 40.52                     | 1.721             |                    |          |
| 9,300.0                                                                                               | 9,300.0               | 9,300.2               | 9,300.0               | 20.9             | 20.1          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 28.8                    | 40.96                     | 1.703             |                    |          |
| 9,400.0                                                                                               | 9,400.0               | 9,400.2               | 9,400.0               | 21.1             | 20.4          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 28.3                    | 41.39                     | 1.685             |                    |          |
| 9,500.0                                                                                               | 9,500.0               | 9,500.2               | 9,500.0               | 21.3             | 20.6          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 27.9                    | 41.83                     | 1.667             |                    |          |
| 9,600.0                                                                                               | 9,600.0               | 9,600.2               | 9,600.0               | 21.5             | 20.8          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 27.5                    | 42.27                     | 1.650             |                    |          |
| 9,700.0                                                                                               | 9,700.0               | 9,700.2               | 9,700.0               | 21.8             | 21.0          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 27.0                    | 42.71                     | 1.633             |                    |          |
| 9,800.0                                                                                               | 9,800.0               | 9,800.2               | 9,800.0               | 22.0             | 21.2          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 26.6                    | 43.14                     | 1.616             |                    |          |
| 9,900.0                                                                                               | 9,900.0               | 9,900.2               | 9,900.0               | 22.2             | 21.4          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 26.2                    | 43.58                     | 1.600             |                    |          |
| 10,000.0                                                                                              | 10,000.0              | 10,000.2              | 10,000.0              | 22.4             | 21.6          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 25.7                    | 44.02                     | 1.584             |                    |          |
| 10,100.0                                                                                              | 10,100.0              | 10,100.2              | 10,100.0              | 22.6             | 21.9          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 25.3                    | 44.46                     | 1.569             |                    |          |
| 10,136.6                                                                                              | 10,136.6              | 10,136.9              | 10,136.6              | 22.7             | 21.9          | 45.78                 | 48.6                                | 50.0         | 69.7                   | 25.1                    | 44.62                     | 1.563             |                    |          |
| 10,200.0                                                                                              | 10,200.0              | 10,196.8              | 10,196.5              | 22.9             | 22.1          | 47.27                 | 48.0                                | 52.0         | 70.9                   | 26.0                    | 44.89                     | 1.579             |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation



# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 3BR - Chalmers 5301 44-24 3BR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 0.0 usft |
| Reference                                                                                             |                       | Offset                |                       | Semi Major Axis  |               |                       | Distance                            |              |                        |                         |                           |                   |                    | Warning  |
| Measured Depth (usft)                                                                                 | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 10,259.5                                                                                              | 10,259.5              | 10,250.0              | 10,249.1              | 23.0             | 22.2          | 52.46                 | 45.8                                | 59.6         | 75.9                   | 30.8                    | 45.14                     | 1.682             |                    |          |
| 10,275.0                                                                                              | 10,275.0              | 10,264.8              | 10,263.5              | 23.0             | 22.2          | -53.99                | 44.9                                | 62.8         | 77.9                   | 32.6                    | 45.24                     | 1.721             |                    |          |
| 10,300.0                                                                                              | 10,300.0              | 10,286.9              | 10,284.9              | 23.1             | 22.3          | -51.22                | 43.3                                | 68.2         | 81.1                   | 35.8                    | 45.30                     | 1.791             |                    |          |
| 10,325.0                                                                                              | 10,324.8              | 10,308.9              | 10,305.8              | 23.1             | 22.3          | -48.75                | 41.5                                | 74.6         | 84.5                   | 39.2                    | 45.31                     | 1.865             |                    |          |
| 10,350.0                                                                                              | 10,349.5              | 10,330.6              | 10,326.3              | 23.2             | 22.4          | -46.56                | 39.4                                | 81.8         | 88.0                   | 42.7                    | 45.27                     | 1.943             |                    |          |
| 10,375.0                                                                                              | 10,373.9              | 10,352.3              | 10,346.2              | 23.2             | 22.5          | -44.62                | 37.1                                | 89.8         | 91.4                   | 46.2                    | 45.17                     | 2.023             |                    |          |
| 10,400.0                                                                                              | 10,398.0              | 10,375.0              | 10,366.7              | 23.3             | 22.5          | -42.81                | 34.3                                | 99.2         | 94.8                   | 49.8                    | 45.01                     | 2.107             |                    |          |
| 10,425.0                                                                                              | 10,421.7              | 10,395.1              | 10,384.5              | 23.3             | 22.6          | -41.37                | 31.7                                | 108.2        | 98.2                   | 53.4                    | 44.79                     | 2.193             |                    |          |
| 10,450.0                                                                                              | 10,445.0              | 10,416.3              | 10,402.7              | 23.4             | 22.7          | -40.03                | 28.7                                | 118.5        | 101.6                  | 57.1                    | 44.50                     | 2.282             |                    |          |
| 10,475.0                                                                                              | 10,467.8              | 10,437.3              | 10,420.4              | 23.5             | 22.7          | -38.85                | 25.5                                | 129.5        | 104.8                  | 60.7                    | 44.15                     | 2.374             |                    |          |
| 10,500.0                                                                                              | 10,490.0              | 10,458.3              | 10,437.5              | 23.5             | 22.8          | -37.82                | 22.1                                | 141.2        | 107.9                  | 64.2                    | 43.75                     | 2.467             |                    |          |
| 10,525.0                                                                                              | 10,511.5              | 10,479.2              | 10,453.9              | 23.6             | 22.9          | -36.92                | 18.5                                | 153.6        | 111.0                  | 67.7                    | 43.30                     | 2.563             |                    |          |
| 10,550.0                                                                                              | 10,532.4              | 10,500.0              | 10,469.7              | 23.7             | 23.0          | -36.14                | 14.8                                | 166.6        | 113.8                  | 71.0                    | 42.80                     | 2.660             |                    |          |
| 10,575.0                                                                                              | 10,552.5              | 10,520.6              | 10,484.8              | 23.7             | 23.1          | -35.48                | 10.8                                | 180.1        | 116.6                  | 74.3                    | 42.27                     | 2.758             |                    |          |
| 10,600.0                                                                                              | 10,571.9              | 10,541.2              | 10,499.2              | 23.8             | 23.2          | -34.91                | 6.7                                 | 194.2        | 119.2                  | 77.5                    | 41.71                     | 2.857             |                    |          |
| 10,625.0                                                                                              | 10,590.3              | 10,561.8              | 10,513.0              | 23.9             | 23.3          | -34.44                | 2.5                                 | 208.9        | 121.6                  | 80.4                    | 41.13                     | 2.956             |                    |          |
| 10,650.0                                                                                              | 10,607.9              | 10,582.2              | 10,526.0              | 24.0             | 23.4          | -34.05                | -1.9                                | 224.0        | 123.8                  | 83.3                    | 40.54                     | 3.054             |                    |          |
| 10,675.0                                                                                              | 10,624.5              | 10,602.6              | 10,538.3              | 24.2             | 23.6          | -33.74                | -6.5                                | 239.6        | 125.9                  | 85.9                    | 39.96                     | 3.150             |                    |          |
| 10,700.0                                                                                              | 10,640.1              | 10,625.0              | 10,551.0              | 24.3             | 23.7          | -33.51                | -11.6                               | 257.3        | 127.8                  | 88.4                    | 39.40                     | 3.243             |                    |          |
| 10,725.0                                                                                              | 10,654.7              | 10,643.3              | 10,560.8              | 24.4             | 23.9          | -33.36                | -15.9                               | 272.2        | 129.4                  | 90.6                    | 38.85                     | 3.332             |                    |          |
| 10,750.0                                                                                              | 10,668.2              | 10,663.6              | 10,570.8              | 24.6             | 24.0          | -33.28                | -20.8                               | 289.1        | 130.9                  | 92.6                    | 38.35                     | 3.414             |                    |          |
| 10,775.0                                                                                              | 10,680.5              | 10,683.8              | 10,580.1              | 24.8             | 24.2          | -33.26                | -25.8                               | 306.3        | 132.2                  | 94.3                    | 37.90                     | 3.489             |                    |          |
| 10,800.0                                                                                              | 10,691.7              | 10,704.0              | 10,588.7              | 25.0             | 24.4          | -33.30                | -31.0                               | 323.9        | 133.3                  | 95.8                    | 37.52                     | 3.554             |                    |          |
| 10,825.0                                                                                              | 10,701.8              | 10,725.0              | 10,596.7              | 25.2             | 24.6          | -33.42                | -36.4                               | 342.6        | 134.2                  | 97.0                    | 37.22                     | 3.607             |                    |          |
| 10,850.0                                                                                              | 10,710.6              | 10,744.4              | 10,603.3              | 25.4             | 24.8          | -33.59                | -41.5                               | 360.1        | 134.9                  | 97.9                    | 37.00                     | 3.647             |                    |          |
| 10,875.0                                                                                              | 10,718.1              | 10,764.6              | 10,609.5              | 25.7             | 25.0          | -33.83                | -46.8                               | 378.5        | 135.4                  | 98.5                    | 36.90                     | 3.671             |                    |          |
| 10,900.0                                                                                              | 10,724.5              | 10,784.8              | 10,614.8              | 25.9             | 25.2          | -34.13                | -52.3                               | 397.3        | 135.7                  | 98.8                    | 36.90                     | 3.679             |                    |          |
| 10,925.0                                                                                              | 10,729.5              | 10,805.0              | 10,619.3              | 26.2             | 25.5          | -34.49                | -57.8                               | 416.2        | 135.9                  | 98.8                    | 37.02                     | 3.670             |                    |          |
| 10,950.0                                                                                              | 10,733.3              | 10,825.0              | 10,622.9              | 26.5             | 25.7          | -34.91                | -63.2                               | 435.0        | 135.8                  | 98.5                    | 37.26                     | 3.643             |                    |          |
| 10,975.0                                                                                              | 10,735.7              | 10,845.5              | 10,625.7              | 26.8             | 26.0          | -35.41                | -68.9                               | 454.5        | 135.5                  | 97.8                    | 37.64                     | 3.599             |                    |          |
| 11,000.0                                                                                              | 10,736.9              | 10,865.7              | 10,627.7              | 27.1             | 26.3          | -35.97                | -74.5                               | 473.9        | 135.0                  | 96.9                    | 38.15                     | 3.540             |                    |          |
| 11,006.2                                                                                              | 10,737.0              | 10,870.8              | 10,628.0              | 27.2             | 26.3          | -36.11                | -75.9                               | 478.7        | 134.9                  | 96.6                    | 38.29                     | 3.523             |                    |          |
| 11,026.3                                                                                              | 10,737.1              | 10,887.1              | 10,628.8              | 27.5             | 26.5          | -36.47                | -80.5                               | 494.4        | 134.7                  | 95.9                    | 38.74                     | 3.476             |                    |          |
| 11,100.0                                                                                              | 10,737.6              | 10,956.6              | 10,629.4              | 28.5             | 27.6          | -37.17                | -99.3                               | 561.2        | 135.7                  | 95.5                    | 40.23                     | 3.374             |                    |          |
| 11,200.0                                                                                              | 10,738.3              | 11,053.7              | 10,630.1              | 30.2             | 29.2          | -38.03                | -122.9                              | 655.5        | 137.3                  | 94.8                    | 42.53                     | 3.230             |                    |          |
| 11,300.0                                                                                              | 10,739.0              | 11,150.8              | 10,630.8              | 32.0             | 31.0          | -38.83                | -143.3                              | 750.4        | 138.9                  | 93.8                    | 45.10                     | 3.081             |                    |          |
| 11,400.0                                                                                              | 10,739.7              | 11,247.8              | 10,631.5              | 33.9             | 32.9          | -39.59                | -160.4                              | 845.8        | 140.5                  | 92.6                    | 47.90                     | 2.932             |                    |          |
| 11,500.0                                                                                              | 10,740.4              | 11,344.7              | 10,632.2              | 36.0             | 34.9          | -40.30                | -174.3                              | 941.7        | 141.9                  | 91.0                    | 50.90                     | 2.789             |                    |          |
| 11,600.0                                                                                              | 10,741.1              | 11,441.5              | 10,632.9              | 38.2             | 37.0          | -40.96                | -185.0                              | 1,037.9      | 143.4                  | 89.3                    | 54.06                     | 2.652             |                    |          |
| 11,700.0                                                                                              | 10,741.8              | 11,538.3              | 10,633.5              | 40.5             | 39.3          | -41.57                | -192.3                              | 1,134.4      | 144.8                  | 87.4                    | 57.34                     | 2.525             |                    |          |
| 11,800.0                                                                                              | 10,742.5              | 11,635.0              | 10,634.2              | 42.9             | 41.5          | -42.14                | -196.4                              | 1,231.0      | 146.1                  | 85.4                    | 60.72                     | 2.406             |                    |          |
| 11,900.0                                                                                              | 10,743.2              | 11,732.4              | 10,634.9              | 45.3             | 43.9          | -42.64                | -197.4                              | 1,328.5      | 147.3                  | 83.1                    | 64.15                     | 2.296             |                    |          |
| 11,928.6                                                                                              | 10,743.4              | 11,761.1              | 10,635.1              | 45.9             | 44.6          | -42.68                | -197.4                              | 1,357.1      | 147.4                  | 82.3                    | 65.06                     | 2.265             |                    |          |
| 12,000.0                                                                                              | 10,743.9              | 11,832.4              | 10,635.6              | 47.7             | 46.3          | -42.68                | -197.4                              | 1,428.5      | 147.4                  | 79.9                    | 67.47                     | 2.184             |                    |          |
| 12,100.0                                                                                              | 10,744.6              | 11,932.4              | 10,636.3              | 50.2             | 48.8          | -42.68                | -197.4                              | 1,528.5      | 147.4                  | 76.5                    | 70.92                     | 2.078             |                    |          |
| 12,200.0                                                                                              | 10,745.3              | 12,032.4              | 10,637.0              | 52.7             | 51.4          | -42.68                | -197.4                              | 1,628.5      | 147.4                  | 73.0                    | 74.44                     | 1.980             |                    |          |
| 12,300.0                                                                                              | 10,746.0              | 12,132.4              | 10,637.7              | 55.3             | 54.0          | -42.68                | -197.4                              | 1,728.5      | 147.4                  | 69.4                    | 78.01                     | 1.889             |                    |          |
| 12,400.0                                                                                              | 10,746.7              | 12,232.4              | 10,638.4              | 57.9             | 56.6          | -42.68                | -197.4                              | 1,828.5      | 147.4                  | 65.7                    | 81.64                     | 1.805             |                    |          |
| 12,500.0                                                                                              | 10,747.4              | 12,332.4              | 10,639.1              | 60.5             | 59.3          | -42.68                | -197.4                              | 1,928.5      | 147.4                  | 62.1                    | 85.32                     | 1.728             |                    |          |
| 12,600.0                                                                                              | 10,748.1              | 12,432.4              | 10,639.8              | 63.2             | 61.9          | -42.68                | -197.4                              | 2,028.5      | 147.4                  | 58.4                    | 89.03                     | 1.655             |                    |          |
| 12,700.0                                                                                              | 10,748.8              | 12,532.4              | 10,640.5              | 65.9             | 64.6          | -42.68                | -197.4                              | 2,128.4      | 147.4                  | 54.6                    | 92.78                     | 1.589             |                    |          |
| 12,800.0                                                                                              | 10,749.5              | 12,632.4              | 10,641.2              | 68.6             | 67.4          | -42.68                | -197.4                              | 2,228.4      | 147.4                  | 50.8                    | 96.56                     | 1.526             |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 3BR - Chalmers 5301 44-24 3BR - Design #1 |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------|-----------------------------|---------------------|------------------|-----------------------------|-------------------------------------------|-----------------|------------------------------|-------------------------------|---------------------------------|----------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                 |                             |                             |                             |                     |                  |                             |                                           |                 |                              |                               |                                 |                      | Offset Well Error: | 0.0 usft |
| Reference                                                                                             |                             | Offset                      |                             | Semi Major Axis     |                  | Highside<br>Toolface<br>(°) | Distance                                  |                 | Between<br>Centres<br>(usft) | Between<br>Ellipses<br>(usft) | Minimum<br>Separation<br>(usft) | Separation<br>Factor | Warning            |          |
| Measured<br>Depth<br>(usft)                                                                           | Vertical<br>Depth<br>(usft) | Measured<br>Depth<br>(usft) | Vertical<br>Depth<br>(usft) | Reference<br>(usft) | Offset<br>(usft) |                             | Offset Wellbore Centre<br>+N/-S<br>(usft) | +E/-W<br>(usft) |                              |                               |                                 |                      |                    |          |
| 12,900.0                                                                                              | 10,750.2                    | 12,732.4                    | 10,641.9                    | 71.3                | 70.1             | -42.68                      | -197.4                                    | 2,328.4         | 147.4                        | 47.0                          | 100.37                          | 1.468 Level 3        | Level 3            |          |
| 13,000.0                                                                                              | 10,750.9                    | 12,832.4                    | 10,642.6                    | 74.1                | 72.8             | -42.68                      | -197.4                                    | 2,428.4         | 147.4                        | 43.2                          | 104.20                          | 1.414 Level 3        | Level 3            |          |
| 13,100.0                                                                                              | 10,751.6                    | 12,932.4                    | 10,643.3                    | 76.8                | 75.6             | -42.68                      | -197.4                                    | 2,528.4         | 147.4                        | 39.3                          | 108.05                          | 1.364 Level 3        | Level 3            |          |
| 13,200.0                                                                                              | 10,752.3                    | 13,032.4                    | 10,644.0                    | 79.6                | 78.4             | -42.68                      | -197.4                                    | 2,628.4         | 147.4                        | 35.5                          | 111.93                          | 1.317 Level 3        | Level 3            |          |
| 13,300.0                                                                                              | 10,753.0                    | 13,132.4                    | 10,644.7                    | 82.4                | 81.2             | -42.68                      | -197.4                                    | 2,728.4         | 147.4                        | 31.6                          | 115.82                          | 1.273 Level 3        | Level 3            |          |
| 13,400.0                                                                                              | 10,753.7                    | 13,232.4                    | 10,645.4                    | 85.2                | 84.0             | -42.68                      | -197.4                                    | 2,828.4         | 147.4                        | 27.7                          | 119.73                          | 1.231 Level 2        |                    |          |
| 13,500.0                                                                                              | 10,754.4                    | 13,332.4                    | 10,646.1                    | 88.0                | 86.8             | -42.68                      | -197.4                                    | 2,928.4         | 147.4                        | 23.7                          | 123.65                          | 1.192 Level 2        |                    |          |
| 13,600.0                                                                                              | 10,755.1                    | 13,432.4                    | 10,646.8                    | 90.8                | 89.6             | -42.68                      | -197.4                                    | 3,028.4         | 147.4                        | 19.8                          | 127.58                          | 1.155 Level 2        |                    |          |
| 13,700.0                                                                                              | 10,755.8                    | 13,532.4                    | 10,647.5                    | 93.6                | 92.4             | -42.68                      | -197.4                                    | 3,128.4         | 147.4                        | 15.9                          | 131.53                          | 1.121 Level 2        |                    |          |
| 13,800.0                                                                                              | 10,756.5                    | 13,632.4                    | 10,648.2                    | 96.4                | 95.2             | -42.68                      | -197.4                                    | 3,228.4         | 147.4                        | 11.9                          | 135.48                          | 1.088 Level 2        |                    |          |
| 13,900.0                                                                                              | 10,757.2                    | 13,732.4                    | 10,648.9                    | 99.2                | 98.1             | -42.68                      | -197.4                                    | 3,328.4         | 147.4                        | 7.9                           | 139.45                          | 1.057 Level 2        |                    |          |
| 14,000.0                                                                                              | 10,757.9                    | 13,832.4                    | 10,649.6                    | 102.1               | 100.9            | -42.68                      | -197.4                                    | 3,428.4         | 147.4                        | 4.0                           | 143.43                          | 1.028 Level 2        |                    |          |
| 14,100.0                                                                                              | 10,758.6                    | 13,932.4                    | 10,650.3                    | 104.9               | 103.8            | -42.68                      | -197.4                                    | 3,528.4         | 147.4                        | 0.0                           | 147.41                          | 1.000 Level 1        |                    |          |
| 14,200.0                                                                                              | 10,759.3                    | 14,032.4                    | 10,651.0                    | 107.7               | 106.6            | -42.68                      | -197.4                                    | 3,628.4         | 147.4                        | -4.0                          | 151.40                          | 0.973 Level 1        |                    |          |
| 14,300.0                                                                                              | 10,760.0                    | 14,132.4                    | 10,651.7                    | 110.6               | 109.5            | -42.68                      | -197.4                                    | 3,728.4         | 147.4                        | -8.0                          | 155.40                          | 0.948 Level 1        |                    |          |
| 14,400.0                                                                                              | 10,760.7                    | 14,232.4                    | 10,652.3                    | 113.4               | 112.3            | -42.68                      | -197.4                                    | 3,828.4         | 147.4                        | -12.0                         | 159.41                          | 0.925 Level 1        |                    |          |
| 14,500.0                                                                                              | 10,761.4                    | 14,332.4                    | 10,653.0                    | 116.3               | 115.2            | -42.68                      | -197.4                                    | 3,928.4         | 147.4                        | -16.0                         | 163.42                          | 0.902 Level 1        |                    |          |
| 14,600.0                                                                                              | 10,762.1                    | 14,432.4                    | 10,653.7                    | 119.2               | 118.1            | -42.68                      | -197.4                                    | 4,028.4         | 147.4                        | -20.0                         | 167.43                          | 0.880 Level 1        |                    |          |
| 14,700.0                                                                                              | 10,762.8                    | 14,532.4                    | 10,654.4                    | 122.0               | 120.9            | -42.68                      | -197.4                                    | 4,128.4         | 147.4                        | -24.1                         | 171.46                          | 0.860 Level 1        |                    |          |
| 14,800.0                                                                                              | 10,763.5                    | 14,632.4                    | 10,655.1                    | 124.9               | 123.8            | -42.68                      | -197.4                                    | 4,228.4         | 147.4                        | -28.1                         | 175.48                          | 0.840 Level 1        |                    |          |
| 14,900.0                                                                                              | 10,764.2                    | 14,732.4                    | 10,655.8                    | 127.8               | 126.7            | -42.68                      | -197.4                                    | 4,328.4         | 147.4                        | -32.1                         | 179.52                          | 0.821 Level 1        |                    |          |
| 15,000.0                                                                                              | 10,764.9                    | 14,832.4                    | 10,656.5                    | 130.6               | 129.5            | -42.68                      | -197.4                                    | 4,428.4         | 147.4                        | -36.2                         | 183.55                          | 0.803 Level 1        |                    |          |
| 15,100.0                                                                                              | 10,765.6                    | 14,932.4                    | 10,657.2                    | 133.5               | 132.4            | -42.68                      | -197.4                                    | 4,528.4         | 147.4                        | -40.2                         | 187.59                          | 0.786 Level 1        |                    |          |
| 15,200.0                                                                                              | 10,766.3                    | 15,032.4                    | 10,657.9                    | 136.4               | 135.3            | -42.68                      | -197.4                                    | 4,628.4         | 147.4                        | -44.2                         | 191.64                          | 0.769 Level 1        |                    |          |
| 15,300.0                                                                                              | 10,767.0                    | 15,132.4                    | 10,658.6                    | 139.3               | 138.2            | -42.68                      | -197.4                                    | 4,728.4         | 147.4                        | -48.3                         | 195.68                          | 0.753 Level 1        |                    |          |
| 15,400.0                                                                                              | 10,767.7                    | 15,232.4                    | 10,659.3                    | 142.1               | 141.1            | -42.68                      | -197.4                                    | 4,828.4         | 147.4                        | -52.3                         | 199.73                          | 0.738 Level 1        |                    |          |
| 15,500.0                                                                                              | 10,768.4                    | 15,332.4                    | 10,660.0                    | 145.0               | 144.0            | -42.68                      | -197.4                                    | 4,928.4         | 147.4                        | -56.4                         | 203.79                          | 0.723 Level 1        |                    |          |
| 15,600.0                                                                                              | 10,769.1                    | 15,432.4                    | 10,660.7                    | 147.9               | 146.8            | -42.68                      | -197.4                                    | 5,028.4         | 147.4                        | -60.5                         | 207.84                          | 0.709 Level 1        |                    |          |
| 15,700.0                                                                                              | 10,769.8                    | 15,532.4                    | 10,661.4                    | 150.8               | 149.7            | -42.68                      | -197.4                                    | 5,128.4         | 147.4                        | -64.5                         | 211.90                          | 0.696 Level 1        |                    |          |
| 15,800.0                                                                                              | 10,770.5                    | 15,632.4                    | 10,662.1                    | 153.7               | 152.6            | -42.68                      | -197.4                                    | 5,228.4         | 147.4                        | -68.6                         | 215.97                          | 0.682 Level 1        |                    |          |
| 15,900.0                                                                                              | 10,771.2                    | 15,732.4                    | 10,662.8                    | 156.6               | 155.5            | -42.68                      | -197.4                                    | 5,328.4         | 147.4                        | -72.6                         | 220.03                          | 0.670 Level 1        |                    |          |
| 16,000.0                                                                                              | 10,771.9                    | 15,832.4                    | 10,663.5                    | 159.5               | 158.4            | -42.68                      | -197.4                                    | 5,428.4         | 147.4                        | -76.7                         | 224.10                          | 0.658 Level 1        |                    |          |
| 16,100.0                                                                                              | 10,772.6                    | 15,932.4                    | 10,664.2                    | 162.4               | 161.3            | -42.68                      | -197.4                                    | 5,528.4         | 147.4                        | -80.8                         | 228.17                          | 0.646 Level 1        |                    |          |
| 16,200.0                                                                                              | 10,773.3                    | 16,032.4                    | 10,664.9                    | 165.2               | 164.2            | -42.68                      | -197.4                                    | 5,628.4         | 147.4                        | -84.9                         | 232.24                          | 0.635 Level 1        |                    |          |
| 16,300.0                                                                                              | 10,774.0                    | 16,132.4                    | 10,665.6                    | 168.1               | 167.1            | -42.68                      | -197.4                                    | 5,728.4         | 147.4                        | -88.9                         | 236.31                          | 0.624 Level 1        |                    |          |
| 16,400.0                                                                                              | 10,774.7                    | 16,232.4                    | 10,666.3                    | 171.0               | 170.0            | -42.68                      | -197.4                                    | 5,828.4         | 147.4                        | -93.0                         | 240.38                          | 0.613 Level 1        |                    |          |
| 16,500.0                                                                                              | 10,775.4                    | 16,332.4                    | 10,667.0                    | 173.9               | 172.9            | -42.68                      | -197.4                                    | 5,928.4         | 147.4                        | -97.1                         | 244.46                          | 0.603 Level 1        |                    |          |
| 16,600.0                                                                                              | 10,776.1                    | 16,432.4                    | 10,667.7                    | 176.8               | 175.8            | -42.68                      | -197.4                                    | 6,028.4         | 147.4                        | -101.2                        | 248.54                          | 0.593 Level 1        |                    |          |
| 16,700.0                                                                                              | 10,776.8                    | 16,532.4                    | 10,668.4                    | 179.7               | 178.7            | -42.68                      | -197.4                                    | 6,128.4         | 147.4                        | -105.2                        | 252.62                          | 0.583 Level 1        |                    |          |
| 16,800.0                                                                                              | 10,777.5                    | 16,632.4                    | 10,669.1                    | 182.6               | 181.6            | -42.68                      | -197.4                                    | 6,228.3         | 147.4                        | -109.3                        | 256.70                          | 0.574 Level 1        |                    |          |
| 16,900.0                                                                                              | 10,778.2                    | 16,732.4                    | 10,669.8                    | 185.5               | 184.5            | -42.68                      | -197.4                                    | 6,328.3         | 147.4                        | -113.4                        | 260.78                          | 0.565 Level 1        |                    |          |
| 17,000.0                                                                                              | 10,778.9                    | 16,832.4                    | 10,670.5                    | 188.4               | 187.4            | -42.68                      | -197.4                                    | 6,428.3         | 147.4                        | -117.5                        | 264.87                          | 0.556 Level 1        |                    |          |
| 17,100.0                                                                                              | 10,779.6                    | 16,932.4                    | 10,671.2                    | 191.3               | 190.3            | -42.68                      | -197.4                                    | 6,528.3         | 147.4                        | -121.6                        | 268.95                          | 0.548 Level 1        |                    |          |
| 17,200.0                                                                                              | 10,780.3                    | 17,032.4                    | 10,671.9                    | 194.3               | 193.2            | -42.68                      | -197.4                                    | 6,628.3         | 147.4                        | -125.7                        | 273.04                          | 0.540 Level 1        |                    |          |
| 17,300.0                                                                                              | 10,780.9                    | 17,132.4                    | 10,672.6                    | 197.2               | 196.1            | -42.68                      | -197.4                                    | 6,728.3         | 147.4                        | -129.7                        | 277.13                          | 0.532 Level 1        |                    |          |
| 17,400.0                                                                                              | 10,781.6                    | 17,232.4                    | 10,673.3                    | 200.1               | 199.1            | -42.68                      | -197.4                                    | 6,828.3         | 147.4                        | -133.8                        | 281.22                          | 0.524 Level 1        |                    |          |
| 17,500.0                                                                                              | 10,782.3                    | 17,332.4                    | 10,674.0                    | 203.0               | 202.0            | -42.68                      | -197.4                                    | 6,928.3         | 147.4                        | -137.9                        | 285.31                          | 0.517 Level 1        |                    |          |
| 17,600.0                                                                                              | 10,783.0                    | 17,432.4                    | 10,674.7                    | 205.9               | 204.9            | -42.68                      | -197.4                                    | 7,028.3         | 147.4                        | -142.0                        | 289.40                          | 0.509 Level 1        |                    |          |
| 17,700.0                                                                                              | 10,783.7                    | 17,532.4                    | 10,675.4                    | 208.8               | 207.8            | -42.68                      | -197.4                                    | 7,128.3         | 147.4                        | -146.1                        | 293.49                          | 0.502 Level 1        |                    |          |
| 17,800.0                                                                                              | 10,784.4                    | 17,632.4                    | 10,676.1                    | 211.7               | 210.7            | -42.68                      | -197.4                                    | 7,228.3         | 147.4                        | -150.2                        | 297.59                          | 0.495 Level 1        |                    |          |
| 17,900.0                                                                                              | 10,785.1                    | 17,732.4                    | 10,676.8                    | 214.6               | 213.6            | -42.68                      | -197.4                                    | 7,328.3         | 147.4                        | -154.3                        | 301.68                          | 0.489 Level 1        |                    |          |
| 18,000.0                                                                                              | 10,785.8                    | 17,832.4                    | 10,677.5                    | 217.5               | 216.5            | -42.68                      | -197.4                                    | 7,428.3         | 147.4                        | -158.4                        | 305.77                          | 0.482 Level 1        |                    |          |

CC - Min centre to center distance or convergent point, SF - min separation factor, ES - min ellipse separation

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

| Offset Design 153N-100W-19/20_Altered - Chalmers 5301 44-24 3BR - Chalmers 5301 44-24 3BR - Design #1 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Site Error: | 0.0 usft |
|-------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------|---------------|-----------------------|-------------------------------------|--------------|------------------------|-------------------------|---------------------------|-------------------|--------------------|----------|
| Survey Program: 0-MWD                                                                                 |                       |                       |                       |                  |               |                       |                                     |              |                        |                         |                           |                   | Offset Well Error: | 0.0 usft |
| Reference                                                                                             |                       | Offset                |                       | Semi Major Axis  |               |                       | Distance                            |              |                        |                         |                           |                   |                    | Warning  |
| Measured Depth (usft)                                                                                 | Vertical Depth (usft) | Measured Depth (usft) | Vertical Depth (usft) | Reference (usft) | Offset (usft) | Highside Toolface (°) | Offset Wellbore Centre +N/-S (usft) | +E/-W (usft) | Between Centres (usft) | Between Ellipses (usft) | Minimum Separation (usft) | Separation Factor |                    |          |
| 18,100.0                                                                                              | 10,786.5              | 17,932.4              | 10,678.2              | 220.4            | 219.4         | -42.68                | -197.4                              | 7,528.3      | 147.4                  | -162.5                  | 309.87                    | 0.476             | Level 1            |          |
| 18,200.0                                                                                              | 10,787.2              | 18,032.4              | 10,678.9              | 223.3            | 222.3         | -42.68                | -197.4                              | 7,628.3      | 147.4                  | -166.6                  | 313.97                    | 0.469             | Level 1            |          |
| 18,300.0                                                                                              | 10,787.9              | 18,132.4              | 10,679.6              | 226.2            | 225.3         | -42.68                | -197.4                              | 7,728.3      | 147.4                  | -170.7                  | 318.07                    | 0.463             | Level 1            |          |
| 18,400.0                                                                                              | 10,788.6              | 18,232.4              | 10,680.3              | 229.2            | 228.2         | -42.68                | -197.4                              | 7,828.3      | 147.4                  | -174.8                  | 322.16                    | 0.457             | Level 1            |          |
| 18,500.0                                                                                              | 10,789.3              | 18,332.4              | 10,681.0              | 232.1            | 231.1         | -42.68                | -197.4                              | 7,928.3      | 147.4                  | -178.9                  | 326.26                    | 0.452             | Level 1            |          |
| 18,600.0                                                                                              | 10,790.0              | 18,432.4              | 10,681.7              | 235.0            | 234.0         | -42.68                | -197.4                              | 8,028.3      | 147.4                  | -183.0                  | 330.36                    | 0.446             | Level 1            |          |
| 18,700.0                                                                                              | 10,790.7              | 18,532.4              | 10,682.4              | 237.9            | 236.9         | -42.68                | -197.4                              | 8,128.3      | 147.4                  | -187.1                  | 334.46                    | 0.441             | Level 1            |          |
| 18,800.0                                                                                              | 10,791.4              | 18,632.4              | 10,683.1              | 240.8            | 239.8         | -42.68                | -197.4                              | 8,228.3      | 147.4                  | -191.2                  | 338.56                    | 0.435             | Level 1            |          |
| 18,900.0                                                                                              | 10,792.1              | 18,732.4              | 10,683.8              | 243.7            | 242.7         | -42.68                | -197.4                              | 8,328.3      | 147.4                  | -195.3                  | 342.67                    | 0.430             | Level 1            |          |
| 19,000.0                                                                                              | 10,792.8              | 18,832.4              | 10,684.5              | 246.6            | 245.7         | -42.68                | -197.4                              | 8,428.3      | 147.4                  | -199.4                  | 346.77                    | 0.425             | Level 1            |          |
| 19,100.0                                                                                              | 10,793.5              | 18,932.4              | 10,685.2              | 249.6            | 248.6         | -42.68                | -197.4                              | 8,528.3      | 147.4                  | -203.5                  | 350.87                    | 0.420             | Level 1            |          |
| 19,200.0                                                                                              | 10,794.2              | 19,032.4              | 10,685.9              | 252.5            | 251.5         | -42.68                | -197.4                              | 8,628.3      | 147.4                  | -207.6                  | 354.98                    | 0.415             | Level 1            |          |
| 19,300.0                                                                                              | 10,794.9              | 19,132.4              | 10,686.6              | 255.4            | 254.4         | -42.68                | -197.4                              | 8,728.3      | 147.4                  | -211.7                  | 359.08                    | 0.410             | Level 1            |          |
| 19,400.0                                                                                              | 10,795.6              | 19,232.4              | 10,687.3              | 258.3            | 257.3         | -42.68                | -197.4                              | 8,828.3      | 147.4                  | -215.8                  | 363.18                    | 0.406             | Level 1            |          |
| 19,500.0                                                                                              | 10,796.3              | 19,332.4              | 10,688.0              | 261.2            | 260.2         | -42.68                | -197.4                              | 8,928.3      | 147.4                  | -219.9                  | 367.29                    | 0.401             | Level 1            |          |
| 19,600.0                                                                                              | 10,797.0              | 19,432.4              | 10,688.7              | 264.1            | 263.2         | -42.68                | -197.4                              | 9,028.3      | 147.4                  | -224.0                  | 371.40                    | 0.397             | Level 1            |          |
| 19,700.0                                                                                              | 10,797.7              | 19,532.4              | 10,689.3              | 267.1            | 266.1         | -42.68                | -197.4                              | 9,128.3      | 147.4                  | -228.1                  | 375.50                    | 0.393             | Level 1            |          |
| 19,800.0                                                                                              | 10,798.4              | 19,632.4              | 10,690.0              | 270.0            | 269.0         | -42.68                | -197.4                              | 9,228.3      | 147.4                  | -232.2                  | 379.61                    | 0.388             | Level 1            |          |
| 19,900.0                                                                                              | 10,799.1              | 19,732.4              | 10,690.7              | 272.9            | 271.9         | -42.68                | -197.4                              | 9,328.3      | 147.4                  | -236.3                  | 383.72                    | 0.384             | Level 1            |          |
| 20,000.0                                                                                              | 10,799.8              | 19,832.4              | 10,691.4              | 275.8            | 274.8         | -42.68                | -197.4                              | 9,428.3      | 147.4                  | -240.4                  | 387.82                    | 0.380             | Level 1            |          |
| 20,100.0                                                                                              | 10,800.5              | 19,932.4              | 10,692.1              | 278.7            | 277.8         | -42.68                | -197.4                              | 9,528.3      | 147.4                  | -244.5                  | 391.93                    | 0.376             | Level 1            |          |
| 20,200.0                                                                                              | 10,801.2              | 20,032.4              | 10,692.8              | 281.6            | 280.7         | -42.68                | -197.4                              | 9,628.3      | 147.4                  | -248.7                  | 396.04                    | 0.372             | Level 1            |          |
| 20,300.0                                                                                              | 10,801.9              | 20,132.4              | 10,693.5              | 284.6            | 283.6         | -42.68                | -197.4                              | 9,728.3      | 147.4                  | -252.8                  | 400.15                    | 0.368             | Level 1            |          |
| 20,400.0                                                                                              | 10,802.6              | 20,232.4              | 10,694.2              | 287.5            | 286.5         | -42.68                | -197.4                              | 9,828.3      | 147.4                  | -256.9                  | 404.26                    | 0.365             | Level 1            |          |
| 20,500.0                                                                                              | 10,803.3              | 20,332.4              | 10,694.9              | 290.4            | 289.4         | -42.68                | -197.4                              | 9,928.3      | 147.4                  | -261.0                  | 408.37                    | 0.361             | Level 1            |          |
| 20,600.0                                                                                              | 10,804.0              | 20,432.4              | 10,695.6              | 293.3            | 292.4         | -42.68                | -197.4                              | 10,028.3     | 147.4                  | -265.1                  | 412.48                    | 0.357             | Level 1            |          |
| 20,700.0                                                                                              | 10,804.7              | 20,532.4              | 10,696.3              | 296.2            | 295.3         | -42.68                | -197.4                              | 10,128.3     | 147.4                  | -269.2                  | 416.59                    | 0.354             | Level 1            |          |
| 20,800.0                                                                                              | 10,805.4              | 20,632.4              | 10,697.0              | 299.2            | 298.2         | -42.68                | -197.4                              | 10,228.3     | 147.4                  | -273.3                  | 420.70                    | 0.350             | Level 1            |          |
| 20,900.0                                                                                              | 10,806.1              | 20,732.4              | 10,697.7              | 302.1            | 301.1         | -42.68                | -197.4                              | 10,328.2     | 147.4                  | -277.4                  | 424.81                    | 0.347             | Level 1            |          |
| 21,000.0                                                                                              | 10,806.8              | 20,832.4              | 10,698.4              | 305.0            | 304.0         | -42.68                | -197.4                              | 10,428.2     | 147.4                  | -281.5                  | 428.92                    | 0.344             | Level 1            |          |
| 21,075.8                                                                                              | 10,807.3              | 20,908.2              | 10,699.0              | 307.2            | 306.3         | -42.68                | -197.4                              | 10,504.0     | 147.4                  | -284.6                  | 432.03                    | 0.341             | Level 1            |          |
| 21,106.5                                                                                              | 10,807.5              | 20,938.2              | 10,699.2              | 308.1            | 307.1         | -42.68                | -197.4                              | 10,534.0     | 147.4                  | -285.9                  | 433.28                    | 0.340             | Level 1, ES, SF    |          |

# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

Reference Depths are relative to well @ 1967.0usft

Offset Depths are relative to Offset Datum

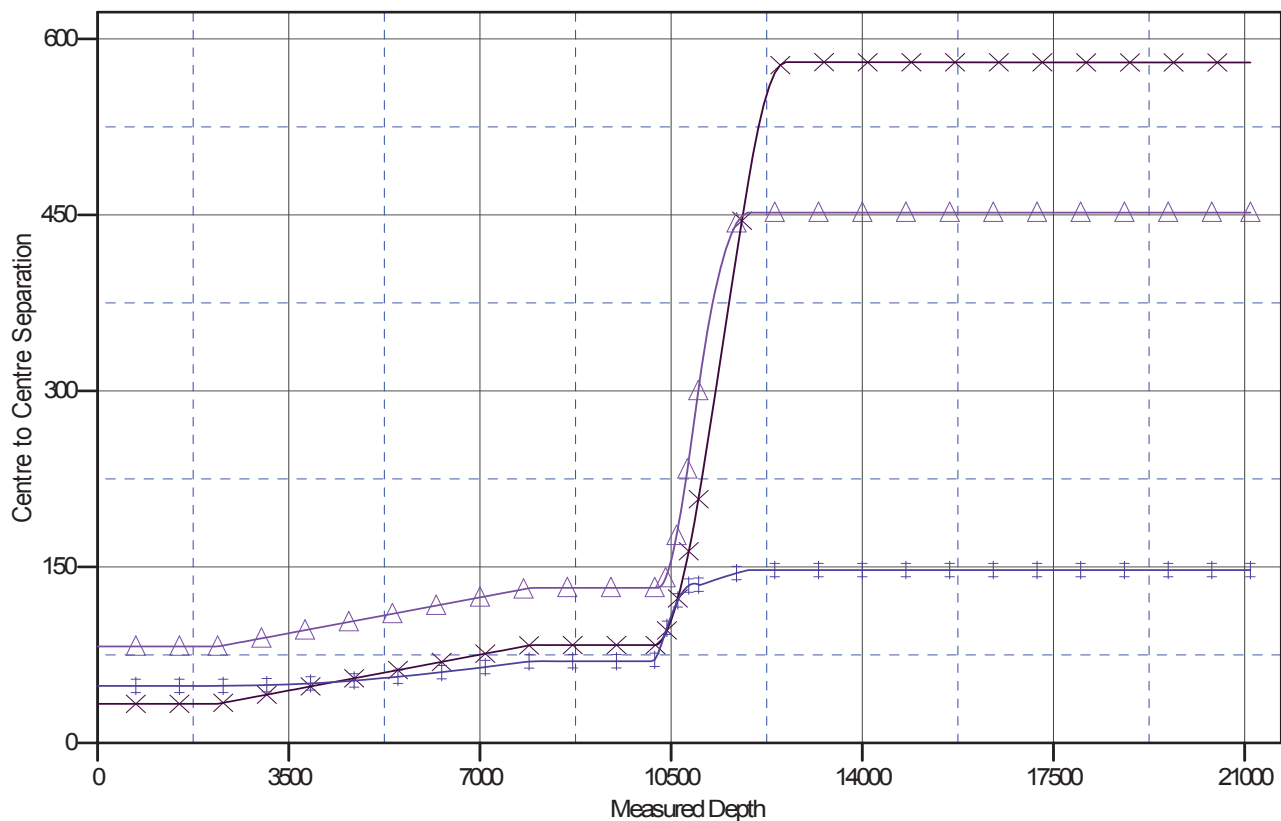
Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Chalmers 5300 44-24 4T2R

Coordinate System is US State Plane 1983, North Dakota Northern Zone

Grid Convergence at Surface is: -2.31°

### Ladder Plot



### LEGEND

Chalmers 5301 44-24 12TXR, Design #1 V0    Chalmers 5301 44-24 2TX, Chalmers #2T, Plan #1 V0    Chalmers 5301 44-24 3BR, Chalmers #3T, Plan #1 V0

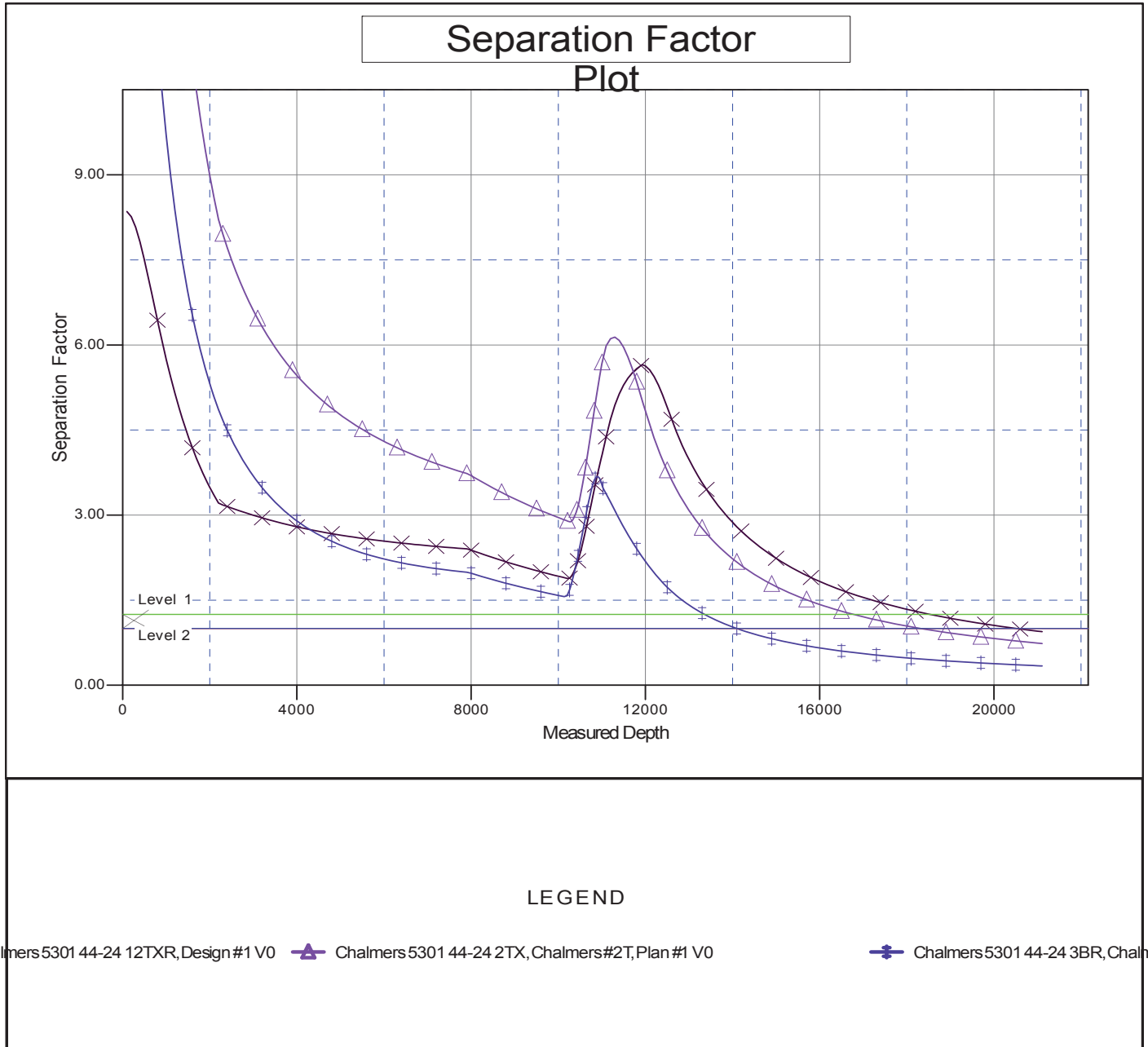
# Ryan Directional Services

## Anticollision Report

|                           |                          |                                     |                               |
|---------------------------|--------------------------|-------------------------------------|-------------------------------|
| <b>Company:</b>           | Oasis                    | <b>Local Co-ordinate Reference:</b> | Well Chalmers 5300 44-24 4T2R |
| <b>Project:</b>           | Indian Hills             | <b>TVD Reference:</b>               | well @ 1967.0usft             |
| <b>Reference Site:</b>    | 153N-100W-19/20_Altered  | <b>MD Reference:</b>                | well @ 1967.0usft             |
| <b>Site Error:</b>        | 0.0 usft                 | <b>North Reference:</b>             | True                          |
| <b>Reference Well:</b>    | Chalmers 5300 44-24 4T2R | <b>Survey Calculation Method:</b>   | Minimum Curvature             |
| <b>Well Error:</b>        | 2.0 usft                 | <b>Output errors are at</b>         | 2.00 sigma                    |
| <b>Reference Wellbore</b> | Chalmers 5300 44-24 4T2R | <b>Database:</b>                    | EDM 5000.1 Single User Db     |
| <b>Reference Design:</b>  | Design #1                | <b>Offset TVD Reference:</b>        | Offset Datum                  |

Reference Depths are relative to well @ 1967.0usft  
 Offset Depths are relative to Offset Datum  
 Central Meridian is 100° 30' 0.000 W

Coordinates are relative to: Chalmers 5300 44-24 4T2R  
 Coordinate System is US State Plane 1983, North Dakota Northern Zone  
 Grid Convergence at Surface is: -2.31°



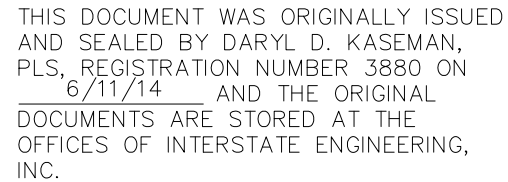


# WELL LOCATION PLAT

"CHALMERS 5301 44-24 3BR"

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE

SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



CHALMERS 5301 44-24 3BR

GROUND ELEV. 1970.0'  
LATITUDE 48°03'20.49"N  
LONGITUDE 103°36'18.55"W  
GPS SURVEY DATUM: NAD 83

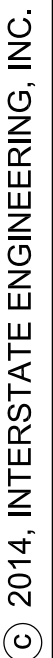
BOTTOM HOLE LOCATION

LATITUDE 48°03'17.16"N  
LONGITUDE 103°33'43.17"W  
GPS SURVEY DATUM: NAD 83

VICINITY MAP

THIS SURVEY AND PLAT IS BEING PROVIDED AT THE REQUEST OF ERIC BAYES OF OASIS PETROLEUM. I CERTIFY THAT THIS PLAT CORRECTLY REPRESENTS WORK PERFORMED BY ME OR UNDER MY SUPERVISION AND IS TRUE AND CORRECT TO THE BEST OF MY KNOWLEDGE AND BELIEF.

REGISTERED LAND SURVEYOR  
DARYL D. KASEMAN  
LS-3880  
DATE: 6/11/14  
NORTH DAKOTA



Drawn By: J.J.S. Project No.: 513-09-235-01  
 Checked By: D.D.K. Date: MAY 2014

**Interstate Engineering, Inc.**  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
[www.interstateeng.com](http://www.interstateeng.com)

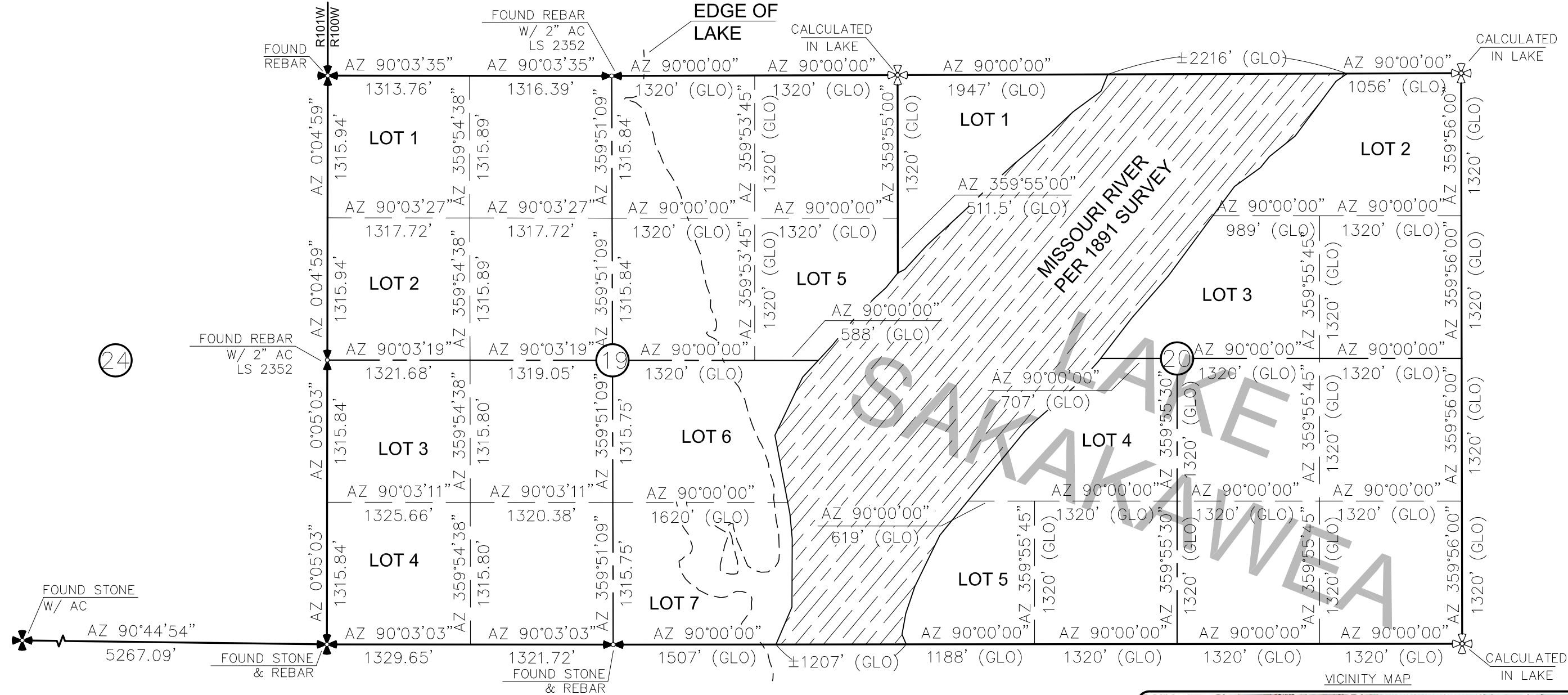
Other offices in Minnesota, North Dakota and South Dakota

Professionals you need, people you trust

1/8  
SHEET NO.



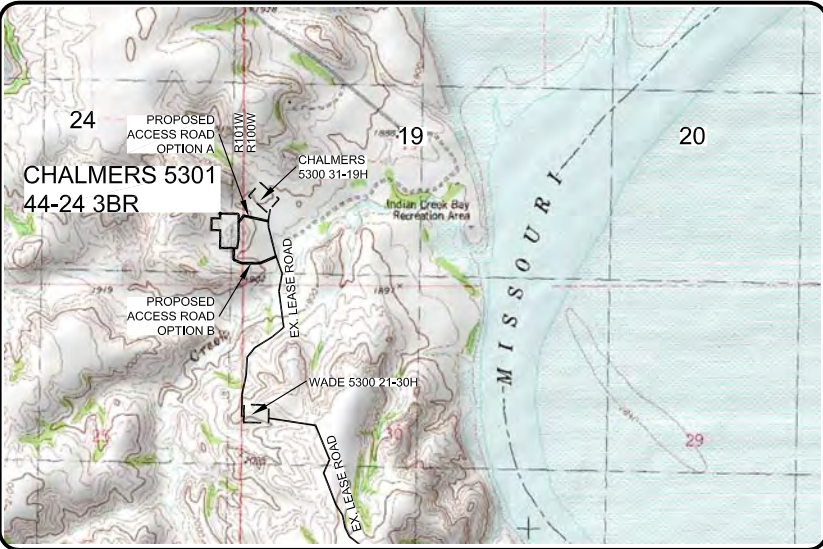
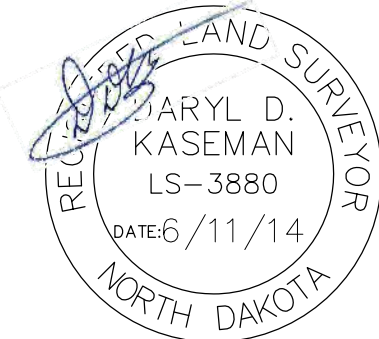
SECTION BREAKDOWN  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"CHALMERS 5301 44-24 3BR"  
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, & SECTION 19 & 20, T153N, R100W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- MONUMENT - RECOVERED
- MONUMENT - NOT RECOVERED

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 6/11/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

ALL AZIMUTHS ARE BASED ON G.P.S. OBSERVATIONS. THE ORIGINAL SURVEY OF THIS AREA FOR THE GENERAL LAND OFFICE (G.L.O.) WAS 1891. THE CORNERS FOUND ARE AS INDICATED AND ALL OTHERS ARE COMPUTED FROM THOSE CORNERS FOUND AND BASED ON G.L.O. DATA. THE MAPPING ANGLE FOR THIS AREA IS APPROXIMATELY -0°03'.

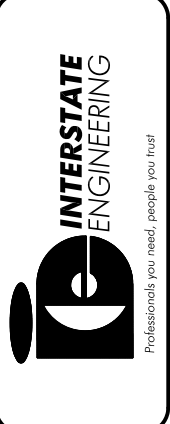


© 2014, INTERSTATE ENGINEERING, INC.

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 3        | 1/20/14 | BHH | CHANGED PAD LAYOUT       |
| REV 4        | 5/09/14 | JJS | ADDED WELL TO PAD        |
| REV 5        | 5/20/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 6        | 6/11/14 | JDM | ADDED DIMENSIONS         |

|                                                                                                                                                        |                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| OASIS PETROLEUM NORTH AMERICA, LLC<br>SECTION BREAKDOWN<br>SECTION 24, T153N, R101W, & SECTIONS 19 & 20, T153N, R100W<br>MCKENZIE COUNTY, NORTH DAKOTA | Project No.: S13-09-23501<br>Date: MAY 2014 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|

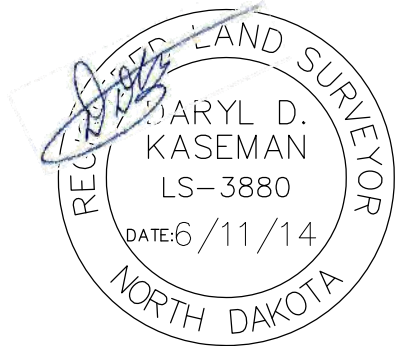
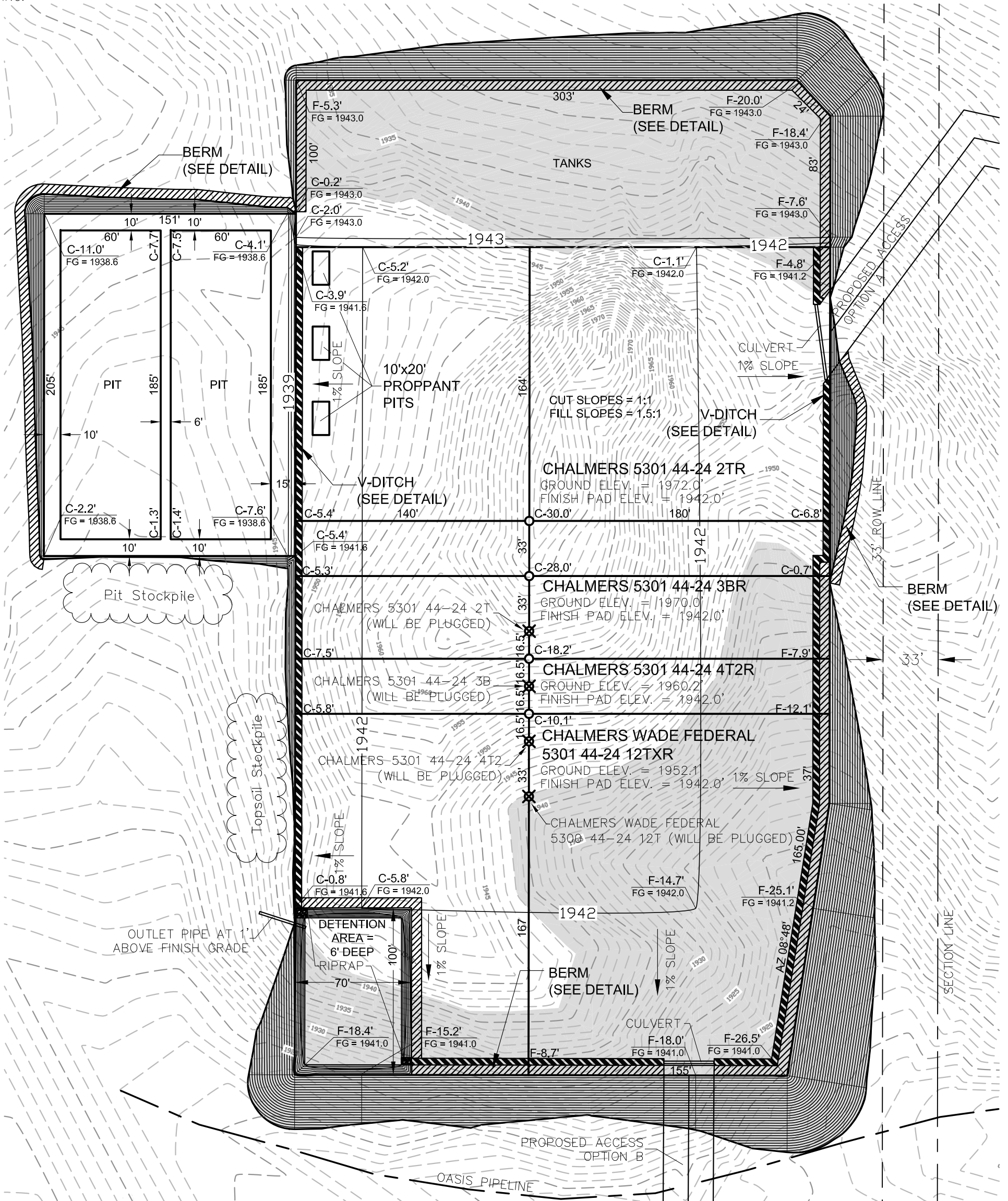
Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota





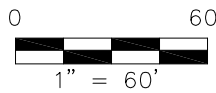
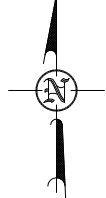
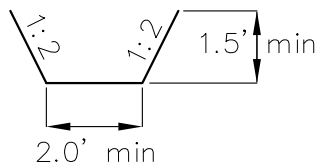
THIS DOCUMENT WAS ORIGINALLY ISSUED  
AND SEALED BY DARYL D. KASEMAN,  
PLS, REGISTRATION NUMBER 3880 ON  
6/11/14 AND THE ORIGINAL  
DOCUMENTS ARE STORED AT THE  
OFFICES OF INTERSTATE ENGINEERING,  
INC.

**PAD LAYOUT**  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"CHALMERS 5301 44-24 3BR"  
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



- BERM
- DITCH
- Proposed Contours
- Original Contours

**V-DITCH DETAIL**



NOTE: All utilities shown are preliminary only, a complete  
utilities location is recommended before construction.

© 2014, INTERSTATE ENGINEERING, INC.

3/8

SHEET NO.



Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
PAD LAYOUT  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.01  
Checked By: D.D.K. Date: MAY 2014

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 3        | 1/20/14 | BHH | CHANGED PAD LAYOUT       |
| REV 4        | 5/09/14 | JJS | ADDED WELL TO PAD        |
| REV 5        | 5/20/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 6        | 6/11/14 | JDM | ADDED DIMENSIONS         |

© 2013 S13-09-235.01 Oasis Petroleum - 2 of 3 Well Wells for Chalmers 5301 44-24 3BR  
31-19H/CAD/CHALMERS 3BR UPDATED 6-11-14.dwg - 6/11/2014 9:23 AM Jackie Miller

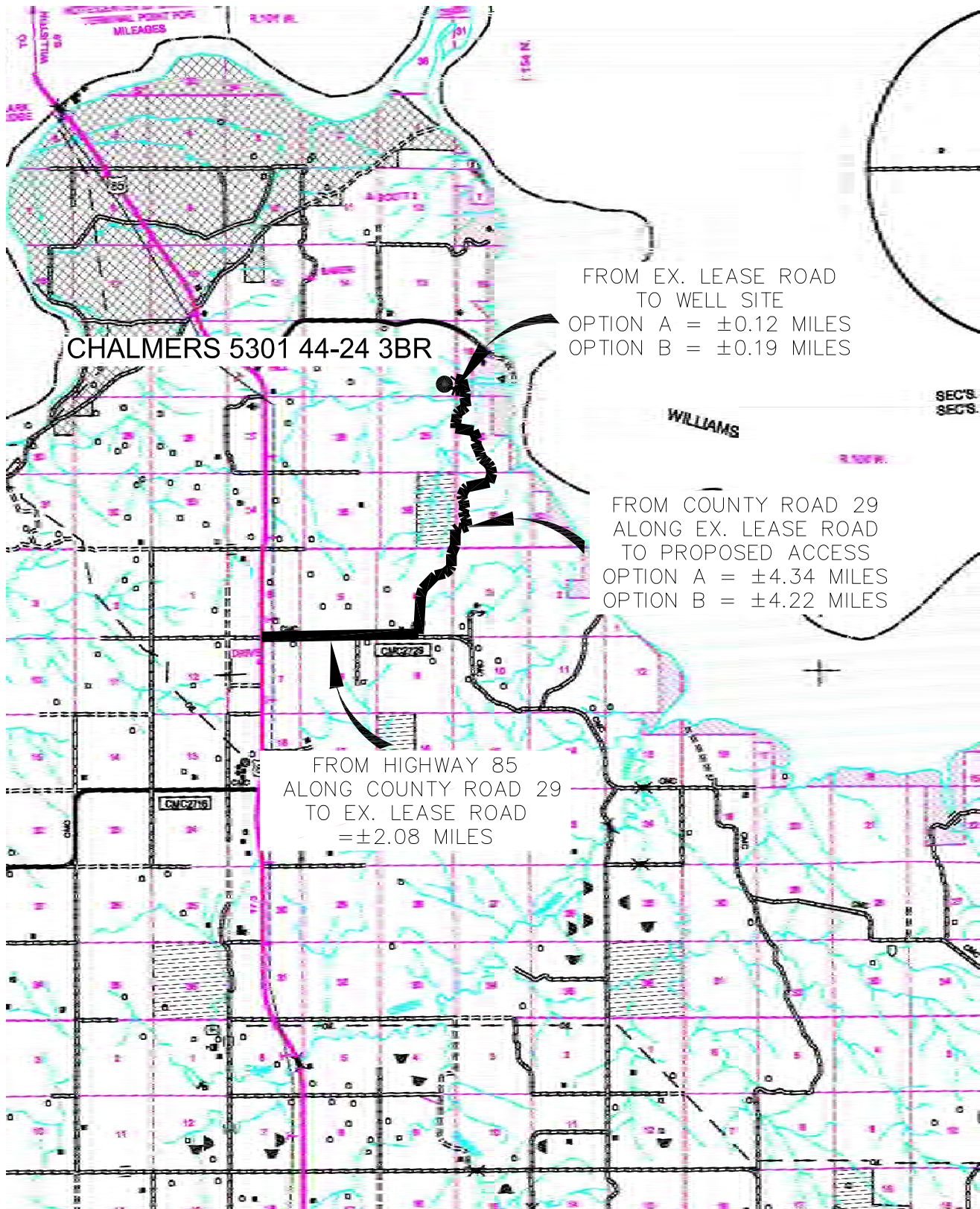


# COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5301 44-24 3BR"

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



CHALMERS 5301 44-24 3BR

FROM EX. LEASE ROAD  
TO WELL SITE  
OPTION A =  $\pm 0.12$  MILES  
OPTION B =  $\pm 0.19$  MILES

FROM COUNTY ROAD 29  
ALONG EX. LEASE ROAD  
TO PROPOSED ACCESS  
OPTION A =  $\pm 4.34$  MILES  
OPTION B =  $\pm 4.22$  MILES

FROM HIGHWAY 85  
ALONG COUNTY ROAD 29  
TO EX. LEASE ROAD  
=  $\pm 2.08$  MILES

**WELL LOCATION SITE QUANTITIES**  
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"CHALMERS 5301 44-24 3BR"  
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

|                                 |                                   |
|---------------------------------|-----------------------------------|
| WELL SITE ELEVATION             | 1970.0                            |
| WELL PAD ELEVATION              | 1942.0                            |
| EXCAVATION PLUS PIT             | 55,136<br><u>9,450</u><br>64,586  |
| EMBANKMENT PLUS SHRINKAGE (30%) | 37,558<br><u>11,267</u><br>48,825 |
| STOCKPILE PIT                   | 9,450                             |
| STOCKPILE TOP SOIL (6")         | 4,770                             |
| BERMS                           | 1,711 LF = 554 CY                 |
| DITCHES                         | 1,029 LF = 157 CY                 |
| DETENTION AREA                  | 1,428 CY                          |
| STOCKPILE MATERIAL              | 2,572                             |
| DISTURBED AREA FROM PAD         | 5.91 ACRES                        |

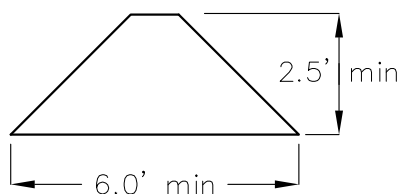
NOTE: ALL QUANTITIES ARE IN CUBIC YARDS (UNLESS NOTED)  
CUT END SLOPES AT 1:1  
FILL END SLOPES AT 1.5:1

WELL SITE LOCATION

926' FSL

245' FEL

BERM DETAIL



DITCH DETAIL



© 2014, INTERSTATE ENGINEERING, INC.

**8/8**



Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.Interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
QUANTITIES  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.01  
Checked By: D.D.K. Date: MAY 2014

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 2        | 12/5/13 | JJS | CHANGED PAD LAYOUT       |
| REV 3        | 1/20/14 | BH  | CHANGED PAD LAYOUT       |
| REV 4        | 5/09/14 | JJS | ADDED WELL TO PAD        |
| REV 5        | 5/20/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 6        | 6/11/14 | JDM | ADDED DIMENSIONS         |

# CROSS SECTIONS

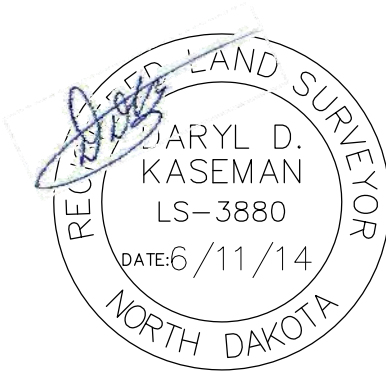
OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5301 44-24 3BR"

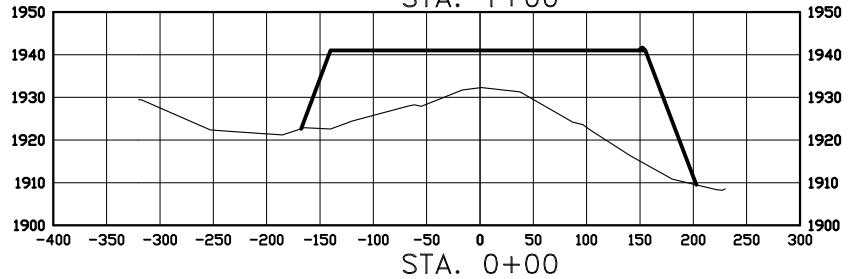
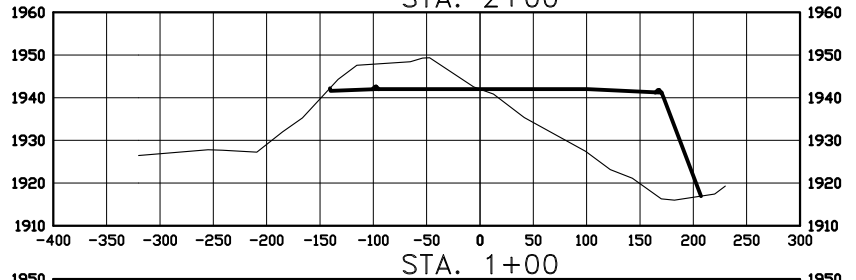
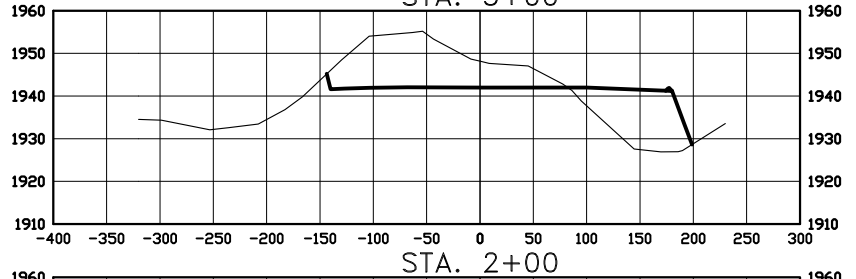
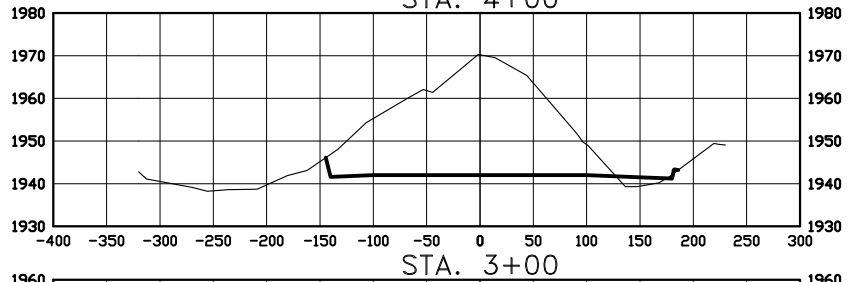
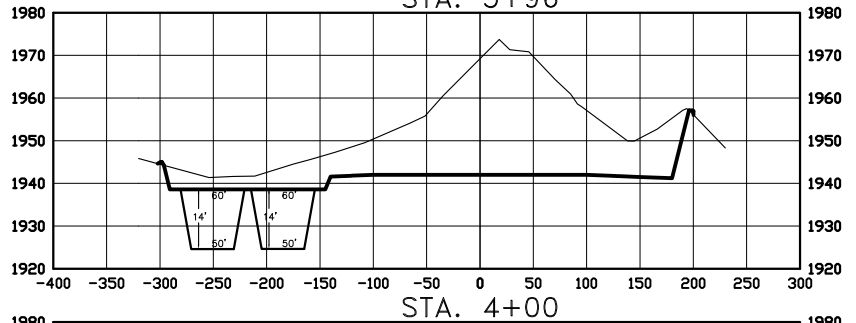
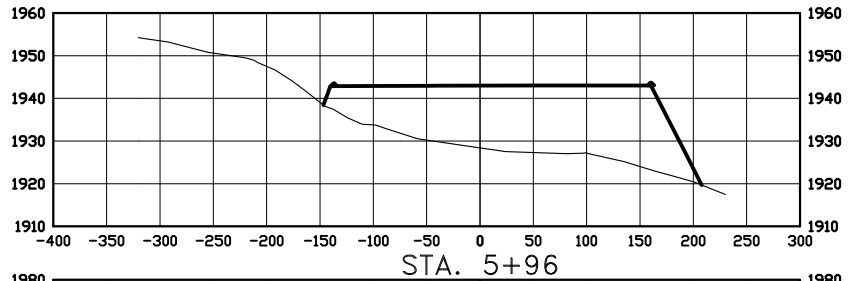
926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE

SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

THIS DOCUMENT WAS ORIGINALLY ISSUED  
AND SEALED BY DARYL D. KASEMAN,  
PLS, REGISTRATION NUMBER 3880 ON  
6/11/14 AND THE ORIGINAL  
DOCUMENTS ARE STORED AT THE  
OFFICES OF INTERSTATE ENGINEERING,  
INC.



SCALE  
HORIZ 1"=180'  
VERT 1"=45'



© 2014, INTERSTATE ENGINEERING, INC.

7/8  
SHEET NO.



Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
PAD CROSS SECTIONS  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235,01  
Checked By: D.D.K. Date: MAY 2014

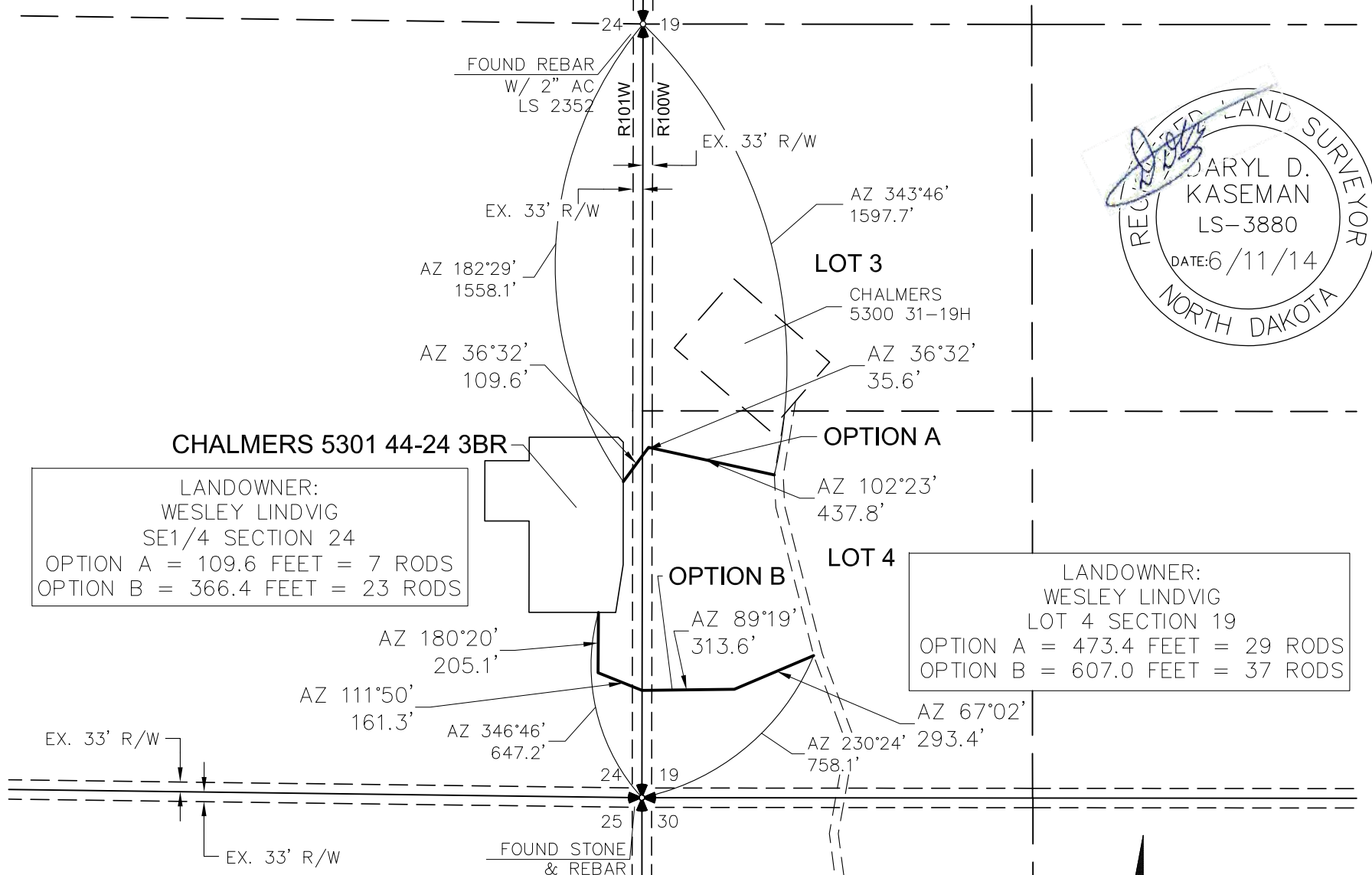
| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 2        | 12/5/13 | JJS | CHANGED PAD LAYOUT       |
| REV 3        | 1/20/14 | BHR | CHANGED PAD LAYOUT       |
| REV 4        | 5/09/14 | JJS | ADDED WELL TO PAD        |
| REV 5        | 5/20/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 6        | 6/11/14 | JDM | ADDED DIMENSIONS         |

# ACCESS APPROACH

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002

"CHALMERS 5301 44-24 3BR"

926 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA

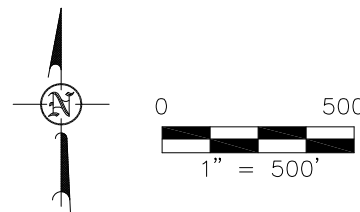


LANDOWNER:  
WESLEY LINDVIG  
SE1/4 SECTION 24  
OPTION A = 109.6 FEET = 7 RODS  
OPTION B = 366.4 FEET = 23 RODS

LANDOWNER:  
WESLEY LINDVIG  
LOT 4 SECTION 19  
OPTION A = 473.4 FEET = 29 RODS  
OPTION B = 607.0 FEET = 37 RODS

THIS DOCUMENT WAS ORIGINALLY ISSUED AND SEALED BY DARYL D. KASEMAN, PLS, REGISTRATION NUMBER 3880 ON 6/11/14 AND THE ORIGINAL DOCUMENTS ARE STORED AT THE OFFICES OF INTERSTATE ENGINEERING, INC.

NOTE: All utilities shown are preliminary only, a complete utilities location is recommended before construction.



| Revision No. | By  | Date    | Description              |
|--------------|-----|---------|--------------------------|
| REV 1        | JLS | 12/5/13 | CHANGED PAD LAYOUT       |
| REV 2        | BH  | 1/20/14 | CHANGED PAD LAYOUT       |
| REV 3        | JLS | 1/20/14 | CHANGED PAD LAYOUT       |
| REV 4        | JLS | 5/29/14 | CHANGED WELL TO PAD      |
| REV 5        | JLS | 5/29/14 | CHANGED NAME, MOVED WELL |
| REV 6        | JLS | 6/11/14 | ADDED DIMENSIONS         |

OASIS PETROLEUM NORTH AMERICA, LLC  
ACCESS APPROACH  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA  
Drawn By: JLS  
Checked By: DDK  
Project No: S13-09-235.01  
Date: MAY 2014

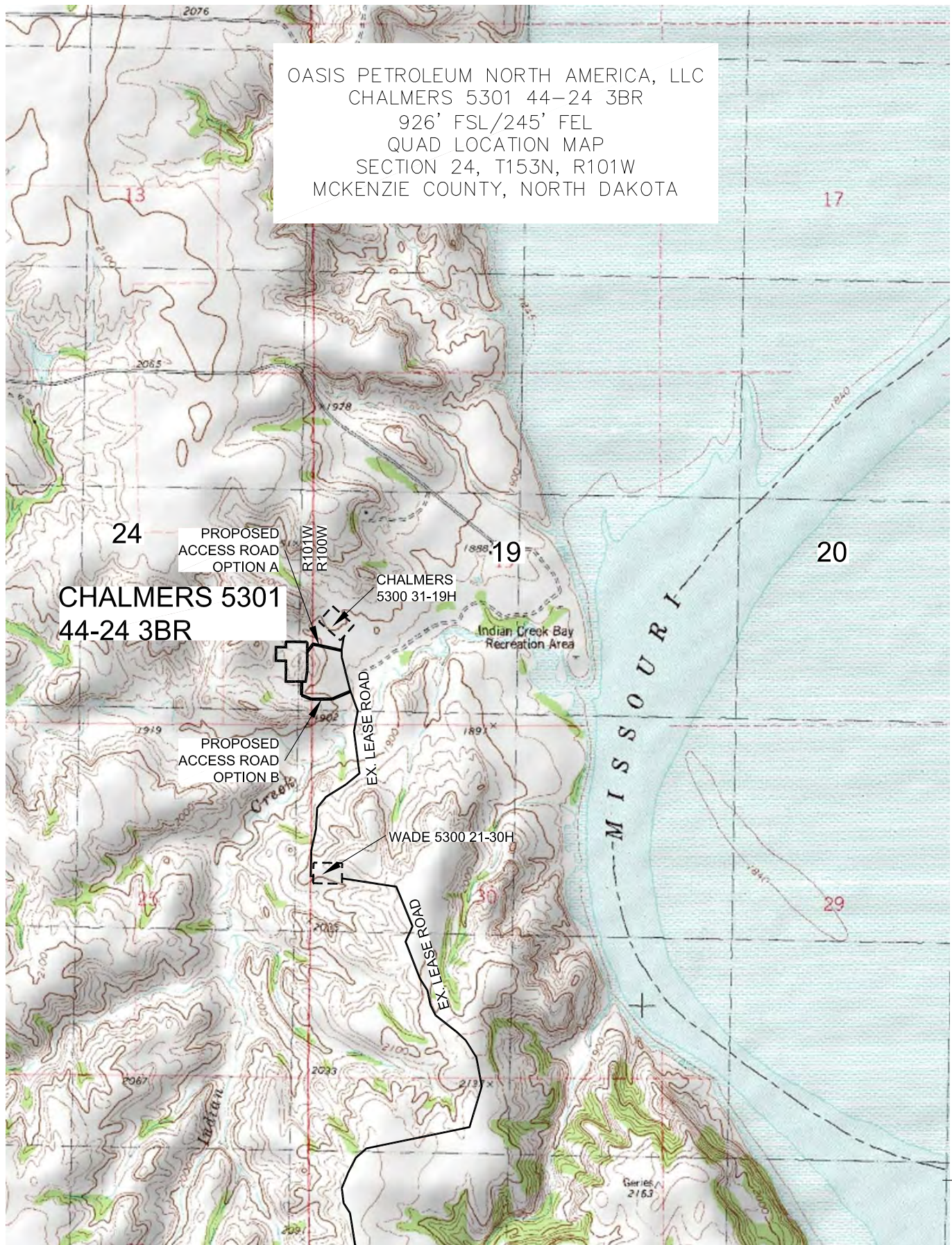
Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota



© 2014, INTERSTATE ENGINEERING, INC.  
4/8  
SHEET NO.



OASIS PETROLEUM NORTH AMERICA, LLC  
 CHALMERS 5301 44-24 3BR  
 926' FSL/245' FEL  
 QUAD LOCATION MAP  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

5/8

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.  
 P.O. Box 648  
 425 East Main Street  
 Sidney, Montana 59270  
 Ph (406) 433-5617  
 Fax (406) 433-5618  
 www.interstateeng.com

Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
 QUAD LOCATION MAP  
 SECTION 24, T153N, R101W  
 MCKENZIE COUNTY, NORTH DAKOTA

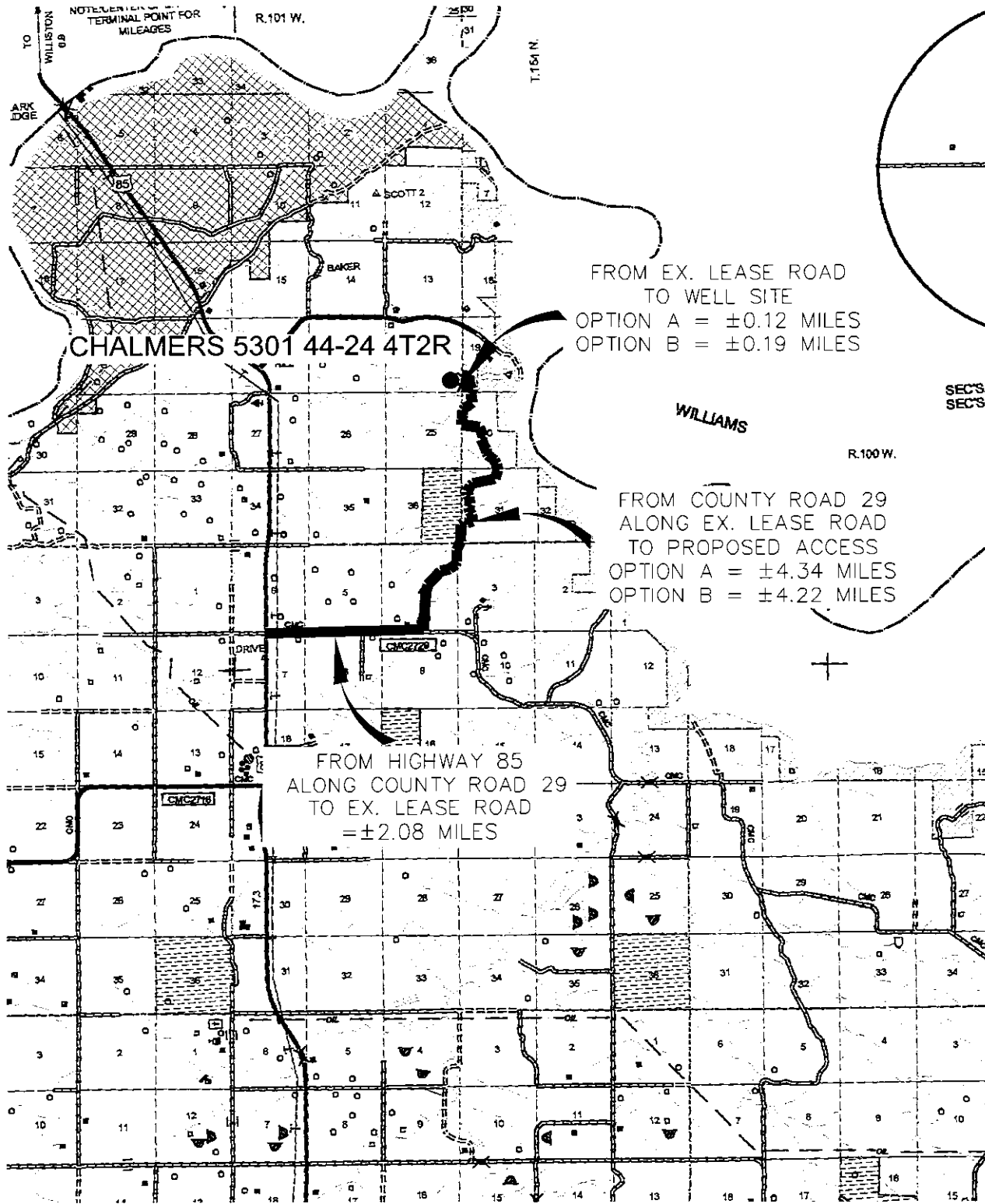
Drawn By: J.J.S. Project No.: S13-09-235.01  
 Checked By: D.D.K. Date: MAY 2014

| Revision No. | Date    | By  | Description              |
|--------------|---------|-----|--------------------------|
| REV 2        | 12/5/13 | JJS | CHANGED PAD LAYOUT       |
| REV 3        | 1/20/14 | BHJ | CHANGED PAD LAYOUT       |
| REV 4        | 5/09/14 | JJS | ADDED WELL TO PAD        |
| REV 5        | 5/20/14 | JJS | CHANGED NAME, MOVED WELL |
| REV 6        | 6/11/14 | JDM | ADDED DIMENSIONS         |

# COUNTY ROAD MAP

OASIS PETROLEUM NORTH AMERICA, LLC  
1001 FANNIN, SUITE 1500, HOUSTON, TX 77002  
"CHALMERS 5301 44-24 4T2R"

877 FEET FROM SOUTH LINE AND 245 FEET FROM EAST LINE  
SECTION 24, T153N, R101W, 5th P.M., MCKENZIE COUNTY, NORTH DAKOTA



© 2014, INTERSTATE ENGINEERING, INC.

SCALE: 1" = 2 MILE

6/8

SHEET NO.



Professionals you need, people you trust

Interstate Engineering, Inc.  
P.O. Box 648  
425 East Main Street  
Sidney, Montana 59270  
Ph (406) 433-5617  
Fax (406) 433-5618  
www.interstateeng.com  
Other offices in Minnesota, North Dakota and South Dakota

OASIS PETROLEUM NORTH AMERICA, LLC  
COUNTY ROAD MAP  
SECTION 24, T153N, R101W  
MCKENZIE COUNTY, NORTH DAKOTA

Drawn By: J.J.S. Project No.: S13-09-235.02  
Checked By: D.D.K. Date: MAY 2014

| Revision No. | Date     | By  | Description              |
|--------------|----------|-----|--------------------------|
| REV 1        | 10/01/13 | JJS | CHANGED WELL NAME        |
| REV 2        | 12/2/13  | JJS | CHANGED PAD LAYOUT       |
| REV 3        | 1/29/14  | GHV | CHANGED PAD LAYOUT       |
| REV 4        | 5/06/14  | JJS | ADDED WELL TO PAD        |
| REV 5        | 3/26/14  | JJS | CHANGED NAME, MOVED WELL |





6/1/2014

Mineral Resources Permit Manager  
North Dakota Industrial Commission  
600 East Boulevard Avenue Dept. 405  
Bismarck, ND 58505-0840

RE: Chalmers 5301 44-24 3BR  
Chalmers 5301 44-24 4T2R  
Request for a legal street address

Dear NDIC:

Oasis Petroleum has requested a physical street address for the Chalmers 5301 44-24 3BR and Chalmers 5301 44-24 4T2R. The request was made to Aaron Chisolm, in McKenzie County. Upon receiving a legal street address, Oasis will submit the address to the NDIC on a Sundry Notice (form 4) pursuant to 43-02-03-28.

Thank you for your consideration.

Respectfully,

A handwritten signature in blue ink, appearing to read "Heather McCowan".

Heather McCowan  
Regulatory Assistant  
Oasis Petroleum North America, LLC